

## JESS Thermodynamic Database v8.9

## Chemical species in reactions with Phosphorus

24-Jan-23

| Charge                                                                                                                      | CAS        | Count | Molecular formula      | Mol. mass |
|-----------------------------------------------------------------------------------------------------------------------------|------------|-------|------------------------|-----------|
| <b>[30]N10</b><br>1,4,7,10,13,16,19,22,25,28-Decaazacyclotriacontane;<br>Decaazacyclotriacontane:1,4,7,10,13,16,19,22,25,28 |            |       |                        |           |
| 0                                                                                                                           | 862-28-2   | 82    | C(20)H(50)N(10)        | 430.684   |
| <b>16HexDiAm</b><br>1,6-Hexanediamine; 1,6-Diaminohexane; Hexamethylenediamine                                              |            |       |                        |           |
| 0                                                                                                                           | 124-09-4   | 15    | C(6)H(16)N(2)          | 116.206   |
| <b>18OctDiAm</b><br>1,8-Octanediamine; 1,8-Diaminooctane; 1,10-Diazadecane                                                  |            |       |                        |           |
| 0                                                                                                                           | 373-44-4   | 16    | C(8)H(20)N(2)          | 144.260   |
| <b>5AMP-2</b><br>Adenosine-5'-monophosphate ion                                                                             |            |       |                        |           |
| -2                                                                                                                          | 6042-43-9  | 105   | C(10)H(12)N(5)O(7)P(1) | 345.209   |
| <b>5IMP-2</b><br>Inosine-5'-monophosphate; Inosinic acid; IMP; I-5'-P                                                       |            |       |                        |           |
| -2                                                                                                                          | 131-99-7   | 34    | C(10)H(11)N(4)O(8)P(1) | 346.193   |
| <b>5NitrSal-2</b><br>5-Nitrosalicylate ion                                                                                  |            |       |                        |           |
| -2                                                                                                                          | ---        | 45    | C(7)H(3)N(1)O(5)       | 181.105   |
| <b>Ac+3</b><br>Actinium(III) ion                                                                                            |            |       |                        |           |
| 3                                                                                                                           | 22541-37-3 | 37    | Ac(1)                  | 227.028   |
| <b>Ac+3_H+1(2)_PO4-3</b>                                                                                                    |            |       |                        |           |
| 2                                                                                                                           | ---        | 2     | Ac(1)H(2)O(4)P(1)      | 324.016   |
| <b>Ag+1</b><br>Silver(I) ion                                                                                                |            |       |                        |           |

|                                                                     |            |      |                                            |         |
|---------------------------------------------------------------------|------------|------|--------------------------------------------|---------|
| 1                                                                   | 14701-21-4 | 1534 | Ag (1)                                     | 107.870 |
| <b>Ag+1_H+1_PO4-3</b>                                               |            |      |                                            |         |
| -1                                                                  | ---        | 2    | Ag (1) H (1) O (4) P (1)                   | 203.850 |
| <b>Ag+1_P2O7-4</b>                                                  |            |      |                                            |         |
| -3                                                                  | ---        | 1    | Ag (1) O (7) P (2)                         | 281.814 |
| <b>Ag+1 (3)_PO4-3_(s)</b><br>Silver(I) phosphate                    |            |      |                                            |         |
| 0                                                                   | 7784-09-0  | 5    | Ag (3) O (4) P (1)                         | 418.582 |
| <b>Ag+1 (4)_P2O7-4_(s)</b>                                          |            |      |                                            |         |
| 0                                                                   | ---        | 1    | Ag (4) O (7) P (2)                         | 605.424 |
| <b>Ag (s)</b><br>Silver; Silver, cubic                              |            |      |                                            |         |
| 0                                                                   | 7440-22-4  | 182  | Ag (1)                                     | 107.870 |
| <b>Ag6Sn4P12Ge6 (s)</b><br>Silver tin germanium phosphorous amalgam |            |      |                                            |         |
| 0                                                                   | ---        | 1    | Ag (6) Ge (6) P (12) Sn (4)                | 1929.53 |
| <b>Al+3</b><br>Aluminium(III) ion; Aluminum(III) ion                |            |      |                                            |         |
| 3                                                                   | 22537-23-1 | 1518 | Al (1)                                     | 26.9820 |
| <b>Al+3_Cys-2_PO4-3</b>                                             |            |      |                                            |         |
| -2                                                                  | ---        | 1    | C (3) H (5) Al (1) N (1) O (6) P (1) S (1) | 241.092 |
| <b>Al+3_H+1_Citric-3_PO4-3</b>                                      |            |      |                                            |         |
| -2                                                                  | ---        | 1    | C (6) H (6) Al (1) O (11) P (1)            | 312.063 |
| <b>Al+3_H+1_Cys-2_PO4-3</b>                                         |            |      |                                            |         |
| -1                                                                  | ---        | 1    | C (3) H (6) Al (1) N (1) O (6) P (1) S (1) | 242.100 |
| <b>Al+3_H+1_NTA-3_PO4-3</b>                                         |            |      |                                            |         |
| -2                                                                  | ---        | 1    | C (6) H (7) Al (1) N (1) O (10) P (1)      | 311.078 |

|                               |     |   |                           |         |
|-------------------------------|-----|---|---------------------------|---------|
| <b>Al+3_H+1_P2O7-4</b>        |     |   |                           |         |
| 0                             | --- | 1 | Al (1) H (1) O (7) P (2)  | 201.934 |
| <b>Al+3_H+1_P2O7-4 (2)</b>    |     |   |                           |         |
| -4                            | --- | 1 | Al (1) H (1) O (14) P (4) | 375.878 |
| <b>Al+3_H+1_P3O10-5</b>       |     |   |                           |         |
| -1                            | --- | 1 | Al (1) H (1) O (10) P (3) | 280.906 |
| <b>Al+3_H+1_P6O18-6</b>       |     |   |                           |         |
| -2                            | --- | 1 | Al (1) H (1) O (18) P (6) | 501.823 |
| <b>Al+3_H+1_PO4-3</b>         |     |   |                           |         |
| 1                             | --- | 2 | Al (1) H (1) O (4) P (1)  | 122.962 |
| <b>Al+3_H+1 (2)_P3O10-5</b>   |     |   |                           |         |
| 0                             | --- | 1 | Al (1) H (2) O (10) P (3) | 281.914 |
| <b>Al+3_H+1 (2)_PO4-3</b>     |     |   |                           |         |
| 2                             | --- | 2 | Al (1) H (2) O (4) P (1)  | 123.970 |
| <b>Al+3_H+1 (3)_PO4-3 (2)</b> |     |   |                           |         |
| 0                             | --- | 1 | Al (1) H (3) O (8) P (2)  | 219.949 |
| <b>Al+3_OH-1_P2O7-4</b>       |     |   |                           |         |
| -2                            | --- | 1 | Al (1) H (1) O (8) P (2)  | 217.933 |
| <b>Al+3_OH-1_P3O10-5</b>      |     |   |                           |         |
| -3                            | --- | 1 | Al (1) H (1) O (11) P (3) | 296.905 |
| <b>Al+3_OH-1_P6O18-6</b>      |     |   |                           |         |
| -4                            | --- | 1 | Al (1) H (1) O (19) P (6) | 517.823 |
| <b>Al+3_OH-1_PO4-3</b>        |     |   |                           |         |
| -1                            | --- | 2 | Al (1) H (1) O (5) P (1)  | 138.961 |

|                                                                       |            |   |                                       |         |
|-----------------------------------------------------------------------|------------|---|---------------------------------------|---------|
| <b>Al+3_P2O7-4</b>                                                    |            |   |                                       |         |
| -1                                                                    | ---        | 1 | Al (1) O (7) P (2)                    | 200.926 |
| <b>Al+3_P2O7-4 (2)</b>                                                |            |   |                                       |         |
| -5                                                                    | ---        | 1 | Al (1) O (14) P (4)                   | 374.870 |
| <b>Al+3_P3O10-5</b>                                                   |            |   |                                       |         |
| -2                                                                    | ---        | 1 | Al (1) O (10) P (3)                   | 279.898 |
| <b>Al+3_P3O10-5 (2)</b>                                               |            |   |                                       |         |
| -7                                                                    | ---        | 1 | Al (1) O (20) P (6)                   | 532.814 |
| <b>Al+3_P6O18-6</b>                                                   |            |   |                                       |         |
| -3                                                                    | ---        | 1 | Al (1) O (18) P (6)                   | 500.815 |
| <b>Al+3_PO4-3</b>                                                     |            |   |                                       |         |
| 0                                                                     | ---        | 2 | Al (1) O (4) P (1)                    | 121.954 |
| <b>Al+3_PO4-3 (s)</b><br>Berlinite; Aluminium phosphate, rhombohedral |            |   |                                       |         |
| 0                                                                     | 13769-32-9 | 2 | Al (1) O (4) P (1)                    | 121.954 |
| <b>Al+3_PO4-3_H2O(2) (s)</b><br>Variscite                             |            |   |                                       |         |
| 0                                                                     | 13824-50-5 | 3 | Al (1) H (4) O (6) P (1)              | 157.984 |
| <b>Al+3_Thr-1_P3O10-5</b>                                             |            |   |                                       |         |
| -3                                                                    | ---        | 1 | C (4) H (8) Al (1) N (1) O (13) P (3) | 398.010 |
| <b>Al+3 (2)_H+1_PO4-3</b>                                             |            |   |                                       |         |
| 4                                                                     | ---        | 2 | Al (2) H (1) O (4) P (1)              | 149.944 |
| <b>Al+3 (2)_H+1 (2)_PO4-3 (2)</b>                                     |            |   |                                       |         |
| 2                                                                     | ---        | 1 | Al (2) H (2) O (8) P (2)              | 245.923 |
| <b>Al+3 (2)_OH-1_PO4-3</b>                                            |            |   |                                       |         |
| 2                                                                     | ---        | 2 | Al (2) H (1) O (5) P (1)              | 165.943 |

|                                                                                                                               |           |     |                                        |         |
|-------------------------------------------------------------------------------------------------------------------------------|-----------|-----|----------------------------------------|---------|
| <b>Al+3 (2) _OH-1 (2) _PO4-3</b>                                                                                              |           |     |                                        |         |
| 1                                                                                                                             | ---       | 3   | Al (2) H (2) O (6) P (1)               | 182.950 |
| <b>Al+3 (2) _OH-1 (3) _PO4-3</b>                                                                                              |           |     |                                        |         |
| 0                                                                                                                             | ---       | 2   | Al (2) H (3) O (7) P (1)               | 199.958 |
| <b>Al+3 (2) _PO4-3</b>                                                                                                        |           |     |                                        |         |
| 3                                                                                                                             | ---       | 4   | Al (2) O (4) P (1)                     | 148.936 |
| <b>Al+3 (2) _PO4-3 (2)</b>                                                                                                    |           |     |                                        |         |
| 0                                                                                                                             | ---       | 1   | Al (2) O (8) P (2)                     | 243.907 |
| <b>Al+3 (3) _H+1 (3) _PO4-3 (3)</b>                                                                                           |           |     |                                        |         |
| 3                                                                                                                             | ---       | 1   | Al (3) H (3) O (12) P (3)              | 368.885 |
| <b>Al+3 (3) _Pb+2 _OH-1 (5) _PO4-3 (2) _H2O_ (s)</b><br>Plumbogummite                                                         |           |     |                                        |         |
| 0                                                                                                                             | ---       | 1   | Al (3) H (7) O (14) P (2) Pb (1)       | 581.141 |
| <b>Al+3 (3) _Pb+2 _OH-1 (6) _SO4-2 _PO4-3_ (s)</b><br>Hinsdalite                                                              |           |     |                                        |         |
| 0                                                                                                                             | ---       | 1   | Al (3) H (6) O (14) P (1) Pb (1) S (1) | 581.219 |
| <b>Al+3 (3) _PO4-3 (3)</b>                                                                                                    |           |     |                                        |         |
| 0                                                                                                                             | ---       | 1   | Al (3) O (12) P (3)                    | 365.861 |
| <b>Al+3 (5) _H+1 (6) _K+1 (3) _PO4-3 (8) _H2O (18) _ (s)</b><br>K-Taranakite                                                  |           |     |                                        |         |
| 0                                                                                                                             | ---       | 1   | Al (5) H (42) K (3) O (50) P (8)       | 1342.30 |
| <b>Al+3 (5) _H+1 (9) _NH3 (3) _PO4-3 (8) _H2O (18) _ (s)</b><br>NH4-Taranakite; Ammonium taranakite; Taranakite ammonium salt |           |     |                                        |         |
| 0                                                                                                                             | ---       | 3   | Al (5) H (54) N (3) O (50) P (8)       | 1279.12 |
| <b>Al (s)</b><br>Aluminium; Aluminium, cubic                                                                                  |           |     |                                        |         |
| 0                                                                                                                             | 7429-90-5 | 216 | Al (1)                                 | 26.9820 |

|                                                     |            |     |                     |         |
|-----------------------------------------------------|------------|-----|---------------------|---------|
| <b>Ala-1</b>                                        |            |     |                     |         |
| Alanine ion; L-Alanine ion; L-2-Aminopropanoate ion |            |     |                     |         |
| -1                                                  | ---        | 802 | C(3)H(6)N(1)O(2)    | 88.0861 |
| <b>Am+3</b>                                         |            |     |                     |         |
| Americium(III) ion                                  |            |     |                     |         |
| 3                                                   | 22541-46-4 | 376 | Am(1)               | 243.000 |
| <b>Am+3_H+1_PO4-3</b>                               |            |     |                     |         |
| 1                                                   | ---        | 3   | Am(1)H(1)O(4)P(1)   | 338.980 |
| <b>Am+3_H+1(10)_PO4-3(5)</b>                        |            |     |                     |         |
| -2                                                  | ---        | 2   | Am(1)H(10)O(20)P(5) | 727.937 |
| <b>Am+3_H+1(2)_PO4-3</b>                            |            |     |                     |         |
| 2                                                   | ---        | 4   | Am(1)H(2)O(4)P(1)   | 339.988 |
| <b>Am+3_H+1(2)_PO4-3(2)</b>                         |            |     |                     |         |
| -1                                                  | ---        | 3   | Am(1)H(2)O(8)P(2)   | 434.959 |
| <b>Am+3_H+1(3)_PO4-3(3)</b>                         |            |     |                     |         |
| -3                                                  | ---        | 2   | Am(1)H(3)O(12)P(3)  | 530.939 |
| <b>Am+3_H+1(4)_PO4-3(2)</b>                         |            |     |                     |         |
| 1                                                   | ---        | 2   | Am(1)H(4)O(8)P(2)   | 436.975 |
| <b>Am+3_H+1(4)_PO4-3(4)</b>                         |            |     |                     |         |
| -5                                                  | ---        | 2   | Am(1)H(4)O(16)P(4)  | 626.918 |
| <b>Am+3_H+1(5)_PO4-3(5)</b>                         |            |     |                     |         |
| -7                                                  | ---        | 2   | Am(1)H(5)O(20)P(5)  | 722.898 |
| <b>Am+3_H+1(6)_PO4-3(3)</b>                         |            |     |                     |         |
| 0                                                   | ---        | 2   | Am(1)H(6)O(12)P(3)  | 533.963 |
| <b>Am+3_H+1(6)_PO4-3(6)</b>                         |            |     |                     |         |
| -9                                                  | ---        | 2   | Am(1)H(6)O(24)P(6)  | 818.877 |

|                                                    |            |     |                            |         |
|----------------------------------------------------|------------|-----|----------------------------|---------|
| <b>Am+3_H+1 (8)_PO4-3 (4)</b>                      |            |     |                            |         |
| -1                                                 | ---        | 2   | Am (1) H (8) O (16) P (4)  | 630.950 |
| <b>Am+3_P3O9-3</b>                                 |            |     |                            |         |
| 0                                                  | ---        | 1   | Am (1) O (9) P (3)         | 479.917 |
| <b>Am+3_PO4-3</b>                                  |            |     |                            |         |
| 0                                                  | ---        | 2   | Am (1) O (4) P (1)         | 337.972 |
| <b>Am+3_PO4-3_ (am. , s)</b>                       |            |     |                            |         |
| 0                                                  | ---        | 1   | Am (1) O (4) P (1)         | 337.972 |
| <b>Am+3_PO4-3_ (s)</b><br>Americium(III) phosphate |            |     |                            |         |
| 0                                                  | 14933-41-6 | 1   | Am (1) O (4) P (1)         | 337.972 |
| <b>Am+3_PO4-3_H2O_ (s)</b>                         |            |     |                            |         |
| 0                                                  | ---        | 1   | Am (1) H (2) O (5) P (1)   | 355.987 |
| <b>Am+3_PO4-3 (2)</b>                              |            |     |                            |         |
| -3                                                 | ---        | 2   | Am (1) O (8) P (2)         | 432.943 |
| <b>Am+3_PO4-3 (3)</b>                              |            |     |                            |         |
| -6                                                 | ---        | 1   | Am (1) O (12) P (3)        | 527.915 |
| <b>Am+3_PO4-3 (4)</b>                              |            |     |                            |         |
| -9                                                 | ---        | 1   | Am (1) O (16) P (4)        | 622.886 |
| <b>Am+4</b><br>Americium(IV) ion                   |            |     |                            |         |
| 4                                                  | 22541-71-5 | 179 | Am (1)                     | 243.000 |
| <b>Am+4_H+1_PO4-3</b>                              |            |     |                            |         |
| 2                                                  | ---        | 2   | Am (1) H (1) O (4) P (1)   | 338.980 |
| <b>Am+4_H+1 (10)_PO4-3 (5)</b>                     |            |     |                            |         |
| -1                                                 | ---        | 2   | Am (1) H (10) O (20) P (5) | 727.937 |

|                                  |            |     |                            |         |
|----------------------------------|------------|-----|----------------------------|---------|
| <b>Am+4 _H+1 (12) _PO4-3 (6)</b> |            |     |                            |         |
| -2                               | ---        | 2   | Am (1) H (12) O (24) P (6) | 824.925 |
| <b>Am+4 _H+1 (2) _PO4-3</b>      |            |     |                            |         |
| 3                                | ---        | 2   | Am (1) H (2) O (4) P (1)   | 339.988 |
| <b>Am+4 _H+1 (2) _PO4-3 (2)</b>  |            |     |                            |         |
| 0                                | ---        | 2   | Am (1) H (2) O (8) P (2)   | 434.959 |
| <b>Am+4 _H+1 (3) _PO4-3 (3)</b>  |            |     |                            |         |
| -2                               | ---        | 2   | Am (1) H (3) O (12) P (3)  | 530.939 |
| <b>Am+4 _H+1 (4) _PO4-3 (2)</b>  |            |     |                            |         |
| 2                                | ---        | 2   | Am (1) H (4) O (8) P (2)   | 436.975 |
| <b>Am+4 _H+1 (4) _PO4-3 (4)</b>  |            |     |                            |         |
| -4                               | ---        | 2   | Am (1) H (4) O (16) P (4)  | 626.918 |
| <b>Am+4 _H+1 (5) _PO4-3 (5)</b>  |            |     |                            |         |
| -6                               | ---        | 2   | Am (1) H (5) O (20) P (5)  | 722.898 |
| <b>Am+4 _H+1 (6) _PO4-3 (3)</b>  |            |     |                            |         |
| 1                                | ---        | 2   | Am (1) H (6) O (12) P (3)  | 533.963 |
| <b>Am+4 _H+1 (6) _PO4-3 (6)</b>  |            |     |                            |         |
| -8                               | ---        | 2   | Am (1) H (6) O (24) P (6)  | 818.877 |
| <b>Am+4 _H+1 (8) _PO4-3 (4)</b>  |            |     |                            |         |
| 0                                | ---        | 2   | Am (1) H (8) O (16) P (4)  | 630.950 |
| <b>Am (s)</b><br>Americium       |            |     |                            |         |
| 0                                | ---        | 51  | Am (1)                     | 243.000 |
| <b>AmO2+1</b><br>Americyl(I) ion |            |     |                            |         |
| 1                                | 22878-02-0 | 130 | Am (1) O (2)               | 274.999 |



|                                                     |            |     |                    |         |
|-----------------------------------------------------|------------|-----|--------------------|---------|
| <b>AmO<sub>2</sub>+1_H+1_PO<sub>4</sub>-3</b>       |            |     |                    |         |
| -1                                                  | ---        | 2   | Am(1)H(1)O(6)P(1)  | 370.978 |
| <b>AmO<sub>2</sub>+1_H+1(2)_PO<sub>4</sub>-3</b>    |            |     |                    |         |
| 0                                                   | ---        | 2   | Am(1)H(2)O(6)P(1)  | 371.986 |
| <b>AmO<sub>2</sub>+1_H+1(2)_PO<sub>4</sub>-3(2)</b> |            |     |                    |         |
| -3                                                  | ---        | 2   | Am(1)H(2)O(10)P(2) | 466.958 |
| <b>AmO<sub>2</sub>+1_H+1(3)_PO<sub>4</sub>-3(3)</b> |            |     |                    |         |
| -5                                                  | ---        | 2   | Am(1)H(3)O(14)P(3) | 562.937 |
| <b>AmO<sub>2</sub>+1_H+1(4)_PO<sub>4</sub>-3(2)</b> |            |     |                    |         |
| -1                                                  | ---        | 2   | Am(1)H(4)O(10)P(2) | 468.974 |
| <b>AmO<sub>2</sub>+1_H+1(4)_PO<sub>4</sub>-3(4)</b> |            |     |                    |         |
| -7                                                  | ---        | 2   | Am(1)H(4)O(18)P(4) | 658.917 |
| <b>AmO<sub>2</sub>+1_H+1(5)_PO<sub>4</sub>-3(5)</b> |            |     |                    |         |
| -9                                                  | ---        | 2   | Am(1)H(5)O(22)P(5) | 754.897 |
| <b>AmO<sub>2</sub>+1_H+1(6)_PO<sub>4</sub>-3(3)</b> |            |     |                    |         |
| -2                                                  | ---        | 2   | Am(1)H(6)O(14)P(3) | 565.961 |
| <b>AmO<sub>2</sub>+1_H+1(8)_PO<sub>4</sub>-3(4)</b> |            |     |                    |         |
| -3                                                  | ---        | 2   | Am(1)H(8)O(18)P(4) | 662.949 |
| <b>AmO<sub>2</sub>+2</b><br>Americyl(II) ion        |            |     |                    |         |
| 2                                                   | 12323-66-9 | 149 | Am(1)O(2)          | 274.999 |
| <b>AmO<sub>2</sub>+2_H+1_PO<sub>4</sub>-3</b>       |            |     |                    |         |
| 0                                                   | ---        | 2   | Am(1)H(1)O(6)P(1)  | 370.978 |
| <b>AmO<sub>2</sub>+2_H+1(2)_PO<sub>4</sub>-3</b>    |            |     |                    |         |
| 1                                                   | ---        | 2   | Am(1)H(2)O(6)P(1)  | 371.986 |

|                                                                                                                                                                         |            |     |                           |         |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----|---------------------------|---------|
| <b>AmO<sub>2</sub>+2 _H+1 (2) _PO<sub>4</sub>-3 (2)</b>                                                                                                                 |            |     |                           |         |
| -2                                                                                                                                                                      | ---        | 2   | Am (1) H (2) O (10) P (2) | 466.958 |
| <b>AmO<sub>2</sub>+2 _H+1 (3) _PO<sub>4</sub>-3 (3)</b>                                                                                                                 |            |     |                           |         |
| -4                                                                                                                                                                      | ---        | 2   | Am (1) H (3) O (14) P (3) | 562.937 |
| <b>AmO<sub>2</sub>+2 _H+1 (4) _PO<sub>4</sub>-3 (2)</b>                                                                                                                 |            |     |                           |         |
| 0                                                                                                                                                                       | ---        | 2   | Am (1) H (4) O (10) P (2) | 468.974 |
| <b>AmO<sub>2</sub>+2 _H+1 (4) _PO<sub>4</sub>-3 (4)</b>                                                                                                                 |            |     |                           |         |
| -6                                                                                                                                                                      | ---        | 2   | Am (1) H (4) O (18) P (4) | 658.917 |
| <b>AmO<sub>2</sub>+2 _H+1 (5) _PO<sub>4</sub>-3 (5)</b>                                                                                                                 |            |     |                           |         |
| -8                                                                                                                                                                      | ---        | 2   | Am (1) H (5) O (22) P (5) | 754.897 |
| <b>AmO<sub>2</sub>+2 _H+1 (6) _PO<sub>4</sub>-3 (3)</b>                                                                                                                 |            |     |                           |         |
| -1                                                                                                                                                                      | ---        | 2   | Am (1) H (6) O (14) P (3) | 565.961 |
| <b>AmO<sub>2</sub>+2 _H+1 (8) _PO<sub>4</sub>-3 (4)</b>                                                                                                                 |            |     |                           |         |
| -2                                                                                                                                                                      | ---        | 2   | Am (1) H (8) O (18) P (4) | 662.949 |
| <b>Arg-1</b><br>Arginate                                                                                                                                                |            |     |                           |         |
| -1                                                                                                                                                                      | ---        | 496 | C (6) H (13) N (4) O (2)  | 173.195 |
| <b>Asn-1</b><br>Asparaginate ion; L-2-Aminobutanedioate 4-amide ion; L-beta-Asparaginate ion;<br>Aspartate beta-amide ion; Altheinate ion; Asparamide ion; Agedoite ion |            |     |                           |         |
| -1                                                                                                                                                                      | ---        | 581 | C (4) H (7) N (2) O (3)   | 131.111 |
| <b>Asp-2</b><br>Aspartate ion; Aminosuccinate ion; Asparagite ion; 2-Aminobutanedioate ion; L-2-Aminobutanedioate ion                                                   |            |     |                           |         |
| -2                                                                                                                                                                      | 63-71-8    | 776 | C (4) H (5) N (1) O (4)   | 131.088 |
| <b>Ba+2</b><br>Barium(II) ion                                                                                                                                           |            |     |                           |         |
| 2                                                                                                                                                                       | 22541-12-4 | 584 | Ba (1)                    | 137.330 |

|                                                        |            |   |                           |         |
|--------------------------------------------------------|------------|---|---------------------------|---------|
| <b>Ba+2_H+1_P3O10-5</b>                                |            |   |                           |         |
| -2                                                     | ---        | 3 | Ba (1) H (1) O (10) P (3) | 391.254 |
| <b>Ba+2_H+1_PO4-3</b>                                  |            |   |                           |         |
| 0                                                      | ---        | 2 | Ba (1) H (1) O (4) P (1)  | 233.310 |
| <b>Ba+2_H+1_PO4-3_(s)</b><br>Barium phosphate, dibasic |            |   |                           |         |
| 0                                                      | 10048-98-3 | 2 | Ba (1) H (1) O (4) P (1)  | 233.310 |
| <b>Ba+2_H+1 (2)_PO4-3</b>                              |            |   |                           |         |
| 1                                                      | ---        | 2 | Ba (1) H (2) O (4) P (1)  | 234.318 |
| <b>Ba+2_H+1 (4)_PO4-3 (2)</b>                          |            |   |                           |         |
| 0                                                      | ---        | 2 | Ba (1) H (4) O (8) P (2)  | 331.305 |
| <b>Ba+2_P2O7-4</b>                                     |            |   |                           |         |
| -2                                                     | ---        | 1 | Ba (1) O (7) P (2)        | 311.274 |
| <b>Ba+2_P2O7-4 (2)</b>                                 |            |   |                           |         |
| -6                                                     | ---        | 1 | Ba (1) O (14) P (4)       | 485.218 |
| <b>Ba+2_P3O10-5</b>                                    |            |   |                           |         |
| -3                                                     | ---        | 3 | Ba (1) O (10) P (3)       | 390.246 |
| <b>Ba+2_P3O10-5 (2)</b>                                |            |   |                           |         |
| -8                                                     | ---        | 1 | Ba (1) O (20) P (6)       | 643.162 |
| <b>Ba+2_P3O9-3</b>                                     |            |   |                           |         |
| -1                                                     | ---        | 1 | Ba (1) O (9) P (3)        | 374.247 |
| <b>Ba+2_P4O12-4</b>                                    |            |   |                           |         |
| -2                                                     | ---        | 1 | Ba (1) O (12) P (4)       | 453.219 |
| <b>Ba+2_PO4-3</b>                                      |            |   |                           |         |
| -1                                                     | ---        | 1 | Ba (1) O (4) P (1)        | 232.302 |

|                                                               |            |     |                           |         |
|---------------------------------------------------------------|------------|-----|---------------------------|---------|
| <b>Ba+2_UO2+2 (2)_PO4-3 (2)_ (s)</b><br>Uranocircite          |            |     |                           |         |
| 0                                                             | ---        | 2   | Ba (1) O (12) P (2) U (2) | 867.329 |
| <b>Ba+2 (2)_P3O10-5 (s)</b>                                   |            |     |                           |         |
| -1                                                            | ---        | 2   | Ba (2) O (10) P (3)       | 527.576 |
| <b>Ba+2 (3)_PO4-3 (2)_ (s)</b><br>Barium phosphate            |            |     |                           |         |
| 0                                                             | ---        | 1   | Ba (3) O (8) P (2)        | 601.933 |
| <b>Ba (s)</b><br>Barium; Barium, cubic; Barium, lambda trans. |            |     |                           |         |
| 0                                                             | 7440-39-3  | 64  | Ba (1)                    | 137.330 |
| <b>bAla-1</b><br>beta-Alanate ion                             |            |     |                           |         |
| -1                                                            | ---        | 498 | C (3) H (6) N (1) O (2)   | 88.0861 |
| <b>Be+2</b><br>Beryllium(II) ion                              |            |     |                           |         |
| 2                                                             | 22537-20-8 | 665 | Be (1)                    | 9.01220 |
| <b>Be+2_H+1_P2O7-4</b>                                        |            |     |                           |         |
| -1                                                            | ---        | 1   | Be (1) H (1) O (7) P (2)  | 183.964 |
| <b>Be+2_H+1_P3O10-5</b>                                       |            |     |                           |         |
| -2                                                            | ---        | 1   | Be (1) H (1) O (10) P (3) | 262.936 |
| <b>Be+2_P2O7-4</b>                                            |            |     |                           |         |
| -2                                                            | ---        | 1   | Be (1) O (7) P (2)        | 182.956 |
| <b>Be+2_P2O7-4 (2)</b>                                        |            |     |                           |         |
| -6                                                            | ---        | 1   | Be (1) O (14) P (4)       | 356.900 |
| <b>Be+2 (2)_P2O7-4</b>                                        |            |     |                           |         |
| 0                                                             | ---        | 1   | Be (2) O (7) P (2)        | 191.968 |

|                                                                                                                                                            |            |      |                                 |         |
|------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|------|---------------------------------|---------|
| <b>Bi+3</b><br>Bismuth(III) ion                                                                                                                            |            |      |                                 |         |
| 3                                                                                                                                                          | 23713-46-4 | 335  | Bi (1)                          | 208.980 |
| <b>Bi+3_PO4-3</b>                                                                                                                                          |            |      |                                 |         |
| 0                                                                                                                                                          | ---        | 1    | Bi (1) O (4) P (1)              | 303.952 |
| <b>Bi+3_PO4-3_(s)</b><br>Bismuth(III) phosphate                                                                                                            |            |      |                                 |         |
| 0                                                                                                                                                          | ---        | 1    | Bi (1) O (4) P (1)              | 303.952 |
| <b>Bipy</b><br>2,2'-Bipyridyl; Bipyridine; 2,2'-Bipyridine; Dipyridyl; 2,2'-Dipyridyl; 2-(2-Pyridyl)pyridine; bipy; bpy; dip; dipy; alpha-alpha'-Bipyridyl |            |      |                                 |         |
| 0                                                                                                                                                          | 366-18-7   | 823  | C (10) H (8) N (2)              | 156.187 |
| <b>Br-1</b><br>Bromide ion                                                                                                                                 |            |      |                                 |         |
| -1                                                                                                                                                         | 24959-67-9 | 1213 | Br (1)                          | 79.9040 |
| <b>Br2(g)</b><br>Bromine gas                                                                                                                               |            |      |                                 |         |
| 0                                                                                                                                                          | ---        | 18   | Br (2)                          | 159.808 |
| <b>Br2(l)</b><br>Bromine, liquid                                                                                                                           |            |      |                                 |         |
| 0                                                                                                                                                          | ---        | 285  | Br (2)                          | 159.808 |
| <b>C(s)</b><br>Carbon; Graphite; Carbon, hexagonal                                                                                                         |            |      |                                 |         |
| 0                                                                                                                                                          | 7440-44-0  | 530  | C (1)                           | 12.0110 |
| <b>Ca+2</b><br>Calcium(II) ion                                                                                                                             |            |      |                                 |         |
| 2                                                                                                                                                          | 14127-61-8 | 2161 | Ca (1)                          | 40.0780 |
| <b>Ca+2_Citric-3_PO4-3</b>                                                                                                                                 |            |      |                                 |         |
| -4                                                                                                                                                         | ---        | 1    | C (6) H (5) Ca (1) O (11) P (1) | 324.151 |
| <b>Ca+2_Cys-2_PO4-3</b>                                                                                                                                    |            |      |                                 |         |

|                                |     |   |                                             |         |
|--------------------------------|-----|---|---------------------------------------------|---------|
| -3                             | --- | 1 | C (3) H (5) Ca (1) N (1) O (6) P (1) S (1)  | 254.188 |
| <b>Ca+2_H+1_Ala-1_PO4-3</b>    |     |   |                                             |         |
| -1                             | --- | 1 | C (3) H (7) Ca (1) N (1) O (6) P (1)        | 224.144 |
| <b>Ca+2_H+1_Arg-1_PO4-3</b>    |     |   |                                             |         |
| -1                             | --- | 1 | C (6) H (14) Ca (1) N (4) O (6) P (1)       | 309.252 |
| <b>Ca+2_H+1_Asn-1_PO4-3</b>    |     |   |                                             |         |
| -1                             | --- | 1 | C (4) H (8) Ca (1) N (2) O (7) P (1)        | 267.169 |
| <b>Ca+2_H+1_Asp-2_PO4-3</b>    |     |   |                                             |         |
| -2                             | --- | 1 | C (4) H (6) Ca (1) N (1) O (8) P (1)        | 267.146 |
| <b>Ca+2_H+1_bAla-1_PO4-3</b>   |     |   |                                             |         |
| -1                             | --- | 1 | C (3) H (7) Ca (1) N (1) O (6) P (1)        | 224.144 |
| <b>Ca+2_H+1_Cis-2_PO4-3</b>    |     |   |                                             |         |
| -2                             | --- | 1 | C (6) H (11) Ca (1) N (2) O (8) P (1) S (2) | 374.334 |
| <b>Ca+2_H+1_Citric-3_PO4-3</b> |     |   |                                             |         |
| -3                             | --- | 2 | C (6) H (6) Ca (1) O (11) P (1)             | 325.159 |
| <b>Ca+2_H+1_Citrul-1_PO4-3</b> |     |   |                                             |         |
| -1                             | --- | 1 | C (6) H (13) Ca (1) N (3) O (7) P (1)       | 310.237 |
| <b>Ca+2_H+1_Cys-2_PO4-3</b>    |     |   |                                             |         |
| -2                             | --- | 1 | C (3) H (6) Ca (1) N (1) O (6) P (1) S (1)  | 255.196 |
| <b>Ca+2_H+1_Gln-1_PO4-3</b>    |     |   |                                             |         |
| -1                             | --- | 1 | C (5) H (10) Ca (1) N (2) O (7) P (1)       | 281.196 |
| <b>Ca+2_H+1_Glu-2_PO4-3</b>    |     |   |                                             |         |
| -2                             | --- | 1 | C (5) H (8) Ca (1) N (1) O (8) P (1)        | 281.172 |
| <b>Ca+2_H+1_Gly-1_PO4-3</b>    |     |   |                                             |         |
| -1                             | --- | 1 | C (2) H (5) Ca (1) N (1) O (6) P (1)        | 210.117 |

|                                |     |   |                                |         |
|--------------------------------|-----|---|--------------------------------|---------|
| <b>Ca+2_H+1_His-1_PO4-3</b>    |     |   |                                |         |
| -1                             | --- | 1 | C(6)H(9)Ca(1)N(3)O(6)P(1)      | 290.206 |
| <b>Ca+2_H+1_Hyp-1_PO4-3</b>    |     |   |                                |         |
| -1                             | --- | 1 | C(5)H(9)Ca(1)N(1)O(7)P(1)      | 266.181 |
| <b>Ca+2_H+1_Ile-1_PO4-3</b>    |     |   |                                |         |
| -1                             | --- | 1 | C(6)H(13)Ca(1)N(1)O(6)P(1)     | 266.224 |
| <b>Ca+2_H+1_Lactic-1_PO4-3</b> |     |   |                                |         |
| -1                             | --- | 1 | C(3)H(6)Ca(1)O(7)P(1)          | 225.128 |
| <b>Ca+2_H+1_Leu-1_PO4-3</b>    |     |   |                                |         |
| -1                             | --- | 1 | C(6)H(13)Ca(1)N(1)O(6)P(1)     | 266.224 |
| <b>Ca+2_H+1_Malic-2_PO4-3</b>  |     |   |                                |         |
| -2                             | --- | 1 | C(4)H(5)Ca(1)O(9)P(1)          | 268.130 |
| <b>Ca+2_H+1_Met-1_PO4-3</b>    |     |   |                                |         |
| -1                             | --- | 1 | C(5)H(11)Ca(1)N(1)O(6)P(1)S(1) | 284.257 |
| <b>Ca+2_H+1_NTA-3_PO4-3</b>    |     |   |                                |         |
| -3                             | --- | 1 | C(6)H(7)Ca(1)N(1)O(10)P(1)     | 324.174 |
| <b>Ca+2_H+1_Oxalic-2_PO4-3</b> |     |   |                                |         |
| -2                             | --- | 1 | C(2)H(1)Ca(1)O(8)P(1)          | 224.077 |
| <b>Ca+2_H+1_P2O7-4</b>         |     |   |                                |         |
| -1                             | --- | 2 | Ca(1)H(1)O(7)P(2)              | 215.030 |
| <b>Ca+2_H+1_P3O10-5</b>        |     |   |                                |         |
| -2                             | --- | 3 | Ca(1)H(1)O(10)P(3)             | 294.002 |
| <b>Ca+2_H+1_P4O13-6</b>        |     |   |                                |         |
| -3                             | --- | 1 | Ca(1)H(1)O(13)P(4)             | 372.974 |

|                                                                                                                                |            |   |                             |         |
|--------------------------------------------------------------------------------------------------------------------------------|------------|---|-----------------------------|---------|
| <b>Ca+2_H+1_Phe-1_PO4-3</b>                                                                                                    |            |   |                             |         |
| -1                                                                                                                             | ---        | 1 | C(9)H(11)Ca(1)N(1)O(6)P(1)  | 300.242 |
| <b>Ca+2_H+1_PO4-3</b>                                                                                                          |            |   |                             |         |
| 0                                                                                                                              | ---        | 4 | Ca(1)H(1)O(4)P(1)           | 136.058 |
| <b>Ca+2_H+1_PO4-3_(s)</b><br>Monetite; Calcium phosphate, dibasic; Calcium hydrogen phosphate                                  |            |   |                             |         |
| 0                                                                                                                              | 7757-93-9  | 4 | Ca(1)H(1)O(4)P(1)           | 136.058 |
| <b>Ca+2_H+1_PO4-3_H2O(2)_(s)</b><br>Brushite; Calcium phosphate dihydrate, dibasic; DCPD; Calcium hydrogen phosphate dihydrate |            |   |                             |         |
| 0                                                                                                                              | 14567-92-1 | 3 | Ca(1)H(5)O(6)P(1)           | 172.088 |
| <b>Ca+2_H+1_Ser-1_PO4-3</b>                                                                                                    |            |   |                             |         |
| -1                                                                                                                             | ---        | 1 | C(3)H(7)Ca(1)N(1)O(7)P(1)   | 240.143 |
| <b>Ca+2_H+1_Succinic-2_PO4-3</b>                                                                                               |            |   |                             |         |
| -2                                                                                                                             | ---        | 1 | C(4)H(5)Ca(1)O(8)P(1)       | 252.131 |
| <b>Ca+2_H+1_Tartaric-2_PO4-3</b>                                                                                               |            |   |                             |         |
| -2                                                                                                                             | ---        | 1 | C(4)H(5)Ca(1)O(10)P(1)      | 284.130 |
| <b>Ca+2_H+1_Thr-1_PO4-3</b>                                                                                                    |            |   |                             |         |
| -1                                                                                                                             | ---        | 1 | C(4)H(9)Ca(1)N(1)O(7)P(1)   | 254.170 |
| <b>Ca+2_H+1_Trp-1_PO4-3</b>                                                                                                    |            |   |                             |         |
| -1                                                                                                                             | ---        | 1 | C(11)H(12)Ca(1)N(2)O(6)P(1) | 339.278 |
| <b>Ca+2_H+1_Val-1_PO4-3</b>                                                                                                    |            |   |                             |         |
| -1                                                                                                                             | ---        | 1 | C(5)H(11)Ca(1)N(1)O(6)P(1)  | 252.197 |
| <b>Ca+2_H+1(2)_Ala-1_PO4-3</b>                                                                                                 |            |   |                             |         |
| 0                                                                                                                              | ---        | 1 | C(3)H(8)Ca(1)N(1)O(6)P(1)   | 225.152 |
| <b>Ca+2_H+1(2)_Arg-1_PO4-3</b>                                                                                                 |            |   |                             |         |



|                                   |     |   |                                  |         |
|-----------------------------------|-----|---|----------------------------------|---------|
| 0                                 | --- | 1 | $C(6)H(15)Ca(1)N(4)O(6)P(1)$     | 310.260 |
| <b>Ca+2_H+1(2)_Asn-1_PO4-3</b>    |     |   |                                  |         |
| 0                                 | --- | 1 | $C(4)H(9)Ca(1)N(2)O(7)P(1)$      | 268.177 |
| <b>Ca+2_H+1(2)_Asp-2_PO4-3</b>    |     |   |                                  |         |
| -1                                | --- | 1 | $C(4)H(7)Ca(1)N(1)O(8)P(1)$      | 268.154 |
| <b>Ca+2_H+1(2)_bAla-1_PO4-3</b>   |     |   |                                  |         |
| 0                                 | --- | 1 | $C(3)H(8)Ca(1)N(1)O(6)P(1)$      | 225.152 |
| <b>Ca+2_H+1(2)_Cis-2_PO4-3</b>    |     |   |                                  |         |
| -1                                | --- | 1 | $C(6)H(12)Ca(1)N(2)O(8)P(1)S(2)$ | 375.342 |
| <b>Ca+2_H+1(2)_Citric-3_PO4-3</b> |     |   |                                  |         |
| -2                                | --- | 1 | $C(6)H(7)Ca(1)O(11)P(1)$         | 326.167 |
| <b>Ca+2_H+1(2)_Citrul-1_PO4-3</b> |     |   |                                  |         |
| 0                                 | --- | 1 | $C(6)H(14)Ca(1)N(3)O(7)P(1)$     | 311.245 |
| <b>Ca+2_H+1(2)_Cys-2_PO4-3</b>    |     |   |                                  |         |
| -1                                | --- | 1 | $C(3)H(7)Ca(1)N(1)O(6)P(1)S(1)$  | 256.204 |
| <b>Ca+2_H+1(2)_Gln-1_PO4-3</b>    |     |   |                                  |         |
| 0                                 | --- | 1 | $C(5)H(11)Ca(1)N(2)O(7)P(1)$     | 282.204 |
| <b>Ca+2_H+1(2)_Glu-2_PO4-3</b>    |     |   |                                  |         |
| -1                                | --- | 1 | $C(5)H(9)Ca(1)N(1)O(8)P(1)$      | 282.180 |
| <b>Ca+2_H+1(2)_Gly-1_PO4-3</b>    |     |   |                                  |         |
| 0                                 | --- | 1 | $C(2)H(6)Ca(1)N(1)O(6)P(1)$      | 211.125 |
| <b>Ca+2_H+1(2)_His-1_PO4-3</b>    |     |   |                                  |         |
| 0                                 | --- | 1 | $C(6)H(10)Ca(1)N(3)O(6)P(1)$     | 291.214 |
| <b>Ca+2_H+1(2)_Hyp-1_PO4-3</b>    |     |   |                                  |         |
| 0                                 | --- | 1 | $C(5)H(10)Ca(1)N(1)O(7)P(1)$     | 267.189 |

|                                |     |   |                                |         |
|--------------------------------|-----|---|--------------------------------|---------|
| <b>Ca+2_H+1(2)_Ile-1_PO4-3</b> |     |   |                                |         |
| 0                              | --- | 1 | C(6)H(14)Ca(1)N(1)O(6)P(1)     | 267.232 |
| <b>Ca+2_H+1(2)_Leu-1_PO4-3</b> |     |   |                                |         |
| 0                              | --- | 1 | C(6)H(14)Ca(1)N(1)O(6)P(1)     | 267.232 |
| <b>Ca+2_H+1(2)_Lys-1_PO4-3</b> |     |   |                                |         |
| 0                              | --- | 1 | C(6)H(15)Ca(1)N(2)O(6)P(1)     | 282.247 |
| <b>Ca+2_H+1(2)_Met-1_PO4-3</b> |     |   |                                |         |
| 0                              | --- | 1 | C(5)H(12)Ca(1)N(1)O(6)P(1)S(1) | 285.265 |
| <b>Ca+2_H+1(2)_Orn-1_PO4-3</b> |     |   |                                |         |
| 0                              | --- | 1 | C(5)H(13)Ca(1)N(2)O(6)P(1)     | 268.220 |
| <b>Ca+2_H+1(2)_P3O10-5</b>     |     |   |                                |         |
| -1                             | --- | 1 | Ca(1)H(2)O(10)P(3)             | 295.010 |
| <b>Ca+2_H+1(2)_Phe-1_PO4-3</b> |     |   |                                |         |
| 0                              | --- | 1 | C(9)H(12)Ca(1)N(1)O(6)P(1)     | 301.249 |
| <b>Ca+2_H+1(2)_PO4-3</b>       |     |   |                                |         |
| 1                              | --- | 4 | Ca(1)H(2)O(4)P(1)              | 137.066 |
| <b>Ca+2_H+1(2)_PO4-3(2)</b>    |     |   |                                |         |
| -2                             | --- | 1 | Ca(1)H(2)O(8)P(2)              | 232.037 |
| <b>Ca+2_H+1(2)_Pro-1_PO4-3</b> |     |   |                                |         |
| 0                              | --- | 1 | C(5)H(10)Ca(1)N(1)O(6)P(1)     | 251.190 |
| <b>Ca+2_H+1(2)_Ser-1_PO4-3</b> |     |   |                                |         |
| 0                              | --- | 1 | C(3)H(8)Ca(1)N(1)O(7)P(1)      | 241.151 |
| <b>Ca+2_H+1(2)_Thr-1_PO4-3</b> |     |   |                                |         |
| 0                              | --- | 1 | C(4)H(10)Ca(1)N(1)O(7)P(1)     | 255.178 |

|                                                                                                                                                    |            |   |                             |         |
|----------------------------------------------------------------------------------------------------------------------------------------------------|------------|---|-----------------------------|---------|
| <b>Ca+2_H+1(2)_Trp-1_PO4-3</b>                                                                                                                     |            |   |                             |         |
| 0                                                                                                                                                  | ---        | 1 | C(11)H(13)Ca(1)N(2)O(6)P(1) | 340.286 |
| <b>Ca+2_H+1(2)_Tyr-2_PO4-3</b>                                                                                                                     |            |   |                             |         |
| -1                                                                                                                                                 | ---        | 1 | C(9)H(11)Ca(1)N(1)O(7)P(1)  | 316.241 |
| <b>Ca+2_H+1(2)_Val-1_PO4-3</b>                                                                                                                     |            |   |                             |         |
| 0                                                                                                                                                  | ---        | 1 | C(5)H(12)Ca(1)N(1)O(6)P(1)  | 253.205 |
| <b>Ca+2_H+1(3)_PO4-3(2)</b>                                                                                                                        |            |   |                             |         |
| -1                                                                                                                                                 | ---        | 4 | Ca(1)H(3)O(8)P(2)           | 233.045 |
| <b>Ca+2_H+1(4)_PO4-3(2)</b>                                                                                                                        |            |   |                             |         |
| 0                                                                                                                                                  | ---        | 2 | Ca(1)H(4)O(8)P(2)           | 234.053 |
| <b>Ca+2_H+1(4)_PO4-3(2)_(s)</b><br>Calcium dihydrogen phosphate anhydrous; Monocalcium phosphate anhydrous; Calcium phosphate dihydrogen anhydrous |            |   |                             |         |
| 0                                                                                                                                                  | ---        | 2 | Ca(1)H(4)O(8)P(2)           | 234.053 |
| <b>Ca+2_H+1(4)_PO4-3(2)_H2O_(s)</b><br>Calcium dihydrogen phosphate monohydrate; MCP                                                               |            |   |                             |         |
| 0                                                                                                                                                  | 10031-30-8 | 2 | Ca(1)H(6)O(9)P(2)           | 252.068 |
| <b>Ca+2_H+1(4)_PO4-3(2)_H2O(2)_(s)</b>                                                                                                             |            |   |                             |         |
| 0                                                                                                                                                  | ---        | 1 | Ca(1)H(8)O(10)P(2)          | 270.084 |
| <b>Ca+2_Na+1_P4O12-4</b>                                                                                                                           |            |   |                             |         |
| -1                                                                                                                                                 | ---        | 1 | Ca(1)Na(1)O(12)P(4)         | 378.957 |
| <b>Ca+2_OH-1_P2O7-4</b>                                                                                                                            |            |   |                             |         |
| -3                                                                                                                                                 | ---        | 2 | Ca(1)H(1)O(8)P(2)           | 231.029 |
| <b>Ca+2_OH-1_P3O10-5</b>                                                                                                                           |            |   |                             |         |
| -4                                                                                                                                                 | ---        | 3 | Ca(1)H(1)O(11)P(3)          | 310.001 |
| <b>Ca+2_P2O7-4</b>                                                                                                                                 |            |   |                             |         |

|                                                                     |     |   |                           |         |
|---------------------------------------------------------------------|-----|---|---------------------------|---------|
| -2                                                                  | --- | 3 | Ca (1) O (7) P (2)        | 214.022 |
| <b>Ca+2_P2O7-4 (2)</b>                                              |     |   |                           |         |
| -6                                                                  | --- | 1 | Ca (1) O (14) P (4)       | 387.966 |
| <b>Ca+2_P3O10-5</b>                                                 |     |   |                           |         |
| -3                                                                  | --- | 4 | Ca (1) O (10) P (3)       | 292.994 |
| <b>Ca+2_P3O10-5 (2)</b>                                             |     |   |                           |         |
| -8                                                                  | --- | 1 | Ca (1) O (20) P (6)       | 545.910 |
| <b>Ca+2_P3O9-3</b>                                                  |     |   |                           |         |
| -1                                                                  | --- | 1 | Ca (1) O (9) P (3)        | 276.995 |
| <b>Ca+2_P4O12-4</b>                                                 |     |   |                           |         |
| -2                                                                  | --- | 3 | Ca (1) O (12) P (4)       | 355.967 |
| <b>Ca+2_P4O12-4 (2)</b>                                             |     |   |                           |         |
| -6                                                                  | --- | 1 | Ca (1) O (24) P (8)       | 671.856 |
| <b>Ca+2_P4O13-6</b>                                                 |     |   |                           |         |
| -4                                                                  | --- | 2 | Ca (1) O (13) P (4)       | 371.966 |
| <b>Ca+2_P6O18-6</b>                                                 |     |   |                           |         |
| -4                                                                  | --- | 1 | Ca (1) O (18) P (6)       | 513.911 |
| <b>Ca+2_P8O24-8</b>                                                 |     |   |                           |         |
| -6                                                                  | --- | 1 | Ca (1) O (24) P (8)       | 671.856 |
| <b>Ca+2_PO4-3</b>                                                   |     |   |                           |         |
| -1                                                                  | --- | 1 | Ca (1) O (4) P (1)        | 135.050 |
| <b>Ca+2_UO2+2 (2)_PO4-3 (2)_ (s)</b><br>Autunite (unhydrated)       |     |   |                           |         |
| 0                                                                   | --- | 2 | Ca (1) O (12) P (2) U (2) | 770.077 |
| <b>Ca+2_UO2+2 (2)_PO4-3 (2)_H2O (3)_ (s)</b><br>Autunite (hydrated) |     |   |                           |         |

|                                                                                                                 |           |   |                             |         |
|-----------------------------------------------------------------------------------------------------------------|-----------|---|-----------------------------|---------|
| 0                                                                                                               | ---       | 1 | Ca(1)H(6)O(15)P(2)U(2)      | 824.123 |
| <b>Ca+2_UO2+2(2)_PO4-3(2)_H2O(7)_ (s)</b>                                                                       |           |   |                             |         |
| 0                                                                                                               | ---       | 1 | Ca(1)H(14)O(19)P(2)U(2)     | 896.184 |
| <b>Ca+2_Zn+2(2)_PO4-3(2)_H2O(2)_ (s)</b><br>Scholzite                                                           |           |   |                             |         |
| 0                                                                                                               | ---       | 1 | Ca(1)H(4)O(10)P(2)Zn(2)     | 396.816 |
| <b>Ca+2(10)_OH-1_CO3-2(0.5)_PO4-3(6)_ (s)</b><br>Dahllite; Hydroxyapatite carbonated; Carbonated hydroxyapatite |           |   |                             |         |
| 0                                                                                                               | ---       | 1 | C(0.5)H(1)Ca(10)O(26.5)P(6) | 1017.62 |
| <b>Ca+2(10)_OH-1(2)_PO4-3(6)</b>                                                                                |           |   |                             |         |
| 0                                                                                                               | ---       | 1 | Ca(10)H(2)O(26)P(6)         | 1004.62 |
| <b>Ca+2(2)_Fe+2_PO4-3(2)_H2O(4)_ (s)</b><br>Anapaite; Calcium ferrous phosphate                                 |           |   |                             |         |
| 0                                                                                                               | ---       | 1 | Ca(2)Fe(1)H(8)O(12)P(2)     | 398.006 |
| <b>Ca+2(2)_H+1(2)_PO4-3(2)</b>                                                                                  |           |   |                             |         |
| 0                                                                                                               | ---       | 2 | Ca(2)H(2)O(8)P(2)           | 272.115 |
| <b>Ca+2(2)_P2O7-4_ (s)</b><br>Calcium pyrophosphate                                                             |           |   |                             |         |
| 0                                                                                                               | 7790-76-3 | 3 | Ca(2)O(7)P(2)               | 254.100 |
| <b>Ca+2(2)_P4O12-4</b>                                                                                          |           |   |                             |         |
| 0                                                                                                               | ---       | 2 | Ca(2)O(12)P(4)              | 396.045 |
| <b>Ca+2(2)_P4O13-6</b>                                                                                          |           |   |                             |         |
| -2                                                                                                              | ---       | 2 | Ca(2)O(13)P(4)              | 412.044 |
| <b>Ca+2(3)_PO4-3(2)_ (alpha, s)</b><br>Calcium phosphate, monoclinic; Calcium phosphate tribasic alpha          |           |   |                             |         |
| 0                                                                                                               | ---       | 2 | Ca(3)O(8)P(2)               | 310.177 |
| <b>Ca+2(3)_PO4-3(2)_ (am., s)</b><br>Calcium phosphate, amorphous; ACP; Amorphous calcium phosphate             |           |   |                             |         |

|                                                                                                                                                                                                                                                                                                                                  |            |   |                         |         |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|---|-------------------------|---------|
| 0                                                                                                                                                                                                                                                                                                                                | ---        | 1 | Ca(3)O(8)P(2)           | 310.177 |
| <b>Ca+2(3)_PO4-3(2)_(beta,s)</b><br>Whitlockite (traditional); Tricalcium orthophosphate; Tricalcium phosphate; Calcium phosphate, tribasic; Oxydapatite; Voelicherite; TCP; Oxydapatit; Calcium phosphate, hexagonal (sic) Barin; Calcium phosphate, rhombohedral Robie & Hemingway, Wikipedia; Calcium phosphate tribasic beta |            |   |                         |         |
| 0                                                                                                                                                                                                                                                                                                                                | 7758-87-4  | 2 | Ca(3)O(8)P(2)           | 310.177 |
| <b>Ca+2(4)_Cd+2_OH-1_PO4-3(3)_(s)</b>                                                                                                                                                                                                                                                                                            |            |   |                         |         |
| 0                                                                                                                                                                                                                                                                                                                                | ---        | 1 | Ca(4)Cd(1)H(1)O(13)P(3) | 574.644 |
| <b>Ca+2(4)_H+1_PO4-3(3)_(s)</b><br>Octacalcium phosphate; Octacalcium phosphate anhydrous                                                                                                                                                                                                                                        |            |   |                         |         |
| 0                                                                                                                                                                                                                                                                                                                                | 13767-12-9 | 1 | Ca(4)H(1)O(12)P(3)      | 446.235 |
| <b>Ca+2(4)_H+1_PO4-3(3)_H2O(2.5)_(s)</b>                                                                                                                                                                                                                                                                                         |            |   |                         |         |
| 0                                                                                                                                                                                                                                                                                                                                | ---        | 1 | Ca(4)H(6)O(14.5)P(3)    | 491.273 |
| <b>Ca+2(4)_H+1_PO4-3(3)_H2O(3)_(s)</b><br>Hydrated octacalcium phosphate, OCP; Octacalcium phosphate trihydrate; OCP                                                                                                                                                                                                             |            |   |                         |         |
| 0                                                                                                                                                                                                                                                                                                                                | 17968-19-3 | 1 | Ca(4)H(7)O(15)P(3)      | 500.281 |
| <b>Ca+2(5)_Br-1_PO4-3(3)_(s)</b>                                                                                                                                                                                                                                                                                                 |            |   |                         |         |
| 0                                                                                                                                                                                                                                                                                                                                | ---        | 1 | Br(1)Ca(5)O(12)P(3)     | 565.209 |
| <b>Ca+2(5)_Cl-1_PO4-3(3)_(s)</b>                                                                                                                                                                                                                                                                                                 |            |   |                         |         |
| 0                                                                                                                                                                                                                                                                                                                                | ---        | 2 | Ca(5)Cl(1)O(12)P(3)     | 520.758 |
| <b>Ca+2(5)_F-1_PO4-3(3)_(s)</b><br>Fluorapatite; Apatite; Pentacalcium triphosphate fluoride, hexagonal; Fluoroapatite                                                                                                                                                                                                           |            |   |                         |         |
| 0                                                                                                                                                                                                                                                                                                                                | 64476-38-6 | 3 | Ca(5)F(1)O(12)P(3)      | 504.303 |
| <b>Ca+2(5)_I-1_PO4-3(3)_(s)</b>                                                                                                                                                                                                                                                                                                  |            |   |                         |         |
| 0                                                                                                                                                                                                                                                                                                                                | ---        | 1 | Ca(5)I(1)O(12)P(3)      | 612.205 |
| <b>Ca+2(5)_OH-1_PO4-3(3)_(s)</b><br>Hydroxyapatite; Calcium orthophosphate, basic; Durapatite; Calcium phosphate hydroxide; Hydroxylapatite; HAP; Pentacalcium triphosphate hydroxide, hexagonal                                                                                                                                 |            |   |                         |         |
| 0                                                                                                                                                                                                                                                                                                                                | 1306-06-5  | 3 | Ca(5)H(1)O(13)P(3)      | 502.312 |

|                                                                                             |            |      |                                            |         |
|---------------------------------------------------------------------------------------------|------------|------|--------------------------------------------|---------|
| <b>Ca+2 (5) _OH-1 _PO4-3 (3) _H2O_ (s)</b><br>Hydroxyapatite monohydrate                    |            |      |                                            |         |
| 0                                                                                           | ---        | 1    | Ca (5) H (3) O (14) P (3)                  | 520.328 |
| <b>Ca+2 (8) _H+1 (2) _PO4-3 (6) _H2O (5) _ (s)</b>                                          |            |      |                                            |         |
| 0                                                                                           | ---        | 1    | Ca (8) H (12) O (29) P (6)                 | 982.546 |
| <b>Ca+2 (9) _Mg+2 _H+1 _PO4-3 (7) _ (s)</b><br>Whitlockite (Internat. Mineralogical Assoc.) |            |      |                                            |         |
| 0                                                                                           | ---        | 1    | Ca (9) H (1) Mg (1) O (28) P (7)           | 1050.82 |
| <b>Ca (s)</b><br>Calcium; Calcium, cubic; Calcium, fcc; Calcium, alpha                      |            |      |                                            |         |
| 0                                                                                           | 7440-70-2  | 184  | Ca (1)                                     | 40.0780 |
| <b>Ca3P2 (s)</b><br>Tricalcium diphosphide; Calcium phosphide                               |            |      |                                            |         |
| 0                                                                                           | 1305-99-3  | 1    | Ca (3) P (2)                               | 182.182 |
| <b>Cd+2</b><br>Cadmium(II) ion                                                              |            |      |                                            |         |
| 2                                                                                           | 22537-48-0 | 3554 | Cd (1)                                     | 112.410 |
| <b>Cd+2 _Cys-2 _PO4-3</b>                                                                   |            |      |                                            |         |
| -3                                                                                          | ---        | 1    | C (3) H (5) Cd (1) N (1) O (6) P (1) S (1) | 326.520 |
| <b>Cd+2 _H+1 _Citric-3 _PO4-3</b>                                                           |            |      |                                            |         |
| -3                                                                                          | ---        | 1    | C (6) H (6) Cd (1) O (11) P (1)            | 397.491 |
| <b>Cd+2 _H+1 _Cys-2 _PO4-3</b>                                                              |            |      |                                            |         |
| -2                                                                                          | ---        | 1    | C (3) H (6) Cd (1) N (1) O (6) P (1) S (1) | 327.528 |
| <b>Cd+2 _H+1 _NTA-3 _PO4-3</b>                                                              |            |      |                                            |         |
| -3                                                                                          | ---        | 1    | C (6) H (7) Cd (1) N (1) O (10) P (1)      | 396.506 |
| <b>Cd+2 _H+1 _P3O10-5</b>                                                                   |            |      |                                            |         |
| -2                                                                                          | ---        | 3    | Cd (1) H (1) O (10) P (3)                  | 366.334 |

|                                  |     |   |                        |         |
|----------------------------------|-----|---|------------------------|---------|
| <b>Cd+2_H+1_PO4-3</b>            |     |   |                        |         |
| 0                                | --- | 2 | Cd(1)H(1)O(4)P(1)      | 208.390 |
| <b>Cd+2_H+1_Tartaric-2_PO4-3</b> |     |   |                        |         |
| -2                               | --- | 1 | C(4)H(5)Cd(1)O(10)P(1) | 356.462 |
| <b>Cd+2_H+1(2)_P2O7-4</b>        |     |   |                        |         |
| 0                                | --- | 1 | Cd(1)H(2)O(7)P(2)      | 288.370 |
| <b>Cd+2_H+1(2)_PO4-3</b>         |     |   |                        |         |
| 1                                | --- | 3 | Cd(1)H(2)O(4)P(1)      | 209.398 |
| <b>Cd+2_H+1(2)_PO4-3(2)</b>      |     |   |                        |         |
| -2                               | --- | 2 | Cd(1)H(2)O(8)P(2)      | 304.369 |
| <b>Cd+2_H+1(3)_PO4-3(2)</b>      |     |   |                        |         |
| -1                               | --- | 2 | Cd(1)H(3)O(8)P(2)      | 305.377 |
| <b>Cd+2_H+1(3)_PO4-3(3)</b>      |     |   |                        |         |
| -4                               | --- | 2 | Cd(1)H(3)O(12)P(3)     | 400.349 |
| <b>Cd+2_H+1(4)_PO4-3(2)</b>      |     |   |                        |         |
| 0                                | --- | 2 | Cd(1)H(4)O(8)P(2)      | 306.385 |
| <b>Cd+2_OH-1_P2O7-4</b>          |     |   |                        |         |
| -3                               | --- | 2 | Cd(1)H(1)O(8)P(2)      | 303.361 |
| <b>Cd+2_OH-1_P3O10-5</b>         |     |   |                        |         |
| -4                               | --- | 2 | Cd(1)H(1)O(11)P(3)     | 382.333 |
| <b>Cd+2_P2O7-4</b>               |     |   |                        |         |
| -2                               | --- | 3 | Cd(1)O(7)P(2)          | 286.354 |
| <b>Cd+2_P2O7-4(2)</b>            |     |   |                        |         |
| -6                               | --- | 1 | Cd(1)O(14)P(4)         | 460.298 |



|                                                                                                  |            |     |                     |         |
|--------------------------------------------------------------------------------------------------|------------|-----|---------------------|---------|
| <b>Cd+2_P3O10-5</b>                                                                              |            |     |                     |         |
| -3                                                                                               | ---        | 3   | Cd(1)O(10)P(3)      | 365.326 |
| <b>Cd+2_P3O9-3</b>                                                                               |            |     |                     |         |
| -1                                                                                               | ---        | 1   | Cd(1)O(9)P(3)       | 349.327 |
| <b>Cd+2_P4O13-6</b>                                                                              |            |     |                     |         |
| -4                                                                                               | ---        | 1   | Cd(1)O(13)P(4)      | 444.298 |
| <b>Cd+2(3)_P3O10-5(4)</b>                                                                        |            |     |                     |         |
| -14                                                                                              | ---        | 1   | Cd(3)O(40)P(12)     | 1348.89 |
| <b>Cd+2(3)_PO4-3(2)_(s)</b><br>Cadmium phosphate                                                 |            |     |                     |         |
| 0                                                                                                | 13477-17-3 | 2   | Cd(3)O(8)P(2)       | 527.173 |
| <b>Cd+2(5)_H+1(2)_PO4-3(4)_H2O(4)_(s)</b>                                                        |            |     |                     |         |
| 0                                                                                                | ---        | 1   | Cd(5)H(10)O(20)P(4) | 1016.01 |
| <b>Cd+2(5)_OH-1_PO4-3(3)_(s)</b><br>Cadmium hydroxyapatite; Hydroxyapatite - cadmium substituted |            |     |                     |         |
| 0                                                                                                | ---        | 2   | Cd(5)H(1)O(13)P(3)  | 863.972 |
| <b>Cd(s)</b><br>Cadmium; Cadmium, hexagonal; Cadmium, gamma                                      |            |     |                     |         |
| 0                                                                                                | 7440-43-9  | 97  | Cd(1)               | 112.410 |
| <b>Ce+2</b>                                                                                      |            |     |                     |         |
| 2                                                                                                | ---        | 7   | Ce(1)               | 140.120 |
| <b>Ce+2_P2O7-4</b>                                                                               |            |     |                     |         |
| -2                                                                                               | ---        | 1   | Ce(1)O(7)P(2)       | 314.064 |
| <b>Ce+3</b><br>Cerium(III) ion                                                                   |            |     |                     |         |
| 3                                                                                                | 18923-26-7 | 627 | Ce(1)               | 140.120 |
| <b>Ce+3_EDTA-4</b>                                                                               |            |     |                     |         |

|                                 |     |    |                                         |         |
|---------------------------------|-----|----|-----------------------------------------|---------|
| -1                              | --- | 41 | C (10) H (12) Ce (1) N (2) O (8)        | 428.334 |
| <b>Ce+3_EDTA-4_P3O10-5</b>      |     |    |                                         |         |
| -6                              | --- | 2  | C (10) H (12) Ce (1) N (2) O (18) P (3) | 681.250 |
| <b>Ce+3_H+1_EDTA-4_P3O10-5</b>  |     |    |                                         |         |
| -5                              | --- | 2  | C (10) H (13) Ce (1) N (2) O (18) P (3) | 682.258 |
| <b>Ce+3_H+1_PO4-3</b>           |     |    |                                         |         |
| 1                               | --- | 2  | Ce (1) H (1) O (4) P (1)                | 236.100 |
| <b>Ce+3_H+1 (2)_P3O10-5 (2)</b> |     |    |                                         |         |
| -5                              | --- | 1  | Ce (1) H (2) O (20) P (6)               | 647.968 |
| <b>Ce+3_H+1 (2)_PO4-3</b>       |     |    |                                         |         |
| 2                               | --- | 2  | Ce (1) H (2) O (4) P (1)                | 237.108 |
| <b>Ce+3_H+1 (2)_PO4-3 (2)</b>   |     |    |                                         |         |
| -1                              | --- | 2  | Ce (1) H (2) O (8) P (2)                | 332.079 |
| <b>Ce+3_P2O7-4</b>              |     |    |                                         |         |
| -1                              | --- | 1  | Ce (1) O (7) P (2)                      | 314.064 |
| <b>Ce+3_P2O7-4_(s)</b>          |     |    |                                         |         |
| -1                              | --- | 1  | Ce (1) O (7) P (2)                      | 314.064 |
| <b>Ce+3_P2O7-4 (2)</b>          |     |    |                                         |         |
| -5                              | --- | 1  | Ce (1) O (14) P (4)                     | 488.008 |
| <b>Ce+3_P3O10-5</b>             |     |    |                                         |         |
| -2                              | --- | 1  | Ce (1) O (10) P (3)                     | 393.036 |
| <b>Ce+3_P3O10-5 (2)</b>         |     |    |                                         |         |
| -7                              | --- | 2  | Ce (1) O (20) P (6)                     | 645.952 |
| <b>Ce+3_PO4-3</b>               |     |    |                                         |         |
| 0                               | --- | 1  | Ce (1) O (4) P (1)                      | 235.092 |

|                                                                                                                                                                                                                                                                             |            |     |                       |         |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----|-----------------------|---------|
| <b>Ce+3_PO4-3_(s)</b><br>Cerium(III) Phosphate                                                                                                                                                                                                                              |            |     |                       |         |
| 0                                                                                                                                                                                                                                                                           | ---        | 1   | Ce(1)O(4)P(1)         | 235.092 |
| <b>Ce+3_PO4-3(2)</b>                                                                                                                                                                                                                                                        |            |     |                       |         |
| -3                                                                                                                                                                                                                                                                          | ---        | 1   | Ce(1)O(8)P(2)         | 330.063 |
| <b>Ce+3(2)_P2O7-4</b>                                                                                                                                                                                                                                                       |            |     |                       |         |
| 2                                                                                                                                                                                                                                                                           | ---        | 1   | Ce(2)O(7)P(2)         | 454.184 |
| <b>Ce+4</b><br>Cerium(IV) ion                                                                                                                                                                                                                                               |            |     |                       |         |
| 4                                                                                                                                                                                                                                                                           | 16065-90-0 | 69  | Ce(1)                 | 140.120 |
| <b>Ce+4_H+1(2)_P2O7-4</b>                                                                                                                                                                                                                                                   |            |     |                       |         |
| 2                                                                                                                                                                                                                                                                           | ---        | 1   | Ce(1)H(2)O(7)P(2)     | 316.080 |
| <b>Ce+4_H2PO2-1</b>                                                                                                                                                                                                                                                         |            |     |                       |         |
| 3                                                                                                                                                                                                                                                                           | ---        | 3   | Ce(1)H(2)O(2)P(1)     | 205.109 |
| <b>Ce+4_H2PO2-1(2)</b>                                                                                                                                                                                                                                                      |            |     |                       |         |
| 2                                                                                                                                                                                                                                                                           | ---        | 2   | Ce(1)H(4)O(4)P(2)     | 270.097 |
| <b>Ce+4_H2PO2-1(3)</b>                                                                                                                                                                                                                                                      |            |     |                       |         |
| 1                                                                                                                                                                                                                                                                           | ---        | 1   | Ce(1)H(6)O(6)P(3)     | 335.086 |
| <b>Ce+4_P2O7-4</b>                                                                                                                                                                                                                                                          |            |     |                       |         |
| 0                                                                                                                                                                                                                                                                           | ---        | 1   | Ce(1)O(7)P(2)         | 314.064 |
| <b>Ce+4_P2O7-4_(s)</b>                                                                                                                                                                                                                                                      |            |     |                       |         |
| 0                                                                                                                                                                                                                                                                           | ---        | 1   | Ce(1)O(7)P(2)         | 314.064 |
| <b>Cis-2</b><br>Cystinate ion; 3,3'-dithiobis(2-aminopropanoate) ion; dicysteinate ion; beta,beta'-dithiodialanate ion; alpha-diamino-beta-dithiolactate ion; beta,beta'-diamino-beta,beta'-dicarboxydiethyl disulfide ion; bis(beta-amino-beta-carboxyethyl) disulfide ion |            |     |                       |         |
| -2                                                                                                                                                                                                                                                                          | 58823-23-7 | 314 | C(6)H(10)N(2)O(4)S(2) | 238.276 |

|                                |            |      |                   |         |
|--------------------------------|------------|------|-------------------|---------|
| <b>Citric-3</b><br>Citrate ion |            |      |                   |         |
| -3                             | 126-44-3   | 1347 | C(6)H(5)O(7)      | 189.102 |
| <b>Citrul-1</b>                |            |      |                   |         |
| -1                             | ---        | 394  | C(6)H(12)N(3)O(3) | 174.180 |
| <b>Cl-1</b><br>Chloride ion    |            |      |                   |         |
| -1                             | 16887-00-6 | 2242 | Cl(1)             | 35.4530 |
| <b>Cl2(g)</b><br>Chlorine gas  |            |      |                   |         |
| 0                              | 7782-50-5  | 589  | Cl(2)             | 70.9060 |
| <b>Cm+3</b><br>Curium(III) ion |            |      |                   |         |
| 3                              | 22541-42-0 | 162  | Cm(1)             | 247.000 |
| <b>Cm+3_H+1(2)_PO4-3</b>       |            |      |                   |         |
| 2                              | ---        | 1    | Cm(1)H(2)O(4)P(1) | 343.988 |
| <b>Cm+3_P3O9-3</b>             |            |      |                   |         |
| 0                              | ---        | 1    | Cm(1)O(9)P(3)     | 483.917 |
| <b>Cm+3_PO4-3</b>              |            |      |                   |         |
| 0                              | ---        | 2    | Cm(1)O(4)P(1)     | 341.972 |
| <b>Cm+3_PO4-3(2)</b>           |            |      |                   |         |
| -3                             | ---        | 1    | Cm(1)O(8)P(2)     | 436.943 |
| <b>Cm+3_PO4-3(3)</b>           |            |      |                   |         |
| -6                             | ---        | 1    | Cm(1)O(12)P(3)    | 531.915 |
| <b>Cm+3_PO4-3(4)</b>           |            |      |                   |         |
| -9                             | ---        | 1    | Cm(1)O(16)P(4)    | 626.886 |
| <b>Cm+4</b>                    |            |      |                   |         |

|                                  |            |      |                              |         |
|----------------------------------|------------|------|------------------------------|---------|
| 4                                | ---        | 14   | Cm(1)                        | 247.000 |
| <b>Co+2</b><br>Cobalt(II) ion    |            |      |                              |         |
| 2                                | 22541-53-3 | 2857 | Co(1)                        | 58.9330 |
| <b>Co+2_Bipy</b>                 |            |      |                              |         |
| 2                                | ---        | 12   | C(10)H(8)Co(1)N(2)           | 215.120 |
| <b>Co+2_H+1_Bipy_PO4-3</b>       |            |      |                              |         |
| 0                                | ---        | 1    | C(10)H(9)Co(1)N(2)O(4)P(1)   | 311.100 |
| <b>Co+2_H+1_HPO3-2</b>           |            |      |                              |         |
| 1                                | ---        | 1    | Co(1)H(2)O(3)P(1)            | 139.921 |
| <b>Co+2_H+1_NTA-3_PO4-3</b>      |            |      |                              |         |
| -3                               | ---        | 1    | C(6)H(7)Co(1)N(1)O(10)P(1)   | 343.029 |
| <b>Co+2_H+1_P2O7-4</b>           |            |      |                              |         |
| -1                               | ---        | 3    | Co(1)H(1)O(7)P(2)            | 233.885 |
| <b>Co+2_H+1_P3O10-5</b>          |            |      |                              |         |
| -2                               | ---        | 3    | Co(1)H(1)O(10)P(3)           | 312.857 |
| <b>Co+2_H+1_P3O10-5(2)</b>       |            |      |                              |         |
| -7                               | ---        | 1    | Co(1)H(1)O(20)P(6)           | 565.773 |
| <b>Co+2_H+1_PO4-3</b>            |            |      |                              |         |
| 0                                | ---        | 3    | Co(1)H(1)O(4)P(1)            | 154.913 |
| <b>Co+2_H+1_PO4-3(s)</b>         |            |      |                              |         |
| 0                                | ---        | 2    | Co(1)H(1)O(4)P(1)            | 154.913 |
| <b>Co+2_H+1(2)_[30]N10_PO4-3</b> |            |      |                              |         |
| 1                                | ---        | 1    | C(20)H(52)Co(1)N(10)O(4)P(1) | 586.604 |
| <b>Co+2_H+1(2)_PO4-3</b>         |            |      |                              |         |

|                                  |     |    |                              |         |
|----------------------------------|-----|----|------------------------------|---------|
| 1                                | --- | 2  | Co(1)H(2)O(4)P(1)            | 155.921 |
| <b>Co+2_H+1(4)_[30]N10_PO4-3</b> |     |    |                              |         |
| 3                                | --- | 1  | C(20)H(54)Co(1)N(10)O(4)P(1) | 588.620 |
| <b>Co+2_H+1(4)_PO4-3(2)</b>      |     |    |                              |         |
| 0                                | --- | 2  | Co(1)H(4)O(8)P(2)            | 252.908 |
| <b>Co+2_HPO3-2</b>               |     |    |                              |         |
| 0                                | --- | 1  | Co(1)H(1)O(3)P(1)            | 138.913 |
| <b>Co+2_NTA-3</b>                |     |    |                              |         |
| -1                               | --- | 33 | C(6)H(6)Co(1)N(1)O(6)        | 247.050 |
| <b>Co+2_P2O7-4</b>               |     |    |                              |         |
| -2                               | --- | 3  | Co(1)O(7)P(2)                | 232.877 |
| <b>Co+2_P2O7-4(2)</b>            |     |    |                              |         |
| -6                               | --- | 1  | Co(1)O(14)P(4)               | 406.821 |
| <b>Co+2_P3O10-5</b>              |     |    |                              |         |
| -3                               | --- | 5  | Co(1)O(10)P(3)               | 311.849 |
| <b>Co+2_P3O10-5(2)</b>           |     |    |                              |         |
| -8                               | --- | 1  | Co(1)O(20)P(6)               | 564.765 |
| <b>Co+2_P3O9-3</b>               |     |    |                              |         |
| -1                               | --- | 1  | Co(1)O(9)P(3)                | 295.850 |
| <b>Co+2_P4O12-4</b>              |     |    |                              |         |
| -2                               | --- | 1  | Co(1)O(12)P(4)               | 374.822 |
| <b>Co+2_P6O18-6</b>              |     |    |                              |         |
| -4                               | --- | 1  | Co(1)O(18)P(6)               | 532.766 |
| <b>Co+2_P8O24-8</b>              |     |    |                              |         |
| -6                               | --- | 1  | Co(1)O(24)P(8)               | 690.711 |

|                                                                             |            |     |                                         |         |
|-----------------------------------------------------------------------------|------------|-----|-----------------------------------------|---------|
| <b>Co+2_PO4-3</b>                                                           |            |     |                                         |         |
| -1                                                                          | ---        | 1   | Co (1) O (4) P (1)                      | 153.905 |
| <b>Co+2_Pyrid2Azo4DiMeAniline_P3O10-5</b>                                   |            |     |                                         |         |
| -3                                                                          | ---        | 1   | C (13) H (14) Co (1) N (4) O (10) P (3) | 538.130 |
| <b>Co+2 (2)_H+1_[30]N10_PO4-3</b>                                           |            |     |                                         |         |
| 2                                                                           | ---        | 1   | C (20) H (51) Co (2) N (10) O (4) P (1) | 644.530 |
| <b>Co+2 (2)_H+1 (2)_ [30]N10_PO4-3</b>                                      |            |     |                                         |         |
| 3                                                                           | ---        | 1   | C (20) H (52) Co (2) N (10) O (4) P (1) | 645.538 |
| <b>Co+2 (2)_H+1 (3)_ [30]N10_PO4-3</b>                                      |            |     |                                         |         |
| 4                                                                           | ---        | 1   | C (20) H (53) Co (2) N (10) O (4) P (1) | 646.546 |
| <b>Co+2 (2)_P2O7-4_H2O (6)_ (s)</b>                                         |            |     |                                         |         |
| 0                                                                           | ---        | 1   | Co (2) H (12) O (13) P (2)              | 399.902 |
| <b>Co+2 (3)_PO4-3 (2)_ (s)</b><br>Cobaltous phosphate; Cobalt(II) phosphate |            |     |                                         |         |
| 0                                                                           | 13455-36-2 | 2   | Co (3) O (8) P (2)                      | 366.742 |
| <b>Co+3</b><br>Cobalt(III) ion                                              |            |     |                                         |         |
| 3                                                                           | 22541-63-5 | 697 | Co (1)                                  | 58.9330 |
| <b>Co+3_EtDiAm (2)_OH-1_PO4-3</b>                                           |            |     |                                         |         |
| -1                                                                          | ---        | 1   | C (4) H (17) Co (1) N (4) O (5) P (1)   | 291.110 |
| <b>Co+3_EtDiAm (2)_PO4-3</b>                                                |            |     |                                         |         |
| 0                                                                           | ---        | 1   | C (4) H (16) Co (1) N (4) O (4) P (1)   | 274.102 |
| <b>Co+3_EtDiAm (3)</b>                                                      |            |     |                                         |         |
| 3                                                                           | ---        | 38  | C (6) H (24) Co (1) N (6)               | 239.230 |
| <b>Co+3_EtDiAm (3)_PO4-3</b>                                                |            |     |                                         |         |
| 0                                                                           | ---        | 1   | C (6) H (24) Co (1) N (6) O (4) P (1)   | 334.201 |

|                                         |     |   |                            |         |
|-----------------------------------------|-----|---|----------------------------|---------|
| <b>Co+3_H+1_EtDiAm(2)_OH-1_PO4-3</b>    |     |   |                            |         |
| 0                                       | --- | 2 | C(4)H(18)Co(1)N(4)O(5)P(1) | 292.118 |
| <b>Co+3_H+1_EtDiAm(2)_PO4-3</b>         |     |   |                            |         |
| 1                                       | --- | 1 | C(4)H(17)Co(1)N(4)O(4)P(1) | 275.110 |
| <b>Co+3_H+1_EtDiAm(2)_PO4-3_H2O</b>     |     |   |                            |         |
| 1                                       | --- | 1 | C(4)H(19)Co(1)N(4)O(5)P(1) | 293.126 |
| <b>Co+3_H+1_EtDiAm(3)_PO4-3</b>         |     |   |                            |         |
| 1                                       | --- | 1 | C(6)H(25)Co(1)N(6)O(4)P(1) | 335.209 |
| <b>Co+3_H+1_EtDiAm(3)_PO4-3(2)</b>      |     |   |                            |         |
| -2                                      | --- | 1 | C(6)H(25)Co(1)N(6)O(8)P(2) | 430.181 |
| <b>Co+3_H+1_NH3(4)_OH-1_PO4-3</b>       |     |   |                            |         |
| 0                                       | --- | 2 | Co(1)H(14)N(4)O(5)P(1)     | 240.042 |
| <b>Co+3_H+1_NH3(4)_OH-1(2)_PO4-3</b>    |     |   |                            |         |
| -1                                      | --- | 1 | Co(1)H(15)N(4)O(6)P(1)     | 257.049 |
| <b>Co+3_H+1_NH3(5)_PO4-3</b>            |     |   |                            |         |
| 1                                       | --- | 3 | Co(1)H(16)N(5)O(4)P(1)     | 240.065 |
| <b>Co+3_H+1(2)_EtDiAm(2)_OH-1_PO4-3</b> |     |   |                            |         |
| 1                                       | --- | 1 | C(4)H(19)Co(1)N(4)O(5)P(1) | 293.126 |
| <b>Co+3_H+1(2)_EtDiAm(2)_PO4-3_H2O</b>  |     |   |                            |         |
| 2                                       | --- | 1 | C(4)H(20)Co(1)N(4)O(5)P(1) | 294.134 |
| <b>Co+3_H+1(2)_NH3(4)_OH-1_PO4-3</b>    |     |   |                            |         |
| 1                                       | --- | 1 | Co(1)H(15)N(4)O(5)P(1)     | 241.050 |
| <b>Co+3_H+1(2)_NH3(4)_OH-1(2)_PO4-3</b> |     |   |                            |         |
| 0                                       | --- | 2 | Co(1)H(16)N(4)O(6)P(1)     | 258.057 |



|                                         |     |    |                              |         |
|-----------------------------------------|-----|----|------------------------------|---------|
| <b>Co+3_H+1(2)_NH3(5)_PO4-3</b>         |     |    |                              |         |
| 2                                       | --- | 3  | Co(1)H(17)N(5)O(4)P(1)       | 241.073 |
| <b>Co+3_H+1(3)_NH3(4)_OH-1(2)_PO4-3</b> |     |    |                              |         |
| 1                                       | --- | 1  | Co(1)H(17)N(4)O(6)P(1)       | 259.065 |
| <b>Co+3_H+1(3)_NH3(5)_PO4-3</b>         |     |    |                              |         |
| 3                                       | --- | 3  | Co(1)H(18)N(5)O(4)P(1)       | 242.081 |
| <b>Co+3_NH3(4)_OH-1_PO4-3</b>           |     |    |                              |         |
| -1                                      | --- | 1  | Co(1)H(13)N(4)O(5)P(1)       | 239.034 |
| <b>Co+3_NH3(5)</b>                      |     |    |                              |         |
| 3                                       | --- | 14 | Co(1)H(15)N(5)               | 144.086 |
| <b>Co+3_NH3(5)_Cl-1</b>                 |     |    |                              |         |
| 2                                       | --- | 21 | Cl(1)Co(1)H(15)N(5)          | 179.539 |
| <b>Co+3_NH3(5)_Cl-1_P2O7-4</b>          |     |    |                              |         |
| -2                                      | --- | 1  | Cl(1)Co(1)H(15)N(5)O(7)P(2)  | 353.482 |
| <b>Co+3_NH3(5)_Cl-1_P3O10-5</b>         |     |    |                              |         |
| -3                                      | --- | 1  | Cl(1)Co(1)H(15)N(5)O(10)P(3) | 432.455 |
| <b>Co+3_NH3(5)_PO4-3</b>                |     |    |                              |         |
| 0                                       | --- | 1  | Co(1)H(15)N(5)O(4)P(1)       | 239.057 |
| <b>Co+3_NH3(6)</b>                      |     |    |                              |         |
| 3                                       | --- | 39 | Co(1)H(18)N(6)               | 161.116 |
| <b>Co+3_NH3(6)_P3O9-3</b>               |     |    |                              |         |
| 0                                       | --- | 1  | Co(1)H(18)N(6)O(9)P(3)       | 398.033 |
| <b>Co+3_NH3(6)_P4O12-4</b>              |     |    |                              |         |
| -1                                      | --- | 1  | Co(1)H(18)N(6)O(12)P(4)      | 477.005 |

|                                                     |            |      |                          |         |
|-----------------------------------------------------|------------|------|--------------------------|---------|
| <b>Co (s)</b><br>Cobalt; Cobalt, hcp; Cobalt, alpha |            |      |                          |         |
| 0                                                   | 7440-48-4  | 133  | Co (1)                   | 58.9330 |
| <b>CO2 (g)</b><br>Carbon dioxide gas                |            |      |                          |         |
| 0                                                   | 124-38-9   | 110  | C (1) O (2)              | 44.0098 |
| <b>Co2P (s)</b><br>Dicobalt phosphide               |            |      |                          |         |
| 0                                                   | ---        | 1    | Co (2) P (1)             | 148.840 |
| <b>CO3-2</b><br>Carbonate ion                       |            |      |                          |         |
| -2                                                  | 3812-32-6  | 1435 | C (1) O (3)              | 60.0092 |
| <b>CoP (s)</b><br>Cobalt monophosphide              |            |      |                          |         |
| 0                                                   | ---        | 1    | Co (1) P (1)             | 89.9070 |
| <b>CoP3 (s)</b><br>Cobalt triphosphide              |            |      |                          |         |
| 0                                                   | ---        | 1    | Co (1) P (3)             | 151.855 |
| <b>Cr+3</b><br>Chromium(III) ion                    |            |      |                          |         |
| 3                                                   | 16065-83-1 | 540  | Cr (1)                   | 51.9960 |
| <b>Cr+3_H+1_PO4-3</b>                               |            |      |                          |         |
| 1                                                   | ---        | 2    | Cr (1) H (1) O (4) P (1) | 147.976 |
| <b>Cr+3_H+1 (2)_PO4-3</b>                           |            |      |                          |         |
| 2                                                   | ---        | 2    | Cr (1) H (2) O (4) P (1) | 148.984 |
| <b>Cr+3_H2PO2-1</b>                                 |            |      |                          |         |
| 2                                                   | ---        | 2    | Cr (1) H (2) O (2) P (1) | 116.985 |
| <b>Cr+3_H2PO2-1 (2)</b>                             |            |      |                          |         |
| 1                                                   | ---        | 1    | Cr (1) H (4) O (4) P (2) | 181.973 |

|                                              |            |     |                            |         |
|----------------------------------------------|------------|-----|----------------------------|---------|
| <b>Cr+3_OH-1_PO4-3</b>                       |            |     |                            |         |
| -1                                           | ---        | 2   | Cr (1) H (1) O (5) P (1)   | 163.975 |
| <b>Cr+3_OH-1_PO4-3 (2)</b>                   |            |     |                            |         |
| -4                                           | ---        | 2   | Cr (1) H (1) O (9) P (2)   | 258.947 |
| <b>Cr+3_OH-1 (2)</b>                         |            |     |                            |         |
| 1                                            | ---        | 22  | Cr (1) H (2) O (2)         | 86.0107 |
| <b>Cr+3_OH-1 (2)_PO4-3</b>                   |            |     |                            |         |
| -2                                           | ---        | 2   | Cr (1) H (2) O (6) P (1)   | 180.982 |
| <b>Cr+3_OH-1 (3)</b>                         |            |     |                            |         |
| 0                                            | ---        | 14  | Cr (1) H (3) O (3)         | 103.018 |
| <b>Cr+3_OH-1 (3)_PO4-3</b>                   |            |     |                            |         |
| -3                                           | ---        | 2   | Cr (1) H (3) O (7) P (1)   | 197.990 |
| <b>CrO3H2PO4-1</b>                           |            |     |                            |         |
| -1                                           | ---        | 1   | Cr (1) H (2) O (7) P (1)   | 196.982 |
| <b>CrO3HPO4-2</b>                            |            |     |                            |         |
| -2                                           | ---        | 2   | Cr (1) H (1) O (7) P (1)   | 195.974 |
| <b>CrO4-2</b><br>Chromate ion                |            |     |                            |         |
| -2                                           | 11104-59-9 | 127 | Cr (1) O (4)               | 115.994 |
| <b>Cs+1</b><br>Caesium(I) ion; Cesium(I) ion |            |     |                            |         |
| 1                                            | 18459-37-5 | 246 | Cs (1)                     | 132.910 |
| <b>Cs+1_H+1_PO4-3</b>                        |            |     |                            |         |
| -1                                           | ---        | 2   | Cs (1) H (1) O (4) P (1)   | 228.890 |
| <b>Cs+1_H+1 (10)_PO4-3 (5)</b>               |            |     |                            |         |
| -4                                           | ---        | 2   | Cs (1) H (10) O (20) P (5) | 617.847 |

|                                |     |   |                            |         |
|--------------------------------|-----|---|----------------------------|---------|
| <b>Cs+1_H+1 (12)_PO4-3 (6)</b> |     |   |                            |         |
| -5                             | --- | 2 | Cs (1) H (12) O (24) P (6) | 714.835 |
| <b>Cs+1_H+1 (2)_PO4-3</b>      |     |   |                            |         |
| 0                              | --- | 2 | Cs (1) H (2) O (4) P (1)   | 229.898 |
| <b>Cs+1_H+1 (2)_PO4-3 (2)</b>  |     |   |                            |         |
| -3                             | --- | 2 | Cs (1) H (2) O (8) P (2)   | 324.869 |
| <b>Cs+1_H+1 (3)_PO4-3 (3)</b>  |     |   |                            |         |
| -5                             | --- | 2 | Cs (1) H (3) O (12) P (3)  | 420.849 |
| <b>Cs+1_H+1 (4)_PO4-3 (2)</b>  |     |   |                            |         |
| -1                             | --- | 2 | Cs (1) H (4) O (8) P (2)   | 326.885 |
| <b>Cs+1_H+1 (4)_PO4-3 (4)</b>  |     |   |                            |         |
| -7                             | --- | 2 | Cs (1) H (4) O (16) P (4)  | 516.828 |
| <b>Cs+1_H+1 (5)_PO4-3 (5)</b>  |     |   |                            |         |
| -9                             | --- | 2 | Cs (1) H (5) O (20) P (5)  | 612.808 |
| <b>Cs+1_H+1 (6)_PO4-3 (3)</b>  |     |   |                            |         |
| -2                             | --- | 2 | Cs (1) H (6) O (12) P (3)  | 423.872 |
| <b>Cs+1_H+1 (6)_PO4-3 (6)</b>  |     |   |                            |         |
| -11                            | --- | 2 | Cs (1) H (6) O (24) P (6)  | 708.787 |
| <b>Cs+1_H+1 (8)_PO4-3 (4)</b>  |     |   |                            |         |
| -3                             | --- | 2 | Cs (1) H (8) O (16) P (4)  | 520.860 |
| <b>Cs+1_P2O7-4</b>             |     |   |                            |         |
| -3                             | --- | 1 | Cs (1) O (7) P (2)         | 306.854 |
| <b>Cs+1_P3O10-5</b>            |     |   |                            |         |
| -4                             | --- | 1 | Cs (1) O (10) P (3)        | 385.826 |

|                                           |            |     |                           |         |
|-------------------------------------------|------------|-----|---------------------------|---------|
| <b>Cs+1_P4O12-4</b>                       |            |     |                           |         |
| -3                                        | ---        | 1   | Cs (1) O (12) P (4)       | 448.799 |
| <b>Cs+1_P6O18-6</b>                       |            |     |                           |         |
| -5                                        | ---        | 1   | Cs (1) O (18) P (6)       | 606.743 |
| <b>Cu+1</b><br>Copper(I) ion; Cuprous ion |            |     |                           |         |
| 1                                         | 17493-86-6 | 491 | Cu (1)                    | 63.5460 |
| <b>Cu+1_H+1 (2)_PO4-3</b>                 |            |     |                           |         |
| 0                                         | ---        | 2   | Cu (1) H (2) O (4) P (1)  | 160.534 |
| <b>Cu+1_H+1 (3)_PO4-3 (2)</b>             |            |     |                           |         |
| -2                                        | ---        | 2   | Cu (1) H (3) O (8) P (2)  | 256.513 |
| <b>Cu+1_H+1 (4)_PO4-3 (2)</b>             |            |     |                           |         |
| -1                                        | ---        | 2   | Cu (1) H (4) O (8) P (2)  | 257.521 |
| <b>Cu+1_P2O7-4 (2)</b>                    |            |     |                           |         |
| -7                                        | ---        | 1   | Cu (1) O (14) P (4)       | 411.434 |
| <b>Cu+1_PH3_Br-1_(s)</b>                  |            |     |                           |         |
| 0                                         | ---        | 1   | Br (1) Cu (1) H (3) P (1) | 177.448 |
| <b>Cu+1_PH3_Cl-1_(s)</b>                  |            |     |                           |         |
| 0                                         | ---        | 1   | Cl (1) Cu (1) H (3) P (1) | 132.997 |
| <b>Cu+1_PH3_I-1_(s)</b>                   |            |     |                           |         |
| 0                                         | ---        | 1   | Cu (1) H (3) I (1) P (1)  | 224.444 |
| <b>Cu+1_PH3 (2)_Br-1_(s)</b>              |            |     |                           |         |
| 0                                         | ---        | 1   | Br (1) Cu (1) H (6) P (2) | 211.446 |
| <b>Cu+1_PH3 (2)_Cl-1_(s)</b>              |            |     |                           |         |
| 0                                         | ---        | 1   | Cl (1) Cu (1) H (6) P (2) | 166.995 |

|                                           |            |      |                            |         |
|-------------------------------------------|------------|------|----------------------------|---------|
| <b>Cu+1_PH3(2)_I-1_(s)</b>                |            |      |                            |         |
| 0                                         | ---        | 1    | Cu(1)H(6)I(1)P(2)          | 258.442 |
| <b>Cu+2</b><br>Copper(II) ion; Cupric ion |            |      |                            |         |
| 2                                         | 15158-11-9 | 9346 | Cu(1)                      | 63.5460 |
| <b>Cu+2_Bipy</b>                          |            |      |                            |         |
| 2                                         | ---        | 155  | C(10)H(8)Cu(1)N(2)         | 219.733 |
| <b>Cu+2_EtDiAm_P2O7-4</b>                 |            |      |                            |         |
| -2                                        | ---        | 1    | C(2)H(8)Cu(1)N(2)O(7)P(2)  | 297.589 |
| <b>Cu+2_EtDiAm_P3O10-5</b>                |            |      |                            |         |
| -3                                        | ---        | 1    | C(2)H(8)Cu(1)N(2)O(10)P(3) | 376.561 |
| <b>Cu+2_H+1_Ala-1_PO4-3</b>               |            |      |                            |         |
| -1                                        | ---        | 1    | C(3)H(7)Cu(1)N(1)O(6)P(1)  | 247.612 |
| <b>Cu+2_H+1_Arg-1_PO4-3</b>               |            |      |                            |         |
| -1                                        | ---        | 1    | C(6)H(14)Cu(1)N(4)O(6)P(1) | 332.720 |
| <b>Cu+2_H+1_Asn-1_PO4-3</b>               |            |      |                            |         |
| -1                                        | ---        | 1    | C(4)H(8)Cu(1)N(2)O(7)P(1)  | 290.637 |
| <b>Cu+2_H+1_Asp-2_PO4-3</b>               |            |      |                            |         |
| -2                                        | ---        | 1    | C(4)H(6)Cu(1)N(1)O(8)P(1)  | 290.614 |
| <b>Cu+2_H+1_bAla-1_PO4-3</b>              |            |      |                            |         |
| -1                                        | ---        | 1    | C(3)H(7)Cu(1)N(1)O(6)P(1)  | 247.612 |
| <b>Cu+2_H+1_Bipy_PO4-3</b>                |            |      |                            |         |
| 0                                         | ---        | 1    | C(10)H(9)Cu(1)N(2)O(4)P(1) | 315.713 |
| <b>Cu+2_H+1_Citrul-1_PO4-3</b>            |            |      |                            |         |
| -1                                        | ---        | 1    | C(6)H(13)Cu(1)N(3)O(7)P(1) | 333.705 |

|                             |     |   |                                |         |
|-----------------------------|-----|---|--------------------------------|---------|
| <b>Cu+2_H+1_Gln-1_PO4-3</b> |     |   |                                |         |
| -1                          | --- | 1 | C(5)H(10)Cu(1)N(2)O(7)P(1)     | 304.664 |
| <b>Cu+2_H+1_Gly-1_PO4-3</b> |     |   |                                |         |
| -1                          | --- | 1 | C(2)H(5)Cu(1)N(1)O(6)P(1)      | 233.585 |
| <b>Cu+2_H+1_His-1_PO4-3</b> |     |   |                                |         |
| -1                          | --- | 1 | C(6)H(9)Cu(1)N(3)O(6)P(1)      | 313.674 |
| <b>Cu+2_H+1_Hyp-1_PO4-3</b> |     |   |                                |         |
| -1                          | --- | 1 | C(5)H(9)Cu(1)N(1)O(7)P(1)      | 289.649 |
| <b>Cu+2_H+1_Ile-1_PO4-3</b> |     |   |                                |         |
| -1                          | --- | 1 | C(6)H(13)Cu(1)N(1)O(6)P(1)     | 289.692 |
| <b>Cu+2_H+1_Leu-1_PO4-3</b> |     |   |                                |         |
| -1                          | --- | 1 | C(6)H(13)Cu(1)N(1)O(6)P(1)     | 289.692 |
| <b>Cu+2_H+1_Met-1_PO4-3</b> |     |   |                                |         |
| -1                          | --- | 1 | C(5)H(11)Cu(1)N(1)O(6)P(1)S(1) | 307.725 |
| <b>Cu+2_H+1_NTA-3_PO4-3</b> |     |   |                                |         |
| -3                          | --- | 1 | C(6)H(7)Cu(1)N(1)O(10)P(1)     | 347.642 |
| <b>Cu+2_H+1_P2O7-4</b>      |     |   |                                |         |
| -1                          | --- | 3 | Cu(1)H(1)O(7)P(2)              | 238.498 |
| <b>Cu+2_H+1_P2O7-4(2)</b>   |     |   |                                |         |
| -5                          | --- | 3 | Cu(1)H(1)O(14)P(4)             | 412.442 |
| <b>Cu+2_H+1_P3O10-5</b>     |     |   |                                |         |
| -2                          | --- | 3 | Cu(1)H(1)O(10)P(3)             | 317.470 |
| <b>Cu+2_H+1_P3O10-5(2)</b>  |     |   |                                |         |
| -7                          | --- | 1 | Cu(1)H(1)O(20)P(6)             | 570.386 |

|                                  |     |   |                              |         |
|----------------------------------|-----|---|------------------------------|---------|
| <b>Cu+2_H+1_P4O13-6</b>          |     |   |                              |         |
| -3                               | --- | 1 | Cu(1)H(1)O(13)P(4)           | 396.442 |
| <b>Cu+2_H+1_P4O13-6(2)</b>       |     |   |                              |         |
| -9                               | --- | 2 | Cu(1)H(1)O(26)P(8)           | 728.330 |
| <b>Cu+2_H+1_Phe-1_PO4-3</b>      |     |   |                              |         |
| -1                               | --- | 1 | C(9)H(11)Cu(1)N(1)O(6)P(1)   | 323.709 |
| <b>Cu+2_H+1_Phenanth_PO4-3</b>   |     |   |                              |         |
| 0                                | --- | 1 | C(12)H(9)Cu(1)N(2)O(4)P(1)   | 339.734 |
| <b>Cu+2_H+1_PO4-3</b>            |     |   |                              |         |
| 0                                | --- | 3 | Cu(1)H(1)O(4)P(1)            | 159.526 |
| <b>Cu+2_H+1_Pro-1_PO4-3</b>      |     |   |                              |         |
| -1                               | --- | 1 | C(5)H(9)Cu(1)N(1)O(6)P(1)    | 273.650 |
| <b>Cu+2_H+1_Ser-1_PO4-3</b>      |     |   |                              |         |
| -1                               | --- | 1 | C(3)H(7)Cu(1)N(1)O(7)P(1)    | 263.611 |
| <b>Cu+2_H+1_Spermine_P3O10-5</b> |     |   |                              |         |
| -2                               | --- | 1 | C(10)H(27)Cu(1)N(4)O(10)P(3) | 519.813 |
| <b>Cu+2_H+1_Thr-1_PO4-3</b>      |     |   |                              |         |
| -1                               | --- | 1 | C(4)H(9)Cu(1)N(1)O(7)P(1)    | 277.638 |
| <b>Cu+2_H+1_Trp-1_PO4-3</b>      |     |   |                              |         |
| -1                               | --- | 1 | C(11)H(12)Cu(1)N(2)O(6)P(1)  | 362.746 |
| <b>Cu+2_H+1_Val-1_PO4-3</b>      |     |   |                              |         |
| -1                               | --- | 1 | C(5)H(11)Cu(1)N(1)O(6)P(1)   | 275.665 |
| <b>Cu+2_H+1(2)_Lys-1_PO4-3</b>   |     |   |                              |         |
| 0                                | --- | 1 | C(6)H(15)Cu(1)N(2)O(6)P(1)   | 305.715 |



|                                     |     |   |                              |         |
|-------------------------------------|-----|---|------------------------------|---------|
| <b>Cu+2_H+1(2)_Orn-1_PO4-3</b>      |     |   |                              |         |
| 0                                   | --- | 1 | C(5)H(13)Cu(1)N(2)O(6)P(1)   | 291.688 |
| <b>Cu+2_H+1(2)_P2O7-4</b>           |     |   |                              |         |
| 0                                   | --- | 2 | Cu(1)H(2)O(7)P(2)            | 239.506 |
| <b>Cu+2_H+1(2)_P2O7-4(2)</b>        |     |   |                              |         |
| -4                                  | --- | 2 | Cu(1)H(2)O(14)P(4)           | 413.450 |
| <b>Cu+2_H+1(2)_P3O10-5</b>          |     |   |                              |         |
| -1                                  | --- | 1 | Cu(1)H(2)O(10)P(3)           | 318.478 |
| <b>Cu+2_H+1(2)_P4O13-6</b>          |     |   |                              |         |
| -2                                  | --- | 1 | Cu(1)H(2)O(13)P(4)           | 397.450 |
| <b>Cu+2_H+1(2)_P4O13-6(2)</b>       |     |   |                              |         |
| -8                                  | --- | 2 | Cu(1)H(2)O(26)P(8)           | 729.338 |
| <b>Cu+2_H+1(2)_PO4-3</b>            |     |   |                              |         |
| 1                                   | --- | 3 | Cu(1)H(2)O(4)P(1)            | 160.534 |
| <b>Cu+2_H+1(2)_PO4-3(2)</b>         |     |   |                              |         |
| -2                                  | --- | 3 | Cu(1)H(2)O(8)P(2)            | 255.505 |
| <b>Cu+2_H+1(2)_Spermine_P3O10-5</b> |     |   |                              |         |
| -1                                  | --- | 1 | C(10)H(28)Cu(1)N(4)O(10)P(3) | 520.821 |
| <b>Cu+2_H+1(2)_Tyr-2_PO4-3</b>      |     |   |                              |         |
| -1                                  | --- | 1 | C(9)H(11)Cu(1)N(1)O(7)P(1)   | 339.709 |
| <b>Cu+2_H+1(3)_P2O7-4(2)</b>        |     |   |                              |         |
| -3                                  | --- | 1 | Cu(1)H(3)O(14)P(4)           | 414.457 |
| <b>Cu+2_H+1(3)_P4O13-6(2)</b>       |     |   |                              |         |
| -7                                  | --- | 2 | Cu(1)H(3)O(26)P(8)           | 730.346 |

|                                      |     |   |                                         |         |
|--------------------------------------|-----|---|-----------------------------------------|---------|
| <b>Cu+2_H+1 (3)_PO4-3 (2)</b>        |     |   |                                         |         |
| -1                                   | --- | 4 | Cu (1) H (3) O (8) P (2)                | 256.513 |
| <b>Cu+2_H+1 (3)_PO4-3 (3)</b>        |     |   |                                         |         |
| -4                                   | --- | 2 | Cu (1) H (3) O (12) P (3)               | 351.485 |
| <b>Cu+2_H+1 (3)_Spermine_P3O10-5</b> |     |   |                                         |         |
| 0                                    | --- | 1 | C (10) H (29) Cu (1) N (4) O (10) P (3) | 521.829 |
| <b>Cu+2_H+1 (4)_P2O7-4 (2)</b>       |     |   |                                         |         |
| -2                                   | --- | 1 | Cu (1) H (4) O (14) P (4)               | 415.465 |
| <b>Cu+2_H+1 (4)_P4O13-6 (2)</b>      |     |   |                                         |         |
| -6                                   | --- | 1 | Cu (1) H (4) O (26) P (8)               | 731.354 |
| <b>Cu+2_H+1 (4)_PO4-3 (2)</b>        |     |   |                                         |         |
| 0                                    | --- | 2 | Cu (1) H (4) O (8) P (2)                | 257.521 |
| <b>Cu+2_H+1 (4)_Spermine_P3O10-5</b> |     |   |                                         |         |
| 1                                    | --- | 1 | C (10) H (30) Cu (1) N (4) O (10) P (3) | 522.837 |
| <b>Cu+2_H+1 (5)_P2O7-4 (2)</b>       |     |   |                                         |         |
| -1                                   | --- | 1 | Cu (1) H (5) O (14) P (4)               | 416.473 |
| <b>Cu+2_H+1 (5)_Spermine_P3O10-5</b> |     |   |                                         |         |
| 2                                    | --- | 1 | C (10) H (31) Cu (1) N (4) O (10) P (3) | 523.845 |
| <b>Cu+2_H2PO2-1</b>                  |     |   |                                         |         |
| 1                                    | --- | 1 | Cu (1) H (2) O (2) P (1)                | 128.535 |
| <b>Cu+2_His-1_P2O7-4</b>             |     |   |                                         |         |
| -3                                   | --- | 1 | C (6) H (8) Cu (1) N (3) O (9) P (2)    | 391.638 |
| <b>Cu+2_His-1_P3O10-5</b>            |     |   |                                         |         |
| -4                                   | --- | 1 | C (6) H (8) Cu (1) N (3) O (12) P (3)   | 470.610 |

|                            |     |    |                         |         |
|----------------------------|-----|----|-------------------------|---------|
| <b>Cu+2_HPO3-2_(s)</b>     |     |    |                         |         |
| 0                          | --- | 1  | Cu(1)H(1)O(3)P(1)       | 143.526 |
| <b>Cu+2_HPO3-2(2)</b>      |     |    |                         |         |
| -2                         | --- | 1  | Cu(1)H(2)O(6)P(2)       | 223.506 |
| <b>Cu+2_NH3_P3O10-5(3)</b> |     |    |                         |         |
| -13                        | --- | 1  | Cu(1)H(3)N(1)O(30)P(9)  | 839.325 |
| <b>Cu+2_NH3_P4O13-6(3)</b> |     |    |                         |         |
| -16                        | --- | 1  | Cu(1)H(3)N(1)O(39)P(12) | 1076.24 |
| <b>Cu+2_NH3(2)_P2O7-4</b>  |     |    |                         |         |
| -2                         | --- | 2  | Cu(1)H(6)N(2)O(7)P(2)   | 271.551 |
| <b>Cu+2_NH3(3)_P2O7-4</b>  |     |    |                         |         |
| -2                         | --- | 1  | Cu(1)H(9)N(3)O(7)P(2)   | 288.581 |
| <b>Cu+2_NH3(3)_P3O10-5</b> |     |    |                         |         |
| -3                         | --- | 1  | Cu(1)H(9)N(3)O(10)P(3)  | 367.554 |
| <b>Cu+2_NH3(3)_P4O13-6</b> |     |    |                         |         |
| -4                         | --- | 1  | Cu(1)H(9)N(3)O(13)P(4)  | 446.526 |
| <b>Cu+2_NH3(4)</b>         |     |    |                         |         |
| 2                          | --- | 13 | Cu(1)H(12)N(4)          | 131.668 |
| <b>Cu+2_NTA-3</b>          |     |    |                         |         |
| -1                         | --- | 56 | C(6)H(6)Cu(1)N(1)O(6)   | 251.663 |
| <b>Cu+2_OH-1_P2O7-4</b>    |     |    |                         |         |
| -3                         | --- | 1  | Cu(1)H(1)O(8)P(2)       | 254.497 |
| <b>Cu+2_OH-1_P3O10-5</b>   |     |    |                         |         |
| -4                         | --- | 1  | Cu(1)H(1)O(11)P(3)      | 333.469 |

|                          |     |   |                           |         |
|--------------------------|-----|---|---------------------------|---------|
| <b>Cu+2_OH-1_P4O13-6</b> |     |   |                           |         |
| -5                       | --- | 1 | Cu (1) H (1) O (14) P (4) | 412.442 |
| <b>Cu+2_P2O7-4</b>       |     |   |                           |         |
| -2                       | --- | 4 | Cu (1) O (7) P (2)        | 237.490 |
| <b>Cu+2_P2O7-4 (2)</b>   |     |   |                           |         |
| -6                       | --- | 4 | Cu (1) O (14) P (4)       | 411.434 |
| <b>Cu+2_P3O10-5</b>      |     |   |                           |         |
| -3                       | --- | 4 | Cu (1) O (10) P (3)       | 316.462 |
| <b>Cu+2_P3O10-5 (2)</b>  |     |   |                           |         |
| -8                       | --- | 2 | Cu (1) O (20) P (6)       | 569.378 |
| <b>Cu+2_P3O10-5 (4)</b>  |     |   |                           |         |
| -18                      | --- | 1 | Cu (1) O (40) P (12)      | 1075.21 |
| <b>Cu+2_P3O9-3</b>       |     |   |                           |         |
| -1                       | --- | 2 | Cu (1) O (9) P (3)        | 300.463 |
| <b>Cu+2_P3O9-3 (2)</b>   |     |   |                           |         |
| -4                       | --- | 1 | Cu (1) O (18) P (6)       | 537.379 |
| <b>Cu+2_P4O12-4</b>      |     |   |                           |         |
| -2                       | --- | 2 | Cu (1) O (12) P (4)       | 379.435 |
| <b>Cu+2_P4O12-4 (2)</b>  |     |   |                           |         |
| -6                       | --- | 2 | Cu (1) O (24) P (8)       | 695.324 |
| <b>Cu+2_P4O13-6</b>      |     |   |                           |         |
| -4                       | --- | 3 | Cu (1) O (13) P (4)       | 395.434 |
| <b>Cu+2_P4O13-6 (2)</b>  |     |   |                           |         |
| -10                      | --- | 2 | Cu (1) O (26) P (8)       | 727.322 |

|                                                                 |            |    |                            |         |
|-----------------------------------------------------------------|------------|----|----------------------------|---------|
| <b>Cu+2_P4O13-6(4)</b>                                          |            |    |                            |         |
| -22                                                             | ---        | 1  | Cu(1)O(52)P(16)            | 1391.10 |
| <b>Cu+2_Pb+2(2)_OH-1(3)_PO4-3_H2O(3)_(s)</b><br>Tsumebite (sic) |            |    |                            |         |
| 0                                                               | ---        | 1  | Cu(1)H(9)O(10)P(1)Pb(2)    | 677.986 |
| <b>Cu+2_Phe-1_P2O7-4</b>                                        |            |    |                            |         |
| -3                                                              | ---        | 1  | C(9)H(10)Cu(1)N(1)O(9)P(2) | 401.674 |
| <b>Cu+2_Phe-1_PO4-3</b>                                         |            |    |                            |         |
| -2                                                              | ---        | 1  | C(9)H(10)Cu(1)N(1)O(6)P(1) | 322.702 |
| <b>Cu+2_Phenanth</b>                                            |            |    |                            |         |
| 2                                                               | ---        | 74 | C(12)H(8)Cu(1)N(2)         | 243.755 |
| <b>Cu+2_PO4-3</b>                                               |            |    |                            |         |
| -1                                                              | ---        | 2  | Cu(1)O(4)P(1)              | 158.518 |
| <b>Cu+2_PO4-3(2)</b>                                            |            |    |                            |         |
| -4                                                              | ---        | 2  | Cu(1)O(8)P(2)              | 253.489 |
| <b>Cu+2_UO2+2(2)_PO4-3(2)_(s)</b><br>Torbernite                 |            |    |                            |         |
| 0                                                               | ---        | 2  | Cu(1)O(12)P(2)U(2)         | 793.545 |
| <b>Cu+2(2)_H+1(2)_PO4-3(2)</b>                                  |            |    |                            |         |
| 0                                                               | ---        | 2  | Cu(2)H(2)O(8)P(2)          | 319.051 |
| <b>Cu+2(2)_OH-1_PO4-3_(s)</b><br>Libethenite                    |            |    |                            |         |
| 0                                                               | ---        | 1  | Cu(2)H(1)O(5)P(1)          | 239.071 |
| <b>Cu+2(2)_P2O7-4_(s)</b>                                       |            |    |                            |         |
| 0                                                               | 15191-80-7 | 1  | Cu(2)O(7)P(2)              | 301.036 |
| <b>Cu+2(2)_P2O7-4_H2O(4)_(s)</b>                                |            |    |                            |         |

|                                                                                                                                                                                                                           |             |     |                                        |         |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|-----|----------------------------------------|---------|
| 0                                                                                                                                                                                                                         | ---         | 1   | $\text{Cu (2) H (8) O (11) P (2)}$     | 373.097 |
| <b>Cu+2 (3) _OH-1 (3) _PO4-3 _ (s)</b><br>Cornetite                                                                                                                                                                       |             |     |                                        |         |
| 0                                                                                                                                                                                                                         | ---         | 1   | $\text{Cu (3) H (3) O (7) P (1)}$      | 336.632 |
| <b>Cu+2 (3) _PO4-3 (2) _ (s)</b><br>Copper phosphate                                                                                                                                                                      |             |     |                                        |         |
| 0                                                                                                                                                                                                                         | 7798-23-4   | 2   | $\text{Cu (3) O (8) P (2)}$            | 380.581 |
| <b>Cu+2 (3) _PO4-3 (2) _H2O (2) _ (s)</b><br>Copper phosphate dihydrate                                                                                                                                                   |             |     |                                        |         |
| 0                                                                                                                                                                                                                         | 10031-48-8  | 2   | $\text{Cu (3) H (4) O (10) P (2)}$     | 416.612 |
| <b>Cu+2 (3) _PO4-3 (2) _H2O (3) _ (s)</b><br>Copper phosphate trihydrate                                                                                                                                                  |             |     |                                        |         |
| 0                                                                                                                                                                                                                         | ---         | 2   | $\text{Cu (3) H (6) O (11) P (2)}$     | 434.627 |
| <b>Cu+2 (5) _OH-1 (4) _PO4-3 (2) _ (s)</b><br>Pseudomalachite                                                                                                                                                             |             |     |                                        |         |
| 0                                                                                                                                                                                                                         | ---         | 1   | $\text{Cu (5) H (4) O (12) P (2)}$     | 575.703 |
| <b>Cu (s)</b><br>Copper; Copper, cubic                                                                                                                                                                                    |             |     |                                        |         |
| 0                                                                                                                                                                                                                         | 7440-50-8   | 254 | $\text{Cu (1)}$                        | 63.5460 |
| <b>Cu3P (s)</b>                                                                                                                                                                                                           |             |     |                                        |         |
| 0                                                                                                                                                                                                                         | ---         | 1   | $\text{Cu (3) P (1)}$                  | 221.612 |
| <b>CuP2 (s)</b>                                                                                                                                                                                                           |             |     |                                        |         |
| 0                                                                                                                                                                                                                         | ---         | 1   | $\text{Cu (1) P (2)}$                  | 125.494 |
| <b>Cys-2</b><br>Cysteinate ion; L-Cysteinate ion; beta-mercaptoalanate ion; 2-amino-3-mercaptopropanoate ion; 2-amino-3-mercaptopropionate ion; alpha-amino-beta-thiolpropionate ion                                      |             |     |                                        |         |
| -2                                                                                                                                                                                                                        | 104170-20-9 | 673 | $\text{C (3) H (5) N (1) O (2) S (1)}$ | 119.138 |
| <b>Dien</b><br>Diethylenetriamine; Iminobis(2-ethylamine); Dien; 1,4,7-Triazaheptane; 3-Azapentane-1,5-diamine; N1-(2-Aminoethyl)-1,2-ethanediamine; 3NH-pd; 2,2-tri; 2,2'-Diaminodiethylamine; 2,2'-Iminobis(ethylamine) |             |     |                                        |         |

|                                    |            |     |                              |         |
|------------------------------------|------------|-----|------------------------------|---------|
| 0                                  | 111-40-0   | 254 | C(4)H(13)N(3)                | 103.167 |
| <b>Dy+3</b><br>Dysprosium(III) ion |            |     |                              |         |
| 3                                  | 22541-21-5 | 592 | Dy(1)                        | 162.500 |
| <b>Dy+3_EDTA-4</b>                 |            |     |                              |         |
| -1                                 | ---        | 14  | C(10)H(12)Dy(1)N(2)O(8)      | 450.714 |
| <b>Dy+3_EDTA-4_P3O10-5</b>         |            |     |                              |         |
| -6                                 | ---        | 2   | C(10)H(12)Dy(1)N(2)O(18)P(3) | 703.630 |
| <b>Dy+3_H+1_EDTA-4_P3O10-5</b>     |            |     |                              |         |
| -5                                 | ---        | 2   | C(10)H(13)Dy(1)N(2)O(18)P(3) | 704.638 |
| <b>Dy+3_H+1_P3O10-5</b>            |            |     |                              |         |
| -1                                 | ---        | 1   | Dy(1)H(1)O(10)P(3)           | 416.424 |
| <b>Dy+3_H+1_PO4-3</b>              |            |     |                              |         |
| 1                                  | ---        | 2   | Dy(1)H(1)O(4)P(1)            | 258.480 |
| <b>Dy+3_H+1(2)_P3O10-5(2)</b>      |            |     |                              |         |
| -5                                 | ---        | 1   | Dy(1)H(2)O(20)P(6)           | 670.348 |
| <b>Dy+3_H+1(2)_PO4-3</b>           |            |     |                              |         |
| 2                                  | ---        | 2   | Dy(1)H(2)O(4)P(1)            | 259.488 |
| <b>Dy+3_H+1(2)_PO4-3(2)</b>        |            |     |                              |         |
| -1                                 | ---        | 2   | Dy(1)H(2)O(8)P(2)            | 354.459 |
| <b>Dy+3_P3O10-5(2)</b>             |            |     |                              |         |
| -7                                 | ---        | 1   | Dy(1)O(20)P(6)               | 668.332 |
| <b>Dy+3_PO4-3</b>                  |            |     |                              |         |
| 0                                  | ---        | 1   | Dy(1)O(4)P(1)                | 257.472 |
| <b>Dy+3_PO4-3_(s)</b>              |            |     |                              |         |

|                                                    |            |      |                              |          |
|----------------------------------------------------|------------|------|------------------------------|----------|
| 0                                                  | ---        | 1    | Dy(1)O(4)P(1)                | 257.472  |
| <b>Dy+3_PO4-3(2)</b>                               |            |      |                              |          |
| -3                                                 | ---        | 1    | Dy(1)O(8)P(2)                | 352.443  |
| <b>Dy+3(2)_P2O7-4</b>                              |            |      |                              |          |
| 2                                                  | ---        | 1    | Dy(2)O(7)P(2)                | 498.944  |
| <b>e-1</b><br>Electron                             |            |      |                              |          |
| -1                                                 | ---        | 3399 | E(1)                         | 0.000000 |
| <b>EDTA-4</b><br>Ethylenediaminetetraacetate anion |            |      |                              |          |
| -4                                                 | ---        | 1457 | C(10)H(12)N(2)O(8)           | 288.214  |
| <b>Er+3</b><br>Erbium(III) ion                     |            |      |                              |          |
| 3                                                  | 18472-30-5 | 620  | Er(1)                        | 167.260  |
| <b>Er+3_EDTA-4</b>                                 |            |      |                              |          |
| -1                                                 | ---        | 15   | C(10)H(12)Er(1)N(2)O(8)      | 455.474  |
| <b>Er+3_EDTA-4_P3O10-5</b>                         |            |      |                              |          |
| -6                                                 | ---        | 2    | C(10)H(12)Er(1)N(2)O(18)P(3) | 708.390  |
| <b>Er+3_H+1_EDTA-4_P3O10-5</b>                     |            |      |                              |          |
| -5                                                 | ---        | 2    | C(10)H(13)Er(1)N(2)O(18)P(3) | 709.398  |
| <b>Er+3_H+1_P3O10-5</b>                            |            |      |                              |          |
| -1                                                 | ---        | 1    | Er(1)H(1)O(10)P(3)           | 421.184  |
| <b>Er+3_H+1_PO4-3</b>                              |            |      |                              |          |
| 1                                                  | ---        | 2    | Er(1)H(1)O(4)P(1)            | 263.240  |
| <b>Er+3_H+1(2)_P3O10-5(2)</b>                      |            |      |                              |          |
| -5                                                 | ---        | 1    | Er(1)H(2)O(20)P(6)           | 675.108  |



|                                                                              |            |     |                         |         |
|------------------------------------------------------------------------------|------------|-----|-------------------------|---------|
| <b>Er+3_H+1(2)_PO4-3</b>                                                     |            |     |                         |         |
| 2                                                                            | ---        | 2   | Er(1)H(2)O(4)P(1)       | 264.248 |
| <b>Er+3_H+1(2)_PO4-3(2)</b>                                                  |            |     |                         |         |
| -1                                                                           | ---        | 2   | Er(1)H(2)O(8)P(2)       | 359.219 |
| <b>Er+3_H2PO2-1</b>                                                          |            |     |                         |         |
| 2                                                                            | ---        | 1   | Er(1)H(2)O(2)P(1)       | 232.249 |
| <b>Er+3_P3O10-5(2)</b>                                                       |            |     |                         |         |
| -7                                                                           | ---        | 1   | Er(1)O(20)P(6)          | 673.092 |
| <b>Er+3_PO4-3</b>                                                            |            |     |                         |         |
| 0                                                                            | ---        | 1   | Er(1)O(4)P(1)           | 262.232 |
| <b>Er+3_PO4-3_(s)</b>                                                        |            |     |                         |         |
| 0                                                                            | ---        | 1   | Er(1)O(4)P(1)           | 262.232 |
| <b>Er+3_PO4-3(2)</b>                                                         |            |     |                         |         |
| -3                                                                           | ---        | 1   | Er(1)O(8)P(2)           | 357.203 |
| <b>Er+3(2)_P2O7-4</b>                                                        |            |     |                         |         |
| 2                                                                            | ---        | 1   | Er(2)O(7)P(2)           | 508.464 |
| <b>EtDiAm</b><br>Ethylenediamine; 1,2-Ethanediamine; 1,2-Diaminoethane; en ! |            |     |                         |         |
| 0                                                                            | 107-15-3   | 879 | C(2)H(8)N(2)            | 60.0989 |
| <b>EtDiAm_P3O10-5</b>                                                        |            |     |                         |         |
| -5                                                                           | ---        | 1   | C(2)H(8)N(2)O(10)P(3)   | 313.015 |
| <b>Eu+3</b><br>Europium(III) ion                                             |            |     |                         |         |
| 3                                                                            | 22541-18-0 | 683 | Eu(1)                   | 151.960 |
| <b>Eu+3_EDTA-4</b>                                                           |            |     |                         |         |
| -1                                                                           | ---        | 14  | C(10)H(12)Eu(1)N(2)O(8) | 440.174 |

|                                |     |   |                              |         |
|--------------------------------|-----|---|------------------------------|---------|
| <b>Eu+3_EDTA-4_P3O10-5</b>     |     |   |                              |         |
| -6                             | --- | 2 | C(10)H(12)Eu(1)N(2)O(18)P(3) | 693.090 |
| <b>Eu+3_H+1_EDTA-4_P3O10-5</b> |     |   |                              |         |
| -5                             | --- | 2 | C(10)H(13)Eu(1)N(2)O(18)P(3) | 694.098 |
| <b>Eu+3_H+1_P3O10-5</b>        |     |   |                              |         |
| -1                             | --- | 1 | Eu(1)H(1)O(10)P(3)           | 405.884 |
| <b>Eu+3_H+1_PO4-3</b>          |     |   |                              |         |
| 1                              | --- | 2 | Eu(1)H(1)O(4)P(1)            | 247.940 |
| <b>Eu+3_H+1(2)_P3O10-5(2)</b>  |     |   |                              |         |
| -5                             | --- | 1 | Eu(1)H(2)O(20)P(6)           | 659.808 |
| <b>Eu+3_H+1(2)_PO4-3</b>       |     |   |                              |         |
| 2                              | --- | 2 | Eu(1)H(2)O(4)P(1)            | 248.948 |
| <b>Eu+3_H+1(2)_PO4-3(2)</b>    |     |   |                              |         |
| -1                             | --- | 2 | Eu(1)H(2)O(8)P(2)            | 343.919 |
| <b>Eu+3_H2PO2-1</b>            |     |   |                              |         |
| 2                              | --- | 1 | Eu(1)H(2)O(2)P(1)            | 216.949 |
| <b>Eu+3_P3O10-5(2)</b>         |     |   |                              |         |
| -7                             | --- | 1 | Eu(1)O(20)P(6)               | 657.792 |
| <b>Eu+3_PO4-3</b>              |     |   |                              |         |
| 0                              | --- | 1 | Eu(1)O(4)P(1)                | 246.932 |
| <b>Eu+3_PO4-3_(s)</b>          |     |   |                              |         |
| 0                              | --- | 1 | Eu(1)O(4)P(1)                | 246.932 |
| <b>Eu+3_PO4-3(2)</b>           |     |   |                              |         |
| -3                             | --- | 1 | Eu(1)O(8)P(2)                | 341.903 |

|                                          |            |      |                            |         |
|------------------------------------------|------------|------|----------------------------|---------|
| <b>Eu+3 (2)_P2O7-4</b>                   |            |      |                            |         |
| 2                                        | ---        | 2    | Eu(2)O(7)P(2)              | 477.864 |
| <b>F-1</b><br>Fluoride ion               |            |      |                            |         |
| -1                                       | 16984-48-8 | 1090 | F(1)                       | 18.9984 |
| <b>F2 (g)</b><br>Fluorine gas            |            |      |                            |         |
| 0                                        | 7782-41-4  | 259  | F(2)                       | 37.9968 |
| <b>Fe+2</b><br>Iron(II) ion; Ferrous ion |            |      |                            |         |
| 2                                        | 15438-31-0 | 1769 | Fe(1)                      | 55.8452 |
| <b>Fe+2_H+1_Ala-1_PO4-3</b>              |            |      |                            |         |
| -1                                       | ---        | 1    | C(3)H(7)Fe(1)N(1)O(6)P(1)  | 239.911 |
| <b>Fe+2_H+1_Arg-1_PO4-3</b>              |            |      |                            |         |
| -1                                       | ---        | 1    | C(6)H(14)Fe(1)N(4)O(6)P(1) | 325.020 |
| <b>Fe+2_H+1_Asn-1_PO4-3</b>              |            |      |                            |         |
| -1                                       | ---        | 1    | C(4)H(8)Fe(1)N(2)O(7)P(1)  | 282.936 |
| <b>Fe+2_H+1_Asp-2_PO4-3</b>              |            |      |                            |         |
| -2                                       | ---        | 1    | C(4)H(6)Fe(1)N(1)O(8)P(1)  | 282.913 |
| <b>Fe+2_H+1_bAla-1_PO4-3</b>             |            |      |                            |         |
| -1                                       | ---        | 1    | C(3)H(7)Fe(1)N(1)O(6)P(1)  | 239.911 |
| <b>Fe+2_H+1_Citric-3_PO4-3</b>           |            |      |                            |         |
| -3                                       | ---        | 1    | C(6)H(6)Fe(1)O(11)P(1)     | 340.926 |
| <b>Fe+2_H+1_Citrul-1_PO4-3</b>           |            |      |                            |         |
| -1                                       | ---        | 1    | C(6)H(13)Fe(1)N(3)O(7)P(1) | 326.004 |
| <b>Fe+2_H+1_Cys-2_PO4-3</b>              |            |      |                            |         |

|                                 |     |   |                                |         |
|---------------------------------|-----|---|--------------------------------|---------|
| -2                              | --- | 1 | C(3)H(6)Fe(1)N(1)O(6)P(1)S(1)  | 270.963 |
| <b>Fe+2_H+1_Gln-1_PO4-3</b>     |     |   |                                |         |
| -1                              | --- | 1 | C(5)H(10)Fe(1)N(2)O(7)P(1)     | 296.963 |
| <b>Fe+2_H+1_Glu-2_PO4-3</b>     |     |   |                                |         |
| -2                              | --- | 1 | C(5)H(8)Fe(1)N(1)O(8)P(1)      | 296.940 |
| <b>Fe+2_H+1_Gly-1_PO4-3</b>     |     |   |                                |         |
| -1                              | --- | 1 | C(2)H(5)Fe(1)N(1)O(6)P(1)      | 225.884 |
| <b>Fe+2_H+1_His-1_PO4-3</b>     |     |   |                                |         |
| -1                              | --- | 1 | C(6)H(9)Fe(1)N(3)O(6)P(1)      | 305.973 |
| <b>Fe+2_H+1_Histamine_PO4-3</b> |     |   |                                |         |
| 0                               | --- | 1 | C(5)H(10)Fe(1)N(3)O(4)P(1)     | 262.971 |
| <b>Fe+2_H+1_Hyp-1_PO4-3</b>     |     |   |                                |         |
| -1                              | --- | 1 | C(5)H(9)Fe(1)N(1)O(7)P(1)      | 281.948 |
| <b>Fe+2_H+1_Ile-1_PO4-3</b>     |     |   |                                |         |
| -1                              | --- | 1 | C(6)H(13)Fe(1)N(1)O(6)P(1)     | 281.992 |
| <b>Fe+2_H+1_Leu-1_PO4-3</b>     |     |   |                                |         |
| -1                              | --- | 1 | C(6)H(13)Fe(1)N(1)O(6)P(1)     | 281.992 |
| <b>Fe+2_H+1_Malic-2_PO4-3</b>   |     |   |                                |         |
| -2                              | --- | 1 | C(4)H(5)Fe(1)O(9)P(1)          | 283.898 |
| <b>Fe+2_H+1_Met-1_PO4-3</b>     |     |   |                                |         |
| -1                              | --- | 1 | C(5)H(11)Fe(1)N(1)O(6)P(1)S(1) | 300.025 |
| <b>Fe+2_H+1_Oxalic-2_PO4-3</b>  |     |   |                                |         |
| -2                              | --- | 1 | C(2)H(1)Fe(1)O(8)P(1)          | 239.844 |
| <b>Fe+2_H+1_P3O9-3</b>          |     |   |                                |         |
| 0                               | --- | 1 | Fe(1)H(1)O(9)P(3)              | 293.770 |

|                                   |     |   |                             |         |
|-----------------------------------|-----|---|-----------------------------|---------|
| <b>Fe+2_H+1_Phe-1_PO4-3</b>       |     |   |                             |         |
| -1                                | --- | 1 | C(9)H(11)Fe(1)N(1)O(6)P(1)  | 316.009 |
| <b>Fe+2_H+1_PO4-3</b>             |     |   |                             |         |
| 0                                 | --- | 3 | Fe(1)H(1)O(4)P(1)           | 151.825 |
| <b>Fe+2_H+1_Pro-1_PO4-3</b>       |     |   |                             |         |
| -1                                | --- | 1 | C(5)H(9)Fe(1)N(1)O(6)P(1)   | 265.949 |
| <b>Fe+2_H+1_Pyruvic-1_PO4-3</b>   |     |   |                             |         |
| -1                                | --- | 1 | C(3)H(4)Fe(1)O(7)P(1)       | 238.880 |
| <b>Fe+2_H+1_Salicylic-2_PO4-3</b> |     |   |                             |         |
| -2                                | --- | 1 | C(7)H(5)Fe(1)O(7)P(1)       | 287.932 |
| <b>Fe+2_H+1_Ser-1_PO4-3</b>       |     |   |                             |         |
| -1                                | --- | 1 | C(3)H(7)Fe(1)N(1)O(7)P(1)   | 255.910 |
| <b>Fe+2_H+1_Succinic-2_PO4-3</b>  |     |   |                             |         |
| -2                                | --- | 1 | C(4)H(5)Fe(1)O(8)P(1)       | 267.898 |
| <b>Fe+2_H+1_Thr-1_PO4-3</b>       |     |   |                             |         |
| -1                                | --- | 1 | C(4)H(9)Fe(1)N(1)O(7)P(1)   | 269.937 |
| <b>Fe+2_H+1_Trp-1_PO4-3</b>       |     |   |                             |         |
| -1                                | --- | 1 | C(11)H(12)Fe(1)N(2)O(6)P(1) | 355.045 |
| <b>Fe+2_H+1_Val-1_PO4-3</b>       |     |   |                             |         |
| -1                                | --- | 1 | C(5)H(11)Fe(1)N(1)O(6)P(1)  | 267.965 |
| <b>Fe+2_H+1(2)_Lys-1_PO4-3</b>    |     |   |                             |         |
| 0                                 | --- | 1 | C(6)H(15)Fe(1)N(2)O(6)P(1)  | 298.014 |
| <b>Fe+2_H+1(2)_Orn-1_PO4-3</b>    |     |   |                             |         |
| 0                                 | --- | 1 | C(5)H(13)Fe(1)N(2)O(6)P(1)  | 283.987 |

|                                                         |     |   |                            |         |
|---------------------------------------------------------|-----|---|----------------------------|---------|
| <b>Fe+2_H+1(2)_P3O10-5</b>                              |     |   |                            |         |
| -1                                                      | --- | 1 | Fe(1)H(2)O(10)P(3)         | 310.777 |
| <b>Fe+2_H+1(2)_PO4-3</b><br>Ferrous dihydroxy phosphate |     |   |                            |         |
| 1                                                       | --- | 2 | Fe(1)H(2)O(4)P(1)          | 152.833 |
| <b>Fe+2_H+1(2)_PO4-3(2)</b>                             |     |   |                            |         |
| -2                                                      | --- | 1 | Fe(1)H(2)O(8)P(2)          | 247.804 |
| <b>Fe+2_H+1(2)_Tyr-2_PO4-3</b>                          |     |   |                            |         |
| -1                                                      | --- | 1 | C(9)H(11)Fe(1)N(1)O(7)P(1) | 332.008 |
| <b>Fe+2_H+1(3)_P3O10-5</b>                              |     |   |                            |         |
| 0                                                       | --- | 1 | Fe(1)H(3)O(10)P(3)         | 311.785 |
| <b>Fe+2_H+1(3)_PO4-3(2)</b>                             |     |   |                            |         |
| -1                                                      | --- | 3 | Fe(1)H(3)O(8)P(2)          | 248.812 |
| <b>Fe+2_H+1(4)_P2O7-4(2)</b>                            |     |   |                            |         |
| -2                                                      | --- | 1 | Fe(1)H(4)O(14)P(4)         | 407.765 |
| <b>Fe+2_H+1(4)_PO4-3(2)</b>                             |     |   |                            |         |
| 0                                                       | --- | 2 | Fe(1)H(4)O(8)P(2)          | 249.820 |
| <b>Fe+2_P3O10-5</b>                                     |     |   |                            |         |
| -3                                                      | --- | 1 | Fe(1)O(10)P(3)             | 308.761 |
| <b>Fe+2_P3O9-3</b>                                      |     |   |                            |         |
| -1                                                      | --- | 1 | Fe(1)O(9)P(3)              | 292.762 |
| <b>Fe+2_UO2+2(2)_PO4-3(2)_(s)</b><br>Bassetite          |     |   |                            |         |
| 0                                                       | --- | 2 | Fe(1)O(12)P(2)U(2)         | 785.844 |
| <b>Fe+2(3)_PO4-3(2)_(s)</b>                             |     |   |                            |         |
| 0                                                       | --- | 1 | Fe(3)O(8)P(2)              | 357.479 |

|                                                                                       |            |      |                                       |         |
|---------------------------------------------------------------------------------------|------------|------|---------------------------------------|---------|
| <b>Fe+2 (3) _PO4-3 (2) _H2O (8) _ (s)</b><br>Vivianite; Ferrous phosphate octahydrate |            |      |                                       |         |
| 0                                                                                     | 14567-67-0 | 1    | Fe (3) H (16) O (16) P (2)            | 501.601 |
| <b>Fe+2 (5) _OH-1 _PO4-3 (3) _ (s)</b><br>Ferrous hydroxy phosphate                   |            |      |                                       |         |
| 0                                                                                     | 53125-82-9 | 1    | Fe (5) H (1) O (13) P (3)             | 581.148 |
| <b>Fe+3</b><br>Iron(III) ion; Ferric ion                                              |            |      |                                       |         |
| 3                                                                                     | 20074-52-6 | 2512 | Fe (1)                                | 55.8452 |
| <b>Fe+3 _Citric-3</b>                                                                 |            |      |                                       |         |
| 0                                                                                     | ---        | 8    | C (6) H (5) Fe (1) O (7)              | 244.947 |
| <b>Fe+3 _Fe+2 (7) _Mg+2 _OH-1 _PO4-3 (6) _H2O (23) _ (s)</b><br>Vivianite (natural)   |            |      |                                       |         |
| 0                                                                                     | ---        | 1    | Fe (8) H (47) Mg (1) O (48) P (6)     | 1472.26 |
| <b>Fe+3 _H+1 _Ala-1 _PO4-3</b>                                                        |            |      |                                       |         |
| 0                                                                                     | ---        | 1    | C (3) H (7) Fe (1) N (1) O (6) P (1)  | 239.911 |
| <b>Fe+3 _H+1 _Arg-1 _PO4-3</b>                                                        |            |      |                                       |         |
| 0                                                                                     | ---        | 1    | C (6) H (14) Fe (1) N (4) O (6) P (1) | 325.020 |
| <b>Fe+3 _H+1 _Asn-1 _PO4-3</b>                                                        |            |      |                                       |         |
| 0                                                                                     | ---        | 1    | C (4) H (8) Fe (1) N (2) O (7) P (1)  | 282.936 |
| <b>Fe+3 _H+1 _Asp-2 _PO4-3</b>                                                        |            |      |                                       |         |
| -1                                                                                    | ---        | 1    | C (4) H (6) Fe (1) N (1) O (8) P (1)  | 282.913 |
| <b>Fe+3 _H+1 _bAla-1 _PO4-3</b>                                                       |            |      |                                       |         |
| 0                                                                                     | ---        | 1    | C (3) H (7) Fe (1) N (1) O (6) P (1)  | 239.911 |
| <b>Fe+3 _H+1 _Citric-3 _PO4-3</b>                                                     |            |      |                                       |         |
| -2                                                                                    | ---        | 4    | C (6) H (6) Fe (1) O (11) P (1)       | 340.926 |
| <b>Fe+3 _H+1 _Citrul-1 _PO4-3</b>                                                     |            |      |                                       |         |

|                                |     |   |                                  |         |
|--------------------------------|-----|---|----------------------------------|---------|
| 0                              | --- | 1 | $C(6)H(13)Fe(1)N(3)O(7)P(1)$     | 326.004 |
| <b>Fe+3_H+1_Gln-1_PO4-3</b>    |     |   |                                  |         |
| 0                              | --- | 1 | $C(5)H(10)Fe(1)N(2)O(7)P(1)$     | 296.963 |
| <b>Fe+3_H+1_Glu-2_PO4-3</b>    |     |   |                                  |         |
| -1                             | --- | 1 | $C(5)H(8)Fe(1)N(1)O(8)P(1)$      | 296.940 |
| <b>Fe+3_H+1_Gly-1_PO4-3</b>    |     |   |                                  |         |
| 0                              | --- | 1 | $C(2)H(5)Fe(1)N(1)O(6)P(1)$      | 225.884 |
| <b>Fe+3_H+1_HPO3-2</b>         |     |   |                                  |         |
| 2                              | --- | 2 | $Fe(1)H(2)O(3)P(1)$              | 136.833 |
| <b>Fe+3_H+1_Hyp-1_PO4-3</b>    |     |   |                                  |         |
| 0                              | --- | 1 | $C(5)H(9)Fe(1)N(1)O(7)P(1)$      | 281.948 |
| <b>Fe+3_H+1_Ile-1_PO4-3</b>    |     |   |                                  |         |
| 0                              | --- | 1 | $C(6)H(13)Fe(1)N(1)O(6)P(1)$     | 281.992 |
| <b>Fe+3_H+1_Lactic-1_PO4-3</b> |     |   |                                  |         |
| 0                              | --- | 1 | $C(3)H(6)Fe(1)O(7)P(1)$          | 240.896 |
| <b>Fe+3_H+1_Leu-1_PO4-3</b>    |     |   |                                  |         |
| 0                              | --- | 1 | $C(6)H(13)Fe(1)N(1)O(6)P(1)$     | 281.992 |
| <b>Fe+3_H+1_Malic-2_PO4-3</b>  |     |   |                                  |         |
| -1                             | --- | 1 | $C(4)H(5)Fe(1)O(9)P(1)$          | 283.898 |
| <b>Fe+3_H+1_Met-1_PO4-3</b>    |     |   |                                  |         |
| 0                              | --- | 1 | $C(5)H(11)Fe(1)N(1)O(6)P(1)S(1)$ | 300.025 |
| <b>Fe+3_H+1_Oxalic-2_PO4-3</b> |     |   |                                  |         |
| -1                             | --- | 1 | $C(2)H(1)Fe(1)O(8)P(1)$          | 239.844 |
| <b>Fe+3_H+1_Phe-1_PO4-3</b>    |     |   |                                  |         |
| 0                              | --- | 1 | $C(9)H(11)Fe(1)N(1)O(6)P(1)$     | 316.009 |



|                                   |     |   |                                |         |
|-----------------------------------|-----|---|--------------------------------|---------|
| <b>Fe+3_H+1_PO4-3</b>             |     |   |                                |         |
| 1                                 | --- | 7 | Fe(1)H(1)O(4)P(1)              | 151.825 |
| <b>Fe+3_H+1_Pro-1_PO4-3</b>       |     |   |                                |         |
| 0                                 | --- | 1 | C(5)H(9)Fe(1)N(1)O(6)P(1)      | 265.949 |
| <b>Fe+3_H+1_Salicylic-2_PO4-3</b> |     |   |                                |         |
| -1                                | --- | 1 | C(7)H(5)Fe(1)O(7)P(1)          | 287.932 |
| <b>Fe+3_H+1_Ser-1_PO4-3</b>       |     |   |                                |         |
| 0                                 | --- | 1 | C(3)H(7)Fe(1)N(1)O(7)P(1)      | 255.910 |
| <b>Fe+3_H+1_Succinic-2_PO4-3</b>  |     |   |                                |         |
| -1                                | --- | 1 | C(4)H(5)Fe(1)O(8)P(1)          | 267.898 |
| <b>Fe+3_H+1_Thr-1_PO4-3</b>       |     |   |                                |         |
| 0                                 | --- | 1 | C(4)H(9)Fe(1)N(1)O(7)P(1)      | 269.937 |
| <b>Fe+3_H+1_Trp-1_PO4-3</b>       |     |   |                                |         |
| 0                                 | --- | 1 | C(11)H(12)Fe(1)N(2)O(6)P(1)    | 355.045 |
| <b>Fe+3_H+1_Val-1_PO4-3</b>       |     |   |                                |         |
| 0                                 | --- | 1 | C(5)H(11)Fe(1)N(1)O(6)P(1)     | 267.965 |
| <b>Fe+3_H+1(2)_Cis-2_PO4-3</b>    |     |   |                                |         |
| 0                                 | --- | 1 | C(6)H(12)Fe(1)N(2)O(8)P(1)S(2) | 391.109 |
| <b>Fe+3_H+1(2)_Citric-3_PO4-3</b> |     |   |                                |         |
| -1                                | --- | 1 | C(6)H(7)Fe(1)O(11)P(1)         | 341.934 |
| <b>Fe+3_H+1(2)_HPO3-2(2)</b>      |     |   |                                |         |
| 1                                 | --- | 2 | Fe(1)H(4)O(6)P(2)              | 217.821 |
| <b>Fe+3_H+1(2)_Lys-1_PO4-3</b>    |     |   |                                |         |
| 1                                 | --- | 1 | C(6)H(15)Fe(1)N(2)O(6)P(1)     | 298.014 |

|                                |     |   |                            |         |
|--------------------------------|-----|---|----------------------------|---------|
| <b>Fe+3_H+1(2)_Orn-1_PO4-3</b> |     |   |                            |         |
| 1                              | --- | 1 | C(5)H(13)Fe(1)N(2)O(6)P(1) | 283.987 |
| <b>Fe+3_H+1(2)_P2O7-4</b>      |     |   |                            |         |
| 1                              | --- | 2 | Fe(1)H(2)O(7)P(2)          | 231.805 |
| <b>Fe+3_H+1(2)_P3O10-5</b>     |     |   |                            |         |
| 0                              | --- | 1 | Fe(1)H(2)O(10)P(3)         | 310.777 |
| <b>Fe+3_H+1(2)_P3O10-5(2)</b>  |     |   |                            |         |
| -5                             | --- | 1 | Fe(1)H(2)O(20)P(6)         | 563.693 |
| <b>Fe+3_H+1(2)_PO4-3</b>       |     |   |                            |         |
| 2                              | --- | 6 | Fe(1)H(2)O(4)P(1)          | 152.833 |
| <b>Fe+3_H+1(2)_PO4-3(2)</b>    |     |   |                            |         |
| -1                             | --- | 3 | Fe(1)H(2)O(8)P(2)          | 247.804 |
| <b>Fe+3_H+1(2)_Tyr-2_PO4-3</b> |     |   |                            |         |
| 0                              | --- | 1 | C(9)H(11)Fe(1)N(1)O(7)P(1) | 332.008 |
| <b>Fe+3_H+1(3)_P2O7-4</b>      |     |   |                            |         |
| 2                              | --- | 1 | Fe(1)H(3)O(7)P(2)          | 232.813 |
| <b>Fe+3_H+1(3)_P3O10-5</b>     |     |   |                            |         |
| 1                              | --- | 1 | Fe(1)H(3)O(10)P(3)         | 311.785 |
| <b>Fe+3_H+1(3)_PO4-3</b>       |     |   |                            |         |
| 3                              | --- | 2 | Fe(1)H(3)O(4)P(1)          | 153.841 |
| <b>Fe+3_H+1(3)_PO4-3(2)</b>    |     |   |                            |         |
| 0                              | --- | 2 | Fe(1)H(3)O(8)P(2)          | 248.812 |
| <b>Fe+3_H+1(4)_P2O7-4(2)</b>   |     |   |                            |         |
| -1                             | --- | 1 | Fe(1)H(4)O(14)P(4)         | 407.765 |

|                              |     |   |                    |         |
|------------------------------|-----|---|--------------------|---------|
| <b>Fe+3_H+1(4)_PO4-3(2)</b>  |     |   |                    |         |
| 1                            | --- | 5 | Fe(1)H(4)O(8)P(2)  | 249.820 |
| <b>Fe+3_H+1(5)_PO4-3(2)</b>  |     |   |                    |         |
| 2                            | --- | 2 | Fe(1)H(5)O(8)P(2)  | 250.828 |
| <b>Fe+3_H+1(6)_P2O7-4(3)</b> |     |   |                    |         |
| -3                           | --- | 1 | Fe(1)H(6)O(21)P(6) | 583.724 |
| <b>Fe+3_H+1(6)_PO4-3(3)</b>  |     |   |                    |         |
| 0                            | --- | 2 | Fe(1)H(6)O(12)P(3) | 346.808 |
| <b>Fe+3_H+1(7)_PO4-3(3)</b>  |     |   |                    |         |
| 1                            | --- | 2 | Fe(1)H(7)O(12)P(3) | 347.816 |
| <b>Fe+3_H2PO2-1</b>          |     |   |                    |         |
| 2                            | --- | 3 | Fe(1)H(2)O(2)P(1)  | 120.834 |
| <b>Fe+3_H2PO2-1(2)</b>       |     |   |                    |         |
| 1                            | --- | 3 | Fe(1)H(4)O(4)P(2)  | 185.823 |
| <b>Fe+3_H2PO2-1(3)</b>       |     |   |                    |         |
| 0                            | --- | 1 | Fe(1)H(6)O(6)P(3)  | 250.811 |
| <b>Fe+3_HPO3-2</b>           |     |   |                    |         |
| 1                            | --- | 2 | Fe(1)H(1)O(3)P(1)  | 135.825 |
| <b>Fe+3_HPO3-2(2)</b>        |     |   |                    |         |
| -1                           | --- | 3 | Fe(1)H(2)O(6)P(2)  | 215.806 |
| <b>Fe+3_HPO3-2(3)</b>        |     |   |                    |         |
| -3                           | --- | 1 | Fe(1)H(3)O(9)P(3)  | 295.786 |
| <b>Fe+3_OH-1_PO4-3</b>       |     |   |                    |         |
| -1                           | --- | 1 | Fe(1)H(1)O(5)P(1)  | 167.824 |

|                                                                                                   |            |   |                          |         |
|---------------------------------------------------------------------------------------------------|------------|---|--------------------------|---------|
| <b>Fe+3_OH-1(2)_PO4-3(2)</b>                                                                      |            |   |                          |         |
| -5                                                                                                | ---        | 1 | Fe(1)H(2)O(10)P(2)       | 279.803 |
| <b>Fe+3_OH-1(2)_PO4-3(3)</b>                                                                      |            |   |                          |         |
| -8                                                                                                | ---        | 1 | Fe(1)H(2)O(14)P(3)       | 374.775 |
| <b>Fe+3_PO4-3</b>                                                                                 |            |   |                          |         |
| 0                                                                                                 | ---        | 2 | Fe(1)O(4)P(1)            | 150.817 |
| <b>Fe+3_PO4-3_(am.,s)</b>                                                                         |            |   |                          |         |
| 0                                                                                                 | ---        | 2 | Fe(1)O(4)P(1)            | 150.817 |
| <b>Fe+3_PO4-3_(s)</b><br>Ferric phosphate                                                         |            |   |                          |         |
| 0                                                                                                 | 10045-86-0 | 2 | Fe(1)O(4)P(1)            | 150.817 |
| <b>Fe+3_PO4-3_H2O(2)_(s)</b><br>Strengite; Ferric phosphate dihydrate; Iron phosphate dihydrate   |            |   |                          |         |
| 0                                                                                                 | 13463-10-0 | 3 | Fe(1)H(4)O(6)P(1)        | 186.847 |
| <b>Fe+3_PO4-3(2)</b>                                                                              |            |   |                          |         |
| -3                                                                                                | ---        | 1 | Fe(1)O(8)P(2)            | 245.788 |
| <b>Fe+3_SCN-1</b>                                                                                 |            |   |                          |         |
| 2                                                                                                 | ---        | 4 | C(1)Fe(1)N(1)S(1)        | 113.923 |
| <b>Fe+3(2)_Ca+2_OH-1(2)_PO4-3(2)_H2O(8)_(s)</b><br>Mitridatite; Calcium ferric hydroxyl phosphate |            |   |                          |         |
| 0                                                                                                 | ---        | 2 | Ca(1)Fe(2)H(18)O(18)P(2) | 519.849 |
| <b>Fe+3(2)_Fe+2_OH-1(2)_PO4-3(2)_(s)</b><br>Limpscombite; Lipscombite                             |            |   |                          |         |
| 0                                                                                                 | 66733-51-5 | 1 | Fe(3)H(2)O(10)P(2)       | 391.494 |
| <b>Fe+3(2)_H+1_PO4-3</b>                                                                          |            |   |                          |         |
| 4                                                                                                 | ---        | 4 | Fe(2)H(1)O(4)P(1)        | 207.670 |
| <b>Fe+3(2)_H+1(3)_PO4-3(2)</b>                                                                    |            |   |                          |         |

|                                                                                              |            |   |                                                               |         |
|----------------------------------------------------------------------------------------------|------------|---|---------------------------------------------------------------|---------|
| 3                                                                                            | ---        | 2 | $\text{Fe}(2)\text{H}(3)\text{O}(8)\text{P}(2)$               | 304.657 |
| <b>Fe+3(2)_Mn+2_OH-1(2)_PO4-3(2)_H2O(8)_(s)</b><br>Laveite / stronzite; Stronzite / laveite  |            |   |                                                               |         |
| 0                                                                                            | ---        | 2 | $\text{Fe}(2)\text{H}(18)\text{Mn}(1)\text{O}(18)\text{P}(2)$ | 534.709 |
| <b>Fe+3(2)_OH-1(2)_PO4-3(3)</b>                                                              |            |   |                                                               |         |
| -5                                                                                           | ---        | 1 | $\text{Fe}(2)\text{H}(2)\text{O}(14)\text{P}(3)$              | 430.620 |
| <b>Fe+3(2)_P2O7-4</b>                                                                        |            |   |                                                               |         |
| 2                                                                                            | ---        | 1 | $\text{Fe}(2)\text{O}(7)\text{P}(2)$                          | 285.634 |
| <b>Fe+3(3)_Fe+2_OH-1(2)_PO4-3(3)_H2O(12)_(s)</b>                                             |            |   |                                                               |         |
| 0                                                                                            | ---        | 2 | $\text{Fe}(4)\text{H}(26)\text{O}(26)\text{P}(3)$             | 758.494 |
| <b>Fe+3(3)_H+1_PO4-3(2)</b>                                                                  |            |   |                                                               |         |
| 4                                                                                            | ---        | 3 | $\text{Fe}(3)\text{H}(1)\text{O}(8)\text{P}(2)$               | 358.487 |
| <b>Fe+3(3)_H+1(2)_PO4-3(3)</b>                                                               |            |   |                                                               |         |
| 2                                                                                            | ---        | 2 | $\text{Fe}(3)\text{H}(2)\text{O}(12)\text{P}(3)$              | 454.466 |
| <b>Fe+3(3)_H+1(4)_PO4-3(3)</b>                                                               |            |   |                                                               |         |
| 4                                                                                            | ---        | 4 | $\text{Fe}(3)\text{H}(4)\text{O}(12)\text{P}(3)$              | 456.482 |
| <b>Fe+3(3)_H+1(6)_PO4-3(4)</b>                                                               |            |   |                                                               |         |
| 3                                                                                            | ---        | 5 | $\text{Fe}(3)\text{H}(6)\text{O}(16)\text{P}(4)$              | 553.470 |
| <b>Fe+3(3)_H+1(8)_PO4-3(5)</b>                                                               |            |   |                                                               |         |
| 2                                                                                            | ---        | 2 | $\text{Fe}(3)\text{H}(8)\text{O}(20)\text{P}(5)$              | 650.457 |
| <b>Fe+3(3)_H+1(9)_PO4-3(5)</b>                                                               |            |   |                                                               |         |
| 3                                                                                            | ---        | 2 | $\text{Fe}(3)\text{H}(9)\text{O}(20)\text{P}(5)$              | 651.465 |
| <b>Fe+3(3)_OH-1(3)_PO4-3(2)_H2O(3)_(s)</b><br>Tinticite; Ferric hydroxy phosphate trihydrate |            |   |                                                               |         |
| 0                                                                                            | 12426-84-5 | 2 | $\text{Fe}(3)\text{H}(9)\text{O}(14)\text{P}(2)$              | 462.547 |

|                                                                                                                 |            |     |                                   |         |
|-----------------------------------------------------------------------------------------------------------------|------------|-----|-----------------------------------|---------|
| <b>Fe+3 (4) _Ca+2 _OH-1 (8) _PO4-3 (2) _H2O (7) _ (s)</b><br>Fouchерite                                         |            |     |                                   |         |
| 0                                                                                                               | ---        | 2   | Ca (1) Fe (4) H (22) O (23) P (2) | 715.568 |
| <b>Fe+3 (4) _Fe+2 _OH-1 (5) _PO4-3 (3) _ (s)</b><br>Rock Bridgite; Rockbridgite; Rockbridgeite; Bridgite (rock) |            |     |                                   |         |
| 0                                                                                                               | 61180-51-6 | 1   | Fe (5) H (5) O (17) P (3)         | 649.178 |
| <b>Fe+3 (4) _OH-1 (3) _PO4-3 (3) _H2O (12) _ (s)</b><br>Cacoxenite                                              |            |     |                                   |         |
| 0                                                                                                               | ---        | 2   | Fe (4) H (27) O (27) P (3)        | 775.501 |
| <b>Fe+3 (4) _P2O7-4 (3) _ (s)</b>                                                                               |            |     |                                   |         |
| 0                                                                                                               | ---        | 1   | Fe (4) O (21) P (6)               | 745.212 |
| <b>Fe+3 (5) _Fe+2 _OH-1 (5) _PO4-3 (4) _H2O (6) _ (s)</b><br>Beraunite                                          |            |     |                                   |         |
| 0                                                                                                               | 12187-94-9 | 1   | Fe (6) H (17) O (27) P (4)        | 908.086 |
| <b>Fe (s)</b><br>Iron; Iron, cubic; Iron, bcc                                                                   |            |     |                                   |         |
| 0                                                                                                               | 7439-89-6  | 184 | Fe (1)                            | 55.8452 |
| <b>Fe3KH8 (PO4) 6 . 6H2O (s)</b>                                                                                |            |     |                                   |         |
| 0                                                                                                               | ---        | 1   | Fe (3) H (20) K (1) O (30) P (6)  | 892.618 |
| <b>Fe3NH4H8 (PO4) 6 . 6H2O (s)</b>                                                                              |            |     |                                   |         |
| 0                                                                                                               | ---        | 1   | Fe (3) H (24) N (1) O (30) P (6)  | 871.559 |
| <b>Ga+3</b><br>Gallium(III) ion                                                                                 |            |     |                                   |         |
| 3                                                                                                               | 22537-33-3 | 472 | Ga (1)                            | 69.7230 |
| <b>Ga+3 _H+1 _P2O7-4</b>                                                                                        |            |     |                                   |         |
| 0                                                                                                               | ---        | 1   | Ga (1) H (1) O (7) P (2)          | 244.675 |
| <b>Ga+3 _H+1 _P3O10-5</b>                                                                                       |            |     |                                   |         |
| -1                                                                                                              | ---        | 1   | Ga (1) H (1) O (10) P (3)         | 323.647 |

|                                            |            |     |                                       |         |
|--------------------------------------------|------------|-----|---------------------------------------|---------|
| <b>Ga+3_H+1_PO4-3</b>                      |            |     |                                       |         |
| 1                                          | ---        | 2   | Ga (1) H (1) O (4) P (1)              | 165.703 |
| <b>Ga+3_H+1 (2)_P3O10-5 (2)</b>            |            |     |                                       |         |
| -5                                         | ---        | 1   | Ga (1) H (2) O (20) P (6)             | 577.571 |
| <b>Ga+3_H+1 (2)_PO4-3</b>                  |            |     |                                       |         |
| 2                                          | ---        | 2   | Ga (1) H (2) O (4) P (1)              | 166.711 |
| <b>Ga+3_H+1 (4)_P3O10-5 (2)</b>            |            |     |                                       |         |
| -3                                         | ---        | 1   | Ga (1) H (4) O (20) P (6)             | 579.587 |
| <b>Ga+3_P3O10-5</b>                        |            |     |                                       |         |
| -2                                         | ---        | 1   | Ga (1) O (10) P (3)                   | 322.639 |
| <b>Ga+3_PO4-3 (s)</b><br>Gallium phosphate |            |     |                                       |         |
| 0                                          | ---        | 1   | Ga (1) O (4) P (1)                    | 164.695 |
| <b>Ga+3_Thr-1_P3O10-5</b>                  |            |     |                                       |         |
| -3                                         | ---        | 1   | C (4) H (8) Ga (1) N (1) O (13) P (3) | 440.751 |
| <b>Ga+3 (2)_H+1_PO4-3</b>                  |            |     |                                       |         |
| 4                                          | ---        | 2   | Ga (2) H (1) O (4) P (1)              | 235.426 |
| <b>Ga (s)</b><br>Gallium                   |            |     |                                       |         |
| 0                                          | 7440-55-3  | 36  | Ga (1)                                | 69.7230 |
| <b>GaP (s)</b><br>Gallium phosphide        |            |     |                                       |         |
| 0                                          | ---        | 1   | Ga (1) P (1)                          | 100.697 |
| <b>Gd+3</b><br>Gadolinium(III) ion         |            |     |                                       |         |
| 3                                          | 22541-19-1 | 790 | Gd (1)                                | 157.253 |
| <b>Gd+3_EDTA-4</b>                         |            |     |                                       |         |

|                                 |     |    |                                         |         |
|---------------------------------|-----|----|-----------------------------------------|---------|
| -1                              | --- | 14 | C (10) H (12) Gd (1) N (2) O (8)        | 445.467 |
| <b>Gd+3_EDTA-4_P3O10-5</b>      |     |    |                                         |         |
| -6                              | --- | 2  | C (10) H (12) Gd (1) N (2) O (18) P (3) | 698.383 |
| <b>Gd+3_H+1_EDTA-4_P3O10-5</b>  |     |    |                                         |         |
| -5                              | --- | 2  | C (10) H (13) Gd (1) N (2) O (18) P (3) | 699.391 |
| <b>Gd+3_H+1_P3O10-5</b>         |     |    |                                         |         |
| -1                              | --- | 1  | Gd (1) H (1) O (10) P (3)               | 411.177 |
| <b>Gd+3_H+1_PO4-3</b>           |     |    |                                         |         |
| 1                               | --- | 2  | Gd (1) H (1) O (4) P (1)                | 253.233 |
| <b>Gd+3_H+1 (2)_P3O10-5 (2)</b> |     |    |                                         |         |
| -5                              | --- | 1  | Gd (1) H (2) O (20) P (6)               | 665.101 |
| <b>Gd+3_H+1 (2)_PO4-3</b>       |     |    |                                         |         |
| 2                               | --- | 2  | Gd (1) H (2) O (4) P (1)                | 254.241 |
| <b>Gd+3_H+1 (2)_PO4-3 (2)</b>   |     |    |                                         |         |
| -1                              | --- | 2  | Gd (1) H (2) O (8) P (2)                | 349.212 |
| <b>Gd+3_P3O10-5 (2)</b>         |     |    |                                         |         |
| -7                              | --- | 1  | Gd (1) O (20) P (6)                     | 663.085 |
| <b>Gd+3_PO4-3</b>               |     |    |                                         |         |
| 0                               | --- | 1  | Gd (1) O (4) P (1)                      | 252.225 |
| <b>Gd+3_PO4-3_(s)</b>           |     |    |                                         |         |
| 0                               | --- | 1  | Gd (1) O (4) P (1)                      | 252.225 |
| <b>Gd+3_PO4-3 (2)</b>           |     |    |                                         |         |
| -3                              | --- | 1  | Gd (1) O (8) P (2)                      | 347.196 |
| <b>Gd+3 (2)_P2O7-4</b>          |     |    |                                         |         |
| 2                               | --- | 1  | Gd (2) O (7) P (2)                      | 488.450 |



|                                                                                                                                          |            |       |                                 |         |
|------------------------------------------------------------------------------------------------------------------------------------------|------------|-------|---------------------------------|---------|
| <b>Ge (s)</b><br>Germanium; Germanium, cubic; Germanium (diamond structure)                                                              |            |       |                                 |         |
| 0                                                                                                                                        | 7440-56-4  | 29    | Ge (1)                          | 72.6300 |
| <b>GeP (s)</b><br>Germanium phosphide                                                                                                    |            |       |                                 |         |
| 0                                                                                                                                        | ---        | 1     | Ge (1) P (1)                    | 103.604 |
| <b>Gln-1</b><br>Glutamate ion; L-Glutamate ion; 2-aminoglutaramate ion; glutamate 5-amide ion; L-2-aminopentanedioate 5-amide ion        |            |       |                                 |         |
| -1                                                                                                                                       | ---        | 536   | C (5) H (9) N (2) O (3)         | 145.138 |
| <b>Glu-2</b><br>Glutamate ion; 2-aminopentanedioate ion; L-Glutamate ion; alpha-aminoglutarate ion; 1-aminopropane-1,3-dicarboxylate ion |            |       |                                 |         |
| -2                                                                                                                                       | 138-18-1   | 650   | C (5) H (7) N (1) O (4)         | 145.115 |
| <b>Gly-1</b><br>Glycinate ion; Aminoacetate ion                                                                                          |            |       |                                 |         |
| -1                                                                                                                                       | ---        | 1356  | C (2) H (4) N (1) O (2)         | 74.0593 |
| <b>H+1</b><br>Hydrogen ion; Proton                                                                                                       |            |       |                                 |         |
| 1                                                                                                                                        | 12408-02-5 | 31679 | H (1)                           | 1.00794 |
| <b>H+1_16HexDiAm_P2O7-4</b>                                                                                                              |            |       |                                 |         |
| -3                                                                                                                                       | ---        | 1     | C (6) H (17) N (2) O (7) P (2)  | 291.158 |
| <b>H+1_16HexDiAm_P3O10-5</b>                                                                                                             |            |       |                                 |         |
| -4                                                                                                                                       | ---        | 1     | C (6) H (17) N (2) O (10) P (3) | 370.130 |
| <b>H+1_18OctDiAm_P2O7-4</b>                                                                                                              |            |       |                                 |         |
| -3                                                                                                                                       | ---        | 1     | C (8) H (21) N (2) O (7) P (2)  | 319.212 |
| <b>H+1_18OctDiAm_P3O10-5</b>                                                                                                             |            |       |                                 |         |
| -4                                                                                                                                       | ---        | 1     | C (8) H (21) N (2) O (10) P (3) | 398.184 |
| <b>H+1_Adenosine-1</b><br>Adenosine; 9-beta-D-Ribofuranosyladenine; Adenine-9-beta-D-ribofuranoside                                      |            |       |                                 |         |

|                                                                                 |         |    |                       |         |
|---------------------------------------------------------------------------------|---------|----|-----------------------|---------|
| 0                                                                               | 58-61-7 | 17 | C(10)H(13)N(5)O(4)    | 267.244 |
| <b>H+1_CrO4-2</b>                                                               |         |    |                       |         |
| -1                                                                              | ---     | 16 | Cr(1)H(1)O(4)         | 117.002 |
| <b>H+1_Dien_P2O7-4</b>                                                          |         |    |                       |         |
| -3                                                                              | ---     | 1  | C(4)H(14)N(3)O(7)P(2) | 278.119 |
| <b>H+1_EtDiAm_P2O7-4</b>                                                        |         |    |                       |         |
| -3                                                                              | ---     | 1  | C(2)H(9)N(2)O(7)P(2)  | 235.051 |
| <b>H+1_EtDiAm_P3O10-5</b>                                                       |         |    |                       |         |
| -4                                                                              | ---     | 1  | C(2)H(9)N(2)O(10)P(3) | 314.023 |
| <b>H+1_H2PO2-1</b>                                                              |         |    |                       |         |
| 0                                                                               | ---     | 9  | H(3)O(2)P(1)          | 65.9966 |
| <b>H+1_Hg+1(2)_PO4-3_(S)</b>                                                    |         |    |                       |         |
| 0                                                                               | ---     | 1  | H(1)Hg(2)O(4)P(1)     | 497.164 |
| <b>H+1_HgCH3+1_HPO3-2</b>                                                       |         |    |                       |         |
| 0                                                                               | ---     | 2  | C(1)H(5)Hg(1)O(3)P(1) | 296.615 |
| <b>H+1_HgCH3+1_PO4-3</b>                                                        |         |    |                       |         |
| -1                                                                              | ---     | 3  | C(1)H(4)Hg(1)O(4)P(1) | 311.606 |
| <b>H+1_HPO3-2</b>                                                               |         |    |                       |         |
| -1                                                                              | ---     | 16 | H(2)O(3)P(1)          | 80.9881 |
| <b>H+1_Inosine-1</b>                                                            |         |    |                       |         |
| Inosine; Hypoxanthine 9-D-ribofuranoside; 9-(beta-D-Ribofuranosyl)-hypoxanthine |         |    |                       |         |
| 0                                                                               | 58-63-9 | 10 | C(10)H(12)N(4)O(5)    | 268.229 |
| <b>H+1_K+1_HPO3-2</b>                                                           |         |    |                       |         |
| 0                                                                               | ---     | 2  | H(2)K(1)O(3)P(1)      | 120.086 |
| <b>H+1_K+1_P2O7-4</b>                                                           |         |    |                       |         |

|                                                                                    |           |   |                    |         |
|------------------------------------------------------------------------------------|-----------|---|--------------------|---------|
| -2                                                                                 | ---       | 2 | H(1)K(1)O(7)P(2)   | 214.050 |
| H+1_K+1_P3O10-5                                                                    |           |   |                    |         |
| -3                                                                                 | ---       | 2 | H(1)K(1)O(10)P(3)  | 293.022 |
| H+1_K+1_P3O8-5                                                                     |           |   |                    |         |
| -3                                                                                 | ---       | 1 | H(1)K(1)O(8)P(3)   | 261.023 |
| H+1_K+1_P4O13-6                                                                    |           |   |                    |         |
| -4                                                                                 | ---       | 1 | H(1)K(1)O(13)P(4)  | 371.994 |
| H+1_K+1_PO4-3                                                                      |           |   |                    |         |
| -1                                                                                 | ---       | 2 | H(1)K(1)O(4)P(1)   | 135.078 |
| H+1_K+1(2)_P2O7-4                                                                  |           |   |                    |         |
| -1                                                                                 | ---       | 1 | H(1)K(2)O(7)P(2)   | 253.148 |
| H+1_K+1(2)_P3O10-5                                                                 |           |   |                    |         |
| -2                                                                                 | ---       | 1 | H(1)K(2)O(10)P(3)  | 332.120 |
| H+1_K+1(2)_PO4-3                                                                   |           |   |                    |         |
| 0                                                                                  | ---       | 2 | H(1)K(2)O(4)P(1)   | 174.176 |
| H+1_K+1(2)_PO4-3(s)<br>Dipotassium hydrogen phosphate; Potassium phosphate dibasic |           |   |                    |         |
| 0                                                                                  | 7758-11-4 | 2 | H(1)K(2)O(4)P(1)   | 174.176 |
| H+1_Li+1_P2O7-4                                                                    |           |   |                    |         |
| -2                                                                                 | ---       | 1 | H(1)Li(1)O(7)P(2)  | 181.893 |
| H+1_Li+1_P3O10-5                                                                   |           |   |                    |         |
| -3                                                                                 | ---       | 1 | H(1)Li(1)O(10)P(3) | 260.865 |
| H+1_Li+1_P3O8-5                                                                    |           |   |                    |         |
| -3                                                                                 | ---       | 1 | H(1)Li(1)O(8)P(3)  | 228.866 |
| H+1_Li+1_P4O13-6                                                                   |           |   |                    |         |

|                             |     |   |                                |         |
|-----------------------------|-----|---|--------------------------------|---------|
| -4                          | --- | 1 | H (1) Li (1) O (13) P (4)      | 339.837 |
| <b>H+1_Li+1_PO4-3</b>       |     |   |                                |         |
| -1                          | --- | 2 | H (1) Li (1) O (4) P (1)       | 102.921 |
| <b>H+1_MeAm_P2O7-4</b>      |     |   |                                |         |
| -3                          | --- | 1 | C (1) H (6) N (1) O (7) P (2)  | 206.009 |
| <b>H+1_MeAm_P3O10-5</b>     |     |   |                                |         |
| -4                          | --- | 1 | C (1) H (6) N (1) O (10) P (3) | 284.981 |
| <b>H+1_Na+1_HPO3-2</b>      |     |   |                                |         |
| 0                           | --- | 2 | H (2) Na (1) O (3) P (1)       | 103.978 |
| <b>H+1_Na+1_P2O7-4</b>      |     |   |                                |         |
| -2                          | --- | 2 | H (1) Na (1) O (7) P (2)       | 197.942 |
| <b>H+1_Na+1_P3O10-5</b>     |     |   |                                |         |
| -3                          | --- | 2 | H (1) Na (1) O (10) P (3)      | 276.914 |
| <b>H+1_Na+1_P3O8-5</b>      |     |   |                                |         |
| -3                          | --- | 1 | H (1) Na (1) O (8) P (3)       | 244.915 |
| <b>H+1_Na+1_P4O13-6</b>     |     |   |                                |         |
| -4                          | --- | 1 | H (1) Na (1) O (13) P (4)      | 355.886 |
| <b>H+1_Na+1_PO4-3</b>       |     |   |                                |         |
| -1                          | --- | 2 | H (1) Na (1) O (4) P (1)       | 118.970 |
| <b>H+1_Na+1 (2)_P2O7-4</b>  |     |   |                                |         |
| -1                          | --- | 1 | H (1) Na (2) O (7) P (2)       | 220.932 |
| <b>H+1_Na+1 (2)_P3O10-5</b> |     |   |                                |         |
| -2                          | --- | 1 | H (1) Na (2) O (10) P (3)      | 299.904 |
| <b>H+1_Na+1 (2)_PO4-3</b>   |     |   |                                |         |
| 0                           | --- | 2 | H (1) Na (2) O (4) P (1)       | 141.960 |

|                                                                             |            |     |                     |         |
|-----------------------------------------------------------------------------|------------|-----|---------------------|---------|
| <b>H+1_Na+1(2)_PO4-3_H2O(12)_(s)</b><br>Disodium hydrogen phosphate hydrate |            |     |                     |         |
| 0                                                                           | ---        | 2   | H(25)Na(2)O(16)P(1) | 358.143 |
| <b>H+1_NH3</b><br>Ammonium ion                                              |            |     |                     |         |
| 1                                                                           | 14798-03-9 | 108 | H(4)N(1)            | 18.0385 |
| <b>H+1_NH3_PO4-3</b>                                                        |            |     |                     |         |
| -2                                                                          | ---        | 3   | H(4)N(1)O(4)P(1)    | 113.010 |
| <b>H+1_NpO2+1_PO4-3</b>                                                     |            |     |                     |         |
| -1                                                                          | ---        | 3   | H(1)Np(1)O(6)P(1)   | 365.026 |
| <b>H+1_P2O7-4</b>                                                           |            |     |                     |         |
| -3                                                                          | 42499-21-8 | 20  | H(1)O(7)P(2)        | 174.952 |
| <b>H+1_P3O10-5</b>                                                          |            |     |                     |         |
| -4                                                                          | ---        | 56  | H(1)O(10)P(3)       | 253.924 |
| <b>H+1_P3O8-5</b>                                                           |            |     |                     |         |
| -4                                                                          | ---        | 2   | H(1)O(8)P(3)        | 221.925 |
| <b>H+1_P3O9-3</b>                                                           |            |     |                     |         |
| -2                                                                          | ---        | 3   | H(1)O(9)P(3)        | 237.925 |
| <b>H+1_P4O12-4</b>                                                          |            |     |                     |         |
| -3                                                                          | ---        | 1   | H(1)O(12)P(4)       | 316.897 |
| <b>H+1_P4O13-6</b>                                                          |            |     |                     |         |
| -5                                                                          | ---        | 10  | H(1)O(13)P(4)       | 332.896 |
| <b>H+1_P6O18-6</b>                                                          |            |     |                     |         |
| -5                                                                          | ---        | 1   | H(1)O(18)P(6)       | 474.841 |
| <b>H+1_PaO2+1_PO4-3</b>                                                     |            |     |                     |         |
| -1                                                                          | ---        | 2   | H(1)O(6)P(1)Pa(1)   | 359.018 |

|                                           |     |     |                        |         |
|-------------------------------------------|-----|-----|------------------------|---------|
| <b>H+1_Pb(CH3)2+2_PO4-3</b>               |     |     |                        |         |
| 0                                         | --- | 1   | C(2)H(7)O(4)P(1)Pb(1)  | 333.249 |
| <b>H+1_Pb(CH3)3+1_PO4-3</b>               |     |     |                        |         |
| -1                                        | --- | 2   | C(3)H(10)O(4)P(1)Pb(1) | 348.284 |
| <b>H+1_Penten_P2O7-4</b>                  |     |     |                        |         |
| -3                                        | --- | 1   | C(10)H(29)N(6)O(7)P(2) | 407.324 |
| <b>H+1_PH3</b>                            |     |     |                        |         |
| 1                                         | --- | 1   | H(4)P(1)               | 35.0058 |
| <b>H+1_PO4-3</b><br>Hydrogenphosphate ion |     |     |                        |         |
| -2                                        | --- | 317 | H(1)O(4)P(1)           | 95.9795 |
| <b>H+1_PuO2+1_PO4-3</b>                   |     |     |                        |         |
| -1                                        | --- | 3   | H(1)O(6)P(1)Pu(1)      | 371.978 |
| <b>H+1_Sn(CH3)2+2_P3O10-5</b>             |     |     |                        |         |
| -2                                        | --- | 2   | C(2)H(7)O(10)P(3)Sn(1) | 402.704 |
| <b>H+1_Tetren_P2O7-4</b>                  |     |     |                        |         |
| -3                                        | --- | 1   | C(8)H(24)N(5)O(7)P(2)  | 364.256 |
| <b>H+1_Tl+1_P2O7-4</b>                    |     |     |                        |         |
| -2                                        | --- | 2   | H(1)O(7)P(2)Tl(1)      | 379.335 |
| <b>H+1_Tl+1_PO4-3</b>                     |     |     |                        |         |
| -1                                        | --- | 2   | H(1)O(4)P(1)Tl(1)      | 300.363 |
| <b>H+1_Trien_P2O7-4</b>                   |     |     |                        |         |
| -3                                        | --- | 1   | C(6)H(19)N(4)O(7)P(2)  | 321.187 |
| <b>H+1_UO2+1_PO4-3</b>                    |     |     |                        |         |
| -1                                        | --- | 2   | H(1)O(6)P(1)U(1)       | 366.007 |

|                                 |     |    |                        |         |
|---------------------------------|-----|----|------------------------|---------|
| <b>H+1_VO2+1_P2O7-4</b>         |     |    |                        |         |
| -2                              | --- | 1  | H(1)O(9)P(2)V(1)       | 257.893 |
| <b>H+1_VO2+1_P3O10-5</b>        |     |    |                        |         |
| -3                              | --- | 1  | H(1)O(12)P(3)V(1)      | 336.865 |
| <b>H+1_VO2+1_PO4-3</b>          |     |    |                        |         |
| -1                              | --- | 1  | H(1)O(6)P(1)V(1)       | 178.920 |
| <b>H+1(2)_13PrDiAm</b>          |     |    |                        |         |
| 2                               | --- | 4  | C(3)H(12)N(2)          | 76.1417 |
| <b>H+1(2)_13PrDiAm_P2O7-4</b>   |     |    |                        |         |
| -2                              | --- | 1  | C(3)H(12)N(2)O(7)P(2)  | 250.086 |
| <b>H+1(2)_14BuDiAm</b>          |     |    |                        |         |
| 2                               | --- | 11 | C(4)H(14)N(2)          | 90.1686 |
| <b>H+1(2)_14BuDiAm_P2O7-4</b>   |     |    |                        |         |
| -2                              | --- | 1  | C(4)H(14)N(2)O(7)P(2)  | 264.112 |
| <b>H+1(2)_16HexDiAm_P2O7-4</b>  |     |    |                        |         |
| -2                              | --- | 1  | C(6)H(18)N(2)O(7)P(2)  | 292.166 |
| <b>H+1(2)_16HexDiAm_P3O10-5</b> |     |    |                        |         |
| -3                              | --- | 1  | C(6)H(18)N(2)O(10)P(3) | 371.138 |
| <b>H+1(2)_18OctDiAm_P2O7-4</b>  |     |    |                        |         |
| -2                              | --- | 1  | C(8)H(22)N(2)O(7)P(2)  | 320.220 |
| <b>H+1(2)_18OctDiAm_P3O10-5</b> |     |    |                        |         |
| -3                              | --- | 1  | C(8)H(22)N(2)O(10)P(3) | 399.192 |
| <b>H+1(2)_Cadaverine</b>        |     |    |                        |         |
| 2                               | --- | 11 | C(5)H(16)N(2)          | 104.195 |

|                                                      |            |   |                                 |         |
|------------------------------------------------------|------------|---|---------------------------------|---------|
| <b>H+1 (2) _Cadaverine_P2O7-4</b>                    |            |   |                                 |         |
| -2                                                   | ---        | 1 | C (5) H (16) N (2) O (7) P (2)  | 278.139 |
| <b>H+1 (2) _Dien_P2O7-4</b>                          |            |   |                                 |         |
| -2                                                   | ---        | 1 | C (4) H (15) N (3) O (7) P (2)  | 279.127 |
| <b>H+1 (2) _EtDiAm_P2O7-4</b>                        |            |   |                                 |         |
| -2                                                   | ---        | 1 | C (2) H (10) N (2) O (7) P (2)  | 236.059 |
| <b>H+1 (2) _EtDiAm_P3O10-5</b>                       |            |   |                                 |         |
| -3                                                   | ---        | 1 | C (2) H (10) N (2) O (10) P (3) | 315.031 |
| <b>H+1 (2) _EtDiAm_PO4-3</b>                         |            |   |                                 |         |
| -1                                                   | ---        | 1 | C (2) H (10) N (2) O (4) P (1)  | 157.086 |
| <b>H+1 (2) _H2PO2-1</b>                              |            |   |                                 |         |
| 1                                                    | ---        | 1 | H (4) O (2) P (1)               | 67.0046 |
| <b>H+1 (2) _HPO3-2</b>                               |            |   |                                 |         |
| 0                                                    | 13598-36-2 | 4 | H (3) O (3) P (1)               | 81.9960 |
| <b>H+1 (2) _HPO3-2_(s)</b><br>Phosphorous acid solid |            |   |                                 |         |
| 0                                                    | ---        | 1 | H (3) O (3) P (1)               | 81.9960 |
| <b>H+1 (2) _HPO3-2 (2)</b>                           |            |   |                                 |         |
| -2                                                   | ---        | 1 | H (4) O (6) P (2)               | 161.976 |
| <b>H+1 (2) _K+1_P2O7-4</b>                           |            |   |                                 |         |
| -1                                                   | ---        | 1 | H (2) K (1) O (7) P (2)         | 215.058 |
| <b>H+1 (2) _K+1_P3O10-5</b>                          |            |   |                                 |         |
| -2                                                   | ---        | 1 | H (2) K (1) O (10) P (3)        | 294.030 |
| <b>H+1 (2) _K+1_PO4-3</b>                            |            |   |                                 |         |
| 0                                                    | ---        | 3 | H (2) K (1) O (4) P (1)         | 136.086 |



|                                                                                                |           |   |                                 |         |
|------------------------------------------------------------------------------------------------|-----------|---|---------------------------------|---------|
| <b>H+1 (2) _K+1_PO4-3_(s)</b><br>Potassium dihydrogen phosphate; Potassium phosphate monobasic |           |   |                                 |         |
| 0                                                                                              | 7778-77-0 | 3 | H (2) K (1) O (4) P (1)         | 136.086 |
| <b>H+1 (2) _K+1 (2)_P3O10-5</b>                                                                |           |   |                                 |         |
| -1                                                                                             | ---       | 1 | H (2) K (2) O (10) P (3)        | 333.128 |
| <b>H+1 (2) _Li+1_PO4-3</b>                                                                     |           |   |                                 |         |
| 0                                                                                              | ---       | 2 | H (2) Li (1) O (4) P (1)        | 103.929 |
| <b>H+1 (2) _MeAm_P2O7-4</b>                                                                    |           |   |                                 |         |
| -2                                                                                             | ---       | 1 | C (1) H (7) N (1) O (7) P (2)   | 207.017 |
| <b>H+1 (2) _MeAm_P3O10-5</b>                                                                   |           |   |                                 |         |
| -3                                                                                             | ---       | 1 | C (1) H (7) N (1) O (10) P (3)  | 285.989 |
| <b>H+1 (2) _MeAm (2)_P2O7-4</b>                                                                |           |   |                                 |         |
| -2                                                                                             | ---       | 1 | C (2) H (12) N (2) O (7) P (2)  | 238.075 |
| <b>H+1 (2) _MeAm (2)_P3O10-5</b>                                                               |           |   |                                 |         |
| -3                                                                                             | ---       | 1 | C (2) H (12) N (2) O (10) P (3) | 317.047 |
| <b>H+1 (2) _Na+1_P2O7-4</b>                                                                    |           |   |                                 |         |
| -1                                                                                             | ---       | 1 | H (2) Na (1) O (7) P (2)        | 198.950 |
| <b>H+1 (2) _Na+1_P3O10-5</b>                                                                   |           |   |                                 |         |
| -2                                                                                             | ---       | 1 | H (2) Na (1) O (10) P (3)       | 277.922 |
| <b>H+1 (2) _Na+1_PO4-3</b>                                                                     |           |   |                                 |         |
| 0                                                                                              | ---       | 3 | H (2) Na (1) O (4) P (1)        | 119.978 |
| <b>H+1 (2) _Na+1 (2)_P3O10-5</b>                                                               |           |   |                                 |         |
| -1                                                                                             | ---       | 1 | H (2) Na (2) O (10) P (3)       | 300.912 |
| <b>H+1 (2) _NH3_PO4-3</b>                                                                      |           |   |                                 |         |
| -1                                                                                             | ---       | 3 | H (5) N (1) O (4) P (1)         | 114.018 |

|                                  |     |    |                                 |         |
|----------------------------------|-----|----|---------------------------------|---------|
| <b>H+1 (2) _NpO2+1_PO4-3</b>     |     |    |                                 |         |
| 0                                | --- | 2  | H (2) Np (1) O (6) P (1)        | 366.034 |
| <b>H+1 (2) _NpO2+1_PO4-3 (2)</b> |     |    |                                 |         |
| -3                               | --- | 2  | H (2) Np (1) O (10) P (2)       | 461.006 |
| <b>H+1 (2) _P2O6-4</b>           |     |    |                                 |         |
| -2                               | --- | 1  | H (2) O (6) P (2)               | 159.960 |
| <b>H+1 (2) _P2O7-4</b>           |     |    |                                 |         |
| -2                               | --- | 23 | H (2) O (7) P (2)               | 175.960 |
| <b>H+1 (2) _P3O10-5</b>          |     |    |                                 |         |
| -3                               | --- | 11 | H (2) O (10) P (3)              | 254.932 |
| <b>H+1 (2) _P3O10-5 (2)</b>      |     |    |                                 |         |
| -8                               | --- | 14 | H (2) O (20) P (6)              | 507.848 |
| <b>H+1 (2) _P3O8-5</b>           |     |    |                                 |         |
| -3                               | --- | 1  | H (2) O (8) P (3)               | 222.933 |
| <b>H+1 (2) _P3O9-3</b>           |     |    |                                 |         |
| -1                               | --- | 1  | H (2) O (9) P (3)               | 238.933 |
| <b>H+1 (2) _P4O13-6</b>          |     |    |                                 |         |
| -4                               | --- | 3  | H (2) O (13) P (4)              | 333.904 |
| <b>H+1 (2) _PaO2+1_PO4-3</b>     |     |    |                                 |         |
| 0                                | --- | 2  | H (2) O (6) P (1) Pa (1)        | 360.026 |
| <b>H+1 (2) _PaO2+1_PO4-3 (2)</b> |     |    |                                 |         |
| -3                               | --- | 2  | H (2) O (10) P (2) Pa (1)       | 454.998 |
| <b>H+1 (2) _Penten_P2O7-4</b>    |     |    |                                 |         |
| -2                               | --- | 1  | C (10) H (30) N (6) O (7) P (2) | 408.332 |

|                                                 |            |     |                                 |         |
|-------------------------------------------------|------------|-----|---------------------------------|---------|
| <b>H+1 (2) _PO3-3</b>                           |            |     |                                 |         |
| -1                                              | ---        | 2   | H (2) O (3) P (1)               | 80.9881 |
| <b>H+1 (2) _PO4-3</b><br>Dihydrogenphospate ion |            |     |                                 |         |
| -1                                              | 14066-20-7 | 284 | H (2) O (4) P (1)               | 96.9875 |
| <b>H+1 (2) _PO4-3 (2)</b>                       |            |     |                                 |         |
| -4                                              | ---        | 2   | H (2) O (8) P (2)               | 191.959 |
| <b>H+1 (2) _PuO2+1 _PO4-3</b>                   |            |     |                                 |         |
| 0                                               | ---        | 1   | H (2) O (6) P (1) Pu (1)        | 372.986 |
| <b>H+1 (2) _PuO2+1 _PO4-3 (2)</b>               |            |     |                                 |         |
| -3                                              | ---        | 1   | H (2) O (10) P (2) Pu (1)       | 467.958 |
| <b>H+1 (2) _Sn (CH3) 2+2 _P3O10-5</b>           |            |     |                                 |         |
| -1                                              | ---        | 2   | C (2) H (8) O (10) P (3) Sn (1) | 403.712 |
| <b>H+1 (2) _Spermidine</b>                      |            |     |                                 |         |
| 2                                               | ---        | 4   | C (7) H (21) N (3)              | 147.264 |
| <b>H+1 (2) _Spermidine _P2O7-4</b>              |            |     |                                 |         |
| -2                                              | ---        | 1   | C (7) H (21) N (3) O (7) P (2)  | 321.208 |
| <b>H+1 (2) _Tetren _P2O7-4</b>                  |            |     |                                 |         |
| -2                                              | ---        | 1   | C (8) H (25) N (5) O (7) P (2)  | 365.264 |
| <b>H+1 (2) _Tetren _PO4-3</b>                   |            |     |                                 |         |
| -1                                              | ---        | 1   | C (8) H (25) N (5) O (4) P (1)  | 286.292 |
| <b>H+1 (2) _Trien _P2O7-4</b>                   |            |     |                                 |         |
| -2                                              | ---        | 1   | C (6) H (20) N (4) O (7) P (2)  | 322.195 |
| <b>H+1 (2) _UO2+1 _PO4-3</b>                    |            |     |                                 |         |
| 0                                               | ---        | 2   | H (2) O (6) P (1) U (1)         | 367.015 |

|                                   |     |    |                                 |         |
|-----------------------------------|-----|----|---------------------------------|---------|
| <b>H+1 (2) _UO2+1_PO4-3 (2)</b>   |     |    |                                 |         |
| -3                                | --- | 2  | H (2) O (10) P (2) U (1)        | 461.987 |
| <b>H+1 (2) _VO2+1_P2O7-4</b>      |     |    |                                 |         |
| -1                                | --- | 1  | H (2) O (9) P (2) V (1)         | 258.901 |
| <b>H+1 (2) _VO2+1_P3O10-5</b>     |     |    |                                 |         |
| -2                                | --- | 1  | H (2) O (12) P (3) V (1)        | 337.873 |
| <b>H+1 (2) _VO2+1_P3O10-5 (2)</b> |     |    |                                 |         |
| -7                                | --- | 2  | H (2) O (22) P (6) V (1)        | 590.789 |
| <b>H+1 (2) _VO2+1_PO4-3</b>       |     |    |                                 |         |
| 0                                 | --- | 1  | H (2) O (6) P (1) V (1)         | 179.928 |
| <b>H+1 (2) _VO2+1_PO4-3 (2)</b>   |     |    |                                 |         |
| -3                                | --- | 1  | H (2) O (10) P (2) V (1)        | 274.900 |
| <b>H+1 (3) _[16]N5</b>            |     |    |                                 |         |
| 3                                 | --- | 11 | C (11) H (30) N (5)             | 232.393 |
| <b>H+1 (3) _[16]N5_PO4-3</b>      |     |    |                                 |         |
| 0                                 | --- | 1  | C (11) H (30) N (5) O (4) P (1) | 327.364 |
| <b>H+1 (3) _[18]N6</b>            |     |    |                                 |         |
| 3                                 | --- | 12 | C (12) H (33) N (6)             | 261.434 |
| <b>H+1 (3) _[18]N6_PO4-3</b>      |     |    |                                 |         |
| 0                                 | --- | 1  | C (12) H (33) N (6) O (4) P (1) | 356.406 |
| <b>H+1 (3) _16HexDiAm_P2O7-4</b>  |     |    |                                 |         |
| -1                                | --- | 1  | C (6) H (19) N (2) O (7) P (2)  | 293.174 |
| <b>H+1 (3) _16HexDiAm_P3O10-5</b> |     |    |                                 |         |
| -2                                | --- | 1  | C (6) H (19) N (2) O (10) P (3) | 372.146 |

|                                 |     |   |                        |         |
|---------------------------------|-----|---|------------------------|---------|
| <b>H+1(3)_18OctDiAm_P2O7-4</b>  |     |   |                        |         |
| -1                              | --- | 1 | C(8)H(23)N(2)O(7)P(2)  | 321.228 |
| <b>H+1(3)_18OctDiAm_P3O10-5</b> |     |   |                        |         |
| -2                              | --- | 1 | C(8)H(23)N(2)O(10)P(3) | 400.200 |
| <b>H+1(3)_Dien_P2O7-4</b>       |     |   |                        |         |
| -1                              | --- | 1 | C(4)H(16)N(3)O(7)P(2)  | 280.135 |
| <b>H+1(3)_EtDiAm_P2O7-4</b>     |     |   |                        |         |
| -1                              | --- | 1 | C(2)H(11)N(2)O(7)P(2)  | 237.067 |
| <b>H+1(3)_EtDiAm_P3O10-5</b>    |     |   |                        |         |
| -2                              | --- | 1 | C(2)H(11)N(2)O(10)P(3) | 316.039 |
| <b>H+1(3)_EtDiAm_PO4-3</b>      |     |   |                        |         |
| 0                               | --- | 1 | C(2)H(11)N(2)O(4)P(1)  | 158.094 |
| <b>H+1(3)_K+1_P3O10-5</b>       |     |   |                        |         |
| -1                              | --- | 1 | H(3)K(1)O(10)P(3)      | 295.038 |
| <b>H+1(3)_MeAm_P2O7-4</b>       |     |   |                        |         |
| -1                              | --- | 1 | C(1)H(8)N(1)O(7)P(2)   | 208.025 |
| <b>H+1(3)_MeAm_P3O10-5</b>      |     |   |                        |         |
| -2                              | --- | 1 | C(1)H(8)N(1)O(10)P(3)  | 286.997 |
| <b>H+1(3)_MeAm(2)_P2O7-4</b>    |     |   |                        |         |
| -1                              | --- | 1 | C(2)H(13)N(2)O(7)P(2)  | 239.082 |
| <b>H+1(3)_Na+1_P3O10-5</b>      |     |   |                        |         |
| -1                              | --- | 1 | H(3)Na(1)O(10)P(3)     | 278.930 |
| <b>H+1(3)_NH3_PO4-3</b>         |     |   |                        |         |
| 0                               | --- | 2 | H(6)N(1)O(4)P(1)       | 115.026 |

|                                                     |     |    |                        |         |
|-----------------------------------------------------|-----|----|------------------------|---------|
| <b>H+1 (3) _NH3_PO4-3_(s)</b>                       |     |    |                        |         |
| 0                                                   | --- | 1  | H(6)N(1)O(4)P(1)       | 115.026 |
| <b>H+1 (3) _NH3(3)_PO4-3_(s)</b>                    |     |    |                        |         |
| 0                                                   | --- | 1  | H(12)N(3)O(4)P(1)      | 149.087 |
| <b>H+1 (3) _NpO2+1_PO4-3(3)</b>                     |     |    |                        |         |
| -5                                                  | --- | 2  | H(3)Np(1)O(14)P(3)     | 556.985 |
| <b>H+1 (3) _P2O7-4</b>                              |     |    |                        |         |
| -1                                                  | --- | 5  | H(3)O(7)P(2)           | 176.968 |
| <b>H+1 (3) _P3O10-5</b>                             |     |    |                        |         |
| -2                                                  | --- | 5  | H(3)O(10)P(3)          | 255.940 |
| <b>H+1 (3) _P4O13-6</b>                             |     |    |                        |         |
| -3                                                  | --- | 2  | H(3)O(13)P(4)          | 334.912 |
| <b>H+1 (3) _PaO2+1_PO4-3(3)</b>                     |     |    |                        |         |
| -5                                                  | --- | 2  | H(3)O(14)P(3)Pa(1)     | 550.977 |
| <b>H+1 (3) _Penten_P2O7-4</b>                       |     |    |                        |         |
| -1                                                  | --- | 1  | C(10)H(31)N(6)O(7)P(2) | 409.340 |
| <b>H+1 (3) _PO2-3</b>                               |     |    |                        |         |
| 0                                                   | --- | 2  | H(3)O(2)P(1)           | 65.9966 |
| <b>H+1 (3) _PO3-3</b>                               |     |    |                        |         |
| 0                                                   | --- | 5  | H(3)O(3)P(1)           | 81.9960 |
| <b>H+1 (3) _PO4-3</b>                               |     |    |                        |         |
| 0                                                   | --- | 64 | H(3)O(4)P(1)           | 97.9954 |
| <b>H+1 (3) _PO4-3_(1)</b><br>Phosphoric acid liquid |     |    |                        |         |
| 0                                                   | --- | 1  | H(3)O(4)P(1)           | 97.9954 |

|                                    |     |   |                        |         |
|------------------------------------|-----|---|------------------------|---------|
| <b>H+1 (3) _PO4-3_(s)</b>          |     |   |                        |         |
| 0                                  | --- | 2 | H(3)O(4)P(1)           | 97.9954 |
| <b>H+1 (3) _PO4-3_H2O(0.5)_(s)</b> |     |   |                        |         |
| 0                                  | --- | 2 | H(4)O(4.5)P(1)         | 107.003 |
| <b>H+1 (3) _PO4-3(2)</b>           |     |   |                        |         |
| -3                                 | --- | 5 | H(3)O(8)P(2)           | 192.967 |
| <b>H+1 (3) _PuO2+1_PO4-3(3)</b>    |     |   |                        |         |
| -5                                 | --- | 1 | H(3)O(14)P(3)Pu(1)     | 563.937 |
| <b>H+1 (3) _Spermidine</b>         |     |   |                        |         |
| 3                                  | --- | 3 | C(7)H(22)N(3)          | 148.272 |
| <b>H+1 (3) _Spermidine_P2O7-4</b>  |     |   |                        |         |
| -1                                 | --- | 1 | C(7)H(22)N(3)O(7)P(2)  | 322.216 |
| <b>H+1 (3) _Spermine</b>           |     |   |                        |         |
| 3                                  | --- | 4 | C(10)H(29)N(4)         | 205.367 |
| <b>H+1 (3) _Spermine_P2O7-4</b>    |     |   |                        |         |
| -1                                 | --- | 1 | C(10)H(29)N(4)O(7)P(2) | 379.311 |
| <b>H+1 (3) _Tetren_P2O7-4</b>      |     |   |                        |         |
| -1                                 | --- | 1 | C(8)H(26)N(5)O(7)P(2)  | 366.272 |
| <b>H+1 (3) _Tetren_PO4-3</b>       |     |   |                        |         |
| 0                                  | --- | 1 | C(8)H(26)N(5)O(4)P(1)  | 287.300 |
| <b>H+1 (3) _Trien_P2O7-4</b>       |     |   |                        |         |
| -1                                 | --- | 1 | C(6)H(21)N(4)O(7)P(2)  | 323.203 |
| <b>H+1 (3) _UO2+1_PO4-3(3)</b>     |     |   |                        |         |
| -5                                 | --- | 2 | H(3)O(14)P(3)U(1)      | 557.966 |

|                                 |     |   |                        |         |
|---------------------------------|-----|---|------------------------|---------|
| <b>H+1(4)_16HexDiAm_P2O7-4</b>  |     |   |                        |         |
| 0                               | --- | 1 | C(6)H(20)N(2)O(7)P(2)  | 294.182 |
| <b>H+1(4)_16HexDiAm_P3O10-5</b> |     |   |                        |         |
| -1                              | --- | 1 | C(6)H(20)N(2)O(10)P(3) | 373.154 |
| <b>H+1(4)_18OctDiAm_P2O7-4</b>  |     |   |                        |         |
| 0                               | --- | 1 | C(8)H(24)N(2)O(7)P(2)  | 322.236 |
| <b>H+1(4)_18OctDiAm_P3O10-5</b> |     |   |                        |         |
| -1                              | --- | 1 | C(8)H(24)N(2)O(10)P(3) | 401.208 |
| <b>H+1(4)_Dien_P2O7-4</b>       |     |   |                        |         |
| 0                               | --- | 1 | C(4)H(17)N(3)O(7)P(2)  | 281.143 |
| <b>H+1(4)_EtDiAm_P2O7-4</b>     |     |   |                        |         |
| 0                               | --- | 1 | C(2)H(12)N(2)O(7)P(2)  | 238.075 |
| <b>H+1(4)_EtDiAm_P3O10-5</b>    |     |   |                        |         |
| -1                              | --- | 1 | C(2)H(12)N(2)O(10)P(3) | 317.047 |
| <b>H+1(4)_EtDiAm_PO4-3</b>      |     |   |                        |         |
| 1                               | --- | 1 | C(2)H(12)N(2)O(4)P(1)  | 159.102 |
| <b>H+1(4)_EtDiAm(2)_P2O7-4</b>  |     |   |                        |         |
| 0                               | --- | 1 | C(4)H(20)N(4)O(7)P(2)  | 298.173 |
| <b>H+1(4)_EtDiAm(2)_P3O10-5</b> |     |   |                        |         |
| -1                              | --- | 1 | C(4)H(20)N(4)O(10)P(3) | 377.146 |
| <b>H+1(4)_MeAm(4)_P2O7-4</b>    |     |   |                        |         |
| 0                               | --- | 1 | C(4)H(24)N(4)O(7)P(2)  | 302.205 |
| <b>H+1(4)_NpO2+1_PO4-3(2)</b>   |     |   |                        |         |
| -1                              | --- | 2 | H(4)Np(1)O(10)P(2)     | 463.022 |



|                                                                  |     |    |                                 |         |
|------------------------------------------------------------------|-----|----|---------------------------------|---------|
| <b>H+1 (4) _NpO2+1_PO4-3 (4)</b>                                 |     |    |                                 |         |
| -7                                                               | --- | 2  | H (4) Np (1) O (18) P (4)       | 652.965 |
| <b>H+1 (4) _P2O6-4</b>                                           |     |    |                                 |         |
| 0                                                                | --- | 2  | H (4) O (6) P (2)               | 161.976 |
| <b>H+1 (4) _P2O7-4</b><br>Pyrophosphoric acid; Diphosphoric acid |     |    |                                 |         |
| 0                                                                | --- | 10 | H (4) O (7) P (2)               | 177.976 |
| <b>H+1 (4) _P3O10-5</b>                                          |     |    |                                 |         |
| -1                                                               | --- | 3  | H (4) O (10) P (3)              | 256.948 |
| <b>H+1 (4) _P4O13-6</b>                                          |     |    |                                 |         |
| -2                                                               | --- | 1  | H (4) O (13) P (4)              | 335.920 |
| <b>H+1 (4) _PaO2+1_PO4-3 (2)</b>                                 |     |    |                                 |         |
| -1                                                               | --- | 2  | H (4) O (10) P (2) Pa (1)       | 457.014 |
| <b>H+1 (4) _PaO2+1_PO4-3 (4)</b>                                 |     |    |                                 |         |
| -7                                                               | --- | 2  | H (4) O (18) P (4) Pa (1)       | 646.957 |
| <b>H+1 (4) _Penten_P2O7-4</b>                                    |     |    |                                 |         |
| 0                                                                | --- | 1  | C (10) H (32) N (6) O (7) P (2) | 410.348 |
| <b>H+1 (4) _PO4-3</b>                                            |     |    |                                 |         |
| 1                                                                | --- | 1  | H (4) O (4) P (1)               | 99.0034 |
| <b>H+1 (4) _PO4-3 (2)</b>                                        |     |    |                                 |         |
| -2                                                               | --- | 5  | H (4) O (8) P (2)               | 193.975 |
| <b>H+1 (4) _PuO2+1_PO4-3 (2)</b>                                 |     |    |                                 |         |
| -1                                                               | --- | 1  | H (4) O (10) P (2) Pu (1)       | 469.974 |
| <b>H+1 (4) _PuO2+1_PO4-3 (4)</b>                                 |     |    |                                 |         |
| -7                                                               | --- | 1  | H (4) O (18) P (4) Pu (1)       | 659.917 |

|                                 |     |   |                        |         |
|---------------------------------|-----|---|------------------------|---------|
| <b>H+1(4)_Spermidine</b>        |     |   |                        |         |
| 4                               | --- | 1 | C(7)H(23)N(3)          | 149.280 |
| <b>H+1(4)_Spermidine_P2O7-4</b> |     |   |                        |         |
| 0                               | --- | 1 | C(7)H(23)N(3)O(7)P(2)  | 323.224 |
| <b>H+1(4)_Spermine</b>          |     |   |                        |         |
| 4                               | --- | 3 | C(10)H(30)N(4)         | 206.375 |
| <b>H+1(4)_Spermine_P2O7-4</b>   |     |   |                        |         |
| 0                               | --- | 1 | C(10)H(30)N(4)O(7)P(2) | 380.319 |
| <b>H+1(4)_Tetren_P2O7-4</b>     |     |   |                        |         |
| 0                               | --- | 1 | C(8)H(27)N(5)O(7)P(2)  | 367.280 |
| <b>H+1(4)_Tetren_PO4-3</b>      |     |   |                        |         |
| 1                               | --- | 1 | C(8)H(27)N(5)O(4)P(1)  | 288.308 |
| <b>H+1(4)_Trien_P2O7-4</b>      |     |   |                        |         |
| 0                               | --- | 1 | C(6)H(22)N(4)O(7)P(2)  | 324.211 |
| <b>H+1(4)_UO2+1_PO4-3(2)</b>    |     |   |                        |         |
| -1                              | --- | 2 | H(4)O(10)P(2)U(1)      | 464.003 |
| <b>H+1(4)_UO2+1_PO4-3(4)</b>    |     |   |                        |         |
| -7                              | --- | 2 | H(4)O(18)P(4)U(1)      | 653.946 |
| <b>H+1(4)_VO2+1_P2O7-4(2)</b>   |     |   |                        |         |
| -3                              | --- | 1 | H(4)O(16)P(4)V(1)      | 434.860 |
| <b>H+1(5)_Dien_P2O7-4</b>       |     |   |                        |         |
| 1                               | --- | 1 | C(4)H(18)N(3)O(7)P(2)  | 282.151 |
| <b>H+1(5)_EtDiAm_P2O7-4</b>     |     |   |                        |         |
| 1                               | --- | 1 | C(2)H(13)N(2)O(7)P(2)  | 239.082 |

|                                                                      |     |   |                                 |         |
|----------------------------------------------------------------------|-----|---|---------------------------------|---------|
| <b>H+1 (5) _EtDiAm(2) _P2O7-4</b>                                    |     |   |                                 |         |
| 1                                                                    | --- | 1 | C (4) H (21) N (4) O (7) P (2)  | 299.181 |
| <b>H+1 (5) _EtDiAm(2) _P3O10-5</b>                                   |     |   |                                 |         |
| 0                                                                    | --- | 1 | C (4) H (21) N (4) O (10) P (3) | 378.154 |
| <b>H+1 (5) _NpO2+1 _PO4-3 (5)</b>                                    |     |   |                                 |         |
| -9                                                                   | --- | 2 | H (5) Np (1) O (22) P (5)       | 748.945 |
| <b>H+1 (5) _P2O7-4</b>                                               |     |   |                                 |         |
| 1                                                                    | --- | 1 | H (5) O (7) P (2)               | 178.984 |
| <b>H+1 (5) _P3O10-5</b><br>Triphosphoric acid; Hydrogen triphosphate |     |   |                                 |         |
| 0                                                                    | --- | 2 | H (5) O (10) P (3)              | 257.956 |
| <b>H+1 (5) _PaO2+1 _PO4-3 (5)</b>                                    |     |   |                                 |         |
| -9                                                                   | --- | 2 | H (5) O (22) P (5) Pa (1)       | 742.937 |
| <b>H+1 (5) _Penten _P2O7-4</b>                                       |     |   |                                 |         |
| 1                                                                    | --- | 1 | C (10) H (33) N (6) O (7) P (2) | 411.356 |
| <b>H+1 (5) _PO4-3 (2)</b>                                            |     |   |                                 |         |
| -1                                                                   | --- | 5 | H (5) O (8) P (2)               | 194.983 |
| <b>H+1 (5) _PuO2+1 _PO4-3 (5)</b>                                    |     |   |                                 |         |
| -9                                                                   | --- | 1 | H (5) O (22) P (5) Pu (1)       | 755.897 |
| <b>H+1 (5) _Tetren _P2O7-4</b>                                       |     |   |                                 |         |
| 1                                                                    | --- | 1 | C (8) H (28) N (5) O (7) P (2)  | 368.288 |
| <b>H+1 (5) _Tetren _PO4-3</b>                                        |     |   |                                 |         |
| 2                                                                    | --- | 1 | C (8) H (28) N (5) O (4) P (1)  | 289.315 |
| <b>H+1 (5) _Trien _P2O7-4</b>                                        |     |   |                                 |         |
| 1                                                                    | --- | 1 | C (6) H (23) N (4) O (7) P (2)  | 325.219 |

|                                  |     |   |                                 |         |
|----------------------------------|-----|---|---------------------------------|---------|
| <b>H+1 (5) _UO2+1_PO4-3 (5)</b>  |     |   |                                 |         |
| -9                               | --- | 2 | H (5) O (22) P (5) U (1)        | 749.926 |
| <b>H+1 (6) _Dien_P2O7-4</b>      |     |   |                                 |         |
| 2                                | --- | 1 | C (4) H (19) N (3) O (7) P (2)  | 283.159 |
| <b>H+1 (6) _NpO2+1_PO4-3 (3)</b> |     |   |                                 |         |
| -2                               | --- | 2 | H (6) Np (1) O (14) P (3)       | 560.009 |
| <b>H+1 (6) _PaO2+1_PO4-3 (3)</b> |     |   |                                 |         |
| -2                               | --- | 2 | H (6) O (14) P (3) Pa (1)       | 554.001 |
| <b>H+1 (6) _Penten_P2O7-4</b>    |     |   |                                 |         |
| 2                                | --- | 1 | C (10) H (34) N (6) O (7) P (2) | 412.364 |
| <b>H+1 (6) _PO4-3 (2)</b>        |     |   |                                 |         |
| 0                                | --- | 3 | H (6) O (8) P (2)               | 195.991 |
| <b>H+1 (6) _PuO2+1_PO4-3 (3)</b> |     |   |                                 |         |
| -2                               | --- | 1 | H (6) O (14) P (3) Pu (1)       | 566.961 |
| <b>H+1 (6) _Tetren_P2O7-4</b>    |     |   |                                 |         |
| 2                                | --- | 1 | C (8) H (29) N (5) O (7) P (2)  | 369.296 |
| <b>H+1 (6) _Tetren_PO4-3</b>     |     |   |                                 |         |
| 3                                | --- | 1 | C (8) H (29) N (5) O (4) P (1)  | 290.323 |
| <b>H+1 (6) _Trien_P2O7-4</b>     |     |   |                                 |         |
| 2                                | --- | 1 | C (6) H (24) N (4) O (7) P (2)  | 326.227 |
| <b>H+1 (6) _UO2+1_PO4-3 (3)</b>  |     |   |                                 |         |
| -2                               | --- | 2 | H (6) O (14) P (3) U (1)        | 560.990 |
| <b>H+1 (7) _Penten_P2O7-4</b>    |     |   |                                 |         |
| 3                                | --- | 1 | C (10) H (35) N (6) O (7) P (2) | 413.372 |

|                                        |            |      |                        |         |
|----------------------------------------|------------|------|------------------------|---------|
| <b>H+1(7)_Tetren_P2O7-4</b>            |            |      |                        |         |
| 3                                      | ---        | 1    | C(8)H(30)N(5)O(7)P(2)  | 370.304 |
| <b>H+1(7)_Tetren_PO4-3</b>             |            |      |                        |         |
| 4                                      | ---        | 1    | C(8)H(30)N(5)O(4)P(1)  | 291.331 |
| <b>H+1(7)_Trien_P2O7-4</b>             |            |      |                        |         |
| 3                                      | ---        | 1    | C(6)H(25)N(4)O(7)P(2)  | 327.235 |
| <b>H+1(8)_Penten_P2O7-4</b>            |            |      |                        |         |
| 4                                      | ---        | 1    | C(10)H(36)N(6)O(7)P(2) | 414.380 |
| <b>H2(g)</b><br>Hydrogen gas; Hydrogen |            |      |                        |         |
| 0                                      | 1333-74-0  | 1564 | H(2)                   | 2.01588 |
| <b>H2Mo11PO39-5</b>                    |            |      |                        |         |
| -5                                     | ---        | 1    | H(2)Mo(11)O(39)P(1)    | 1712.55 |
| <b>H2Mo5P2O23-4</b>                    |            |      |                        |         |
| -4                                     | ---        | 1    | H(2)Mo(5)O(23)P(2)     | 911.760 |
| <b>H2Mo9PO33-5</b>                     |            |      |                        |         |
| -5                                     | ---        | 1    | H(2)Mo(9)O(33)P(1)     | 1424.63 |
| <b>H2O</b><br>Water                    |            |      |                        |         |
| 0                                      | 7732-18-5  | 5947 | H(2)O(1)               | 18.0153 |
| <b>H2PO2-1</b><br>Hypophosphite ion    |            |      |                        |         |
| -1                                     | 15460-68-1 | 27   | H(2)O(2)P(1)           | 64.9887 |
| <b>H3Mo9PO33-4</b>                     |            |      |                        |         |
| -4                                     | ---        | 1    | H(3)Mo(9)O(33)P(1)     | 1425.64 |
| <b>H3PO2(g)</b>                        |            |      |                        |         |

|                                |            |     |                           |         |
|--------------------------------|------------|-----|---------------------------|---------|
| 0                              | ---        | 1   | H (3) O (2) P (1)         | 65.9966 |
| <b>H4Mo9PO33-3</b>             |            |     |                           |         |
| -3                             | ---        | 1   | H (4) Mo (9) O (33) P (1) | 1426.64 |
| <b>H5Mo9PO34-4</b>             |            |     |                           |         |
| -4                             | ---        | 1   | H (5) Mo (9) O (34) P (1) | 1443.65 |
| <b>H6Mo9PO34-3</b>             |            |     |                           |         |
| -3                             | ---        | 1   | H (6) Mo (9) O (34) P (1) | 1444.66 |
| <b>Hf+4</b><br>Hafnium(IV) ion |            |     |                           |         |
| 4                              | 22541-25-9 | 148 | Hf (1)                    | 178.492 |
| <b>Hf+4_H+1 (3)_PO4-3 (3)</b>  |            |     |                           |         |
| -2                             | ---        | 1   | H (3) Hf (1) O (12) P (3) | 466.431 |
| <b>Hg+1</b><br>Mercury(I) ion  |            |     |                           |         |
| 1                              | 22542-11-6 | 55  | Hg (1)                    | 200.592 |
| <b>Hg+1_OH-1_P2O7-4</b>        |            |     |                           |         |
| -4                             | ---        | 1   | H (1) Hg (1) O (8) P (2)  | 391.543 |
| <b>Hg+1_OH-1_P3O10-5</b>       |            |     |                           |         |
| -5                             | ---        | 1   | H (1) Hg (1) O (11) P (3) | 470.515 |
| <b>Hg+1_P3O10-5 (2)</b>        |            |     |                           |         |
| -9                             | ---        | 1   | Hg (1) O (20) P (6)       | 706.424 |
| <b>Hg+1 (2)_OH-1_P2O7-4</b>    |            |     |                           |         |
| -3                             | ---        | 1   | H (1) Hg (2) O (8) P (2)  | 592.135 |
| <b>Hg+1 (2)_P2O7-4</b>         |            |     |                           |         |
| -2                             | ---        | 1   | Hg (2) O (7) P (2)        | 575.128 |
| <b>Hg+1 (2)_P2O7-4 (2)</b>     |            |     |                           |         |

|                                                       |            |      |                   |         |
|-------------------------------------------------------|------------|------|-------------------|---------|
| -6                                                    | ---        | 1    | Hg(2)O(14)P(4)    | 749.072 |
| <b>Hg+2</b><br>Mercury(II) ion; Mercuric ion          |            |      |                   |         |
| 2                                                     | 14302-87-5 | 1203 | Hg(1)             | 200.592 |
| <b>Hg+2_H+1_PO4-3</b>                                 |            |      |                   |         |
| 0                                                     | ---        | 2    | H(1)Hg(1)O(4)P(1) | 296.572 |
| <b>Hg+2_H+1_PO4-3(s)</b><br>Mercury phosphate dibasic |            |      |                   |         |
| 0                                                     | 7782-66-3  | 2    | H(1)Hg(1)O(4)P(1) | 296.572 |
| <b>Hg+2_OH-1_P2O7-4</b>                               |            |      |                   |         |
| -3                                                    | ---        | 1    | H(1)Hg(1)O(8)P(2) | 391.543 |
| <b>Hg+2_P2O7-4</b>                                    |            |      |                   |         |
| -2                                                    | ---        | 1    | Hg(1)O(7)P(2)     | 374.536 |
| <b>Hg+2_P2O7-4(2)</b>                                 |            |      |                   |         |
| -6                                                    | ---        | 1    | Hg(1)O(14)P(4)    | 548.480 |
| <b>Hg+2_PO4-3</b>                                     |            |      |                   |         |
| -1                                                    | ---        | 1    | Hg(1)O(4)P(1)     | 295.564 |
| <b>Hg+2(2)_H+1_P2O7-4</b>                             |            |      |                   |         |
| 1                                                     | ---        | 1    | H(1)Hg(2)O(7)P(2) | 576.136 |
| <b>Hg+2(2)_OH-1_P2O7-4</b>                            |            |      |                   |         |
| -1                                                    | ---        | 1    | H(1)Hg(2)O(8)P(2) | 592.135 |
| <b>Hg+2(2)_OH-1(2)_P2O7-4</b>                         |            |      |                   |         |
| -2                                                    | ---        | 1    | H(2)Hg(2)O(9)P(2) | 609.143 |
| <b>Hg+2(3)_OH-1(3)_PO4-3(s)</b>                       |            |      |                   |         |
| 0                                                     | ---        | 2    | H(3)Hg(3)O(7)P(1) | 747.770 |
| <b>Hg+2(3)_PO4-3(2)_ (s)</b>                          |            |      |                   |         |

|                                       |            |     |                           |         |
|---------------------------------------|------------|-----|---------------------------|---------|
| 0                                     | ---        | 2   | Hg (3) O (8) P (2)        | 791.719 |
| <b>Hg (1)</b><br>Mercury; Quicksilver |            |     |                           |         |
| 0                                     | 7439-97-6  | 160 | Hg (1)                    | 200.592 |
| <b>Hg2+2</b><br>Mercurous ion         |            |     |                           |         |
| 2                                     | 12596-26-8 | 78  | Hg (2)                    | 401.184 |
| <b>Hg2+2_H+1_P3O10-5</b>              |            |     |                           |         |
| -2                                    | ---        | 1   | H (1) Hg (2) O (10) P (3) | 655.108 |
| <b>Hg2+2_H+1_PO4-3_(s)</b>            |            |     |                           |         |
| 0                                     | 7782-67-4  | 3   | H (1) Hg (2) O (4) P (1)  | 497.164 |
| <b>Hg2+2_OH-1_P2O7-4</b>              |            |     |                           |         |
| -3                                    | ---        | 1   | H (1) Hg (2) O (8) P (2)  | 592.135 |
| <b>Hg2+2_OH-1_P3O10-5</b>             |            |     |                           |         |
| -4                                    | ---        | 1   | H (1) Hg (2) O (11) P (3) | 671.107 |
| <b>Hg2+2_OH-1 (2)_P2O7-4</b>          |            |     |                           |         |
| -4                                    | ---        | 1   | H (2) Hg (2) O (9) P (2)  | 609.143 |
| <b>Hg2+2_OH-1 (2)_P3O10-5</b>         |            |     |                           |         |
| -5                                    | ---        | 1   | H (2) Hg (2) O (12) P (3) | 688.115 |
| <b>Hg2+2_P2O7-4</b>                   |            |     |                           |         |
| -2                                    | ---        | 1   | Hg (2) O (7) P (2)        | 575.128 |
| <b>Hg2+2_P2O7-4 (2)</b>               |            |     |                           |         |
| -6                                    | ---        | 1   | Hg (2) O (14) P (4)       | 749.072 |
| <b>Hg2+2_P3O10-5</b>                  |            |     |                           |         |
| -3                                    | ---        | 1   | Hg (2) O (10) P (3)       | 654.100 |
| <b>Hg2+2_P3O10-5 (2)</b>              |            |     |                           |         |



|                                                                                                                                                                  |            |      |                         |         |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|------|-------------------------|---------|
| -8                                                                                                                                                               | ---        | 1    | Hg(2)O(20)P(6)          | 907.016 |
| <b>HgCH3+1</b><br>Methyl-mercury ion; Methylmercury ion                                                                                                          |            |      |                         |         |
| 1                                                                                                                                                                | 22967-92-6 | 182  | C(1)H(3)Hg(1)           | 215.627 |
| <b>HgCH3+1_HPO3-2</b>                                                                                                                                            |            |      |                         |         |
| -1                                                                                                                                                               | ---        | 1    | C(1)H(4)Hg(1)O(3)P(1)   | 295.607 |
| <b>HIMDA-2</b>                                                                                                                                                   |            |      |                         |         |
| -2                                                                                                                                                               | ---        | 201  | C(6)H(9)N(1)O(5)        | 175.141 |
| <b>His-1</b><br>Histidinate ion; L-Histidinate ion; alpha-amino-4-imidazolepropionate ion;<br>glyoxaline-5-alanate ion; L-2-amino-3-(4-imidazolyl)propanoate ion |            |      |                         |         |
| -1                                                                                                                                                               | 26302-81-8 | 1164 | C(6)H(8)N(3)O(2)        | 154.148 |
| <b>Histamine</b><br>Histamine; 4-(2-Aminoethyl)imidazole; 1H-Imidazole-4-ethanamine; 2-(4-Imidazolyl)ethylamine                                                  |            |      |                         |         |
| 0                                                                                                                                                                | 51-45-6    | 531  | C(5)H(9)N(3)            | 111.147 |
| <b>HMo11PO39-6</b>                                                                                                                                               |            |      |                         |         |
| -6                                                                                                                                                               | ---        | 1    | H(1)Mo(11)O(39)P(1)     | 1711.54 |
| <b>HMo5P2O23-5</b>                                                                                                                                               |            |      |                         |         |
| -5                                                                                                                                                               | ---        | 1    | H(1)Mo(5)O(23)P(2)      | 910.752 |
| <b>HMo9PO33-6</b>                                                                                                                                                |            |      |                         |         |
| -6                                                                                                                                                               | ---        | 1    | H(1)Mo(9)O(33)P(1)      | 1423.62 |
| <b>Ho+3</b><br>Holmium(III) ion                                                                                                                                  |            |      |                         |         |
| 3                                                                                                                                                                | 22541-22-6 | 537  | Ho(1)                   | 164.930 |
| <b>Ho+3_EDTA-4</b>                                                                                                                                               |            |      |                         |         |
| -1                                                                                                                                                               | ---        | 14   | C(10)H(12)Ho(1)N(2)O(8) | 453.144 |
| <b>Ho+3_EDTA-4_P3O10-5</b>                                                                                                                                       |            |      |                         |         |

|                                                        |     |   |                              |         |
|--------------------------------------------------------|-----|---|------------------------------|---------|
| -6                                                     | --- | 2 | C(10)H(12)Ho(1)N(2)O(18)P(3) | 706.060 |
| <b>Ho+3_H+1_EDTA-4_P3O10-5</b>                         |     |   |                              |         |
| -5                                                     | --- | 2 | C(10)H(13)Ho(1)N(2)O(18)P(3) | 707.068 |
| <b>Ho+3_H+1_PO4-3</b>                                  |     |   |                              |         |
| 1                                                      | --- | 3 | H(1)Ho(1)O(4)P(1)            | 260.910 |
| <b>Ho+3_H+1(2)_P3O10-5(2)</b>                          |     |   |                              |         |
| -5                                                     | --- | 1 | H(2)Ho(1)O(20)P(6)           | 672.778 |
| <b>Ho+3_H+1(2)_PO4-3</b>                               |     |   |                              |         |
| 2                                                      | --- | 2 | H(2)Ho(1)O(4)P(1)            | 261.918 |
| <b>Ho+3_H+1(2)_PO4-3(2)</b>                            |     |   |                              |         |
| -1                                                     | --- | 3 | H(2)Ho(1)O(8)P(2)            | 356.889 |
| <b>Ho+3_P3O10-5(2)</b>                                 |     |   |                              |         |
| -7                                                     | --- | 1 | Ho(1)O(20)P(6)               | 670.762 |
| <b>Ho+3_PO4-3</b>                                      |     |   |                              |         |
| 0                                                      | --- | 2 | Ho(1)O(4)P(1)                | 259.902 |
| <b>Ho+3_PO4-3_(s)</b>                                  |     |   |                              |         |
| 0                                                      | --- | 2 | Ho(1)O(4)P(1)                | 259.902 |
| <b>Ho+3_PO4-3_H2O_(s)</b>                              |     |   |                              |         |
| 0                                                      | --- | 1 | H(2)Ho(1)O(5)P(1)            | 277.917 |
| <b>Ho+3_PO4-3(2)</b>                                   |     |   |                              |         |
| -3                                                     | --- | 2 | Ho(1)O(8)P(2)                | 354.873 |
| <b>Ho+3(2)_P2O7-4</b>                                  |     |   |                              |         |
| 2                                                      | --- | 1 | Ho(2)O(7)P(2)                | 503.804 |
| <b>HPO3-2</b><br>Hydrogen phosphite ion; Phosphite ion |     |   |                              |         |

|                                                                                                                                                                                                 |            |     |                    |         |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----|--------------------|---------|
| -2                                                                                                                                                                                              | 15477-76-6 | 42  | H(1)O(3)P(1)       | 79.9801 |
| <b>Hyp-1</b><br>Hydroxyprolinate ion; L-Hydroxyprolinate ion; 4-hydroxy-2-pyrrolidinecarboxylate ion; L-4-hydroxypyrrolidine-2-carboxylate ion                                                  |            |     |                    |         |
| -1                                                                                                                                                                                              | ---        | 402 | C(5)H(8)N(1)O(3)   | 130.123 |
| <b>I2(s)</b><br>Iodine; Iodine, orthorhombic                                                                                                                                                    |            |     |                    |         |
| 0                                                                                                                                                                                               | 7553-56-2  | 244 | I(2)               | 253.800 |
| <b>Ile-1</b><br>Isoleucinate ion; L-Isoleucinate ion; 2-amino-3-methylvalerate ion; alpha-amino-beta-methylvalerate ion; 2-amino-3-methylpentanoate ion; ILeu; L-2-amino-3-methylpentanoate ion |            |     |                    |         |
| -1                                                                                                                                                                                              | 57031-97-7 | 428 | C(6)H(12)N(1)O(2)  | 130.167 |
| <b>In+3</b><br>Indium(III) ion                                                                                                                                                                  |            |     |                    |         |
| 3                                                                                                                                                                                               | 22537-49-1 | 523 | In(1)              | 114.820 |
| <b>In+3_H+1_P2O7-4</b>                                                                                                                                                                          |            |     |                    |         |
| 0                                                                                                                                                                                               | ---        | 1   | H(1)In(1)O(7)P(2)  | 289.772 |
| <b>In+3_H+1(2)_PO4-3</b>                                                                                                                                                                        |            |     |                    |         |
| 2                                                                                                                                                                                               | ---        | 1   | H(2)In(1)O(4)P(1)  | 211.808 |
| <b>In+3_H+1(4)_P3O10-5(2)</b>                                                                                                                                                                   |            |     |                    |         |
| -3                                                                                                                                                                                              | ---        | 1   | H(4)In(1)O(20)P(6) | 624.684 |
| <b>In+3_PO4-3_(s)</b>                                                                                                                                                                           |            |     |                    |         |
| 0                                                                                                                                                                                               | ---        | 1   | In(1)O(4)P(1)      | 209.792 |
| <b>In+3(2)_H+1_PO4-3</b>                                                                                                                                                                        |            |     |                    |         |
| 4                                                                                                                                                                                               | ---        | 1   | H(1)In(2)O(4)P(1)  | 325.620 |
| <b>In(s)</b><br>Indium                                                                                                                                                                          |            |     |                    |         |
| 0                                                                                                                                                                                               | 7440-74-6  | 41  | In(1)              | 114.820 |

|                                    |            |     |                         |         |
|------------------------------------|------------|-----|-------------------------|---------|
| <b>InP (s)</b><br>Indium phosphide |            |     |                         |         |
| 0                                  | ---        | 1   | In (1) P (1)            | 145.794 |
| <b>K+1</b><br>Potassium(I) ion     |            |     |                         |         |
| 1                                  | 24203-36-9 | 462 | K (1)                   | 39.0980 |
| <b>K+1_HPO3-2</b>                  |            |     |                         |         |
| -1                                 | ---        | 1   | H (1) K (1) O (3) P (1) | 119.078 |
| <b>K+1_P2O7-4</b>                  |            |     |                         |         |
| -3                                 | ---        | 1   | K (1) O (7) P (2)       | 213.042 |
| <b>K+1_P3O10-5</b>                 |            |     |                         |         |
| -4                                 | ---        | 1   | K (1) O (10) P (3)      | 292.014 |
| <b>K+1_P3O8-5</b>                  |            |     |                         |         |
| -4                                 | ---        | 1   | K (1) O (8) P (3)       | 260.015 |
| <b>K+1_P4O12-4</b>                 |            |     |                         |         |
| -3                                 | ---        | 1   | K (1) O (12) P (4)      | 354.987 |
| <b>K+1_P4O13-6</b>                 |            |     |                         |         |
| -5                                 | ---        | 1   | K (1) O (13) P (4)      | 370.986 |
| <b>K+1_P6O18-6</b>                 |            |     |                         |         |
| -5                                 | ---        | 1   | K (1) O (18) P (6)      | 512.931 |
| <b>K+1_PO4-3</b>                   |            |     |                         |         |
| -2                                 | ---        | 1   | K (1) O (4) P (1)       | 134.070 |
| <b>K+1 (2)_P2O7-4</b>              |            |     |                         |         |
| -2                                 | ---        | 1   | K (2) O (7) P (2)       | 252.140 |
| <b>K+1 (2)_P3O10-5</b>             |            |     |                         |         |
| -3                                 | ---        | 1   | K (2) O (10) P (3)      | 331.112 |

|                                                                                |            |     |                              |         |
|--------------------------------------------------------------------------------|------------|-----|------------------------------|---------|
| <b>K+1 (2) _P4O12-4</b>                                                        |            |     |                              |         |
| -2                                                                             | ---        | 1   | K(2)O(12)P(4)                | 394.085 |
| <b>K+1 (2) _PO4-3</b>                                                          |            |     |                              |         |
| -1                                                                             | ---        | 1   | K(2)O(4)P(1)                 | 173.168 |
| <b>K+1 (3) _PO4-3 (s)</b><br>Potassium phosphate; Potassium phosphate tribasic |            |     |                              |         |
| 0                                                                              | 7778-53-2  | 2   | K(3)O(4)P(1)                 | 212.266 |
| <b>K(s)</b><br>Potassium; Potassium, cubic                                     |            |     |                              |         |
| 0                                                                              | 7440-09-7  | 121 | K(1)                         | 39.0980 |
| <b>La+3</b><br>Lanthanum(III) ion                                              |            |     |                              |         |
| 3                                                                              | 16096-89-2 | 882 | La(1)                        | 138.910 |
| <b>La+3_EDTA-4</b>                                                             |            |     |                              |         |
| -1                                                                             | ---        | 44  | C(10)H(12)La(1)N(2)O(8)      | 427.124 |
| <b>La+3_EDTA-4_P3O10-5</b>                                                     |            |     |                              |         |
| -6                                                                             | ---        | 2   | C(10)H(12)La(1)N(2)O(18)P(3) | 680.040 |
| <b>La+3_H+1_EDTA-4_P3O10-5</b>                                                 |            |     |                              |         |
| -5                                                                             | ---        | 2   | C(10)H(13)La(1)N(2)O(18)P(3) | 681.048 |
| <b>La+3_H+1_P2O7-4</b>                                                         |            |     |                              |         |
| 0                                                                              | ---        | 2   | H(1)La(1)O(7)P(2)            | 313.862 |
| <b>La+3_H+1_P3O10-5</b>                                                        |            |     |                              |         |
| -1                                                                             | ---        | 2   | H(1)La(1)O(10)P(3)           | 392.834 |
| <b>La+3_H+1_P4O13-6</b>                                                        |            |     |                              |         |
| -2                                                                             | ---        | 2   | H(1)La(1)O(13)P(4)           | 471.806 |
| <b>La+3_H+1_PO4-3</b>                                                          |            |     |                              |         |

|                                 |     |   |                           |         |
|---------------------------------|-----|---|---------------------------|---------|
| 1                               | --- | 2 | H (1) La (1) O (4) P (1)  | 234.890 |
| <b>La+3_H+1 (2)_P3O10-5 (2)</b> |     |   |                           |         |
| -5                              | --- | 1 | H (2) La (1) O (20) P (6) | 646.758 |
| <b>La+3_H+1 (2)_PO4-3</b>       |     |   |                           |         |
| 2                               | --- | 2 | H (2) La (1) O (4) P (1)  | 235.898 |
| <b>La+3_H+1 (2)_PO4-3 (2)</b>   |     |   |                           |         |
| -1                              | --- | 2 | H (2) La (1) O (8) P (2)  | 330.869 |
| <b>La+3_P2O7-4</b>              |     |   |                           |         |
| -1                              | --- | 2 | La (1) O (7) P (2)        | 312.854 |
| <b>La+3_P2O7-4 (2)</b>          |     |   |                           |         |
| -5                              | --- | 1 | La (1) O (14) P (4)       | 486.798 |
| <b>La+3_P3O10-5</b>             |     |   |                           |         |
| -2                              | --- | 2 | La (1) O (10) P (3)       | 391.826 |
| <b>La+3_P3O10-5 (2)</b>         |     |   |                           |         |
| -7                              | --- | 1 | La (1) O (20) P (6)       | 644.742 |
| <b>La+3_P3O9-3</b>              |     |   |                           |         |
| 0                               | --- | 1 | La (1) O (9) P (3)        | 375.827 |
| <b>La+3_P4O12-4</b>             |     |   |                           |         |
| -1                              | --- | 1 | La (1) O (12) P (4)       | 454.799 |
| <b>La+3_P4O13-6</b>             |     |   |                           |         |
| -3                              | --- | 2 | La (1) O (13) P (4)       | 470.798 |
| <b>La+3_PO4-3</b>               |     |   |                           |         |
| 0                               | --- | 1 | La (1) O (4) P (1)        | 233.882 |
| <b>La+3_PO4-3_(s)</b>           |     |   |                           |         |
| 0                               | --- | 2 | La (1) O (4) P (1)        | 233.882 |

|                                                                                                                                                                            |            |     |                   |         |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----|-------------------|---------|
| <b>La+3_PO4-3(2)</b>                                                                                                                                                       |            |     |                   |         |
| -3                                                                                                                                                                         | ---        | 1   | La(1)O(8)P(2)     | 328.853 |
| <b>La+3(2)_P2O7-4</b>                                                                                                                                                      |            |     |                   |         |
| 2                                                                                                                                                                          | ---        | 1   | La(2)O(7)P(2)     | 451.764 |
| <b>Lactic-1</b><br>Lactate ion                                                                                                                                             |            |     |                   |         |
| -1                                                                                                                                                                         | ---        | 643 | C(3)H(5)O(3)      | 89.0709 |
| <b>Leu-1</b><br>Leucinate ion; L-Leucinate ion; 2-amino-4-methylvalerate ion; alpha-aminoisocaproate ion; 2-amino-4-methylpentanoate ion; L-2-amino-4-methylpentanoate ion |            |     |                   |         |
| -1                                                                                                                                                                         | 17332-93-3 | 511 | C(6)H(12)N(1)O(2) | 130.167 |
| <b>Li+1</b><br>Lithium(I) ion                                                                                                                                              |            |     |                   |         |
| 1                                                                                                                                                                          | 17341-24-1 | 246 | Li(1)             | 6.94100 |
| <b>Li+1_P2O7-4</b>                                                                                                                                                         |            |     |                   |         |
| -3                                                                                                                                                                         | ---        | 1   | Li(1)O(7)P(2)     | 180.885 |
| <b>Li+1_P3O10-5</b>                                                                                                                                                        |            |     |                   |         |
| -4                                                                                                                                                                         | ---        | 1   | Li(1)O(10)P(3)    | 259.857 |
| <b>Li+1_P3O8-5</b>                                                                                                                                                         |            |     |                   |         |
| -4                                                                                                                                                                         | ---        | 1   | Li(1)O(8)P(3)     | 227.858 |
| <b>Li+1_P4O12-4</b>                                                                                                                                                        |            |     |                   |         |
| -3                                                                                                                                                                         | ---        | 1   | Li(1)O(12)P(4)    | 322.830 |
| <b>Li+1_P4O13-6</b>                                                                                                                                                        |            |     |                   |         |
| -5                                                                                                                                                                         | ---        | 1   | Li(1)O(13)P(4)    | 338.829 |
| <b>Li+1_P6O18-6</b>                                                                                                                                                        |            |     |                   |         |
| -5                                                                                                                                                                         | ---        | 1   | Li(1)O(18)P(6)    | 480.774 |
| <b>Li+1_PO4-3</b>                                                                                                                                                          |            |     |                   |         |

|                                                                                                                                       |            |     |                          |         |
|---------------------------------------------------------------------------------------------------------------------------------------|------------|-----|--------------------------|---------|
| -2                                                                                                                                    | ---        | 1   | Li (1) O (4) P (1)       | 101.913 |
| <b>Li+1 (3) _PO4-3 (s)</b><br>Lithium phosphate                                                                                       |            |     |                          |         |
| 0                                                                                                                                     | ---        | 2   | Li (3) O (4) P (1)       | 115.795 |
| <b>Li (s)</b><br>Lithium; Lithium, cubic                                                                                              |            |     |                          |         |
| 0                                                                                                                                     | 7439-93-2  | 62  | Li (1)                   | 6.94100 |
| <b>Lu+3</b><br>Lutetium(III) ion                                                                                                      |            |     |                          |         |
| 3                                                                                                                                     | 22541-24-8 | 493 | Lu (1)                   | 174.970 |
| <b>Lu+3 _H+1 _PO4-3</b>                                                                                                               |            |     |                          |         |
| 1                                                                                                                                     | ---        | 2   | H (1) Lu (1) O (4) P (1) | 270.950 |
| <b>Lu+3 _H+1 (2) _PO4-3</b>                                                                                                           |            |     |                          |         |
| 2                                                                                                                                     | ---        | 2   | H (2) Lu (1) O (4) P (1) | 271.958 |
| <b>Lu+3 _H+1 (2) _PO4-3 (2)</b>                                                                                                       |            |     |                          |         |
| -1                                                                                                                                    | ---        | 2   | H (2) Lu (1) O (8) P (2) | 366.929 |
| <b>Lu+3 _PO4-3</b>                                                                                                                    |            |     |                          |         |
| 0                                                                                                                                     | ---        | 1   | Lu (1) O (4) P (1)       | 269.942 |
| <b>Lu+3 _PO4-3 (s)</b>                                                                                                                |            |     |                          |         |
| 0                                                                                                                                     | ---        | 1   | Lu (1) O (4) P (1)       | 269.942 |
| <b>Lu+3 _PO4-3 (2)</b>                                                                                                                |            |     |                          |         |
| -3                                                                                                                                    | ---        | 1   | Lu (1) O (8) P (2)       | 364.913 |
| <b>Lu+3 (2) _P2O7-4</b>                                                                                                               |            |     |                          |         |
| 2                                                                                                                                     | ---        | 1   | Lu (2) O (7) P (2)       | 523.884 |
| <b>Lys-1</b><br>Lysinate ion; L-Lysinate ion; 2,6-diaminohexanoate ion; alpha,epsilon-diaminocaproate ion; L-2,6-diaminohexanoate ion |            |     |                          |         |
| -1                                                                                                                                    | 17781-81-6 | 611 | C (6) H (13) N (2) O (2) | 145.181 |



|                                                                                                                                                                                                                                                                                                       |            |      |                            |         |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|------|----------------------------|---------|
| <b>Malic-2</b><br>Malate ion                                                                                                                                                                                                                                                                          |            |      |                            |         |
| -2                                                                                                                                                                                                                                                                                                    | 149-61-1   | 682  | C(4)H(4)O(5)               | 132.073 |
| <b>MeAm</b><br>Methylamine; Aminomethane                                                                                                                                                                                                                                                              |            |      |                            |         |
| 0                                                                                                                                                                                                                                                                                                     | 74-89-5    | 61   | C(1)H(5)N(1)               | 31.0574 |
| <b>Met-1</b><br>Methioninate ion; L-Methioninate ion; 2-amino-4-(methylthio)butyrate ion; alpha-amino-gamma-methylmercaptobutyrate ion; 2-amino-4-methylthiobutanoate ion; L-2-amino-4-(methylthio)butanoate ion; gamma-methylthio-alpha-aminobutyrate ion; Meonine ion; Methilalanin ion; Neston ion |            |      |                            |         |
| -1                                                                                                                                                                                                                                                                                                    | 93375-49-6 | 527  | C(5)H(10)N(1)O(2)S(1)      | 148.200 |
| <b>Mg+2</b><br>Magnesium(II) ion                                                                                                                                                                                                                                                                      |            |      |                            |         |
| 2                                                                                                                                                                                                                                                                                                     | 22537-22-0 | 2035 | Mg(1)                      | 24.3050 |
| <b>Mg+2_H+1_Ala-1_PO4-3</b>                                                                                                                                                                                                                                                                           |            |      |                            |         |
| -1                                                                                                                                                                                                                                                                                                    | ---        | 1    | C(3)H(7)Mg(1)N(1)O(6)P(1)  | 208.371 |
| <b>Mg+2_H+1_Arg-1_PO4-3</b>                                                                                                                                                                                                                                                                           |            |      |                            |         |
| -1                                                                                                                                                                                                                                                                                                    | ---        | 1    | C(6)H(14)Mg(1)N(4)O(6)P(1) | 293.479 |
| <b>Mg+2_H+1_Asn-1_PO4-3</b>                                                                                                                                                                                                                                                                           |            |      |                            |         |
| -1                                                                                                                                                                                                                                                                                                    | ---        | 1    | C(4)H(8)Mg(1)N(2)O(7)P(1)  | 251.396 |
| <b>Mg+2_H+1_Asp-2_PO4-3</b>                                                                                                                                                                                                                                                                           |            |      |                            |         |
| -2                                                                                                                                                                                                                                                                                                    | ---        | 1    | C(4)H(6)Mg(1)N(1)O(8)P(1)  | 251.373 |
| <b>Mg+2_H+1_bAla-1_PO4-3</b>                                                                                                                                                                                                                                                                          |            |      |                            |         |
| -1                                                                                                                                                                                                                                                                                                    | ---        | 1    | C(3)H(7)Mg(1)N(1)O(6)P(1)  | 208.371 |
| <b>Mg+2_H+1_Citric-3_PO4-3</b>                                                                                                                                                                                                                                                                        |            |      |                            |         |
| -3                                                                                                                                                                                                                                                                                                    | ---        | 1    | C(6)H(6)Mg(1)O(11)P(1)     | 309.386 |
| <b>Mg+2_H+1_Citrul-1_PO4-3</b>                                                                                                                                                                                                                                                                        |            |      |                            |         |
| -1                                                                                                                                                                                                                                                                                                    | ---        | 1    | C(6)H(13)Mg(1)N(3)O(7)P(1) | 294.464 |

|                                                               |     |   |                                             |         |
|---------------------------------------------------------------|-----|---|---------------------------------------------|---------|
| <b>Mg+2_H+1_Gln-1_PO4-3</b>                                   |     |   |                                             |         |
| -1                                                            | --- | 1 | C (5) H (10) Mg (1) N (2) O (7) P (1)       | 265.423 |
| <b>Mg+2_H+1_Glu-2_PO4-3</b>                                   |     |   |                                             |         |
| -2                                                            | --- | 1 | C (5) H (8) Mg (1) N (1) O (8) P (1)        | 265.399 |
| <b>Mg+2_H+1_Gly-1_PO4-3</b>                                   |     |   |                                             |         |
| -1                                                            | --- | 1 | C (2) H (5) Mg (1) N (1) O (6) P (1)        | 194.344 |
| <b>Mg+2_H+1_His-1_PO4-3</b>                                   |     |   |                                             |         |
| -1                                                            | --- | 1 | C (6) H (9) Mg (1) N (3) O (6) P (1)        | 274.433 |
| <b>Mg+2_H+1_Hyp-1_PO4-3</b>                                   |     |   |                                             |         |
| -1                                                            | --- | 1 | C (5) H (9) Mg (1) N (1) O (7) P (1)        | 250.408 |
| <b>Mg+2_H+1_Ile-1_PO4-3</b>                                   |     |   |                                             |         |
| -1                                                            | --- | 1 | C (6) H (13) Mg (1) N (1) O (6) P (1)       | 250.451 |
| <b>Mg+2_H+1_Lactic-1_PO4-3</b>                                |     |   |                                             |         |
| -1                                                            | --- | 1 | C (3) H (6) Mg (1) O (7) P (1)              | 209.355 |
| <b>Mg+2_H+1_Leu-1_PO4-3</b>                                   |     |   |                                             |         |
| -1                                                            | --- | 1 | C (6) H (13) Mg (1) N (1) O (6) P (1)       | 250.451 |
| <b>Mg+2_H+1_Malic-2_PO4-3</b>                                 |     |   |                                             |         |
| -2                                                            | --- | 1 | C (4) H (5) Mg (1) O (9) P (1)              | 252.357 |
| <b>Mg+2_H+1_Met-1_PO4-3</b>                                   |     |   |                                             |         |
| -1                                                            | --- | 1 | C (5) H (11) Mg (1) N (1) O (6) P (1) S (1) | 268.484 |
| <b>Mg+2_H+1_NH3_PO4-3</b>                                     |     |   |                                             |         |
| 0                                                             | --- | 1 | H (4) Mg (1) N (1) O (4) P (1)              | 137.315 |
| <b>Mg+2_H+1_NH3_PO4-3_(s)</b><br>Magnesium ammonium phosphate |     |   |                                             |         |
| 0                                                             | --- | 2 | H (4) Mg (1) N (1) O (4) P (1)              | 137.315 |

|                                                                                            |            |   |                            |         |
|--------------------------------------------------------------------------------------------|------------|---|----------------------------|---------|
| <b>Mg+2_H+1_NH3_PO4-3_H2O_(s)</b><br>Dittmarite; Magnesium ammonium phosphate monohydrate  |            |   |                            |         |
| 0                                                                                          | ---        | 1 | H(6)Mg(1)N(1)O(5)P(1)      | 155.330 |
| <b>Mg+2_H+1_NH3_PO4-3_H2O(6)_(s)</b><br>Struvite; Magnesium ammonium phosphate hexahydrate |            |   |                            |         |
| 0                                                                                          | 15490-91-2 | 3 | H(16)Mg(1)N(1)O(10)P(1)    | 245.407 |
| <b>Mg+2_H+1_Oxalic-2_PO4-3</b>                                                             |            |   |                            |         |
| -2                                                                                         | ---        | 1 | C(2)H(1)Mg(1)O(8)P(1)      | 208.304 |
| <b>Mg+2_H+1_P2O7-4</b>                                                                     |            |   |                            |         |
| -1                                                                                         | ---        | 3 | H(1)Mg(1)O(7)P(2)          | 199.257 |
| <b>Mg+2_H+1_P3O10-5</b>                                                                    |            |   |                            |         |
| -2                                                                                         | ---        | 3 | H(1)Mg(1)O(10)P(3)         | 278.229 |
| <b>Mg+2_H+1_P3O10-5(2)</b>                                                                 |            |   |                            |         |
| -7                                                                                         | ---        | 1 | H(1)Mg(1)O(20)P(6)         | 531.145 |
| <b>Mg+2_H+1_P4O13-6</b>                                                                    |            |   |                            |         |
| -3                                                                                         | ---        | 1 | H(1)Mg(1)O(13)P(4)         | 357.201 |
| <b>Mg+2_H+1_Phe-1_PO4-3</b>                                                                |            |   |                            |         |
| -1                                                                                         | ---        | 1 | C(9)H(11)Mg(1)N(1)O(6)P(1) | 284.468 |
| <b>Mg+2_H+1_PO4-3</b>                                                                      |            |   |                            |         |
| 0                                                                                          | ---        | 4 | H(1)Mg(1)O(4)P(1)          | 120.285 |
| <b>Mg+2_H+1_PO4-3_(s)</b><br>Magnesium hydrogen phosphate                                  |            |   |                            |         |
| 0                                                                                          | 7757-86-0  | 1 | H(1)Mg(1)O(4)P(1)          | 120.285 |
| <b>Mg+2_H+1_PO4-3_H2O(2)_(s)</b><br>Magnesium phosphate dihydrate                          |            |   |                            |         |
| 0                                                                                          | ---        | 1 | H(5)Mg(1)O(6)P(1)          | 156.315 |

|                                                                                         |           |   |                             |         |
|-----------------------------------------------------------------------------------------|-----------|---|-----------------------------|---------|
| <b>Mg+2_H+1_PO4-3_H2O(3)_ (s)</b><br>Newberyite; Dibasic magnesium phosphate trihydrate |           |   |                             |         |
| 0                                                                                       | 7782-75-4 | 3 | H(7)Mg(1)O(7)P(1)           | 174.330 |
| <b>Mg+2_H+1_Salicylic-2_PO4-3</b>                                                       |           |   |                             |         |
| -2                                                                                      | ---       | 1 | C(7)H(5)Mg(1)O(7)P(1)       | 256.392 |
| <b>Mg+2_H+1_Ser-1_PO4-3</b>                                                             |           |   |                             |         |
| -1                                                                                      | ---       | 1 | C(3)H(7)Mg(1)N(1)O(7)P(1)   | 224.370 |
| <b>Mg+2_H+1_Succinic-2_PO4-3</b>                                                        |           |   |                             |         |
| -2                                                                                      | ---       | 1 | C(4)H(5)Mg(1)O(8)P(1)       | 236.358 |
| <b>Mg+2_H+1_Thr-1_PO4-3</b>                                                             |           |   |                             |         |
| -1                                                                                      | ---       | 1 | C(4)H(9)Mg(1)N(1)O(7)P(1)   | 238.397 |
| <b>Mg+2_H+1_Trp-1_PO4-3</b>                                                             |           |   |                             |         |
| -1                                                                                      | ---       | 1 | C(11)H(12)Mg(1)N(2)O(6)P(1) | 323.505 |
| <b>Mg+2_H+1_Val-1_PO4-3</b>                                                             |           |   |                             |         |
| -1                                                                                      | ---       | 1 | C(5)H(11)Mg(1)N(1)O(6)P(1)  | 236.424 |
| <b>Mg+2_H+1(2)_Ala-1_PO4-3</b>                                                          |           |   |                             |         |
| 0                                                                                       | ---       | 1 | C(3)H(8)Mg(1)N(1)O(6)P(1)   | 209.379 |
| <b>Mg+2_H+1(2)_Arg-1_PO4-3</b>                                                          |           |   |                             |         |
| 0                                                                                       | ---       | 1 | C(6)H(15)Mg(1)N(4)O(6)P(1)  | 294.487 |
| <b>Mg+2_H+1(2)_Asn-1_PO4-3</b>                                                          |           |   |                             |         |
| 0                                                                                       | ---       | 1 | C(4)H(9)Mg(1)N(2)O(7)P(1)   | 252.404 |
| <b>Mg+2_H+1(2)_Asp-2_PO4-3</b>                                                          |           |   |                             |         |
| -1                                                                                      | ---       | 1 | C(4)H(7)Mg(1)N(1)O(8)P(1)   | 252.381 |
| <b>Mg+2_H+1(2)_bAla-1_PO4-3</b>                                                         |           |   |                             |         |
| 0                                                                                       | ---       | 1 | C(3)H(8)Mg(1)N(1)O(6)P(1)   | 209.379 |

|                                   |     |   |                                |         |
|-----------------------------------|-----|---|--------------------------------|---------|
| <b>Mg+2_H+1(2)_Cis-2_PO4-3</b>    |     |   |                                |         |
| -1                                | --- | 1 | C(6)H(12)Mg(1)N(2)O(8)P(1)S(2) | 359.569 |
| <b>Mg+2_H+1(2)_Citrul-1_PO4-3</b> |     |   |                                |         |
| 0                                 | --- | 1 | C(6)H(14)Mg(1)N(3)O(7)P(1)     | 295.472 |
| <b>Mg+2_H+1(2)_Cys-2_PO4-3</b>    |     |   |                                |         |
| -1                                | --- | 1 | C(3)H(7)Mg(1)N(1)O(6)P(1)S(1)  | 240.431 |
| <b>Mg+2_H+1(2)_Gln-1_PO4-3</b>    |     |   |                                |         |
| 0                                 | --- | 1 | C(5)H(11)Mg(1)N(2)O(7)P(1)     | 266.431 |
| <b>Mg+2_H+1(2)_Glu-2_PO4-3</b>    |     |   |                                |         |
| -1                                | --- | 1 | C(5)H(9)Mg(1)N(1)O(8)P(1)      | 266.407 |
| <b>Mg+2_H+1(2)_Gly-1_PO4-3</b>    |     |   |                                |         |
| 0                                 | --- | 1 | C(2)H(6)Mg(1)N(1)O(6)P(1)      | 195.352 |
| <b>Mg+2_H+1(2)_His-1_PO4-3</b>    |     |   |                                |         |
| 0                                 | --- | 1 | C(6)H(10)Mg(1)N(3)O(6)P(1)     | 275.441 |
| <b>Mg+2_H+1(2)_Hyp-1_PO4-3</b>    |     |   |                                |         |
| 0                                 | --- | 1 | C(5)H(10)Mg(1)N(1)O(7)P(1)     | 251.416 |
| <b>Mg+2_H+1(2)_Ile-1_PO4-3</b>    |     |   |                                |         |
| 0                                 | --- | 1 | C(6)H(14)Mg(1)N(1)O(6)P(1)     | 251.459 |
| <b>Mg+2_H+1(2)_Leu-1_PO4-3</b>    |     |   |                                |         |
| 0                                 | --- | 1 | C(6)H(14)Mg(1)N(1)O(6)P(1)     | 251.459 |
| <b>Mg+2_H+1(2)_Lys-1_PO4-3</b>    |     |   |                                |         |
| 0                                 | --- | 1 | C(6)H(15)Mg(1)N(2)O(6)P(1)     | 266.474 |
| <b>Mg+2_H+1(2)_Met-1_PO4-3</b>    |     |   |                                |         |
| 0                                 | --- | 1 | C(5)H(12)Mg(1)N(1)O(6)P(1)S(1) | 269.492 |

|                                  |     |   |                                        |         |
|----------------------------------|-----|---|----------------------------------------|---------|
| <b>Mg+2_H+1 (2) _Orn-1_PO4-3</b> |     |   |                                        |         |
| 0                                | --- | 1 | C (5) H (13) Mg (1) N (2) O (6) P (1)  | 252.447 |
| <b>Mg+2_H+1 (2) _P2O7-4</b>      |     |   |                                        |         |
| 0                                | --- | 1 | H (2) Mg (1) O (7) P (2)               | 200.265 |
| <b>Mg+2_H+1 (2) _Phe-1_PO4-3</b> |     |   |                                        |         |
| 0                                | --- | 1 | C (9) H (12) Mg (1) N (1) O (6) P (1)  | 285.476 |
| <b>Mg+2_H+1 (2) _PO4-3</b>       |     |   |                                        |         |
| 1                                | --- | 4 | H (2) Mg (1) O (4) P (1)               | 121.293 |
| <b>Mg+2_H+1 (2) _PO4-3 (2)</b>   |     |   |                                        |         |
| -2                               | --- | 1 | H (2) Mg (1) O (8) P (2)               | 216.264 |
| <b>Mg+2_H+1 (2) _Pro-1_PO4-3</b> |     |   |                                        |         |
| 0                                | --- | 1 | C (5) H (10) Mg (1) N (1) O (6) P (1)  | 235.417 |
| <b>Mg+2_H+1 (2) _Ser-1_PO4-3</b> |     |   |                                        |         |
| 0                                | --- | 1 | C (3) H (8) Mg (1) N (1) O (7) P (1)   | 225.378 |
| <b>Mg+2_H+1 (2) _Thr-1_PO4-3</b> |     |   |                                        |         |
| 0                                | --- | 1 | C (4) H (10) Mg (1) N (1) O (7) P (1)  | 239.405 |
| <b>Mg+2_H+1 (2) _Trp-1_PO4-3</b> |     |   |                                        |         |
| 0                                | --- | 1 | C (11) H (13) Mg (1) N (2) O (6) P (1) | 324.513 |
| <b>Mg+2_H+1 (2) _Tyr-2_PO4-3</b> |     |   |                                        |         |
| -1                               | --- | 1 | C (9) H (11) Mg (1) N (1) O (7) P (1)  | 300.468 |
| <b>Mg+2_H+1 (2) _Val-1_PO4-3</b> |     |   |                                        |         |
| 0                                | --- | 1 | C (5) H (12) Mg (1) N (1) O (6) P (1)  | 237.432 |
| <b>Mg+2_H+1 (3) _PO4-3 (2)</b>   |     |   |                                        |         |
| -1                               | --- | 4 | H (3) Mg (1) O (8) P (2)               | 217.272 |

|                                                                                                                                     |     |   |                            |         |
|-------------------------------------------------------------------------------------------------------------------------------------|-----|---|----------------------------|---------|
| <b>Mg+2_H+1(4)_PO4-3(2)</b>                                                                                                         |     |   |                            |         |
| 0                                                                                                                                   | --- | 2 | H(4)Mg(1)O(8)P(2)          | 218.280 |
| <b>Mg+2_K+1_PO4-3</b>                                                                                                               |     |   |                            |         |
| 0                                                                                                                                   | --- | 1 | K(1)Mg(1)O(4)P(1)          | 158.375 |
| <b>Mg+2_K+1_PO4-3_H2O(6)_(s)</b><br>Magnesium potassium phosphate hexahydrate; Potassium struvite; K-Struvite; Struvite (potassium) |     |   |                            |         |
| 0                                                                                                                                   | --- | 1 | H(12)K(1)Mg(1)O(10)P(1)    | 266.466 |
| <b>Mg+2_OH-1_P2O7-4</b>                                                                                                             |     |   |                            |         |
| -3                                                                                                                                  | --- | 2 | H(1)Mg(1)O(8)P(2)          | 215.256 |
| <b>Mg+2_OH-1_P3O10-5</b>                                                                                                            |     |   |                            |         |
| -4                                                                                                                                  | --- | 3 | H(1)Mg(1)O(11)P(3)         | 294.228 |
| <b>Mg+2_Oxine-1_P3O10-5</b>                                                                                                         |     |   |                            |         |
| -4                                                                                                                                  | --- | 1 | C(9)H(6)Mg(1)N(1)O(11)P(3) | 421.374 |
| <b>Mg+2_P2O7-4</b>                                                                                                                  |     |   |                            |         |
| -2                                                                                                                                  | --- | 4 | Mg(1)O(7)P(2)              | 198.249 |
| <b>Mg+2_P2O7-4(2)</b>                                                                                                               |     |   |                            |         |
| -6                                                                                                                                  | --- | 2 | Mg(1)O(14)P(4)             | 372.193 |
| <b>Mg+2_P3O10-5</b>                                                                                                                 |     |   |                            |         |
| -3                                                                                                                                  | --- | 7 | Mg(1)O(10)P(3)             | 277.221 |
| <b>Mg+2_P3O10-5(2)</b>                                                                                                              |     |   |                            |         |
| -8                                                                                                                                  | --- | 2 | Mg(1)O(20)P(6)             | 530.137 |
| <b>Mg+2_P3O9-3</b>                                                                                                                  |     |   |                            |         |
| -1                                                                                                                                  | --- | 1 | Mg(1)O(9)P(3)              | 261.222 |
| <b>Mg+2_P4O12-4</b>                                                                                                                 |     |   |                            |         |
| -2                                                                                                                                  | --- | 1 | Mg(1)O(12)P(4)             | 340.194 |

|                                                                                 |            |   |                           |         |
|---------------------------------------------------------------------------------|------------|---|---------------------------|---------|
| <b>Mg+2_P4O12-4 (2)</b>                                                         |            |   |                           |         |
| -6                                                                              | ---        | 1 | Mg (1) O (24) P (8)       | 656.083 |
| <b>Mg+2_P4O13-6</b>                                                             |            |   |                           |         |
| -4                                                                              | ---        | 2 | Mg (1) O (13) P (4)       | 356.193 |
| <b>Mg+2_P6O18-6</b>                                                             |            |   |                           |         |
| -4                                                                              | ---        | 1 | Mg (1) O (18) P (6)       | 498.138 |
| <b>Mg+2_PO4-3</b>                                                               |            |   |                           |         |
| -1                                                                              | ---        | 1 | Mg (1) O (4) P (1)        | 119.277 |
| <b>Mg+2_UO2+2 (2)_PO4-3 (2)_ (s)</b><br>Saleeite                                |            |   |                           |         |
| 0                                                                               | ---        | 2 | Mg (1) O (12) P (2) U (2) | 754.304 |
| <b>Mg+2 (2)_F-1_PO4-3 (s)</b><br>Wagnerite                                      |            |   |                           |         |
| 0                                                                               | 11088-75-8 | 1 | F (1) Mg (2) O (4) P (1)  | 162.580 |
| <b>Mg+2 (2)_H+1 (2)_PO4-3 (2)</b>                                               |            |   |                           |         |
| 0                                                                               | ---        | 2 | H (2) Mg (2) O (8) P (2)  | 240.569 |
| <b>Mg+2 (2)_P2O7-4</b>                                                          |            |   |                           |         |
| 0                                                                               | ---        | 1 | Mg (2) O (7) P (2)        | 222.554 |
| <b>Mg+2 (2)_P3O10-5</b>                                                         |            |   |                           |         |
| -1                                                                              | ---        | 1 | Mg (2) O (10) P (3)       | 301.526 |
| <b>Mg+2 (2)_P4O13-6</b>                                                         |            |   |                           |         |
| -2                                                                              | ---        | 2 | Mg (2) O (13) P (4)       | 380.498 |
| <b>Mg+2 (3)_PO4-3 (2)_ (s)</b><br>Magnesium phosphate; Magnesium orthophosphate |            |   |                           |         |
| 0                                                                               | 7757-87-1  | 2 | Mg (3) O (8) P (2)        | 262.858 |
| <b>Mg+2 (3)_PO4-3 (2)_H2O (22)_ (s)</b><br>Cattiite; Trimagnesium phosphate     |            |   |                           |         |



|                                                                                                   |            |      |                                        |         |
|---------------------------------------------------------------------------------------------------|------------|------|----------------------------------------|---------|
| 0                                                                                                 | ---        | 3    | H (44) Mg (3) O (30) P (2)             | 659.194 |
| <b>Mg+2 (3) _PO4-3 (2) _H2O (8) _ (s)</b><br>Bobierite; Magnesium phosphate octahydrate           |            |      |                                        |         |
| 0                                                                                                 | 13446-23-6 | 3    | H (16) Mg (3) O (16) P (2)             | 406.980 |
| <b>Mg+2 (5) _F-1 _PO4-3 (3) _ (s)</b><br>Magnesium fluoroapatite                                  |            |      |                                        |         |
| 0                                                                                                 | 11088-76-9 | 1    | F (1) Mg (5) O (12) P (3)              | 425.438 |
| <b>Mg+2 (5) _OH-1 _PO4-3 (3) _ (s)</b><br>Magnesium hydroxyapatite; Magnesium hydroxide phosphate |            |      |                                        |         |
| 0                                                                                                 | 11089-13-7 | 1    | H (1) Mg (5) O (13) P (3)              | 423.447 |
| <b>Mg (s)</b><br>Magnesium; Magnesium, hexagonal                                                  |            |      |                                        |         |
| 0                                                                                                 | 7439-95-4  | 154  | Mg (1)                                 | 24.3050 |
| <b>Mn+2</b><br>Manganese(II) ion; Manganous ion                                                   |            |      |                                        |         |
| 2                                                                                                 | 16397-91-4 | 1956 | Mn (1)                                 | 54.9380 |
| <b>Mn+2 _Bipy</b>                                                                                 |            |      |                                        |         |
| 2                                                                                                 | ---        | 15   | C (10) H (8) Mn (1) N (2)              | 211.125 |
| <b>Mn+2 _Bipy _P3O10-5</b>                                                                        |            |      |                                        |         |
| -3                                                                                                | ---        | 2    | C (10) H (8) Mn (1) N (2) O (10) P (3) | 464.041 |
| <b>Mn+2 _H+1 _Ala-1 _PO4-3</b>                                                                    |            |      |                                        |         |
| -1                                                                                                | ---        | 1    | C (3) H (7) Mn (1) N (1) O (6) P (1)   | 239.004 |
| <b>Mn+2 _H+1 _Arg-1 _PO4-3</b>                                                                    |            |      |                                        |         |
| -1                                                                                                | ---        | 1    | C (6) H (14) Mn (1) N (4) O (6) P (1)  | 324.112 |
| <b>Mn+2 _H+1 _Asn-1 _PO4-3</b>                                                                    |            |      |                                        |         |
| -1                                                                                                | ---        | 1    | C (4) H (8) Mn (1) N (2) O (7) P (1)   | 282.029 |
| <b>Mn+2 _H+1 _Asp-2 _PO4-3</b>                                                                    |            |      |                                        |         |
| -2                                                                                                | ---        | 1    | C (4) H (6) Mn (1) N (1) O (8) P (1)   | 282.006 |

|                                 |     |   |                               |         |
|---------------------------------|-----|---|-------------------------------|---------|
| <b>Mn+2_H+1_bAla-1_PO4-3</b>    |     |   |                               |         |
| -1                              | --- | 1 | C(3)H(7)Mn(1)N(1)O(6)P(1)     | 239.004 |
| <b>Mn+2_H+1_Bipy_P3O10-5</b>    |     |   |                               |         |
| -2                              | --- | 1 | C(10)H(9)Mn(1)N(2)O(10)P(3)   | 465.049 |
| <b>Mn+2_H+1_Citric-3_PO4-3</b>  |     |   |                               |         |
| -3                              | --- | 1 | C(6)H(6)Mn(1)O(11)P(1)        | 340.019 |
| <b>Mn+2_H+1_Citrul-1_PO4-3</b>  |     |   |                               |         |
| -1                              | --- | 1 | C(6)H(13)Mn(1)N(3)O(7)P(1)    | 325.097 |
| <b>Mn+2_H+1_Cys-2_PO4-3</b>     |     |   |                               |         |
| -2                              | --- | 1 | C(3)H(6)Mn(1)N(1)O(6)P(1)S(1) | 270.056 |
| <b>Mn+2_H+1_Gln-1_PO4-3</b>     |     |   |                               |         |
| -1                              | --- | 1 | C(5)H(10)Mn(1)N(2)O(7)P(1)    | 296.056 |
| <b>Mn+2_H+1_Glu-2_PO4-3</b>     |     |   |                               |         |
| -2                              | --- | 1 | C(5)H(8)Mn(1)N(1)O(8)P(1)     | 296.032 |
| <b>Mn+2_H+1_Gly-1_PO4-3</b>     |     |   |                               |         |
| -1                              | --- | 1 | C(2)H(5)Mn(1)N(1)O(6)P(1)     | 224.977 |
| <b>Mn+2_H+1_His-1_PO4-3</b>     |     |   |                               |         |
| -1                              | --- | 1 | C(6)H(9)Mn(1)N(3)O(6)P(1)     | 305.066 |
| <b>Mn+2_H+1_Histamine_PO4-3</b> |     |   |                               |         |
| 0                               | --- | 1 | C(5)H(10)Mn(1)N(3)O(4)P(1)    | 262.064 |
| <b>Mn+2_H+1_Hyp-1_PO4-3</b>     |     |   |                               |         |
| -1                              | --- | 1 | C(5)H(9)Mn(1)N(1)O(7)P(1)     | 281.041 |
| <b>Mn+2_H+1_Ile-1_PO4-3</b>     |     |   |                               |         |
| -1                              | --- | 1 | C(6)H(13)Mn(1)N(1)O(6)P(1)    | 281.084 |

|                                                          |            |   |                                |         |
|----------------------------------------------------------|------------|---|--------------------------------|---------|
| <b>Mn+2_H+1_Lactic-1_PO4-3</b>                           |            |   |                                |         |
| -1                                                       | ---        | 1 | C(3)H(6)Mn(1)O(7)P(1)          | 239.988 |
| <b>Mn+2_H+1_Leu-1_PO4-3</b>                              |            |   |                                |         |
| -1                                                       | ---        | 1 | C(6)H(13)Mn(1)N(1)O(6)P(1)     | 281.084 |
| <b>Mn+2_H+1_Malic-2_PO4-3</b>                            |            |   |                                |         |
| -2                                                       | ---        | 1 | C(4)H(5)Mn(1)O(9)P(1)          | 282.990 |
| <b>Mn+2_H+1_Met-1_PO4-3</b>                              |            |   |                                |         |
| -1                                                       | ---        | 1 | C(5)H(11)Mn(1)N(1)O(6)P(1)S(1) | 299.117 |
| <b>Mn+2_H+1_Oxalic-2_PO4-3</b>                           |            |   |                                |         |
| -2                                                       | ---        | 1 | C(2)H(1)Mn(1)O(8)P(1)          | 238.937 |
| <b>Mn+2_H+1_P2O7-4</b>                                   |            |   |                                |         |
| -1                                                       | ---        | 1 | H(1)Mn(1)O(7)P(2)              | 229.890 |
| <b>Mn+2_H+1_P3O10-5</b>                                  |            |   |                                |         |
| -2                                                       | ---        | 3 | H(1)Mn(1)O(10)P(3)             | 308.862 |
| <b>Mn+2_H+1_P3O10-5(2)</b>                               |            |   |                                |         |
| -7                                                       | ---        | 1 | H(1)Mn(1)O(20)P(6)             | 561.778 |
| <b>Mn+2_H+1_Phe-1_PO4-3</b>                              |            |   |                                |         |
| -1                                                       | ---        | 1 | C(9)H(11)Mn(1)N(1)O(6)P(1)     | 315.101 |
| <b>Mn+2_H+1_PO4-3</b>                                    |            |   |                                |         |
| 0                                                        | ---        | 3 | H(1)Mn(1)O(4)P(1)              | 150.918 |
| <b>Mn+2_H+1_PO4-3(s)</b><br>Manganese phosphate, dibasic |            |   |                                |         |
| 0                                                        | 51349-94-1 | 3 | H(1)Mn(1)O(4)P(1)              | 150.918 |
| <b>Mn+2_H+1_Pro-1_PO4-3</b>                              |            |   |                                |         |
| -1                                                       | ---        | 1 | C(5)H(9)Mn(1)N(1)O(6)P(1)      | 265.042 |

|                                   |     |   |                                |         |
|-----------------------------------|-----|---|--------------------------------|---------|
| <b>Mn+2_H+1_Salicylic-2_PO4-3</b> |     |   |                                |         |
| -2                                | --- | 1 | C(7)H(5)Mn(1)O(7)P(1)          | 287.025 |
| <b>Mn+2_H+1_Ser-1_PO4-3</b>       |     |   |                                |         |
| -1                                | --- | 1 | C(3)H(7)Mn(1)N(1)O(7)P(1)      | 255.003 |
| <b>Mn+2_H+1_Succinic-2_PO4-3</b>  |     |   |                                |         |
| -2                                | --- | 1 | C(4)H(5)Mn(1)O(8)P(1)          | 266.991 |
| <b>Mn+2_H+1_Thr-1_PO4-3</b>       |     |   |                                |         |
| -1                                | --- | 1 | C(4)H(9)Mn(1)N(1)O(7)P(1)      | 269.030 |
| <b>Mn+2_H+1_Trp-1_PO4-3</b>       |     |   |                                |         |
| -1                                | --- | 1 | C(11)H(12)Mn(1)N(2)O(6)P(1)    | 354.138 |
| <b>Mn+2_H+1_Val-1_PO4-3</b>       |     |   |                                |         |
| -1                                | --- | 1 | C(5)H(11)Mn(1)N(1)O(6)P(1)     | 267.057 |
| <b>Mn+2_H+1(2)_Cis-2_PO4-3</b>    |     |   |                                |         |
| -1                                | --- | 1 | C(6)H(12)Mn(1)N(2)O(8)P(1)S(2) | 390.202 |
| <b>Mn+2_H+1(2)_Lys-1_PO4-3</b>    |     |   |                                |         |
| 0                                 | --- | 1 | C(6)H(15)Mn(1)N(2)O(6)P(1)     | 297.107 |
| <b>Mn+2_H+1(2)_Orn-1_PO4-3</b>    |     |   |                                |         |
| 0                                 | --- | 1 | C(5)H(13)Mn(1)N(2)O(6)P(1)     | 283.080 |
| <b>Mn+2_H+1(2)_P2O7-4</b>         |     |   |                                |         |
| 0                                 | --- | 1 | H(2)Mn(1)O(7)P(2)              | 230.898 |
| <b>Mn+2_H+1(2)_PO4-3</b>          |     |   |                                |         |
| 1                                 | --- | 3 | H(2)Mn(1)O(4)P(1)              | 151.926 |
| <b>Mn+2_H+1(2)_PO4-3(2)</b>       |     |   |                                |         |
| -2                                | --- | 2 | H(2)Mn(1)O(8)P(2)              | 246.897 |

|                                                                                     |            |   |                            |         |
|-------------------------------------------------------------------------------------|------------|---|----------------------------|---------|
| <b>Mn+2_H+1(2)_Tyr-2_PO4-3</b>                                                      |            |   |                            |         |
| -1                                                                                  | ---        | 1 | C(9)H(11)Mn(1)N(1)O(7)P(1) | 331.101 |
| <b>Mn+2_H+1(4)_P2O7-4(2)</b>                                                        |            |   |                            |         |
| -2                                                                                  | ---        | 1 | H(4)Mn(1)O(14)P(4)         | 406.857 |
| <b>Mn+2_H+1(4)_PO4-3(2)</b>                                                         |            |   |                            |         |
| 0                                                                                   | ---        | 3 | H(4)Mn(1)O(8)P(2)          | 248.913 |
| <b>Mn+2_Oxine-1_P3O10-5</b>                                                         |            |   |                            |         |
| -4                                                                                  | ---        | 1 | C(9)H(6)Mn(1)N(1)O(11)P(3) | 452.007 |
| <b>Mn+2_P2O7-4</b>                                                                  |            |   |                            |         |
| -2                                                                                  | ---        | 2 | Mn(1)O(7)P(2)              | 228.882 |
| <b>Mn+2_P3O10-5</b>                                                                 |            |   |                            |         |
| -3                                                                                  | ---        | 5 | Mn(1)O(10)P(3)             | 307.854 |
| <b>Mn+2_P3O10-5(2)</b>                                                              |            |   |                            |         |
| -8                                                                                  | ---        | 1 | Mn(1)O(20)P(6)             | 560.770 |
| <b>Mn+2_P3O9-3</b>                                                                  |            |   |                            |         |
| -1                                                                                  | ---        | 1 | Mn(1)O(9)P(3)              | 291.855 |
| <b>Mn+2_P4O12-4</b>                                                                 |            |   |                            |         |
| -2                                                                                  | ---        | 1 | Mn(1)O(12)P(4)             | 370.827 |
| <b>Mn+2(2)_P2O7-4(s)</b>                                                            |            |   |                            |         |
| 0                                                                                   | ---        | 2 | Mn(2)O(7)P(2)              | 283.820 |
| <b>Mn+2(3)_PO4-3(2)(s)</b><br>Manganese phosphate                                   |            |   |                            |         |
| 0                                                                                   | 14154-09-7 | 1 | Mn(3)O(8)P(2)              | 354.757 |
| <b>Mn+2(3)_PO4-3(2)_H2O(3)(s)</b><br>Reddingite; Manganese(II) phosphate trihydrate |            |   |                            |         |
| 0                                                                                   | ---        | 1 | H(6)Mn(3)O(11)P(2)         | 408.803 |

|                                    |            |    |                                        |         |
|------------------------------------|------------|----|----------------------------------------|---------|
| <b>Mn+3</b>                        |            |    |                                        |         |
| Manganese(III) ion; Manganic ion   |            |    |                                        |         |
| 3                                  | 14546-48-6 | 86 | Mn (1)                                 | 54.9380 |
| <b>Mn+3_H+1_P2O7-4</b>             |            |    |                                        |         |
| 0                                  | ---        | 1  | H (1) Mn (1) O (7) P (2)               | 229.890 |
| <b>Mn+3_H+1_PO4-3</b>              |            |    |                                        |         |
| 1                                  | ---        | 2  | H (1) Mn (1) O (4) P (1)               | 150.918 |
| <b>Mn+3_H+1 (2)_HIMDA-2_P2O7-4</b> |            |    |                                        |         |
| -1                                 | ---        | 1  | C (6) H (11) Mn (1) N (1) O (12) P (2) | 406.039 |
| <b>Mn+3_H+1 (2)_P2O7-4</b>         |            |    |                                        |         |
| 1                                  | ---        | 3  | H (2) Mn (1) O (7) P (2)               | 230.898 |
| <b>Mn+3_H+1 (2)_PO4-3</b>          |            |    |                                        |         |
| 2                                  | ---        | 2  | H (2) Mn (1) O (4) P (1)               | 151.926 |
| <b>Mn+3_H+1 (4)_P2O7-4 (2)</b>     |            |    |                                        |         |
| -1                                 | ---        | 2  | H (4) Mn (1) O (14) P (4)              | 406.857 |
| <b>Mn+3_H+1 (4)_PO4-3 (2)</b>      |            |    |                                        |         |
| 1                                  | ---        | 2  | H (4) Mn (1) O (8) P (2)               | 248.913 |
| <b>Mn+3_H+1 (5)_P2O7-4 (2)</b>     |            |    |                                        |         |
| 0                                  | ---        | 1  | H (5) Mn (1) O (14) P (4)              | 407.865 |
| <b>Mn+3_H+1 (6)_P2O7-4 (3)</b>     |            |    |                                        |         |
| -3                                 | ---        | 1  | H (6) Mn (1) O (21) P (6)              | 582.817 |
| <b>Mn+3_P2O7-4</b>                 |            |    |                                        |         |
| -1                                 | ---        | 1  | Mn (1) O (7) P (2)                     | 228.882 |
| <b>Mn+3_P2O7-4 (2)</b>             |            |    |                                        |         |
| -5                                 | ---        | 1  | Mn (1) O (14) P (4)                    | 402.826 |

|                                                                      |            |     |                      |         |
|----------------------------------------------------------------------|------------|-----|----------------------|---------|
| <b>Mn (s)</b><br>Manganese; Manganese, alpha; Manganese, complex bcc |            |     |                      |         |
| 0                                                                    | 7439-96-5  | 72  | Mn (1)               | 54.9380 |
| <b>Mn<sub>2</sub>P (s)</b><br>Dimanganese phosphide                  |            |     |                      |         |
| 0                                                                    | ---        | 1   | Mn (2) P (1)         | 140.850 |
| <b>MnP (s)</b><br>Manganese monophosphide                            |            |     |                      |         |
| 0                                                                    | ---        | 1   | Mn (1) P (1)         | 85.9120 |
| <b>MnP<sub>3</sub> (s)</b><br>Manganese triphosphide                 |            |     |                      |         |
| 0                                                                    | ---        | 1   | Mn (1) P (3)         | 147.860 |
| <b>Mo<sub>11</sub>PO<sub>39</sub>-7</b>                              |            |     |                      |         |
| -7                                                                   | ---        | 1   | Mo (11) O (39) P (1) | 1710.53 |
| <b>Mo<sub>12</sub>PO<sub>40</sub>-3</b>                              |            |     |                      |         |
| -3                                                                   | ---        | 1   | Mo (12) O (40) P (1) | 1822.49 |
| <b>Mo<sub>18</sub>P<sub>2</sub>O<sub>62</sub>-6</b>                  |            |     |                      |         |
| -6                                                                   | ---        | 1   | Mo (18) O (62) P (2) | 2781.23 |
| <b>Mo<sub>5</sub>P<sub>2</sub>O<sub>23</sub>-6</b>                   |            |     |                      |         |
| -6                                                                   | ---        | 1   | Mo (5) O (23) P (2)  | 909.744 |
| <b>MoO<sub>4</sub>-2</b><br>Molybdate(VI) ion                        |            |     |                      |         |
| -2                                                                   | 14259-85-9 | 281 | Mo (1) O (4)         | 159.960 |
| <b>N<sub>2</sub> (g)</b><br>Nitrogen gas; Nitrogen                   |            |     |                      |         |
| 0                                                                    | 7727-37-9  | 566 | N (2)                | 28.0134 |
| <b>Na+1</b><br>Sodium(I) ion                                         |            |     |                      |         |
| 1                                                                    | 17341-25-2 | 617 | Na (1)               | 22.9900 |

|                                                                                      |     |   |                                |         |
|--------------------------------------------------------------------------------------|-----|---|--------------------------------|---------|
| <b>Na+1_H2PO2-1</b>                                                                  |     |   |                                |         |
| 0                                                                                    | --- | 1 | H (2) Na (1) O (2) P (1)       | 87.9787 |
| <b>Na+1_HPO3-2</b>                                                                   |     |   |                                |         |
| -1                                                                                   | --- | 1 | H (1) Na (1) O (3) P (1)       | 102.970 |
| <b>Na+1_P2O7-4</b>                                                                   |     |   |                                |         |
| -3                                                                                   | --- | 2 | Na (1) O (7) P (2)             | 196.934 |
| <b>Na+1_P2O7-4 (2)</b>                                                               |     |   |                                |         |
| -7                                                                                   | --- | 1 | Na (1) O (14) P (4)            | 370.878 |
| <b>Na+1_P3O10-5</b>                                                                  |     |   |                                |         |
| -4                                                                                   | --- | 1 | Na (1) O (10) P (3)            | 275.906 |
| <b>Na+1_P3O8-5</b>                                                                   |     |   |                                |         |
| -4                                                                                   | --- | 1 | Na (1) O (8) P (3)             | 243.907 |
| <b>Na+1_P3O9-3</b>                                                                   |     |   |                                |         |
| -2                                                                                   | --- | 1 | Na (1) O (9) P (3)             | 259.907 |
| <b>Na+1_P4O12-4</b>                                                                  |     |   |                                |         |
| -3                                                                                   | --- | 1 | Na (1) O (12) P (4)            | 338.879 |
| <b>Na+1_P4O13-6</b>                                                                  |     |   |                                |         |
| -5                                                                                   | --- | 1 | Na (1) O (13) P (4)            | 354.878 |
| <b>Na+1_P6O18-6</b>                                                                  |     |   |                                |         |
| -5                                                                                   | --- | 1 | Na (1) O (18) P (6)            | 496.823 |
| <b>Na+1_PO4-3</b>                                                                    |     |   |                                |         |
| -2                                                                                   | --- | 1 | Na (1) O (4) P (1)             | 117.962 |
| <b>Na+1 (2)_MePhos-2_(s)</b><br>Disodium methylphosphonate; Sodium methylphosphonate |     |   |                                |         |
| 0                                                                                    | --- | 2 | C (1) H (3) Na (2) O (3) P (1) | 139.987 |



|                                                                                                                                          |            |     |                                         |         |
|------------------------------------------------------------------------------------------------------------------------------------------|------------|-----|-----------------------------------------|---------|
| <b>Na+1 (2) _P2O7-4</b>                                                                                                                  |            |     |                                         |         |
| -2                                                                                                                                       | ---        | 1   | Na (2) O (7) P (2)                      | 219.924 |
| <b>Na+1 (2) _P3O10-5</b>                                                                                                                 |            |     |                                         |         |
| -3                                                                                                                                       | ---        | 1   | Na (2) O (10) P (3)                     | 298.896 |
| <b>Na+1 (2) _P4O12-4</b>                                                                                                                 |            |     |                                         |         |
| -2                                                                                                                                       | ---        | 1   | Na (2) O (12) P (4)                     | 361.869 |
| <b>Na+1 (2) _PO4-3</b>                                                                                                                   |            |     |                                         |         |
| -1                                                                                                                                       | ---        | 1   | Na (2) O (4) P (1)                      | 140.952 |
| <b>Na+1 (3) _PO4-3_(s)</b><br>Sodium phosphate                                                                                           |            |     |                                         |         |
| 0                                                                                                                                        | ---        | 2   | Na (3) O (4) P (1)                      | 163.942 |
| <b>Na+1 (4) _P2O7-4_(s)</b><br>Sodium pyrophosphate; Tetrasodium pyrophosphate anhydrous; Tetrasodium diphosphate;<br>Sodium diphosphate |            |     |                                         |         |
| 0                                                                                                                                        | 7722-88-5  | 1   | Na (4) O (7) P (2)                      | 265.904 |
| <b>Na (s)</b><br>Sodium; Sodium, cubic                                                                                                   |            |     |                                         |         |
| 0                                                                                                                                        | 7440-23-5  | 209 | Na (1)                                  | 22.9900 |
| <b>NaZr2P3O12 (s)</b><br>Sodium dizirconium triphosphorus duodecaoxide                                                                   |            |     |                                         |         |
| 0                                                                                                                                        | ---        | 1   | Na (1) O (12) P (3) Zr (2)              | 490.389 |
| <b>Nd+3</b><br>Neodymium(III) ion                                                                                                        |            |     |                                         |         |
| 3                                                                                                                                        | 14913-52-1 | 800 | Nd (1)                                  | 144.240 |
| <b>Nd+3 _EDTA-4</b>                                                                                                                      |            |     |                                         |         |
| -1                                                                                                                                       | ---        | 47  | C (10) H (12) N (2) Nd (1) O (8)        | 432.454 |
| <b>Nd+3 _EDTA-4 _P3O10-5</b>                                                                                                             |            |     |                                         |         |
| -6                                                                                                                                       | ---        | 2   | C (10) H (12) N (2) Nd (1) O (18) P (3) | 685.370 |

|                                |     |   |                              |         |
|--------------------------------|-----|---|------------------------------|---------|
| <b>Nd+3_H+1_EDTA-4_P3O10-5</b> |     |   |                              |         |
| -5                             | --- | 2 | C(10)H(13)N(2)Nd(1)O(18)P(3) | 686.378 |
| <b>Nd+3_H+1_P3O10-5</b>        |     |   |                              |         |
| -1                             | --- | 1 | H(1)Nd(1)O(10)P(3)           | 398.164 |
| <b>Nd+3_H+1_PO4-3</b>          |     |   |                              |         |
| 1                              | --- | 2 | H(1)Nd(1)O(4)P(1)            | 240.220 |
| <b>Nd+3_H+1(2)_P3O10-5(2)</b>  |     |   |                              |         |
| -5                             | --- | 1 | H(2)Nd(1)O(20)P(6)           | 652.088 |
| <b>Nd+3_H+1(2)_PO4-3</b>       |     |   |                              |         |
| 2                              | --- | 2 | H(2)Nd(1)O(4)P(1)            | 241.228 |
| <b>Nd+3_H+1(2)_PO4-3(2)</b>    |     |   |                              |         |
| -1                             | --- | 2 | H(2)Nd(1)O(8)P(2)            | 336.199 |
| <b>Nd+3_H2PO2-1</b>            |     |   |                              |         |
| 2                              | --- | 1 | H(2)Nd(1)O(2)P(1)            | 209.229 |
| <b>Nd+3_P2O7-4</b>             |     |   |                              |         |
| -1                             | --- | 1 | Nd(1)O(7)P(2)                | 318.184 |
| <b>Nd+3_P3O10-5(2)</b>         |     |   |                              |         |
| -7                             | --- | 1 | Nd(1)O(20)P(6)               | 650.072 |
| <b>Nd+3_PO4-3</b>              |     |   |                              |         |
| 0                              | --- | 1 | Nd(1)O(4)P(1)                | 239.212 |
| <b>Nd+3_PO4-3(s)</b>           |     |   |                              |         |
| 0                              | --- | 1 | Nd(1)O(4)P(1)                | 239.212 |
| <b>Nd+3_PO4-3(2)</b>           |     |   |                              |         |
| -3                             | --- | 1 | Nd(1)O(8)P(2)                | 334.183 |

|                                  |            |      |                                       |         |
|----------------------------------|------------|------|---------------------------------------|---------|
| <b>Nd+3 (2) _P2O7-4</b>          |            |      |                                       |         |
| 2                                | ---        | 1    | Nd (2) O (7) P (2)                    | 462.424 |
| <b>NH3</b><br>Ammonia, aqueous   |            |      |                                       |         |
| 0                                | ---        | 1462 | H (3) N (1)                           | 17.0305 |
| <b>Ni+2</b><br>Nickel (II) ion   |            |      |                                       |         |
| 2                                | 14701-22-5 | 5108 | Ni (1)                                | 58.6930 |
| <b>Ni+2 _5NitrSal-2 _P3O10-5</b> |            |      |                                       |         |
| -5                               | ---        | 1    | C (7) H (3) N (1) Ni (1) O (15) P (3) | 492.714 |
| <b>Ni+2 _H+1 _HPO3-2</b>         |            |      |                                       |         |
| 1                                | ---        | 2    | H (2) Ni (1) O (3) P (1)              | 139.681 |
| <b>Ni+2 _H+1 _NTA-3 _PO4-3</b>   |            |      |                                       |         |
| -3                               | ---        | 1    | C (6) H (7) N (1) Ni (1) O (10) P (1) | 342.789 |
| <b>Ni+2 _H+1 _P2O7-4</b>         |            |      |                                       |         |
| -1                               | ---        | 4    | H (1) Ni (1) O (7) P (2)              | 233.645 |
| <b>Ni+2 _H+1 _P3O10-5</b>        |            |      |                                       |         |
| -2                               | ---        | 3    | H (1) Ni (1) O (10) P (3)             | 312.617 |
| <b>Ni+2 _H+1 _P3O10-5 (2)</b>    |            |      |                                       |         |
| -7                               | ---        | 1    | H (1) Ni (1) O (20) P (6)             | 565.533 |
| <b>Ni+2 _H+1 _PO4-3</b>          |            |      |                                       |         |
| 0                                | ---        | 4    | H (1) Ni (1) O (4) P (1)              | 154.673 |
| <b>Ni+2 _H+1 (2) _PO4-3</b>      |            |      |                                       |         |
| 1                                | ---        | 2    | H (2) Ni (1) O (4) P (1)              | 155.681 |
| <b>Ni+2 _HPO3-2</b>              |            |      |                                       |         |
| 0                                | ---        | 1    | H (1) Ni (1) O (3) P (1)              | 138.673 |

|                                           |     |    |                              |         |
|-------------------------------------------|-----|----|------------------------------|---------|
| <b>Ni+2_NTA-3</b>                         |     |    |                              |         |
| -1                                        | --- | 45 | C(6)H(6)N(1)Ni(1)O(6)        | 246.810 |
| <b>Ni+2_OH-1_PO4-3</b>                    |     |    |                              |         |
| -2                                        | --- | 1  | H(1)Ni(1)O(5)P(1)            | 170.672 |
| <b>Ni+2_P2O7-4</b>                        |     |    |                              |         |
| -2                                        | --- | 4  | Ni(1)O(7)P(2)                | 232.637 |
| <b>Ni+2_P2O7-4(2)</b>                     |     |    |                              |         |
| -6                                        | --- | 2  | Ni(1)O(14)P(4)               | 406.581 |
| <b>Ni+2_P3O10-5</b>                       |     |    |                              |         |
| -3                                        | --- | 6  | Ni(1)O(10)P(3)               | 311.609 |
| <b>Ni+2_P3O10-5(2)</b>                    |     |    |                              |         |
| -8                                        | --- | 1  | Ni(1)O(20)P(6)               | 564.525 |
| <b>Ni+2_P3O9-3</b>                        |     |    |                              |         |
| -1                                        | --- | 1  | Ni(1)O(9)P(3)                | 295.610 |
| <b>Ni+2_P4O12-4</b>                       |     |    |                              |         |
| -2                                        | --- | 1  | Ni(1)O(12)P(4)               | 374.582 |
| <b>Ni+2_P4O12-4(2)</b>                    |     |    |                              |         |
| -6                                        | --- | 1  | Ni(1)O(24)P(8)               | 690.471 |
| <b>Ni+2_P4O13-6</b>                       |     |    |                              |         |
| -4                                        | --- | 1  | Ni(1)O(13)P(4)               | 390.581 |
| <b>Ni+2_PO4-3</b>                         |     |    |                              |         |
| -1                                        | --- | 2  | Ni(1)O(4)P(1)                | 153.665 |
| <b>Ni+2_Pyrid2Azo4DiMeAniline_P3O10-5</b> |     |    |                              |         |
| -3                                        | --- | 1  | C(13)H(14)N(4)Ni(1)O(10)P(3) | 537.890 |

|                                             |            |     |                            |         |
|---------------------------------------------|------------|-----|----------------------------|---------|
| <b>Ni+2 (2) _P2O7-4</b>                     |            |     |                            |         |
| 0                                           | ---        | 1   | Ni (2) O (7) P (2)         | 291.330 |
| <b>Ni+2 (2) _P2O7-4 (s)</b>                 |            |     |                            |         |
| 0                                           | ---        | 1   | Ni (2) O (7) P (2)         | 291.330 |
| <b>Ni+2 (2) _P2O7-4_H2O(6) (s)</b>          |            |     |                            |         |
| 0                                           | ---        | 1   | H (12) Ni (2) O (13) P (2) | 399.422 |
| <b>Ni+2 (3) _PO4-3 (2) (s)</b>              |            |     |                            |         |
| 0                                           | ---        | 3   | Ni (3) O (8) P (2)         | 366.022 |
| <b>Ni (s)</b><br>Nickel; Nickel, fcc        |            |     |                            |         |
| 0                                           | 7440-02-0  | 148 | Ni (1)                     | 58.6930 |
| <b>Ni2P (s)</b><br>Dinickel phosphide       |            |     |                            |         |
| 0                                           | ---        | 1   | Ni (2) P (1)               | 148.360 |
| <b>Ni3P (s)</b><br>Trinickel phosphide      |            |     |                            |         |
| 0                                           | ---        | 1   | Ni (3) P (1)               | 207.053 |
| <b>Ni5P2 (s)</b><br>Pentanickel diphosphide |            |     |                            |         |
| 0                                           | ---        | 1   | Ni (5) P (2)               | 355.413 |
| <b>Np+3</b><br>Neptunium(III) ion           |            |     |                            |         |
| 3                                           | 21377-65-1 | 193 | Np (1)                     | 237.048 |
| <b>Np+3_H+1_PO4-3</b>                       |            |     |                            |         |
| 1                                           | ---        | 3   | H (1) Np (1) O (4) P (1)   | 333.028 |
| <b>Np+3_H+1 (10) _PO4-3 (5)</b>             |            |     |                            |         |
| -2                                          | ---        | 2   | H (10) Np (1) O (20) P (5) | 721.985 |

|                                      |     |   |                            |         |
|--------------------------------------|-----|---|----------------------------|---------|
| <b>Np+3_H+1 (12) _PO4-3 (6)</b>      |     |   |                            |         |
| -3                                   | --- | 2 | H (12) Np (1) O (24) P (6) | 818.973 |
| <b>Np+3_H+1 (2) _PO4-3</b>           |     |   |                            |         |
| 2                                    | --- | 3 | H (2) Np (1) O (4) P (1)   | 334.036 |
| <b>Np+3_H+1 (2) _PO4-3 (2)</b>       |     |   |                            |         |
| -1                                   | --- | 3 | H (2) Np (1) O (8) P (2)   | 429.007 |
| <b>Np+3_H+1 (2) _PO4-3 (2) _ (s)</b> |     |   |                            |         |
| -1                                   | --- | 1 | H (2) Np (1) O (8) P (2)   | 429.007 |
| <b>Np+3_H+1 (3) _PO4-3 (3)</b>       |     |   |                            |         |
| -3                                   | --- | 2 | H (3) Np (1) O (12) P (3)  | 524.987 |
| <b>Np+3_H+1 (4) _PO4-3 (2)</b>       |     |   |                            |         |
| 1                                    | --- | 2 | H (4) Np (1) O (8) P (2)   | 431.023 |
| <b>Np+3_H+1 (4) _PO4-3 (4)</b>       |     |   |                            |         |
| -5                                   | --- | 2 | H (4) Np (1) O (16) P (4)  | 620.966 |
| <b>Np+3_H+1 (5) _PO4-3 (5)</b>       |     |   |                            |         |
| -7                                   | --- | 2 | H (5) Np (1) O (20) P (5)  | 716.946 |
| <b>Np+3_H+1 (6) _PO4-3 (3)</b>       |     |   |                            |         |
| 0                                    | --- | 3 | H (6) Np (1) O (12) P (3)  | 528.010 |
| <b>Np+3_H+1 (6) _PO4-3 (6)</b>       |     |   |                            |         |
| -9                                   | --- | 2 | H (6) Np (1) O (24) P (6)  | 812.925 |
| <b>Np+3_H+1 (8) _PO4-3 (4)</b>       |     |   |                            |         |
| -1                                   | --- | 2 | H (8) Np (1) O (16) P (4)  | 624.998 |
| <b>Np+3_PO4-3</b>                    |     |   |                            |         |
| 0                                    | --- | 2 | Np (1) O (4) P (1)         | 332.020 |

|                                      |            |     |                           |         |
|--------------------------------------|------------|-----|---------------------------|---------|
| <b>Np+3_PO4-3 (2)</b>                |            |     |                           |         |
| -3                                   | ---        | 2   | Np (1) O (8) P (2)        | 426.991 |
| <b>Np+3_PO4-3 (3)</b>                |            |     |                           |         |
| -6                                   | ---        | 1   | Np (1) O (12) P (3)       | 521.963 |
| <b>Np+4</b><br>Neptunium(IV) ion     |            |     |                           |         |
| 4                                    | 22578-82-1 | 252 | Np (1)                    | 237.048 |
| <b>Np+4_H+1_PO4-3</b>                |            |     |                           |         |
| 2                                    | ---        | 1   | H (1) Np (1) O (4) P (1)  | 333.028 |
| <b>Np+4_H+1 (2)_PO4-3</b>            |            |     |                           |         |
| 3                                    | ---        | 1   | H (2) Np (1) O (4) P (1)  | 334.036 |
| <b>Np+4_H+1 (2)_PO4-3 (2)</b>        |            |     |                           |         |
| 0                                    | ---        | 1   | H (2) Np (1) O (8) P (2)  | 429.007 |
| <b>Np+4_H+1 (3)_PO4-3 (3)</b>        |            |     |                           |         |
| -2                                   | ---        | 1   | H (3) Np (1) O (12) P (3) | 524.987 |
| <b>Np+4_H+1 (4)_PO4-3 (2)</b>        |            |     |                           |         |
| 2                                    | ---        | 1   | H (4) Np (1) O (8) P (2)  | 431.023 |
| <b>NpO2+1</b><br>Neptunium oxide ion |            |     |                           |         |
| 1                                    | 21057-99-8 | 274 | Np (1) O (2)              | 269.047 |
| <b>NpO2+1_PO4-3</b>                  |            |     |                           |         |
| -2                                   | ---        | 1   | Np (1) O (6) P (1)        | 364.018 |
| <b>NpO2+2</b><br>Neptunyl ion        |            |     |                           |         |
| 2                                    | 18973-22-3 | 216 | Np (1) O (2)              | 269.047 |
| <b>NpO2+2_H+1_PO4-3</b>              |            |     |                           |         |

|                                                                                                                               |            |     |                           |         |
|-------------------------------------------------------------------------------------------------------------------------------|------------|-----|---------------------------|---------|
| 0                                                                                                                             | ---        | 3   | H (1) Np (1) O (6) P (1)  | 365.026 |
| <b>NpO<sub>2</sub>+2_H+1_PO<sub>4</sub>-3_(s)</b>                                                                             |            |     |                           |         |
| 0                                                                                                                             | ---        | 2   | H (1) Np (1) O (6) P (1)  | 365.026 |
| <b>NpO<sub>2</sub>+2_H+1 (2)_PO<sub>4</sub>-3</b>                                                                             |            |     |                           |         |
| 1                                                                                                                             | ---        | 3   | H (2) Np (1) O (6) P (1)  | 366.034 |
| <b>NpO<sub>2</sub>+2_H+1 (2)_PO<sub>4</sub>-3 (2)</b>                                                                         |            |     |                           |         |
| -2                                                                                                                            | ---        | 2   | H (2) Np (1) O (10) P (2) | 461.006 |
| <b>NpO<sub>2</sub>+2_H+1 (3)_PO<sub>4</sub>-3 (3)</b>                                                                         |            |     |                           |         |
| -4                                                                                                                            | ---        | 2   | H (3) Np (1) O (14) P (3) | 556.985 |
| <b>NpO<sub>2</sub>+2_H+1 (4)_PO<sub>4</sub>-3 (2)</b>                                                                         |            |     |                           |         |
| 0                                                                                                                             | ---        | 2   | H (4) Np (1) O (10) P (2) | 463.022 |
| <b>NpO<sub>2</sub>+2_H+1 (4)_PO<sub>4</sub>-3 (4)</b>                                                                         |            |     |                           |         |
| -6                                                                                                                            | ---        | 2   | H (4) Np (1) O (18) P (4) | 652.965 |
| <b>NpO<sub>2</sub>+2_H+1 (5)_PO<sub>4</sub>-3 (5)</b>                                                                         |            |     |                           |         |
| -8                                                                                                                            | ---        | 2   | H (5) Np (1) O (22) P (5) | 748.945 |
| <b>NpO<sub>2</sub>+2_H+1 (6)_PO<sub>4</sub>-3 (3)</b>                                                                         |            |     |                           |         |
| -1                                                                                                                            | ---        | 2   | H (6) Np (1) O (14) P (3) | 560.009 |
| <b>NpO<sub>2</sub>+2_H+1 (8)_PO<sub>4</sub>-3 (4)</b>                                                                         |            |     |                           |         |
| -2                                                                                                                            | ---        | 2   | H (8) Np (1) O (18) P (4) | 656.997 |
| <b>NpO<sub>2</sub>+2 (3)_PO<sub>4</sub>-3 (2)_(s)</b>                                                                         |            |     |                           |         |
| 0                                                                                                                             | ---        | 1   | Np (3) O (14) P (2)       | 997.084 |
| <b>NTA-3</b><br>Nitrilotriacetate ion; N,N-bis(carboxymethyl)glycinate ion; Triglycollamate ion;<br>Triglycinate ion; NTA ion |            |     |                           |         |
| -3                                                                                                                            | 28528-44-1 | 751 | C (6) H (6) N (1) O (6)   | 188.117 |



|                                                                                                                                             |            |      |                   |         |
|---------------------------------------------------------------------------------------------------------------------------------------------|------------|------|-------------------|---------|
| <b>O2 (g)</b><br>Oxygen gas; Oxygen                                                                                                         |            |      |                   |         |
| 0                                                                                                                                           | 7782-44-7  | 2559 | O(2)              | 31.9988 |
| <b>OH-1</b><br>Hydroxide ion                                                                                                                |            |      |                   |         |
| -1                                                                                                                                          | 14280-30-9 | 6558 | H(1)O(1)          | 17.0073 |
| <b>Orn-1</b><br>Ornithinate ion; L-Ornithinate ion; alpha,delta-diaminovalerate ion; 2,5-diaminopentanoate ion; L-2,5-diaminopentanoate ion |            |      |                   |         |
| -1                                                                                                                                          | 17781-80-5 | 505  | C(5)H(11)N(2)O(2) | 131.155 |
| <b>Os (s)</b><br>Osmium                                                                                                                     |            |      |                   |         |
| 0                                                                                                                                           | 7440-04-2  | 20   | Os(1)             | 190.233 |
| <b>OsP2 (s)</b><br>Osmium diphosphide                                                                                                       |            |      |                   |         |
| 0                                                                                                                                           | ---        | 1    | Os(1)P(2)         | 252.181 |
| <b>Oxalic-2</b><br>Oxalate ion                                                                                                              |            |      |                   |         |
| -2                                                                                                                                          | 338-70-5   | 1241 | C(2)O(4)          | 88.0196 |
| <b>Oxine-1</b><br>8-Hydroxyquinolate ion; Oxine ion                                                                                         |            |      |                   |         |
| -1                                                                                                                                          | 16582-16-4 | 162  | C(9)H(6)N(1)O(1)  | 144.153 |
| <b>P (g)</b>                                                                                                                                |            |      |                   |         |
| 0                                                                                                                                           | ---        | 1    | P(1)              | 30.9740 |
| <b>P (red, s)</b><br>Phosphorus, red; Phosphorus, triclinic                                                                                 |            |      |                   |         |
| 0                                                                                                                                           | ---        | 3    | P(1)              | 30.9740 |
| <b>P (s)</b><br>Phosphorus; Phosphorus, white; Phosphorous, alpha; Phosphorous, cubic                                                       |            |      |                   |         |
| 0                                                                                                                                           | 7723-14-0  | 166  | P(1)              | 30.9740 |

|                                                                          |            |     |              |         |
|--------------------------------------------------------------------------|------------|-----|--------------|---------|
| <b>P2 (g)</b>                                                            |            |     |              |         |
| 0                                                                        | ---        | 1   | P (2)        | 61.9480 |
| <b>P2H4 (g)</b>                                                          |            |     |              |         |
| 0                                                                        | ---        | 2   | H (4) P (2)  | 65.9798 |
| <b>P2O7-4</b><br>Pyrophosphate ion; Pyrophosphate anion; Diphosphate ion |            |     |              |         |
| -4                                                                       | 14000-31-8 | 285 | O (7) P (2)  | 173.944 |
| <b>P3O10-5</b><br>Triphosphate ion; Tripolyphosphate ion                 |            |     |              |         |
| -5                                                                       | 14127-68-5 | 283 | O (10) P (3) | 252.916 |
| <b>P3O8-5</b>                                                            |            |     |              |         |
| -5                                                                       | ---        | 8   | O (8) P (3)  | 220.917 |
| <b>P3O9-3</b><br>Trimetaphosphate ion                                    |            |     |              |         |
| -3                                                                       | ---        | 22  | O (9) P (3)  | 236.917 |
| <b>P4 (g)</b>                                                            |            |     |              |         |
| 0                                                                        | ---        | 1   | P (4)        | 123.896 |
| <b>P4H2</b>                                                              |            |     |              |         |
| 0                                                                        | ---        | 2   | H (2) P (4)  | 125.912 |
| <b>P4H2 (g)</b>                                                          |            |     |              |         |
| 0                                                                        | ---        | 1   | H (2) P (4)  | 125.912 |
| <b>P4O10 (hex. , s)</b>                                                  |            |     |              |         |
| 0                                                                        | ---        | 1   | O (10) P (4) | 283.890 |
| <b>P4O12-4</b><br>Tetrametaphosphate ion                                 |            |     |              |         |
| -4                                                                       | 17121-12-9 | 29  | O (12) P (4) | 315.889 |

|                                                            |            |     |                     |         |
|------------------------------------------------------------|------------|-----|---------------------|---------|
| <b>P4O13-6</b><br>Tetraphosphate ion                       |            |     |                     |         |
| -6                                                         | 16132-64-2 | 40  | O(13)P(4)           | 331.888 |
| <b>P4O6(s)</b>                                             |            |     |                     |         |
| 0                                                          | ---        | 1   | O(6)P(4)            | 219.892 |
| <b>P6O18-6</b><br>Hexametaphosphate                        |            |     |                     |         |
| -6                                                         | ---        | 14  | O(18)P(6)           | 473.833 |
| <b>P8O24-8</b>                                             |            |     |                     |         |
| -8                                                         | ---        | 4   | O(24)P(8)           | 631.778 |
| <b>Pa+4</b><br>Protoactinium(IV) ion; Protactinium(IV) ion |            |     |                     |         |
| 4                                                          | 22537-59-3 | 162 | Pa(1)               | 231.040 |
| <b>Pa+4_H+1_PO4-3</b>                                      |            |     |                     |         |
| 2                                                          | ---        | 2   | H(1)O(4)P(1)Pa(1)   | 327.020 |
| <b>Pa+4_H+1(10)_PO4-3(5)</b>                               |            |     |                     |         |
| -1                                                         | ---        | 2   | H(10)O(20)P(5)Pa(1) | 715.977 |
| <b>Pa+4_H+1(12)_PO4-3(6)</b>                               |            |     |                     |         |
| -2                                                         | ---        | 2   | H(12)O(24)P(6)Pa(1) | 812.965 |
| <b>Pa+4_H+1(2)_PO4-3</b>                                   |            |     |                     |         |
| 3                                                          | ---        | 2   | H(2)O(4)P(1)Pa(1)   | 328.028 |
| <b>Pa+4_H+1(2)_PO4-3(2)</b>                                |            |     |                     |         |
| 0                                                          | ---        | 2   | H(2)O(8)P(2)Pa(1)   | 422.999 |
| <b>Pa+4_H+1(3)_PO4-3(3)</b>                                |            |     |                     |         |
| -2                                                         | ---        | 2   | H(3)O(12)P(3)Pa(1)  | 518.979 |
| <b>Pa+4_H+1(4)_PO4-3(2)</b>                                |            |     |                     |         |

|                                 |            |      |                                            |         |
|---------------------------------|------------|------|--------------------------------------------|---------|
| 2                               | ---        | 2    | H (4) O (8) P (2) Pa (1)                   | 425.015 |
| <b>Pa+4 _H+1 (4) _PO4-3 (4)</b> |            |      |                                            |         |
| -4                              | ---        | 2    | H (4) O (16) P (4) Pa (1)                  | 614.958 |
| <b>Pa+4 _H+1 (5) _PO4-3 (5)</b> |            |      |                                            |         |
| -6                              | ---        | 2    | H (5) O (20) P (5) Pa (1)                  | 710.938 |
| <b>Pa+4 _H+1 (6) _PO4-3 (3)</b> |            |      |                                            |         |
| 1                               | ---        | 2    | H (6) O (12) P (3) Pa (1)                  | 522.002 |
| <b>Pa+4 _H+1 (6) _PO4-3 (6)</b> |            |      |                                            |         |
| -8                              | ---        | 2    | H (6) O (24) P (6) Pa (1)                  | 806.917 |
| <b>Pa+4 _H+1 (8) _PO4-3 (4)</b> |            |      |                                            |         |
| 0                               | ---        | 2    | H (8) O (16) P (4) Pa (1)                  | 618.990 |
| <b>PaO2+1</b>                   |            |      |                                            |         |
| 1                               | ---        | 108  | O (2) Pa (1)                               | 263.039 |
| <b>Pb+2</b><br>Lead(II) ion     |            |      |                                            |         |
| 2                               | 14280-50-3 | 2437 | Pb (1)                                     | 207.200 |
| <b>Pb+2 _Cys-2 _PO4-3</b>       |            |      |                                            |         |
| -3                              | ---        | 1    | C (3) H (5) N (1) O (6) P (1) Pb (1) S (1) | 421.310 |
| <b>Pb+2 _H+1 _Ala-1 _PO4-3</b>  |            |      |                                            |         |
| -1                              | ---        | 1    | C (3) H (7) N (1) O (6) P (1) Pb (1)       | 391.266 |
| <b>Pb+2 _H+1 _Arg-1 _PO4-3</b>  |            |      |                                            |         |
| -1                              | ---        | 1    | C (6) H (14) N (4) O (6) P (1) Pb (1)      | 476.374 |
| <b>Pb+2 _H+1 _Asn-1 _PO4-3</b>  |            |      |                                            |         |
| -1                              | ---        | 1    | C (4) H (8) N (2) O (7) P (1) Pb (1)       | 434.291 |
| <b>Pb+2 _H+1 _Asp-2 _PO4-3</b>  |            |      |                                            |         |

|                                   |     |   |                                   |         |
|-----------------------------------|-----|---|-----------------------------------|---------|
| -2                                | --- | 1 | $C(4)H(6)N(1)O(8)P(1)Pb(1)$       | 434.268 |
| <b>Pb+2_H+1_bAla-1_PO4-3</b>      |     |   |                                   |         |
| -1                                | --- | 1 | $C(3)H(7)N(1)O(6)P(1)Pb(1)$       | 391.266 |
| <b>Pb+2_H+1_Citric-3_PO4-3</b>    |     |   |                                   |         |
| -3                                | --- | 2 | $C(6)H(6)O(11)P(1)Pb(1)$          | 492.281 |
| <b>Pb+2_H+1_Citrul-1_PO4-3</b>    |     |   |                                   |         |
| -1                                | --- | 1 | $C(6)H(13)N(3)O(7)P(1)Pb(1)$      | 477.359 |
| <b>Pb+2_H+1_Cys-2_NTA-3_PO4-3</b> |     |   |                                   |         |
| -5                                | --- | 1 | $C(9)H(12)N(2)O(12)P(1)Pb(1)S(1)$ | 610.435 |
| <b>Pb+2_H+1_Cys-2_PO4-3</b>       |     |   |                                   |         |
| -2                                | --- | 2 | $C(3)H(6)N(1)O(6)P(1)Pb(1)S(1)$   | 422.318 |
| <b>Pb+2_H+1_Gln-1_PO4-3</b>       |     |   |                                   |         |
| -1                                | --- | 1 | $C(5)H(10)N(2)O(7)P(1)Pb(1)$      | 448.318 |
| <b>Pb+2_H+1_Glu-2_PO4-3</b>       |     |   |                                   |         |
| -2                                | --- | 1 | $C(5)H(8)N(1)O(8)P(1)Pb(1)$       | 448.294 |
| <b>Pb+2_H+1_Gly-1_PO4-3</b>       |     |   |                                   |         |
| -1                                | --- | 1 | $C(2)H(5)N(1)O(6)P(1)Pb(1)$       | 377.239 |
| <b>Pb+2_H+1_His-1_PO4-3</b>       |     |   |                                   |         |
| -1                                | --- | 1 | $C(6)H(9)N(3)O(6)P(1)Pb(1)$       | 457.328 |
| <b>Pb+2_H+1_Ile-1_PO4-3</b>       |     |   |                                   |         |
| -1                                | --- | 1 | $C(6)H(13)N(1)O(6)P(1)Pb(1)$      | 433.346 |
| <b>Pb+2_H+1_Lactic-1_PO4-3</b>    |     |   |                                   |         |
| -1                                | --- | 1 | $C(3)H(6)O(7)P(1)Pb(1)$           | 392.250 |
| <b>Pb+2_H+1_Leu-1_PO4-3</b>       |     |   |                                   |         |
| -1                                | --- | 1 | $C(6)H(13)N(1)O(6)P(1)Pb(1)$      | 433.346 |

|                                                     |            |   |                                |         |
|-----------------------------------------------------|------------|---|--------------------------------|---------|
| <b>Pb+2_H+1_Malic-2_PO4-3</b>                       |            |   |                                |         |
| -2                                                  | ---        | 1 | C(4)H(5)O(9)P(1)Pb(1)          | 435.252 |
| <b>Pb+2_H+1_Met-1_PO4-3</b>                         |            |   |                                |         |
| -1                                                  | ---        | 1 | C(5)H(11)N(1)O(6)P(1)Pb(1)S(1) | 451.380 |
| <b>Pb+2_H+1_NTA-3_PO4-3</b>                         |            |   |                                |         |
| -3                                                  | ---        | 1 | C(6)H(7)N(1)O(10)P(1)Pb(1)     | 491.296 |
| <b>Pb+2_H+1_Oxalic-2_PO4-3</b>                      |            |   |                                |         |
| -2                                                  | ---        | 1 | C(2)H(1)O(8)P(1)Pb(1)          | 391.199 |
| <b>Pb+2_H+1_P3O10-5</b>                             |            |   |                                |         |
| -2                                                  | ---        | 1 | H(1)O(10)P(3)Pb(1)             | 461.124 |
| <b>Pb+2_H+1_Phe-1_PO4-3</b>                         |            |   |                                |         |
| -1                                                  | ---        | 1 | C(9)H(11)N(1)O(6)P(1)Pb(1)     | 467.364 |
| <b>Pb+2_H+1_PO4-3</b>                               |            |   |                                |         |
| 0                                                   | ---        | 2 | H(1)O(4)P(1)Pb(1)              | 303.180 |
| <b>Pb+2_H+1_PO4-3_(s)</b><br>Lead phosphate dibasic |            |   |                                |         |
| 0                                                   | 15845-52-0 | 4 | H(1)O(4)P(1)Pb(1)              | 303.180 |
| <b>Pb+2_H+1_Pro-1_PO4-3</b>                         |            |   |                                |         |
| -1                                                  | ---        | 1 | C(5)H(9)N(1)O(6)P(1)Pb(1)      | 417.304 |
| <b>Pb+2_H+1_Pyruvic-1_PO4-3</b>                     |            |   |                                |         |
| -1                                                  | ---        | 1 | C(3)H(4)O(7)P(1)Pb(1)          | 390.235 |
| <b>Pb+2_H+1_Ser-1_PO4-3</b>                         |            |   |                                |         |
| -1                                                  | ---        | 1 | C(3)H(7)N(1)O(7)P(1)Pb(1)      | 407.265 |
| <b>Pb+2_H+1_Succinic-2_PO4-3</b>                    |            |   |                                |         |
| -2                                                  | ---        | 1 | C(4)H(5)O(8)P(1)Pb(1)          | 419.253 |

|                                |     |   |                                |         |
|--------------------------------|-----|---|--------------------------------|---------|
| <b>Pb+2_H+1_Thr-1_PO4-3</b>    |     |   |                                |         |
| -1                             | --- | 1 | C(4)H(9)N(1)O(7)P(1)Pb(1)      | 421.292 |
| <b>Pb+2_H+1_Trp-1_PO4-3</b>    |     |   |                                |         |
| -1                             | --- | 1 | C(11)H(12)N(2)O(6)P(1)Pb(1)    | 506.400 |
| <b>Pb+2_H+1_Val-1_PO4-3</b>    |     |   |                                |         |
| -1                             | --- | 1 | C(5)H(11)N(1)O(6)P(1)Pb(1)     | 419.320 |
| <b>Pb+2_H+1(2)_Cis-2_PO4-3</b> |     |   |                                |         |
| -1                             | --- | 1 | C(6)H(12)N(2)O(8)P(1)Pb(1)S(2) | 542.464 |
| <b>Pb+2_H+1(2)_Cys-2_PO4-3</b> |     |   |                                |         |
| -1                             | --- | 1 | C(3)H(7)N(1)O(6)P(1)Pb(1)S(1)  | 423.326 |
| <b>Pb+2_H+1(2)_His-1_PO4-3</b> |     |   |                                |         |
| 0                              | --- | 1 | C(6)H(10)N(3)O(6)P(1)Pb(1)     | 458.336 |
| <b>Pb+2_H+1(2)_Lys-1_PO4-3</b> |     |   |                                |         |
| 0                              | --- | 1 | C(6)H(15)N(2)O(6)P(1)Pb(1)     | 449.369 |
| <b>Pb+2_H+1(2)_Orn-1_PO4-3</b> |     |   |                                |         |
| 0                              | --- | 1 | C(5)H(13)N(2)O(6)P(1)Pb(1)     | 435.342 |
| <b>Pb+2_H+1(2)_P2O7-4</b>      |     |   |                                |         |
| 0                              | --- | 3 | H(2)O(7)P(2)Pb(1)              | 383.160 |
| <b>Pb+2_H+1(2)_PO4-3</b>       |     |   |                                |         |
| 1                              | --- | 2 | H(2)O(4)P(1)Pb(1)              | 304.188 |
| <b>Pb+2_H+1(2)_PO4-3(2)</b>    |     |   |                                |         |
| -2                             | --- | 2 | H(2)O(8)P(2)Pb(1)              | 399.159 |
| <b>Pb+2_H+1(2)_Tyr-2_PO4-3</b> |     |   |                                |         |
| -1                             | --- | 1 | C(9)H(11)N(1)O(7)P(1)Pb(1)     | 483.363 |

|                                    |            |   |                           |         |
|------------------------------------|------------|---|---------------------------|---------|
| <b>Pb+2_H+1 (3)_P2O7-4</b>         |            |   |                           |         |
| 1                                  | ---        | 2 | H (3) O (7) P (2) Pb (1)  | 384.168 |
| <b>Pb+2_H+1 (3)_PO4-3 (3)</b>      |            |   |                           |         |
| -4                                 | ---        | 2 | H (3) O (12) P (3) Pb (1) | 495.139 |
| <b>Pb+2_H+1 (4)_PO4-3 (2)</b>      |            |   |                           |         |
| 0                                  | ---        | 2 | H (4) O (8) P (2) Pb (1)  | 401.175 |
| <b>Pb+2_H+1 (4)_PO4-3 (2)_ (s)</b> |            |   |                           |         |
| 0                                  | 16180-04-4 | 2 | H (4) O (8) P (2) Pb (1)  | 401.175 |
| <b>Pb+2_H+1 (4)_PO4-3 (4)</b>      |            |   |                           |         |
| -6                                 | ---        | 2 | H (4) O (16) P (4) Pb (1) | 591.118 |
| <b>Pb+2_H+1 (6)_PO4-3 (3)</b>      |            |   |                           |         |
| -1                                 | ---        | 2 | H (6) O (12) P (3) Pb (1) | 498.162 |
| <b>Pb+2_H+1 (8)_PO4-3 (4)</b>      |            |   |                           |         |
| -2                                 | ---        | 2 | H (8) O (16) P (4) Pb (1) | 595.150 |
| <b>Pb+2_P2O7-4</b>                 |            |   |                           |         |
| -2                                 | ---        | 2 | O (7) P (2) Pb (1)        | 381.144 |
| <b>Pb+2_P2O7-4 (2)</b>             |            |   |                           |         |
| -6                                 | ---        | 1 | O (14) P (4) Pb (1)       | 555.088 |
| <b>Pb+2_P3O10-5</b>                |            |   |                           |         |
| -3                                 | ---        | 1 | O (10) P (3) Pb (1)       | 460.116 |
| <b>Pb+2_P3O10-5 (2)</b>            |            |   |                           |         |
| -8                                 | ---        | 1 | O (20) P (6) Pb (1)       | 713.032 |
| <b>Pb+2_P3O10-5 (3)</b>            |            |   |                           |         |
| -13                                | ---        | 1 | O (30) P (9) Pb (1)       | 965.948 |



|                                                                                                                            |            |    |                            |         |
|----------------------------------------------------------------------------------------------------------------------------|------------|----|----------------------------|---------|
| <b>Pb+2_P3O9-3</b>                                                                                                         |            |    |                            |         |
| -1                                                                                                                         | ---        | 1  | O (9) P (3) Pb (1)         | 444.117 |
| <b>Pb+2_PO4-3</b>                                                                                                          |            |    |                            |         |
| -1                                                                                                                         | ---        | 1  | O (4) P (1) Pb (1)         | 302.172 |
| <b>Pb+2_UO2+2 (2)_PO4-3 (2)_ (s)</b><br>Przhevalskite                                                                      |            |    |                            |         |
| 0                                                                                                                          | ---        | 2  | O (12) P (2) Pb (1) U (2)  | 937.199 |
| <b>Pb+2 (3)_P3O10-5 (4)</b>                                                                                                |            |    |                            |         |
| -14                                                                                                                        | ---        | 1  | O (40) P (12) Pb (3)       | 1633.26 |
| <b>Pb+2 (3)_PO4-3 (2)_ (s)</b><br>Lead phosphate                                                                           |            |    |                            |         |
| 0                                                                                                                          | 7446-27-7  | 1  | O (8) P (2) Pb (3)         | 811.543 |
| <b>Pb+2 (5)_Br-1_PO4-3 (3)_ (s)</b><br>Lead bromide phosphate                                                              |            |    |                            |         |
| 0                                                                                                                          | 12514-34-0 | 1  | Br (1) O (12) P (3) Pb (5) | 1400.82 |
| <b>Pb+2 (5)_Cl-1_PO4-3 (3)_ (s)</b><br>Chloropyromorphite; Pyromorphite; Cl-pyromorphite                                   |            |    |                            |         |
| 0                                                                                                                          | 12190-77-1 | 1  | Cl (1) O (12) P (3) Pb (5) | 1356.37 |
| <b>Pb+2 (5)_F-1_PO4-3 (3)_ (s)</b><br>Lead fluorapatite; Apatite lead                                                      |            |    |                            |         |
| 0                                                                                                                          | 12528-20-0 | 1  | F (1) O (12) P (3) Pb (5)  | 1339.91 |
| <b>Pb+2 (5)_OH-1_PO4-3 (3)_ (s)</b><br>Lead hydroxyapatite; Hydroxyapatite lead; Hydroxypyromorphite; Hydrous pyromorphite |            |    |                            |         |
| 0                                                                                                                          | 12530-18-6 | 2  | H (1) O (13) P (3) Pb (5)  | 1337.92 |
| <b>Pb (CH3) 2+2</b><br>Dimethyllead ion                                                                                    |            |    |                            |         |
| 2                                                                                                                          | ---        | 10 | C (2) H (6) Pb (1)         | 237.270 |
| <b>Pb (CH3) 3+1</b><br>Trimethyllead ion; Trimethyllead(IV) ion; Trimethyl lead                                            |            |    |                            |         |

|                                                     |            |     |                                       |         |
|-----------------------------------------------------|------------|-----|---------------------------------------|---------|
| 1                                                   | ---        | 22  | C (3) H (9) Pb (1)                    | 252.305 |
| <b>Pb (s)</b><br>Lead; Lead, cubic                  |            |     |                                       |         |
| 0                                                   | 7439-92-1  | 127 | Pb (1)                                | 207.200 |
| <b>PBr3 (g)</b><br>Phosphorous tribromide gas       |            |     |                                       |         |
| 0                                                   | ---        | 1   | Br (3) P (1)                          | 270.686 |
| <b>PBr3 (l)</b><br>Phosphorous tribromide           |            |     |                                       |         |
| 0                                                   | ---        | 1   | Br (3) P (1)                          | 270.686 |
| <b>PCl3 (g)</b><br>Phosphorus trichloride gas       |            |     |                                       |         |
| 0                                                   | ---        | 1   | Cl (3) P (1)                          | 137.333 |
| <b>PCl3 (l)</b><br>Phosphorus trichloride (liquid)  |            |     |                                       |         |
| 0                                                   | 7719-12-2  | 1   | Cl (3) P (1)                          | 137.333 |
| <b>PCl5 (g)</b><br>Phosphorus pentachloride gas     |            |     |                                       |         |
| 0                                                   | ---        | 1   | Cl (5) P (1)                          | 208.239 |
| <b>PCl5 (s)</b><br>Phosphorus pentachloride (solid) |            |     |                                       |         |
| 0                                                   | 10026-13-8 | 1   | Cl (5) P (1)                          | 208.239 |
| <b>Pd+2</b><br>Palladium(II) ion                    |            |     |                                       |         |
| 2                                                   | 16065-88-6 | 567 | Pd (1)                                | 106.420 |
| <b>Pd+2_Dien_PO4-3</b>                              |            |     |                                       |         |
| -1                                                  | ---        | 1   | C (4) H (13) N (3) O (4) P (1) Pd (1) | 304.559 |
| <b>Pd+2_H+1_PO4-3</b>                               |            |     |                                       |         |
| 0                                                   | ---        | 2   | H (1) O (4) P (1) Pd (1)              | 202.400 |

|                                                                                                                                                                                          |            |     |                           |         |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----|---------------------------|---------|
| <b>Pd+2_H+1 (2)_PO4-3</b>                                                                                                                                                                |            |     |                           |         |
| 1                                                                                                                                                                                        | ---        | 2   | H (2) O (4) P (1) Pd (1)  | 203.408 |
| <b>Pd+2_H+1 (2)_PO4-3 (2)</b>                                                                                                                                                            |            |     |                           |         |
| -2                                                                                                                                                                                       | ---        | 2   | H (2) O (8) P (2) Pd (1)  | 298.379 |
| <b>Pd+2_H+1 (3)_PO4-3 (3)</b>                                                                                                                                                            |            |     |                           |         |
| -4                                                                                                                                                                                       | ---        | 2   | H (3) O (12) P (3) Pd (1) | 394.359 |
| <b>Pd+2_H+1 (4)_PO4-3 (2)</b>                                                                                                                                                            |            |     |                           |         |
| 0                                                                                                                                                                                        | ---        | 2   | H (4) O (8) P (2) Pd (1)  | 300.395 |
| <b>Pd+2_H+1 (4)_PO4-3 (4)</b>                                                                                                                                                            |            |     |                           |         |
| -6                                                                                                                                                                                       | ---        | 2   | H (4) O (16) P (4) Pd (1) | 490.338 |
| <b>Pd+2_H+1 (6)_PO4-3 (3)</b>                                                                                                                                                            |            |     |                           |         |
| -1                                                                                                                                                                                       | ---        | 2   | H (6) O (12) P (3) Pd (1) | 397.382 |
| <b>Pd+2_H+1 (8)_PO4-3 (4)</b>                                                                                                                                                            |            |     |                           |         |
| -2                                                                                                                                                                                       | ---        | 2   | H (8) O (16) P (4) Pd (1) | 494.370 |
| <b>Penten</b><br>Penten; Ethylenedinitrilotetrakis (2-ethylamine) ; N,N,N',N'-Tetrakis (2-aminoethyl)ethylenediamine; 1,2-Diamino-N,N,N',N'-tetra (2-aminoethyl)ethane                   |            |     |                           |         |
| 0                                                                                                                                                                                        | ---        | 44  | C (10) H (28) N (6)       | 232.373 |
| <b>PH3</b><br>Phosphine                                                                                                                                                                  |            |     |                           |         |
| 0                                                                                                                                                                                        | 7803-51-2  | 8   | H (3) P (1)               | 33.9978 |
| <b>PH3 (g)</b><br>Phosphine gas                                                                                                                                                          |            |     |                           |         |
| 0                                                                                                                                                                                        | ---        | 8   | H (3) P (1)               | 33.9978 |
| <b>Phe-1</b><br>Phenylalanate ion; L-Phenylalanate ion; beta-phenylalanate ion; alpha-aminohydrocinnamate ion; alpha-amino-beta-phenylpropionate ion; L-2-amino-3,5-phenylpropanoate ion |            |     |                           |         |
| -1                                                                                                                                                                                       | 19701-97-4 | 562 | C (9) H (10) N (1) O (2)  | 164.184 |

|                                              |            |      |                          |         |
|----------------------------------------------|------------|------|--------------------------|---------|
| <b>Pm+3</b><br>Promethium(III) ion           |            |      |                          |         |
| 3                                            | 22541-16-8 | 84   | Pm (1)                   | 145.000 |
| <b>Pm+3_H+1_PO4-3</b>                        |            |      |                          |         |
| 1                                            | ---        | 2    | H (1) O (4) P (1) Pm (1) | 240.980 |
| <b>Pm+3_H+1 (2)_PO4-3</b>                    |            |      |                          |         |
| 2                                            | ---        | 2    | H (2) O (4) P (1) Pm (1) | 241.988 |
| <b>Pm+3_H+1 (2)_PO4-3 (2)</b>                |            |      |                          |         |
| -1                                           | ---        | 2    | H (2) O (8) P (2) Pm (1) | 336.959 |
| <b>Pm+3_P3O9-3</b>                           |            |      |                          |         |
| 0                                            | ---        | 1    | O (9) P (3) Pm (1)       | 381.917 |
| <b>Pm+3_PO4-3</b>                            |            |      |                          |         |
| 0                                            | ---        | 1    | O (4) P (1) Pm (1)       | 239.972 |
| <b>Pm+3_PO4-3 (2)</b>                        |            |      |                          |         |
| -3                                           | ---        | 1    | O (8) P (2) Pm (1)       | 334.943 |
| <b>PO4-3</b><br>Phosphate ion; Phosphate ion |            |      |                          |         |
| -3                                           | 14265-44-2 | 1594 | O (4) P (1)              | 94.9716 |
| <b>POCl3(g)</b><br>Phosphoryl chloride gas   |            |      |                          |         |
| 0                                            | ---        | 1    | Cl (3) O (1) P (1)       | 153.332 |
| <b>POCl3(l)</b><br>Phosphoryl chloride       |            |      |                          |         |
| 0                                            | ---        | 1    | Cl (3) O (1) P (1)       | 153.332 |
| <b>Pr+3</b><br>Praseodymium(III) ion         |            |      |                          |         |
| 3                                            | 22541-14-6 | 722  | Pr (1)                   | 140.910 |

|                                |     |    |                              |         |
|--------------------------------|-----|----|------------------------------|---------|
| <b>Pr+3_EDTA-4</b>             |     |    |                              |         |
| -1                             | --- | 47 | C(10)H(12)N(2)O(8)Pr(1)      | 429.124 |
| <b>Pr+3_EDTA-4_P3O10-5</b>     |     |    |                              |         |
| -6                             | --- | 2  | C(10)H(12)N(2)O(18)P(3)Pr(1) | 682.040 |
| <b>Pr+3_H+1_EDTA-4_P3O10-5</b> |     |    |                              |         |
| -5                             | --- | 2  | C(10)H(13)N(2)O(18)P(3)Pr(1) | 683.048 |
| <b>Pr+3_H+1_P3O10-5</b>        |     |    |                              |         |
| -1                             | --- | 1  | H(1)O(10)P(3)Pr(1)           | 394.834 |
| <b>Pr+3_H+1_PO4-3</b>          |     |    |                              |         |
| 1                              | --- | 2  | H(1)O(4)P(1)Pr(1)            | 236.890 |
| <b>Pr+3_H+1(2)_P3O10-5(2)</b>  |     |    |                              |         |
| -5                             | --- | 2  | H(2)O(20)P(6)Pr(1)           | 648.758 |
| <b>Pr+3_H+1(2)_PO4-3</b>       |     |    |                              |         |
| 2                              | --- | 2  | H(2)O(4)P(1)Pr(1)            | 237.898 |
| <b>Pr+3_H+1(2)_PO4-3(2)</b>    |     |    |                              |         |
| -1                             | --- | 2  | H(2)O(8)P(2)Pr(1)            | 332.869 |
| <b>Pr+3_H2PO2-1</b>            |     |    |                              |         |
| 2                              | --- | 1  | H(2)O(2)P(1)Pr(1)            | 205.899 |
| <b>Pr+3_P3O10-5(2)</b>         |     |    |                              |         |
| -7                             | --- | 1  | O(20)P(6)Pr(1)               | 646.742 |
| <b>Pr+3_PO4-3</b>              |     |    |                              |         |
| 0                              | --- | 1  | O(4)P(1)Pr(1)                | 235.882 |
| <b>Pr+3_PO4-3(s)</b>           |     |    |                              |         |
| 0                              | --- | 1  | O(4)P(1)Pr(1)                | 235.882 |

|                                                                                                               |            |     |                            |         |
|---------------------------------------------------------------------------------------------------------------|------------|-----|----------------------------|---------|
| <b>Pr+3_PO4-3 (2)</b>                                                                                         |            |     |                            |         |
| -3                                                                                                            | ---        | 1   | O (8) P (2) Pr (1)         | 330.853 |
| <b>Pro-1</b><br>Prolinate ion; L-Prolinate ion; 2-pyrrolidinecarboxylate ion; L-pyrrolidine-2-carboxylate ion |            |     |                            |         |
| -1                                                                                                            | 17781-82-7 | 494 | C (5) H (8) N (1) O (2)    | 114.124 |
| <b>Pu+3</b><br>Plutonium(III) ion                                                                             |            |     |                            |         |
| 3                                                                                                             | 22541-70-4 | 222 | Pu (1)                     | 244.000 |
| <b>Pu+3_H+1_PO4-3</b>                                                                                         |            |     |                            |         |
| 1                                                                                                             | ---        | 3   | H (1) O (4) P (1) Pu (1)   | 339.980 |
| <b>Pu+3_H+1 (10)_PO4-3 (5)</b>                                                                                |            |     |                            |         |
| -2                                                                                                            | ---        | 1   | H (10) O (20) P (5) Pu (1) | 728.937 |
| <b>Pu+3_H+1 (12)_PO4-3 (6)</b>                                                                                |            |     |                            |         |
| -3                                                                                                            | ---        | 1   | H (12) O (24) P (6) Pu (1) | 825.925 |
| <b>Pu+3_H+1 (2)_PO4-3</b>                                                                                     |            |     |                            |         |
| 2                                                                                                             | ---        | 3   | H (2) O (4) P (1) Pu (1)   | 340.988 |
| <b>Pu+3_H+1 (2)_PO4-3 (2)</b>                                                                                 |            |     |                            |         |
| -1                                                                                                            | ---        | 2   | H (2) O (8) P (2) Pu (1)   | 435.959 |
| <b>Pu+3_H+1 (3)_PO4-3 (3)</b>                                                                                 |            |     |                            |         |
| -3                                                                                                            | ---        | 1   | H (3) O (12) P (3) Pu (1)  | 531.939 |
| <b>Pu+3_H+1 (4)_PO4-3 (2)</b>                                                                                 |            |     |                            |         |
| 1                                                                                                             | ---        | 1   | H (4) O (8) P (2) Pu (1)   | 437.975 |
| <b>Pu+3_H+1 (4)_PO4-3 (4)</b>                                                                                 |            |     |                            |         |
| -5                                                                                                            | ---        | 1   | H (4) O (16) P (4) Pu (1)  | 627.918 |
| <b>Pu+3_H+1 (5)_PO4-3 (5)</b>                                                                                 |            |     |                            |         |

|                                                   |            |     |                            |         |
|---------------------------------------------------|------------|-----|----------------------------|---------|
| -7                                                | ---        | 1   | H (5) O (20) P (5) Pu (1)  | 723.898 |
| <b>Pu+3_H+1 (6)_PO4-3 (3)</b>                     |            |     |                            |         |
| 0                                                 | ---        | 1   | H (6) O (12) P (3) Pu (1)  | 534.962 |
| <b>Pu+3_H+1 (6)_PO4-3 (6)</b>                     |            |     |                            |         |
| -9                                                | ---        | 1   | H (6) O (24) P (6) Pu (1)  | 819.877 |
| <b>Pu+3_H+1 (8)_PO4-3 (4)</b>                     |            |     |                            |         |
| -1                                                | ---        | 1   | H (8) O (16) P (4) Pu (1)  | 631.950 |
| <b>Pu+3_PO4-3</b>                                 |            |     |                            |         |
| 0                                                 | ---        | 2   | O (4) P (1) Pu (1)         | 338.972 |
| <b>Pu+3_PO4-3 (s)</b><br>Plutonium(III) phosphate |            |     |                            |         |
| 0                                                 | ---        | 1   | O (4) P (1) Pu (1)         | 338.972 |
| <b>Pu+3_PO4-3 (2)</b>                             |            |     |                            |         |
| -3                                                | ---        | 2   | O (8) P (2) Pu (1)         | 433.943 |
| <b>Pu+3_PO4-3 (3)</b>                             |            |     |                            |         |
| -6                                                | ---        | 1   | O (12) P (3) Pu (1)        | 528.915 |
| <b>Pu+3_PO4-3 (4)</b>                             |            |     |                            |         |
| -9                                                | ---        | 1   | O (16) P (4) Pu (1)        | 623.886 |
| <b>Pu+4</b><br>Plutonium(IV) ion                  |            |     |                            |         |
| 4                                                 | 22541-44-2 | 378 | Pu (1)                     | 244.000 |
| <b>Pu+4_H+1_PO4-3</b>                             |            |     |                            |         |
| 2                                                 | ---        | 2   | H (1) O (4) P (1) Pu (1)   | 339.980 |
| <b>Pu+4_H+1 (10)_PO4-3 (5)</b>                    |            |     |                            |         |
| -1                                                | ---        | 1   | H (10) O (20) P (5) Pu (1) | 728.937 |
| <b>Pu+4_H+1 (12)_PO4-3 (6)</b>                    |            |     |                            |         |

|                                                                           |            |   |                            |         |
|---------------------------------------------------------------------------|------------|---|----------------------------|---------|
| -2                                                                        | ---        | 1 | H (12) O (24) P (6) Pu (1) | 825.925 |
| <b>Pu+4 _H+1 (2) _PO4-3</b>                                               |            |   |                            |         |
| 3                                                                         | ---        | 2 | H (2) O (4) P (1) Pu (1)   | 340.988 |
| <b>Pu+4 _H+1 (2) _PO4-3 (2)</b>                                           |            |   |                            |         |
| 0                                                                         | ---        | 2 | H (2) O (8) P (2) Pu (1)   | 435.959 |
| <b>Pu+4 _H+1 (2) _PO4-3 (2) _ (am. , s)</b>                               |            |   |                            |         |
| 0                                                                         | ---        | 1 | H (2) O (8) P (2) Pu (1)   | 435.959 |
| <b>Pu+4 _H+1 (2) _PO4-3 (2) _ (s)</b><br>Plutonium(IV) hydrogen phosphate |            |   |                            |         |
| 0                                                                         | 20190-42-5 | 1 | H (2) O (8) P (2) Pu (1)   | 435.959 |
| <b>Pu+4 _H+1 (3) _PO4-3 (3)</b>                                           |            |   |                            |         |
| -2                                                                        | ---        | 2 | H (3) O (12) P (3) Pu (1)  | 531.939 |
| <b>Pu+4 _H+1 (4) _PO4-3 (2)</b>                                           |            |   |                            |         |
| 2                                                                         | ---        | 2 | H (4) O (8) P (2) Pu (1)   | 437.975 |
| <b>Pu+4 _H+1 (4) _PO4-3 (4)</b>                                           |            |   |                            |         |
| -4                                                                        | ---        | 2 | H (4) O (16) P (4) Pu (1)  | 627.918 |
| <b>Pu+4 _H+1 (5) _PO4-3 (5)</b>                                           |            |   |                            |         |
| -6                                                                        | ---        | 2 | H (5) O (20) P (5) Pu (1)  | 723.898 |
| <b>Pu+4 _H+1 (6) _PO4-3 (3)</b>                                           |            |   |                            |         |
| 1                                                                         | ---        | 1 | H (6) O (12) P (3) Pu (1)  | 534.962 |
| <b>Pu+4 _H+1 (6) _PO4-3 (6)</b>                                           |            |   |                            |         |
| -8                                                                        | ---        | 1 | H (6) O (24) P (6) Pu (1)  | 819.877 |
| <b>Pu+4 _H+1 (8) _PO4-3 (4)</b>                                           |            |   |                            |         |
| 0                                                                         | ---        | 1 | H (8) O (16) P (4) Pu (1)  | 631.950 |
| <b>PuO2+1</b><br>Plutonium(V) ion                                         |            |   |                            |         |



|                                                             |            |     |                           |         |
|-------------------------------------------------------------|------------|-----|---------------------------|---------|
| 1                                                           | ---        | 137 | O (2) Pu (1)              | 275.999 |
| <b>PuO<sub>2</sub>+2</b><br>Plutonyl ion; Plutonium(VI) ion |            |     |                           |         |
| 2                                                           | 22853-00-5 | 226 | O (2) Pu (1)              | 275.999 |
| <b>PuO<sub>2</sub>+2_H+1_PO<sub>4</sub>-3</b>               |            |     |                           |         |
| 0                                                           | ---        | 1   | H (1) O (6) P (1) Pu (1)  | 371.978 |
| <b>PuO<sub>2</sub>+2_H+1_PO<sub>4</sub>-3 (s)</b>           |            |     |                           |         |
| 0                                                           | ---        | 2   | H (1) O (6) P (1) Pu (1)  | 371.978 |
| <b>PuO<sub>2</sub>+2_H+1 (2)_PO<sub>4</sub>-3</b>           |            |     |                           |         |
| 1                                                           | ---        | 2   | H (2) O (6) P (1) Pu (1)  | 372.986 |
| <b>PuO<sub>2</sub>+2_H+1 (2)_PO<sub>4</sub>-3 (2)</b>       |            |     |                           |         |
| -2                                                          | ---        | 1   | H (2) O (10) P (2) Pu (1) | 467.958 |
| <b>PuO<sub>2</sub>+2_H+1 (3)_PO<sub>4</sub>-3 (3)</b>       |            |     |                           |         |
| -4                                                          | ---        | 1   | H (3) O (14) P (3) Pu (1) | 563.937 |
| <b>PuO<sub>2</sub>+2_H+1 (4)_PO<sub>4</sub>-3 (2)</b>       |            |     |                           |         |
| 0                                                           | ---        | 1   | H (4) O (10) P (2) Pu (1) | 469.974 |
| <b>PuO<sub>2</sub>+2_H+1 (4)_PO<sub>4</sub>-3 (4)</b>       |            |     |                           |         |
| -6                                                          | ---        | 1   | H (4) O (18) P (4) Pu (1) | 659.917 |
| <b>PuO<sub>2</sub>+2_H+1 (5)_PO<sub>4</sub>-3 (5)</b>       |            |     |                           |         |
| -8                                                          | ---        | 1   | H (5) O (22) P (5) Pu (1) | 755.897 |
| <b>PuO<sub>2</sub>+2_H+1 (6)_PO<sub>4</sub>-3 (3)</b>       |            |     |                           |         |
| -1                                                          | ---        | 1   | H (6) O (14) P (3) Pu (1) | 566.961 |
| <b>PuO<sub>2</sub>+2_H+1 (8)_PO<sub>4</sub>-3 (4)</b>       |            |     |                           |         |
| -2                                                          | ---        | 1   | H (8) O (18) P (4) Pu (1) | 663.949 |
| <b>Pyrid2Azo4DiMeAniline</b>                                |            |     |                           |         |

|                                                                                                               |            |     |                            |         |
|---------------------------------------------------------------------------------------------------------------|------------|-----|----------------------------|---------|
| Pyridine-2-azo-4-dimethylaniline; 4-(2-Pyridylazo)-N,N-dimethylaniline; 2[4-(Dimethylamino)phenylazo]pyridine |            |     |                            |         |
| 0                                                                                                             | 13103-75-8 | 29  | C (13) H (14) N (4)        | 226.281 |
| <b>Pyruvic-1</b><br>Pyruvate ion                                                                              |            |     |                            |         |
| -1                                                                                                            | 57-60-3    | 296 | C (3) H (3) O (3)          | 87.0550 |
| <b>Ra+2</b><br>Radium(II) ion                                                                                 |            |     |                            |         |
| 2                                                                                                             | 22541-36-2 | 131 | Ra (1)                     | 226.025 |
| <b>Ra+2_H+1_PO4-3</b>                                                                                         |            |     |                            |         |
| 0                                                                                                             | ---        | 1   | H (1) O (4) P (1) Ra (1)   | 322.005 |
| <b>Ra+2_H+1 (10)_PO4-3 (5)</b>                                                                                |            |     |                            |         |
| -3                                                                                                            | ---        | 1   | H (10) O (20) P (5) Ra (1) | 710.962 |
| <b>Ra+2_H+1 (12)_PO4-3 (6)</b>                                                                                |            |     |                            |         |
| -4                                                                                                            | ---        | 1   | H (12) O (24) P (6) Ra (1) | 807.950 |
| <b>Ra+2_H+1 (2)_PO4-3</b>                                                                                     |            |     |                            |         |
| 1                                                                                                             | ---        | 1   | H (2) O (4) P (1) Ra (1)   | 323.013 |
| <b>Ra+2_H+1 (2)_PO4-3 (2)</b>                                                                                 |            |     |                            |         |
| -2                                                                                                            | ---        | 1   | H (2) O (8) P (2) Ra (1)   | 417.984 |
| <b>Ra+2_H+1 (3)_PO4-3 (3)</b>                                                                                 |            |     |                            |         |
| -4                                                                                                            | ---        | 1   | H (3) O (12) P (3) Ra (1)  | 513.964 |
| <b>Ra+2_H+1 (4)_PO4-3 (2)</b>                                                                                 |            |     |                            |         |
| 0                                                                                                             | ---        | 1   | H (4) O (8) P (2) Ra (1)   | 420.000 |
| <b>Ra+2_H+1 (4)_PO4-3 (4)</b>                                                                                 |            |     |                            |         |
| -6                                                                                                            | ---        | 1   | H (4) O (16) P (4) Ra (1)  | 609.943 |
| <b>Ra+2_H+1 (5)_PO4-3 (5)</b>                                                                                 |            |     |                            |         |
| -8                                                                                                            | ---        | 1   | H (5) O (20) P (5) Ra (1)  | 705.923 |

|                                                   |            |     |                           |         |
|---------------------------------------------------|------------|-----|---------------------------|---------|
| <b>Ra+2_H+1 (6)_PO4-3 (3)</b>                     |            |     |                           |         |
| -1                                                | ---        | 1   | H (6) O (12) P (3) Ra (1) | 516.987 |
| <b>Ra+2_H+1 (6)_PO4-3 (6)</b>                     |            |     |                           |         |
| -10                                               | ---        | 1   | H (6) O (24) P (6) Ra (1) | 801.902 |
| <b>Ra+2_H+1 (8)_PO4-3 (4)</b>                     |            |     |                           |         |
| -2                                                | ---        | 1   | H (8) O (16) P (4) Ra (1) | 613.975 |
| <b>Rb+1</b><br>Rubidium(I) ion                    |            |     |                           |         |
| 1                                                 | 22537-38-8 | 125 | Rb (1)                    | 85.4680 |
| <b>Rb+1_P4O12-4</b>                               |            |     |                           |         |
| -3                                                | ---        | 1   | O (12) P (4) Rb (1)       | 401.357 |
| <b>Rb+1_P6O18-6</b>                               |            |     |                           |         |
| -5                                                | ---        | 1   | O (18) P (6) Rb (1)       | 559.301 |
| <b>Salicylic-2</b><br>Salicylate ion              |            |     |                           |         |
| -2                                                | 16887-56-2 | 558 | C (7) H (4) O (3)         | 136.107 |
| <b>Sc+3</b><br>Scandium(III) ion                  |            |     |                           |         |
| 3                                                 | 22537-29-7 | 196 | Sc (1)                    | 44.9560 |
| <b>Sc+3_H+1 (2)_PO4-3</b>                         |            |     |                           |         |
| 2                                                 | ---        | 2   | H (2) O (4) P (1) Sc (1)  | 141.944 |
| <b>Sc+3 (2)_H+1_PO4-3</b>                         |            |     |                           |         |
| 4                                                 | ---        | 2   | H (1) O (4) P (1) Sc (2)  | 185.892 |
| <b>SCN-1</b><br>Thiocyanate ion; NCS-1 equivalent |            |     |                           |         |
| -1                                                | 302-04-5   | 785 | C (1) N (1) S (1)         | 58.0777 |
| <b>Ser-1</b>                                      |            |     |                           |         |

|                                                                                                                                                                                                    |            |     |                              |         |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----|------------------------------|---------|
| Serinate ion; L-Serinate ion; 2-amino-3-hydroxypropionate ion; beta-hydroxyalanate ion; 2-amino-3-hydroxypropanoate ion; alpha-amino-beta-hydroxypropionate ion; L-2-amino-3-hydroxypropanoate ion |            |     |                              |         |
| -1                                                                                                                                                                                                 | 17807-54-4 | 708 | C(3)H(6)N(1)O(3)             | 104.086 |
| <b>Sm+3</b><br>Samarium(III) ion                                                                                                                                                                   |            |     |                              |         |
| 3                                                                                                                                                                                                  | 22541-17-9 | 732 | Sm(1)                        | 150.362 |
| <b>Sm+3_EDTA-4</b>                                                                                                                                                                                 |            |     |                              |         |
| -1                                                                                                                                                                                                 | ---        | 42  | C(10)H(12)N(2)O(8)Sm(1)      | 438.576 |
| <b>Sm+3_EDTA-4_P3O10-5</b>                                                                                                                                                                         |            |     |                              |         |
| -6                                                                                                                                                                                                 | ---        | 1   | C(10)H(12)N(2)O(18)P(3)Sm(1) | 691.492 |
| <b>Sm+3_H+1_EDTA-4_P3O10-5</b>                                                                                                                                                                     |            |     |                              |         |
| -5                                                                                                                                                                                                 | ---        | 2   | C(10)H(13)N(2)O(18)P(3)Sm(1) | 692.500 |
| <b>Sm+3_H+1_P3O10-5</b>                                                                                                                                                                            |            |     |                              |         |
| -1                                                                                                                                                                                                 | ---        | 1   | H(1)O(10)P(3)Sm(1)           | 404.286 |
| <b>Sm+3_H+1_PO4-3</b>                                                                                                                                                                              |            |     |                              |         |
| 1                                                                                                                                                                                                  | ---        | 3   | H(1)O(4)P(1)Sm(1)            | 246.342 |
| <b>Sm+3_H+1(2)_P3O10-5(2)</b>                                                                                                                                                                      |            |     |                              |         |
| -5                                                                                                                                                                                                 | ---        | 2   | H(2)O(20)P(6)Sm(1)           | 658.210 |
| <b>Sm+3_H+1(2)_PO4-3</b>                                                                                                                                                                           |            |     |                              |         |
| 2                                                                                                                                                                                                  | ---        | 2   | H(2)O(4)P(1)Sm(1)            | 247.350 |
| <b>Sm+3_H+1(2)_PO4-3(2)</b>                                                                                                                                                                        |            |     |                              |         |
| -1                                                                                                                                                                                                 | ---        | 3   | H(2)O(8)P(2)Sm(1)            | 342.321 |
| <b>Sm+3_P3O10-5(2)</b>                                                                                                                                                                             |            |     |                              |         |
| -7                                                                                                                                                                                                 | ---        | 1   | O(20)P(6)Sm(1)               | 656.194 |
| <b>Sm+3_PO4-3</b>                                                                                                                                                                                  |            |     |                              |         |
| 0                                                                                                                                                                                                  | ---        | 2   | O(4)P(1)Sm(1)                | 245.334 |

|                               |            |     |                     |         |
|-------------------------------|------------|-----|---------------------|---------|
| <b>Sm+3_PO4-3_(s)</b>         |            |     |                     |         |
| 0                             | ---        | 2   | O(4)P(1)Sm(1)       | 245.334 |
| <b>Sm+3_PO4-3_H2O_(s)</b>     |            |     |                     |         |
| 0                             | ---        | 1   | H(2)O(5)P(1)Sm(1)   | 263.349 |
| <b>Sm+3_PO4-3(2)</b>          |            |     |                     |         |
| -3                            | ---        | 2   | O(8)P(2)Sm(1)       | 340.305 |
| <b>Sm+3(2)_P2O7-4</b>         |            |     |                     |         |
| 2                             | ---        | 1   | O(7)P(2)Sm(2)       | 474.668 |
| <b>Sn+2</b><br>Tin(II) ion    |            |     |                     |         |
| 2                             | 22541-90-8 | 337 | Sn(1)               | 118.710 |
| <b>Sn+2_H+1_P2O7-4</b>        |            |     |                     |         |
| -1                            | ---        | 1   | H(1)O(7)P(2)Sn(1)   | 293.662 |
| <b>Sn+2_H+1_P2O7-4(2)</b>     |            |     |                     |         |
| -5                            | ---        | 1   | H(1)O(14)P(4)Sn(1)  | 467.606 |
| <b>Sn+2_H+1_PO4-3</b>         |            |     |                     |         |
| 0                             | ---        | 1   | H(1)O(4)P(1)Sn(1)   | 214.690 |
| <b>Sn+2_H+1(-1)_P2O7-4(2)</b> |            |     |                     |         |
| -7                            | ---        | 1   | H(-1)O(14)P(4)Sn(1) | 465.590 |
| <b>Sn+2_H+1(-2)_P2O7-4</b>    |            |     |                     |         |
| -4                            | ---        | 1   | H(-2)O(7)P(2)Sn(1)  | 290.638 |
| <b>Sn+2_H+1(2)_P2O7-4</b>     |            |     |                     |         |
| 0                             | ---        | 1   | H(2)O(7)P(2)Sn(1)   | 294.670 |
| <b>Sn+2_H+1(2)_P2O7-4(2)</b>  |            |     |                     |         |
| -4                            | ---        | 1   | H(2)O(14)P(4)Sn(1)  | 468.614 |

|                                 |     |   |                           |         |
|---------------------------------|-----|---|---------------------------|---------|
| <b>Sn+2_H+1 (2) _PO4-3</b>      |     |   |                           |         |
| 1                               | --- | 1 | H (2) O (4) P (1) Sn (1)  | 215.698 |
| <b>Sn+2_H+1 (2) _PO4-3 (2)</b>  |     |   |                           |         |
| -2                              | --- | 1 | H (2) O (8) P (2) Sn (1)  | 310.669 |
| <b>Sn+2_H+1 (3) _P2O7-4 (2)</b> |     |   |                           |         |
| -3                              | --- | 1 | H (3) O (14) P (4) Sn (1) | 469.621 |
| <b>Sn+2_H+1 (3) _PO4-3 (3)</b>  |     |   |                           |         |
| -4                              | --- | 1 | H (3) O (12) P (3) Sn (1) | 406.649 |
| <b>Sn+2_H+1 (4) _P2O7-4 (2)</b> |     |   |                           |         |
| -2                              | --- | 1 | H (4) O (14) P (4) Sn (1) | 470.629 |
| <b>Sn+2_H+1 (4) _PO4-3 (2)</b>  |     |   |                           |         |
| 0                               | --- | 1 | H (4) O (8) P (2) Sn (1)  | 312.685 |
| <b>Sn+2_H+1 (4) _PO4-3 (4)</b>  |     |   |                           |         |
| -6                              | --- | 1 | H (4) O (16) P (4) Sn (1) | 502.628 |
| <b>Sn+2_H+1 (6) _PO4-3 (3)</b>  |     |   |                           |         |
| -1                              | --- | 1 | H (6) O (12) P (3) Sn (1) | 409.672 |
| <b>Sn+2_H+1 (8) _PO4-3 (4)</b>  |     |   |                           |         |
| -2                              | --- | 1 | H (8) O (16) P (4) Sn (1) | 506.660 |
| <b>Sn+2_OH-1_P2O7-4</b>         |     |   |                           |         |
| -3                              | --- | 1 | H (1) O (8) P (2) Sn (1)  | 309.661 |
| <b>Sn+2_P2O7-4</b>              |     |   |                           |         |
| -2                              | --- | 1 | O (7) P (2) Sn (1)        | 292.654 |
| <b>Sn+2_P2O7-4 (2)</b>          |     |   |                           |         |
| -6                              | --- | 1 | O (14) P (4) Sn (1)       | 466.598 |

|                                                                             |            |     |                            |         |
|-----------------------------------------------------------------------------|------------|-----|----------------------------|---------|
| <b>Sn+2_P2O7-4 (3)</b>                                                      |            |     |                            |         |
| -10                                                                         | ---        | 1   | O (21) P (6) Sn (1)        | 640.541 |
| <b>Sn+2_PO4-3</b>                                                           |            |     |                            |         |
| -1                                                                          | ---        | 1   | O (4) P (1) Sn (1)         | 213.682 |
| <b>Sn+2 (2)_P2O7-4 (s)</b><br>Tin(II) pyrophosphate; Stannous pyrophosphate |            |     |                            |         |
| 0                                                                           | 15578-26-4 | 1   | O (7) P (2) Sn (2)         | 411.364 |
| <b>Sn+4</b><br>Tin+4 ion                                                    |            |     |                            |         |
| 4                                                                           | ---        | 196 | Sn (1)                     | 118.710 |
| <b>Sn+4_H+1_P2O7-4</b>                                                      |            |     |                            |         |
| 1                                                                           | ---        | 1   | H (1) O (7) P (2) Sn (1)   | 293.662 |
| <b>Sn+4_H+1_P2O7-4 (2)</b>                                                  |            |     |                            |         |
| -3                                                                          | ---        | 1   | H (1) O (14) P (4) Sn (1)  | 467.606 |
| <b>Sn+4_H+1_PO4-3</b>                                                       |            |     |                            |         |
| 2                                                                           | ---        | 1   | H (1) O (4) P (1) Sn (1)   | 214.690 |
| <b>Sn+4_H+1 (-1)_P2O7-4 (2)</b>                                             |            |     |                            |         |
| -5                                                                          | ---        | 1   | H (-1) O (14) P (4) Sn (1) | 465.590 |
| <b>Sn+4_H+1 (10)_PO4-3 (5)</b>                                              |            |     |                            |         |
| -1                                                                          | ---        | 1   | H (10) O (20) P (5) Sn (1) | 603.647 |
| <b>Sn+4_H+1 (12)_PO4-3 (6)</b>                                              |            |     |                            |         |
| -2                                                                          | ---        | 1   | H (12) O (24) P (6) Sn (1) | 700.635 |
| <b>Sn+4_H+1 (2)_PO4-3</b>                                                   |            |     |                            |         |
| 3                                                                           | ---        | 1   | H (2) O (4) P (1) Sn (1)   | 215.698 |
| <b>Sn+4_H+1 (2)_PO4-3 (2)</b>                                               |            |     |                            |         |
| 0                                                                           | ---        | 1   | H (2) O (8) P (2) Sn (1)   | 310.669 |

|                                         |            |     |                                 |         |
|-----------------------------------------|------------|-----|---------------------------------|---------|
| <b>Sn+4 _H+1 (3) _PO4-3 (3)</b>         |            |     |                                 |         |
| -2                                      | ---        | 1   | H (3) O (12) P (3) Sn (1)       | 406.649 |
| <b>Sn+4 _H+1 (4) _PO4-3 (2)</b>         |            |     |                                 |         |
| 2                                       | ---        | 1   | H (4) O (8) P (2) Sn (1)        | 312.685 |
| <b>Sn+4 _H+1 (4) _PO4-3 (4)</b>         |            |     |                                 |         |
| -4                                      | ---        | 1   | H (4) O (16) P (4) Sn (1)       | 502.628 |
| <b>Sn+4 _H+1 (5) _PO4-3 (5)</b>         |            |     |                                 |         |
| -6                                      | ---        | 1   | H (5) O (20) P (5) Sn (1)       | 598.608 |
| <b>Sn+4 _H+1 (6) _PO4-3 (3)</b>         |            |     |                                 |         |
| 1                                       | ---        | 1   | H (6) O (12) P (3) Sn (1)       | 409.672 |
| <b>Sn+4 _H+1 (6) _PO4-3 (6)</b>         |            |     |                                 |         |
| -8                                      | ---        | 1   | H (6) O (24) P (6) Sn (1)       | 694.587 |
| <b>Sn+4 _H+1 (8) _PO4-3 (4)</b>         |            |     |                                 |         |
| 0                                       | ---        | 1   | H (8) O (16) P (4) Sn (1)       | 506.660 |
| <b>Sn+4 _P2O7-4</b>                     |            |     |                                 |         |
| 0                                       | ---        | 1   | O (7) P (2) Sn (1)              | 292.654 |
| <b>Sn+4 _P2O7-4 (2)</b>                 |            |     |                                 |         |
| -4                                      | ---        | 1   | O (14) P (4) Sn (1)             | 466.598 |
| <b>Sn (CH3) 2+2</b><br>Dimethyl tin ion |            |     |                                 |         |
| 2                                       | 16408-14-3 | 113 | C (2) H (6) Sn (1)              | 148.780 |
| <b>Sn (CH3) 2+2 _P3O10-5</b>            |            |     |                                 |         |
| -3                                      | ---        | 2   | C (2) H (6) O (10) P (3) Sn (1) | 401.696 |
| <b>Sn (CH3) 2+2 _P3O10-5 (2)</b>        |            |     |                                 |         |
| -8                                      | ---        | 1   | C (2) H (6) O (20) P (6) Sn (1) | 654.612 |



|                                                                                                                                                                                                                                              |            |      |                                  |         |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|------|----------------------------------|---------|
| <b>Sn (CH<sub>3</sub>)<sub>2</sub>+2 (2) _P3O10-5</b>                                                                                                                                                                                        |            |      |                                  |         |
| -1                                                                                                                                                                                                                                           | ---        | 2    | C (4) H (12) O (10) P (3) Sn (2) | 550.475 |
| <b>Sn (s)</b><br>Tin; Tin, white; Tin, tetragonal; Tin, beta                                                                                                                                                                                 |            |      |                                  |         |
| 0                                                                                                                                                                                                                                            | 7440-31-5  | 87   | Sn (1)                           | 118.710 |
| <b>SO<sub>4</sub>-2</b><br>Sulfate ion; Sulphate ion                                                                                                                                                                                         |            |      |                                  |         |
| -2                                                                                                                                                                                                                                           | 14808-79-8 | 1270 | O (4) S (1)                      | 96.0576 |
| <b>Spermine</b><br>Spermine; N,N'-bis(3-Aminopropyl)tetramethylenediamine; N,N'-bis(3-Aminopropyl)-1,4-butanediamine; Gerontine; Musculamine; Neuridine; 4,9-Diazadodecane-1,12-diamine; 4,9NH-ddd; 3,4,3-tet; 1,5,10,14-Tetraazatetradecane |            |      |                                  |         |
| 0                                                                                                                                                                                                                                            | 71-44-3    | 31   | C (10) H (26) N (4)              | 202.343 |
| <b>Sr+2</b><br>Strontium(II) ion                                                                                                                                                                                                             |            |      |                                  |         |
| 2                                                                                                                                                                                                                                            | 22537-39-9 | 693  | Sr (1)                           | 87.6200 |
| <b>Sr+2 _H+1 _P3O10-5</b>                                                                                                                                                                                                                    |            |      |                                  |         |
| -2                                                                                                                                                                                                                                           | ---        | 3    | H (1) O (10) P (3) Sr (1)        | 341.544 |
| <b>Sr+2 _H+1 _P4O13-6</b>                                                                                                                                                                                                                    |            |      |                                  |         |
| -3                                                                                                                                                                                                                                           | ---        | 2    | H (1) O (13) P (4) Sr (1)        | 420.516 |
| <b>Sr+2 _H+1 _PO4-3</b>                                                                                                                                                                                                                      |            |      |                                  |         |
| 0                                                                                                                                                                                                                                            | ---        | 2    | H (1) O (4) P (1) Sr (1)         | 183.600 |
| <b>Sr+2 _H+1 _PO4-3 (s)</b>                                                                                                                                                                                                                  |            |      |                                  |         |
| 0                                                                                                                                                                                                                                            | ---        | 3    | H (1) O (4) P (1) Sr (1)         | 183.600 |
| <b>Sr+2 _H+1 (10) _PO4-3 (5)</b>                                                                                                                                                                                                             |            |      |                                  |         |
| -3                                                                                                                                                                                                                                           | ---        | 2    | H (10) O (20) P (5) Sr (1)       | 572.557 |
| <b>Sr+2 _H+1 (2) _PO4-3</b>                                                                                                                                                                                                                  |            |      |                                  |         |
| 1                                                                                                                                                                                                                                            | ---        | 2    | H (2) O (4) P (1) Sr (1)         | 184.608 |

|                                |     |   |                           |         |
|--------------------------------|-----|---|---------------------------|---------|
| <b>Sr+2_H+1 (2) _PO4-3 (2)</b> |     |   |                           |         |
| -2                             | --- | 2 | H (2) O (8) P (2) Sr (1)  | 279.579 |
| <b>Sr+2_H+1 (3) _PO4-3 (3)</b> |     |   |                           |         |
| -4                             | --- | 2 | H (3) O (12) P (3) Sr (1) | 375.559 |
| <b>Sr+2_H+1 (4) _PO4-3 (2)</b> |     |   |                           |         |
| 0                              | --- | 2 | H (4) O (8) P (2) Sr (1)  | 281.595 |
| <b>Sr+2_H+1 (4) _PO4-3 (4)</b> |     |   |                           |         |
| -6                             | --- | 2 | H (4) O (16) P (4) Sr (1) | 471.538 |
| <b>Sr+2_H+1 (5) _PO4-3 (5)</b> |     |   |                           |         |
| -8                             | --- | 2 | H (5) O (20) P (5) Sr (1) | 567.518 |
| <b>Sr+2_H+1 (6) _PO4-3 (3)</b> |     |   |                           |         |
| -1                             | --- | 2 | H (6) O (12) P (3) Sr (1) | 378.582 |
| <b>Sr+2_H+1 (6) _PO4-3 (6)</b> |     |   |                           |         |
| -10                            | --- | 2 | H (6) O (24) P (6) Sr (1) | 663.497 |
| <b>Sr+2_H+1 (8) _PO4-3 (4)</b> |     |   |                           |         |
| -2                             | --- | 2 | H (8) O (16) P (4) Sr (1) | 475.570 |
| <b>Sr+2_OH-1_P2O7-4</b>        |     |   |                           |         |
| -3                             | --- | 2 | H (1) O (8) P (2) Sr (1)  | 278.571 |
| <b>Sr+2_OH-1_P3O10-5</b>       |     |   |                           |         |
| -4                             | --- | 2 | H (1) O (11) P (3) Sr (1) | 357.543 |
| <b>Sr+2_OH-1_PO4-3</b>         |     |   |                           |         |
| -2                             | --- | 1 | H (1) O (5) P (1) Sr (1)  | 199.599 |
| <b>Sr+2_P2O7-4</b>             |     |   |                           |         |
| -2                             | --- | 3 | O (7) P (2) Sr (1)        | 261.564 |

|                                                       |            |   |                           |         |
|-------------------------------------------------------|------------|---|---------------------------|---------|
| <b>Sr+2_P3O10-5</b>                                   |            |   |                           |         |
| -3                                                    | ---        | 3 | O (10) P (3) Sr (1)       | 340.536 |
| <b>Sr+2_P3O9-3</b>                                    |            |   |                           |         |
| -1                                                    | ---        | 1 | O (9) P (3) Sr (1)        | 324.537 |
| <b>Sr+2_P4O12-4</b>                                   |            |   |                           |         |
| -2                                                    | ---        | 1 | O (12) P (4) Sr (1)       | 403.509 |
| <b>Sr+2_P4O13-6</b>                                   |            |   |                           |         |
| -4                                                    | ---        | 2 | O (13) P (4) Sr (1)       | 419.508 |
| <b>Sr+2_P6O18-6</b>                                   |            |   |                           |         |
| -4                                                    | ---        | 1 | O (18) P (6) Sr (1)       | 561.453 |
| <b>Sr+2_P8O24-8</b>                                   |            |   |                           |         |
| -6                                                    | ---        | 1 | O (24) P (8) Sr (1)       | 719.398 |
| <b>Sr+2_PO4-3</b>                                     |            |   |                           |         |
| -1                                                    | ---        | 1 | O (4) P (1) Sr (1)        | 182.592 |
| <b>Sr+2_UO2+2 (2)_PO4-3 (2)_ (s)</b><br>Sr-Autunite   |            |   |                           |         |
| 0                                                     | ---        | 2 | O (12) P (2) Sr (1) U (2) | 817.619 |
| <b>Sr+2 (2)_P2O7-4 (s)</b>                            |            |   |                           |         |
| 0                                                     | 13812-81-2 | 2 | O (7) P (2) Sr (2)        | 349.184 |
| <b>Sr+2 (2)_P4O12-4</b>                               |            |   |                           |         |
| 0                                                     | ---        | 1 | O (12) P (4) Sr (2)       | 491.129 |
| <b>Sr+2 (2)_P4O13-6</b>                               |            |   |                           |         |
| -2                                                    | ---        | 2 | O (13) P (4) Sr (2)       | 507.128 |
| <b>Sr+2 (3)_PO4-3 (2)_ (s)</b><br>Strontium phosphate |            |   |                           |         |
| 0                                                     | 7446-28-8  | 1 | O (8) P (2) Sr (3)        | 452.803 |

|                                                               |            |     |                              |         |
|---------------------------------------------------------------|------------|-----|------------------------------|---------|
| <b>Sr(s)</b><br>Strontium; Strontium, cubic; Strontium, alpha |            |     |                              |         |
| 0                                                             | 7440-24-6  | 55  | Sr(1)                        | 87.6200 |
| <b>Succinic-2</b><br>Succinate ion                            |            |     |                              |         |
| -2                                                            | 56-14-4    | 635 | C(4)H(4)O(4)                 | 116.073 |
| <b>Tartaric-2</b><br>Tartrate ion                             |            |     |                              |         |
| -2                                                            | ---        | 393 | C(4)H(4)O(6)                 | 148.072 |
| <b>Tb+3</b><br>Terbium(III) ion                               |            |     |                              |         |
| 3                                                             | 22541-20-4 | 549 | Tb(1)                        | 158.930 |
| <b>Tb+3_EDTA-4</b>                                            |            |     |                              |         |
| -1                                                            | ---        | 21  | C(10)H(12)N(2)O(8)Tb(1)      | 447.144 |
| <b>Tb+3_EDTA-4_P3O10-5</b>                                    |            |     |                              |         |
| -6                                                            | ---        | 2   | C(10)H(12)N(2)O(18)P(3)Tb(1) | 700.060 |
| <b>Tb+3_H+1_EDTA-4_P3O10-5</b>                                |            |     |                              |         |
| -5                                                            | ---        | 2   | C(10)H(13)N(2)O(18)P(3)Tb(1) | 701.068 |
| <b>Tb+3_H+1_PO4-3</b>                                         |            |     |                              |         |
| 1                                                             | ---        | 1   | H(1)O(4)P(1)Tb(1)            | 254.910 |
| <b>Tb+3_H+1(2)_P3O10-5(2)</b>                                 |            |     |                              |         |
| -5                                                            | ---        | 1   | H(2)O(20)P(6)Tb(1)           | 666.778 |
| <b>Tb+3_H+1(2)_PO4-3</b>                                      |            |     |                              |         |
| 2                                                             | ---        | 1   | H(2)O(4)P(1)Tb(1)            | 255.918 |
| <b>Tb+3_H+1(2)_PO4-3(2)</b>                                   |            |     |                              |         |
| -1                                                            | ---        | 1   | H(2)O(8)P(2)Tb(1)            | 350.889 |
| <b>Tb+3_P3O10-5(2)</b>                                        |            |     |                              |         |

|                                    |            |     |                            |         |
|------------------------------------|------------|-----|----------------------------|---------|
| -7                                 | ---        | 1   | O (20) P (6) Tb (1)        | 664.762 |
| <b>Tb+3_PO4-3</b>                  |            |     |                            |         |
| 0                                  | ---        | 1   | O (4) P (1) Tb (1)         | 253.902 |
| <b>Tb+3_PO4-3_(s)</b>              |            |     |                            |         |
| 0                                  | ---        | 1   | O (4) P (1) Tb (1)         | 253.902 |
| <b>Tb+3_PO4-3(2)</b>               |            |     |                            |         |
| -3                                 | ---        | 1   | O (8) P (2) Tb (1)         | 348.873 |
| <b>Tb+3(2)_P2O7-4</b>              |            |     |                            |         |
| 2                                  | ---        | 1   | O (7) P (2) Tb (2)         | 491.804 |
| <b>Tc+3</b><br>Technetium(III) ion |            |     |                            |         |
| 3                                  | 36640-33-2 | 128 | Tc (1)                     | 98.0000 |
| <b>Tc+3_H+1_PO4-3</b>              |            |     |                            |         |
| 1                                  | ---        | 1   | H (1) O (4) P (1) Tc (1)   | 193.980 |
| <b>Tc+3_H+1(10)_PO4-3(5)</b>       |            |     |                            |         |
| -2                                 | ---        | 1   | H (10) O (20) P (5) Tc (1) | 582.937 |
| <b>Tc+3_H+1(12)_PO4-3(6)</b>       |            |     |                            |         |
| -3                                 | ---        | 1   | H (12) O (24) P (6) Tc (1) | 679.925 |
| <b>Tc+3_H+1(2)_PO4-3</b>           |            |     |                            |         |
| 2                                  | ---        | 1   | H (2) O (4) P (1) Tc (1)   | 194.988 |
| <b>Tc+3_H+1(2)_PO4-3(2)</b>        |            |     |                            |         |
| -1                                 | ---        | 1   | H (2) O (8) P (2) Tc (1)   | 289.959 |
| <b>Tc+3_H+1(3)_PO4-3(3)</b>        |            |     |                            |         |
| -3                                 | ---        | 1   | H (3) O (12) P (3) Tc (1)  | 385.939 |
| <b>Tc+3_H+1(4)_PO4-3(2)</b>        |            |     |                            |         |

|                                      |             |     |                            |         |
|--------------------------------------|-------------|-----|----------------------------|---------|
| 1                                    | ---         | 1   | H (4) O (8) P (2) Tc (1)   | 291.975 |
| <b>Tc+3_H+1 (4) _PO4-3 (4)</b>       |             |     |                            |         |
| -5                                   | ---         | 1   | H (4) O (16) P (4) Tc (1)  | 481.918 |
| <b>Tc+3_H+1 (5) _PO4-3 (5)</b>       |             |     |                            |         |
| -7                                   | ---         | 1   | H (5) O (20) P (5) Tc (1)  | 577.898 |
| <b>Tc+3_H+1 (6) _PO4-3 (3)</b>       |             |     |                            |         |
| 0                                    | ---         | 1   | H (6) O (12) P (3) Tc (1)  | 388.962 |
| <b>Tc+3_H+1 (6) _PO4-3 (6)</b>       |             |     |                            |         |
| -9                                   | ---         | 1   | H (6) O (24) P (6) Tc (1)  | 673.877 |
| <b>Tc+3_H+1 (8) _PO4-3 (4)</b>       |             |     |                            |         |
| -1                                   | ---         | 1   | H (8) O (16) P (4) Tc (1)  | 485.950 |
| <b>TcO+2</b><br>Technetium oxide ion |             |     |                            |         |
| 2                                    | 114113-02-9 | 128 | O (1) Tc (1)               | 113.999 |
| <b>TcO+2_H+1 _PO4-3</b>              |             |     |                            |         |
| 0                                    | ---         | 1   | H (1) O (5) P (1) Tc (1)   | 209.979 |
| <b>TcO+2_H+1 (10) _PO4-3 (5)</b>     |             |     |                            |         |
| -3                                   | ---         | 1   | H (10) O (21) P (5) Tc (1) | 598.937 |
| <b>TcO+2_H+1 (2) _PO4-3</b>          |             |     |                            |         |
| 1                                    | ---         | 1   | H (2) O (5) P (1) Tc (1)   | 210.987 |
| <b>TcO+2_H+1 (2) _PO4-3 (2)</b>      |             |     |                            |         |
| -2                                   | ---         | 1   | H (2) O (9) P (2) Tc (1)   | 305.959 |
| <b>TcO+2_H+1 (3) _PO4-3 (3)</b>      |             |     |                            |         |
| -4                                   | ---         | 1   | H (3) O (13) P (3) Tc (1)  | 401.938 |
| <b>TcO+2_H+1 (4) _PO4-3 (2)</b>      |             |     |                            |         |

|                                                                                                                                                                                                               |            |     |                            |         |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----|----------------------------|---------|
| 0                                                                                                                                                                                                             | ---        | 1   | H (4) O (9) P (2) Tc (1)   | 307.974 |
| <b>TcO+2_H+1 (4)_PO4-3 (4)</b>                                                                                                                                                                                |            |     |                            |         |
| -6                                                                                                                                                                                                            | ---        | 1   | H (4) O (17) P (4) Tc (1)  | 497.918 |
| <b>TcO+2_H+1 (5)_PO4-3 (5)</b>                                                                                                                                                                                |            |     |                            |         |
| -8                                                                                                                                                                                                            | ---        | 1   | H (5) O (21) P (5) Tc (1)  | 593.897 |
| <b>TcO+2_H+1 (6)_PO4-3 (3)</b>                                                                                                                                                                                |            |     |                            |         |
| -1                                                                                                                                                                                                            | ---        | 1   | H (6) O (13) P (3) Tc (1)  | 404.962 |
| <b>TcO+2_H+1 (8)_PO4-3 (4)</b>                                                                                                                                                                                |            |     |                            |         |
| -2                                                                                                                                                                                                            | ---        | 1   | H (8) O (17) P (4) Tc (1)  | 501.949 |
| <b>Tetren</b><br>Tetraethylenepentamine; 1,4,7,10,13-Penta-azatridecane; TEPA; 2,2,2,2-pent; 3,6,9-Triazaundecane-1,11-diamine; Tetren; N1-(2-Aminoethyl)-N2-[2-[(2-aminoethyl)amino]ethyl]-1,2-ethanediamine |            |     |                            |         |
| 0                                                                                                                                                                                                             | 112-57-2   | 93  | C (8) H (23) N (5)         | 189.304 |
| <b>Th+4</b><br>Thorium(IV) ion                                                                                                                                                                                |            |     |                            |         |
| 4                                                                                                                                                                                                             | 16065-92-2 | 659 | Th (1)                     | 232.040 |
| <b>Th+4_H+1_PO4-3</b>                                                                                                                                                                                         |            |     |                            |         |
| 2                                                                                                                                                                                                             | ---        | 2   | H (1) O (4) P (1) Th (1)   | 328.020 |
| <b>Th+4_H+1 (10)_PO4-3 (5)</b>                                                                                                                                                                                |            |     |                            |         |
| -1                                                                                                                                                                                                            | ---        | 1   | H (10) O (20) P (5) Th (1) | 716.977 |
| <b>Th+4_H+1 (12)_PO4-3 (6)</b>                                                                                                                                                                                |            |     |                            |         |
| -2                                                                                                                                                                                                            | ---        | 1   | H (12) O (24) P (6) Th (1) | 813.965 |
| <b>Th+4_H+1 (2)_PO4-3</b>                                                                                                                                                                                     |            |     |                            |         |
| 3                                                                                                                                                                                                             | ---        | 3   | H (2) O (4) P (1) Th (1)   | 329.028 |
| <b>Th+4_H+1 (2)_PO4-3 (2)</b>                                                                                                                                                                                 |            |     |                            |         |
| 0                                                                                                                                                                                                             | ---        | 2   | H (2) O (8) P (2) Th (1)   | 423.999 |

|                                                                    |     |   |                     |         |
|--------------------------------------------------------------------|-----|---|---------------------|---------|
| <b>Th+4_H+1(2)_PO4-3(2)_ (s)</b><br>Thorium dihydrogen diphosphate |     |   |                     |         |
| 0                                                                  | --- | 1 | H(2)O(8)P(2)Th(1)   | 423.999 |
| <b>Th+4_H+1(2)_PO4-3(2)_H2O(4)_ (s)</b>                            |     |   |                     |         |
| 0                                                                  | --- | 1 | H(10)O(12)P(2)Th(1) | 496.060 |
| <b>Th+4_H+1(3)_PO4-3</b>                                           |     |   |                     |         |
| 4                                                                  | --- | 2 | H(3)O(4)P(1)Th(1)   | 330.035 |
| <b>Th+4_H+1(3)_PO4-3(3)</b>                                        |     |   |                     |         |
| -2                                                                 | --- | 2 | H(3)O(12)P(3)Th(1)  | 519.979 |
| <b>Th+4_H+1(4)_PO4-3(2)</b>                                        |     |   |                     |         |
| 2                                                                  | --- | 3 | H(4)O(8)P(2)Th(1)   | 426.015 |
| <b>Th+4_H+1(4)_PO4-3(4)</b>                                        |     |   |                     |         |
| -4                                                                 | --- | 1 | H(4)O(16)P(4)Th(1)  | 615.958 |
| <b>Th+4_H+1(5)_PO4-3(2)</b>                                        |     |   |                     |         |
| 3                                                                  | --- | 1 | H(5)O(8)P(2)Th(1)   | 427.023 |
| <b>Th+4_H+1(5)_PO4-3(5)</b>                                        |     |   |                     |         |
| -6                                                                 | --- | 1 | H(5)O(20)P(5)Th(1)  | 711.938 |
| <b>Th+4_H+1(6)_PO4-3(3)</b>                                        |     |   |                     |         |
| 1                                                                  | --- | 1 | H(6)O(12)P(3)Th(1)  | 523.002 |
| <b>Th+4_H+1(6)_PO4-3(6)</b>                                        |     |   |                     |         |
| -8                                                                 | --- | 1 | H(6)O(24)P(6)Th(1)  | 807.917 |
| <b>Th+4_H+1(8)_PO4-3(4)</b>                                        |     |   |                     |         |
| 0                                                                  | --- | 1 | H(8)O(16)P(4)Th(1)  | 619.990 |
| <b>Th+4_P2O7-4</b>                                                 |     |   |                     |         |
| 0                                                                  | --- | 2 | O(7)P(2)Th(1)       | 405.984 |



|                                                                                                                                                                                           |            |     |                          |         |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----|--------------------------|---------|
| <b>Th+4_P2O7-4 (2)</b>                                                                                                                                                                    |            |     |                          |         |
| -4                                                                                                                                                                                        | ---        | 1   | O (14) P (4) Th (1)      | 579.928 |
| <b>Th+4 (3)_PO4-3 (4)_ (s)</b>                                                                                                                                                            |            |     |                          |         |
| 0                                                                                                                                                                                         | ---        | 1   | O (16) P (4) Th (3)      | 1076.01 |
| <b>Th (s)</b><br>Thorium; Thorium, cubic; Thorium, alpha                                                                                                                                  |            |     |                          |         |
| 0                                                                                                                                                                                         | 7440-29-1  | 45  | Th (1)                   | 232.040 |
| <b>Th3P4 (s)</b><br>Trithorium tetrphosphide                                                                                                                                              |            |     |                          |         |
| 0                                                                                                                                                                                         | ---        | 1   | P (4) Th (3)             | 820.016 |
| <b>ThP (s)</b><br>Thorium monophosphide                                                                                                                                                   |            |     |                          |         |
| 0                                                                                                                                                                                         | ---        | 1   | P (1) Th (1)             | 263.014 |
| <b>Thr-1</b><br>Threoninate ion; L-Threoninate ion; 2-amino-3-hydroxybutyrate ion; alpha-amino-beta-hydroxybutyrate ion; 2-amino-3-hydroxybutanoate ion; L-2-amino-3-hydroxybutanoate ion |            |     |                          |         |
| -1                                                                                                                                                                                        | 68199-29-1 | 602 | C (4) H (8) N (1) O (3)  | 118.112 |
| <b>Ti+4_OH-1 (3)_PO4-3</b>                                                                                                                                                                |            |     |                          |         |
| -2                                                                                                                                                                                        | ---        | 1   | H (3) O (7) P (1) Ti (1) | 193.861 |
| <b>Ti+4_OH-1 (4)_PO4-3</b>                                                                                                                                                                |            |     |                          |         |
| -3                                                                                                                                                                                        | ---        | 1   | H (4) O (8) P (1) Ti (1) | 210.868 |
| <b>TiO2 (Rutile, s)</b><br>Rutile; Titanium dioxide, tetragonal rutile; Titanium dioxide, rutile                                                                                          |            |     |                          |         |
| 0                                                                                                                                                                                         | 1317-80-2  | 5   | O (2) Ti (1)             | 79.8658 |
| <b>Tl+1</b><br>Thalium(I) ion; Thallium(I) ion; Thallous ion                                                                                                                              |            |     |                          |         |
| 1                                                                                                                                                                                         | 22537-56-0 | 254 | Tl (1)                   | 204.383 |
| <b>Tl+1_P2O7-4</b>                                                                                                                                                                        |            |     |                          |         |
| -3                                                                                                                                                                                        | ---        | 1   | O (7) P (2) Tl (1)       | 378.327 |

|                                 |            |     |                                         |         |
|---------------------------------|------------|-----|-----------------------------------------|---------|
| <b>Tl+1_P2O7-4 (2)</b>          |            |     |                                         |         |
| -7                              | ---        | 1   | O (14) P (4) Tl (1)                     | 552.271 |
| <b>Tl+1_P3O10-5</b>             |            |     |                                         |         |
| -4                              | ---        | 1   | O (10) P (3) Tl (1)                     | 457.299 |
| <b>Tl+1_P3O10-5 (2)</b>         |            |     |                                         |         |
| -9                              | ---        | 1   | O (20) P (6) Tl (1)                     | 710.215 |
| <b>Tl+1_PO4-3</b>               |            |     |                                         |         |
| -2                              | ---        | 1   | O (4) P (1) Tl (1)                      | 299.355 |
| <b>Tm+3</b><br>Thulium(III) ion |            |     |                                         |         |
| 3                               | 22541-23-7 | 441 | Tm (1)                                  | 168.930 |
| <b>Tm+3_EDTA-4</b>              |            |     |                                         |         |
| -1                              | ---        | 13  | C (10) H (12) N (2) O (8) Tm (1)        | 457.144 |
| <b>Tm+3_EDTA-4_P3O10-5</b>      |            |     |                                         |         |
| -6                              | ---        | 2   | C (10) H (12) N (2) O (18) P (3) Tm (1) | 710.060 |
| <b>Tm+3_H+1_EDTA-4_P3O10-5</b>  |            |     |                                         |         |
| -5                              | ---        | 2   | C (10) H (13) N (2) O (18) P (3) Tm (1) | 711.068 |
| <b>Tm+3_H+1_PO4-3</b>           |            |     |                                         |         |
| 1                               | ---        | 1   | H (1) O (4) P (1) Tm (1)                | 264.910 |
| <b>Tm+3_H+1 (2)_P3O10-5 (2)</b> |            |     |                                         |         |
| -5                              | ---        | 1   | H (2) O (20) P (6) Tm (1)               | 676.778 |
| <b>Tm+3_H+1 (2)_PO4-3</b>       |            |     |                                         |         |
| 2                               | ---        | 1   | H (2) O (4) P (1) Tm (1)                | 265.918 |
| <b>Tm+3_H+1 (2)_PO4-3 (2)</b>   |            |     |                                         |         |
| -1                              | ---        | 1   | H (2) O (8) P (2) Tm (1)                | 360.889 |

|                                                                                                                                                                                                                                                                                                                    |            |     |                           |         |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----|---------------------------|---------|
| <b>Tm+3_P3O10-5 (2)</b>                                                                                                                                                                                                                                                                                            |            |     |                           |         |
| -7                                                                                                                                                                                                                                                                                                                 | ---        | 1   | O (20) P (6) Tm (1)       | 674.762 |
| <b>Tm+3_PO4-3</b>                                                                                                                                                                                                                                                                                                  |            |     |                           |         |
| 0                                                                                                                                                                                                                                                                                                                  | ---        | 1   | O (4) P (1) Tm (1)        | 263.902 |
| <b>Tm+3_PO4-3_(s)</b>                                                                                                                                                                                                                                                                                              |            |     |                           |         |
| 0                                                                                                                                                                                                                                                                                                                  | ---        | 1   | O (4) P (1) Tm (1)        | 263.902 |
| <b>Tm+3_PO4-3 (2)</b>                                                                                                                                                                                                                                                                                              |            |     |                           |         |
| -3                                                                                                                                                                                                                                                                                                                 | ---        | 1   | O (8) P (2) Tm (1)        | 358.873 |
| <b>Trien</b><br>Triethylenetetramine; 1,4,7,10-Tetraazadecane; N,N'-bis(2-aminoethyl)-1,2-ethanediamine; Trien; 2,2,2-tet; 3,6-Diazaoctane-1,8-diamine; Ethylenebis(2'-aminoethylamine); TETA (Triethylenetetramine); 3,6NH-od; 1,2-Diamino-N,N'-di(2-aminoethyl)ethane; N1,N2-Bis(2-aminoethyl)-1,2-ethanediamine |            |     |                           |         |
| 0                                                                                                                                                                                                                                                                                                                  | 112-24-3   | 155 | C (6) H (18) N (4)        | 146.236 |
| <b>Trp-1</b><br>Tryptophanate ion; L-Tryptophanate ion; L-alpha-aminoindole-3-propionate ion; L-alpha-amino-3-indolepropionate ion; 2-amino-3-indolylpropanoate ion; L-beta-3-indolylalanate ion; L-2-amino-3-(3-indolyl)propanoate ion                                                                            |            |     |                           |         |
| -1                                                                                                                                                                                                                                                                                                                 | 26302-80-7 | 506 | C (11) H (11) N (2) O (2) | 203.221 |
| <b>Tyr-2</b><br>Tyrosinate ion; L-Tyrosinate ion; beta-(p-hydroxyphenyl)alanate ion; alpha-amino-p-hydroxyhydrocinnamate ion; L-2-amino-3-(4-hydroxyphenyl)propanoate ion                                                                                                                                          |            |     |                           |         |
| -2                                                                                                                                                                                                                                                                                                                 | 26302-79-4 | 590 | C (9) H (9) N (1) O (3)   | 179.175 |
| <b>U+3</b><br>Uranium(III) ion                                                                                                                                                                                                                                                                                     |            |     |                           |         |
| 3                                                                                                                                                                                                                                                                                                                  | 22578-81-0 | 35  | U (1)                     | 238.029 |
| <b>U+3_H+1 (2)_PO4-3</b>                                                                                                                                                                                                                                                                                           |            |     |                           |         |
| 2                                                                                                                                                                                                                                                                                                                  | ---        | 1   | H (2) O (4) P (1) U (1)   | 335.017 |
| <b>U+3_H+1 (4)_PO4-3 (2)</b>                                                                                                                                                                                                                                                                                       |            |     |                           |         |
| 1                                                                                                                                                                                                                                                                                                                  | ---        | 1   | H (4) O (8) P (2) U (1)   | 432.004 |

|                                                  |            |     |                        |         |
|--------------------------------------------------|------------|-----|------------------------|---------|
| <b>U+3_H+1(6)_PO4-3(3)</b>                       |            |     |                        |         |
| 0                                                | ---        | 1   | H(6)O(12)P(3)U(1)      | 528.992 |
| <b>U+4</b><br>Uranium(IV) ion                    |            |     |                        |         |
| 4                                                | 16089-60-4 | 405 | U(1)                   | 238.029 |
| <b>U+4_Ca+2_PO4-3(2)_H2O(2)_(s)</b><br>Ningyoite |            |     |                        |         |
| 0                                                | ---        | 1   | Ca(1)H(4)O(10)P(2)U(1) | 504.081 |
| <b>U+4_H+1_PO4-3</b>                             |            |     |                        |         |
| 2                                                | ---        | 3   | H(1)O(4)P(1)U(1)       | 334.009 |
| <b>U+4_H+1(10)_PO4-3(5)</b>                      |            |     |                        |         |
| -1                                               | ---        | 2   | H(10)O(20)P(5)U(1)     | 722.966 |
| <b>U+4_H+1(12)_PO4-3(6)</b>                      |            |     |                        |         |
| -2                                               | ---        | 2   | H(12)O(24)P(6)U(1)     | 819.954 |
| <b>U+4_H+1(2)_PO4-3</b>                          |            |     |                        |         |
| 3                                                | ---        | 2   | H(2)O(4)P(1)U(1)       | 335.017 |
| <b>U+4_H+1(2)_PO4-3(2)</b>                       |            |     |                        |         |
| 0                                                | ---        | 3   | H(2)O(8)P(2)U(1)       | 429.988 |
| <b>U+4_H+1(2)_PO4-3(2)_H2O(4)_(s)</b>            |            |     |                        |         |
| 0                                                | ---        | 2   | H(10)O(12)P(2)U(1)     | 502.049 |
| <b>U+4_H+1(3)_PO4-3(3)</b>                       |            |     |                        |         |
| -2                                               | ---        | 3   | H(3)O(12)P(3)U(1)      | 525.968 |
| <b>U+4_H+1(4)_PO4-3(2)</b>                       |            |     |                        |         |
| 2                                                | ---        | 2   | H(4)O(8)P(2)U(1)       | 432.004 |
| <b>U+4_H+1(4)_PO4-3(4)</b>                       |            |     |                        |         |
| -4                                               | ---        | 3   | H(4)O(16)P(4)U(1)      | 621.947 |

|                                                                |            |      |                   |         |
|----------------------------------------------------------------|------------|------|-------------------|---------|
| <b>U+4_H+1(5)_PO4-3(5)</b>                                     |            |      |                   |         |
| -6                                                             | ---        | 2    | H(5)O(20)P(5)U(1) | 717.927 |
| <b>U+4_H+1(6)_PO4-3(3)</b>                                     |            |      |                   |         |
| 1                                                              | ---        | 2    | H(6)O(12)P(3)U(1) | 528.992 |
| <b>U+4_H+1(6)_PO4-3(6)</b>                                     |            |      |                   |         |
| -8                                                             | ---        | 2    | H(6)O(24)P(6)U(1) | 813.906 |
| <b>U+4_H+1(8)_PO4-3(4)</b>                                     |            |      |                   |         |
| 0                                                              | ---        | 2    | H(8)O(16)P(4)U(1) | 625.979 |
| <b>U+4_P2O7-4</b>                                              |            |      |                   |         |
| 0                                                              | ---        | 1    | O(7)P(2)U(1)      | 411.973 |
| <b>U+4(3)_PO4-3(4)_(s)</b>                                     |            |      |                   |         |
| 0                                                              | ---        | 1    | O(16)P(4)U(3)     | 1093.97 |
| <b>U(s)</b><br>Uranium; Uranium, orthorhombic; Uranium, alpha  |            |      |                   |         |
| 0                                                              | 7440-61-1  | 263  | U(1)              | 238.029 |
| <b>UO2+1</b><br>Uranium dioxide ion                            |            |      |                   |         |
| 1                                                              | 21294-41-7 | 136  | O(2)U(1)          | 270.028 |
| <b>UO2+2</b><br>Uranyl ion; U(VI) cation; Uranate              |            |      |                   |         |
| 2                                                              | 16637-16-4 | 1268 | O(2)U(1)          | 270.028 |
| <b>UO2+2_H+1_PO4-3</b>                                         |            |      |                   |         |
| 0                                                              | ---        | 5    | H(1)O(6)P(1)U(1)  | 366.007 |
| <b>UO2+2_H+1_PO4-3_(s)</b>                                     |            |      |                   |         |
| 0                                                              | ---        | 2    | H(1)O(6)P(1)U(1)  | 366.007 |
| <b>UO2+2_H+1_PO4-3_H2O(3)_(s)</b><br>Uranyl Hydrogen Phosphate |            |      |                   |         |

|                                                                            |     |   |                           |         |
|----------------------------------------------------------------------------|-----|---|---------------------------|---------|
| 0                                                                          | --- | 1 | H (7) O (9) P (1) U (1)   | 420.053 |
| UO <sub>2</sub> +2_H+1_PO <sub>4</sub> -3_H <sub>2</sub> O (4)_(s)         |     |   |                           |         |
| 0                                                                          | --- | 1 | H (9) O (10) P (1) U (1)  | 438.069 |
| UO <sub>2</sub> +2_H+1 (2)_PO <sub>4</sub> -3                              |     |   |                           |         |
| 1                                                                          | --- | 6 | H (2) O (6) P (1) U (1)   | 367.015 |
| UO <sub>2</sub> +2_H+1 (2)_PO <sub>4</sub> -3 (2)                          |     |   |                           |         |
| -2                                                                         | --- | 4 | H (2) O (10) P (2) U (1)  | 461.987 |
| UO <sub>2</sub> +2_H+1 (3)_PO <sub>4</sub> -3                              |     |   |                           |         |
| 2                                                                          | --- | 5 | H (3) O (6) P (1) U (1)   | 368.023 |
| UO <sub>2</sub> +2_H+1 (3)_PO <sub>4</sub> -3 (3)                          |     |   |                           |         |
| -4                                                                         | --- | 1 | H (3) O (14) P (3) U (1)  | 557.966 |
| UO <sub>2</sub> +2_H+1 (4)_PO <sub>4</sub> -3 (2)                          |     |   |                           |         |
| 0                                                                          | --- | 5 | H (4) O (10) P (2) U (1)  | 464.003 |
| UO <sub>2</sub> +2_H+1 (4)_PO <sub>4</sub> -3 (2)_H <sub>2</sub> O (3)_(s) |     |   |                           |         |
| 0                                                                          | --- | 1 | H (10) O (13) P (2) U (1) | 518.049 |
| UO <sub>2</sub> +2_H+1 (4)_PO <sub>4</sub> -3 (4)                          |     |   |                           |         |
| -6                                                                         | --- | 1 | H (4) O (18) P (4) U (1)  | 653.946 |
| UO <sub>2</sub> +2_H+1 (5)_PO <sub>4</sub> -3 (2)                          |     |   |                           |         |
| 1                                                                          | --- | 4 | H (5) O (10) P (2) U (1)  | 465.011 |
| UO <sub>2</sub> +2_H+1 (5)_PO <sub>4</sub> -3 (5)                          |     |   |                           |         |
| -8                                                                         | --- | 1 | H (5) O (22) P (5) U (1)  | 749.926 |
| UO <sub>2</sub> +2_H+1 (6)_PO <sub>4</sub> -3 (3)                          |     |   |                           |         |
| -1                                                                         | --- | 4 | H (6) O (14) P (3) U (1)  | 560.990 |
| UO <sub>2</sub> +2_H+1 (7)_PO <sub>4</sub> -3 (3)                          |     |   |                           |         |
| 0                                                                          | --- | 2 | H (7) O (14) P (3) U (1)  | 561.998 |

|                                                                                                                                              |            |    |                                |         |
|----------------------------------------------------------------------------------------------------------------------------------------------|------------|----|--------------------------------|---------|
| <b>UO<sub>2</sub>+2 _H+1 (8) _PO<sub>4</sub>-3 (4)</b>                                                                                       |            |    |                                |         |
| -2                                                                                                                                           | ---        | 2  | H (8) O (18) P (4) U (1)       | 657.978 |
| <b>UO<sub>2</sub>+2 _PO<sub>4</sub>-3</b>                                                                                                    |            |    |                                |         |
| -1                                                                                                                                           | ---        | 2  | O (6) P (1) U (1)              | 364.999 |
| <b>UO<sub>2</sub>+2 (2) _H+1 (2) _NH<sub>3</sub> (2) _PO<sub>4</sub>-3 (2) _ (s)</b><br>Uramphite                                            |            |    |                                |         |
| 0                                                                                                                                            | ---        | 1  | H (8) N (2) O (12) P (2) U (2) | 766.076 |
| <b>UO<sub>2</sub>+2 (2) _H+1 (2) _PO<sub>4</sub>-3 (2) _ (s)</b><br>H-autunite                                                               |            |    |                                |         |
| 0                                                                                                                                            | ---        | 3  | H (2) O (12) P (2) U (2)       | 732.015 |
| <b>UO<sub>2</sub>+2 (2) _K+1 (2) _PO<sub>4</sub>-3 (2) _ (s)</b><br>K-autunite                                                               |            |    |                                |         |
| 0                                                                                                                                            | ---        | 2  | K (2) O (12) P (2) U (2)       | 808.195 |
| <b>UO<sub>2</sub>+2 (2) _Na+1 (2) _PO<sub>4</sub>-3 (2) _ (s)</b><br>Na-autunite                                                             |            |    |                                |         |
| 0                                                                                                                                            | ---        | 2  | Na (2) O (12) P (2) U (2)      | 775.979 |
| <b>UO<sub>2</sub>+2 (3) _PO<sub>4</sub>-3 (2) _ (s)</b><br>Uranyl phosphate (dehydrated); Uranyl orthophosphate (dehydrated)                 |            |    |                                |         |
| 0                                                                                                                                            | ---        | 3  | O (14) P (2) U (3)             | 1000.03 |
| <b>UO<sub>2</sub>+2 (3) _PO<sub>4</sub>-3 (2) _H<sub>2</sub>O (4) _ (s)</b><br>Uranyl phosphate (hydrated); Uranyl orthophosphate (hydrated) |            |    |                                |         |
| 0                                                                                                                                            | ---        | 2  | H (8) O (18) P (2) U (3)       | 1072.09 |
| <b>UO<sub>2</sub>+2 (5) _H+1 (2) _P<sub>2</sub>O<sub>7</sub>-4</b>                                                                           |            |    |                                |         |
| 8                                                                                                                                            | ---        | 2  | H (2) O (17) P (2) U (5)       | 1526.10 |
| <b>V+2</b><br>Vanadium(II) ion; Vanadous ion                                                                                                 |            |    |                                |         |
| 2                                                                                                                                            | 15121-26-3 | 40 | V (1)                          | 50.9420 |
| <b>V+2 _H+1 _HPO<sub>3</sub>-2</b>                                                                                                           |            |    |                                |         |
| 1                                                                                                                                            | ---        | 2  | H (2) O (3) P (1) V (1)        | 131.930 |

|                                                                               |            |     |                   |         |
|-------------------------------------------------------------------------------|------------|-----|-------------------|---------|
| <b>V+2_H+1(2)_HPO3-2(2)</b>                                                   |            |     |                   |         |
| 0                                                                             | ---        | 1   | H(4)O(6)P(2)V(1)  | 212.918 |
| <b>V+2_HPO3-2</b>                                                             |            |     |                   |         |
| 0                                                                             | ---        | 1   | H(1)O(3)P(1)V(1)  | 130.922 |
| <b>V(s)</b><br>Vanadium; Vanadium, cubic; Vanadium, bcc                       |            |     |                   |         |
| 0                                                                             | 7440-62-2  | 170 | V(1)              | 50.9420 |
| <b>Val-1</b><br>L-Valinate ion; Valinate ion; L-2-Amino-3-methylbutanoate ion |            |     |                   |         |
| -1                                                                            | ---        | 582 | C(5)H(10)N(1)O(2) | 116.140 |
| <b>VO+2</b><br>Vanadyl ion; Oxovanadium ion; Vanadium(IV) ion (vanadyl)       |            |     |                   |         |
| 2                                                                             | 20644-97-7 | 659 | O(1)V(1)          | 66.9414 |
| <b>VO+2_H+1_HPO3-2</b>                                                        |            |     |                   |         |
| 1                                                                             | ---        | 2   | H(2)O(4)P(1)V(1)  | 147.930 |
| <b>VO+2_H+1_P3O10-5</b>                                                       |            |     |                   |         |
| -2                                                                            | ---        | 2   | H(1)O(11)P(3)V(1) | 320.865 |
| <b>VO+2_H+1_PO4-3</b>                                                         |            |     |                   |         |
| 0                                                                             | ---        | 1   | H(1)O(5)P(1)V(1)  | 162.921 |
| <b>VO+2_H+1(2)_HPO3-2(2)</b>                                                  |            |     |                   |         |
| 0                                                                             | ---        | 1   | H(4)O(7)P(2)V(1)  | 228.918 |
| <b>VO+2_H+1(2)_P2O7-4</b>                                                     |            |     |                   |         |
| 0                                                                             | ---        | 1   | H(2)O(8)P(2)V(1)  | 242.901 |
| <b>VO+2_H+1(2)_P3O10-5</b>                                                    |            |     |                   |         |
| -1                                                                            | ---        | 1   | H(2)O(11)P(3)V(1) | 321.873 |
| <b>VO+2_H+1(2)_PO4-3</b>                                                      |            |     |                   |         |



|                                                                                 |            |     |                        |         |
|---------------------------------------------------------------------------------|------------|-----|------------------------|---------|
| 1                                                                               | ---        | 2   | H(2)O(5)P(1)V(1)       | 163.929 |
| <b>VO+2_H+1(3)_PO4-3(2)</b>                                                     |            |     |                        |         |
| -1                                                                              | ---        | 1   | H(3)O(9)P(2)V(1)       | 259.908 |
| <b>VO+2_H+1(4)_PO4-3(2)</b>                                                     |            |     |                        |         |
| 0                                                                               | ---        | 2   | H(4)O(9)P(2)V(1)       | 260.916 |
| <b>VO+2_HPO3-2</b>                                                              |            |     |                        |         |
| 0                                                                               | ---        | 1   | H(1)O(4)P(1)V(1)       | 146.922 |
| <b>VO+2_P3O10-5</b>                                                             |            |     |                        |         |
| -3                                                                              | ---        | 1   | O(11)P(3)V(1)          | 319.857 |
| <b>VO+2(3)_PO4-3(2)</b>                                                         |            |     |                        |         |
| 0                                                                               | ---        | 1   | O(11)P(2)V(3)          | 390.767 |
| <b>VO+2(3)_PO4-3(2)_(s)</b>                                                     |            |     |                        |         |
| 0                                                                               | ---        | 2   | O(11)P(2)V(3)          | 390.767 |
| <b>VO2+1</b><br>Dioxovanadium ion; Pervanadyl ion; Vanadium(V) ion (pervanadyl) |            |     |                        |         |
| 1                                                                               | 18252-79-4 | 143 | O(2)V(1)               | 82.9408 |
| <b>VO2+1_H2PO2-1</b>                                                            |            |     |                        |         |
| 0                                                                               | ---        | 1   | H(2)O(4)P(1)V(1)       | 147.930 |
| <b>VO4-3</b><br>Orthovanadate ion; Vanadate(V) ion; Vanadium Oxide ion          |            |     |                        |         |
| -3                                                                              | 14333-18-7 | 161 | O(4)V(1)               | 114.940 |
| <b>Y+3</b><br>Yttrium(III) ion                                                  |            |     |                        |         |
| 3                                                                               | 22537-40-2 | 495 | Y(1)                   | 88.9060 |
| <b>Y+3_EDTA-4</b>                                                               |            |     |                        |         |
| -1                                                                              | ---        | 18  | C(10)H(12)N(2)O(8)Y(1) | 377.120 |

|                                   |            |     |                             |         |
|-----------------------------------|------------|-----|-----------------------------|---------|
| <b>Y+3_EDTA-4_P3O10-5</b>         |            |     |                             |         |
| -6                                | ---        | 2   | C(10)H(12)N(2)O(18)P(3)Y(1) | 630.036 |
| <b>Y+3_H+1_EDTA-4_P3O10-5</b>     |            |     |                             |         |
| -5                                | ---        | 2   | C(10)H(13)N(2)O(18)P(3)Y(1) | 631.044 |
| <b>Y+3_H+1_P3O10-5</b>            |            |     |                             |         |
| -1                                | ---        | 1   | H(1)O(10)P(3)Y(1)           | 342.830 |
| <b>Y+3_H+1(2)_P3O10-5(2)</b>      |            |     |                             |         |
| -5                                | ---        | 3   | H(2)O(20)P(6)Y(1)           | 596.754 |
| <b>Y+3_H+1(2)_PO4-3</b>           |            |     |                             |         |
| 2                                 | ---        | 2   | H(2)O(4)P(1)Y(1)            | 185.894 |
| <b>Y+3_P3O10-5(2)</b>             |            |     |                             |         |
| -7                                | ---        | 1   | O(20)P(6)Y(1)               | 594.738 |
| <b>Yb+3</b><br>Ytterbium(III) ion |            |     |                             |         |
| 3                                 | 18923-27-8 | 594 | Yb(1)                       | 173.050 |
| <b>Yb+3_H+1_P3O10-5</b>           |            |     |                             |         |
| -1                                | ---        | 1   | H(1)O(10)P(3)Yb(1)          | 426.974 |
| <b>Yb+3_H+1_PO4-3</b>             |            |     |                             |         |
| 1                                 | ---        | 1   | H(1)O(4)P(1)Yb(1)           | 269.030 |
| <b>Yb+3_H+1(2)_PO4-3</b>          |            |     |                             |         |
| 2                                 | ---        | 1   | H(2)O(4)P(1)Yb(1)           | 270.038 |
| <b>Yb+3_H+1(2)_PO4-3(2)</b>       |            |     |                             |         |
| -1                                | ---        | 1   | H(2)O(8)P(2)Yb(1)           | 365.009 |
| <b>Yb+3_P2O7-4</b>                |            |     |                             |         |
| -1                                | ---        | 1   | O(7)P(2)Yb(1)               | 346.994 |

|                                |            |      |                                            |         |
|--------------------------------|------------|------|--------------------------------------------|---------|
| <b>Yb+3_P2O7-4 (2)</b>         |            |      |                                            |         |
| -5                             | ---        | 1    | O (14) P (4) Yb (1)                        | 520.938 |
| <b>Yb+3_PO4-3</b>              |            |      |                                            |         |
| 0                              | ---        | 2    | O (4) P (1) Yb (1)                         | 268.022 |
| <b>Yb+3_PO4-3_(s)</b>          |            |      |                                            |         |
| 0                              | ---        | 1    | O (4) P (1) Yb (1)                         | 268.022 |
| <b>Yb+3_PO4-3 (2)</b>          |            |      |                                            |         |
| -3                             | ---        | 1    | O (8) P (2) Yb (1)                         | 362.993 |
| <b>Yb+3 (2)_P2O7-4</b>         |            |      |                                            |         |
| 2                              | ---        | 1    | O (7) P (2) Yb (2)                         | 520.044 |
| <b>Yb+3 (4)_P2O7-4 (3)_(s)</b> |            |      |                                            |         |
| 0                              | ---        | 1    | O (21) P (6) Yb (4)                        | 1214.03 |
| <b>Zn+2</b><br>Zinc(II) ion    |            |      |                                            |         |
| 2                              | 23713-49-7 | 4977 | Zn (1)                                     | 65.3820 |
| <b>Zn+2_Bipy</b>               |            |      |                                            |         |
| 2                              | ---        | 54   | C (10) H (8) N (2) Zn (1)                  | 221.569 |
| <b>Zn+2_Cys-2_PO4-3</b>        |            |      |                                            |         |
| -3                             | ---        | 1    | C (3) H (5) N (1) O (6) P (1) S (1) Zn (1) | 279.492 |
| <b>Zn+2_H+1_Ala-1_PO4-3</b>    |            |      |                                            |         |
| -1                             | ---        | 1    | C (3) H (7) N (1) O (6) P (1) Zn (1)       | 249.448 |
| <b>Zn+2_H+1_Arg-1_PO4-3</b>    |            |      |                                            |         |
| -1                             | ---        | 1    | C (6) H (14) N (4) O (6) P (1) Zn (1)      | 334.556 |
| <b>Zn+2_H+1_Asn-1_PO4-3</b>    |            |      |                                            |         |
| -1                             | ---        | 1    | C (4) H (8) N (2) O (7) P (1) Zn (1)       | 292.473 |

|                                 |     |   |                               |         |
|---------------------------------|-----|---|-------------------------------|---------|
| <b>Zn+2_H+1_Asp-2_PO4-3</b>     |     |   |                               |         |
| -2                              | --- | 1 | C(4)H(6)N(1)O(8)P(1)Zn(1)     | 292.450 |
| <b>Zn+2_H+1_bAla-1_PO4-3</b>    |     |   |                               |         |
| -1                              | --- | 1 | C(3)H(7)N(1)O(6)P(1)Zn(1)     | 249.448 |
| <b>Zn+2_H+1_Bipy_PO4-3</b>      |     |   |                               |         |
| 0                               | --- | 1 | C(10)H(9)N(2)O(4)P(1)Zn(1)    | 317.549 |
| <b>Zn+2_H+1_Citric-3_PO4-3</b>  |     |   |                               |         |
| -3                              | --- | 2 | C(6)H(6)O(11)P(1)Zn(1)        | 350.463 |
| <b>Zn+2_H+1_Citrul-1_PO4-3</b>  |     |   |                               |         |
| -1                              | --- | 1 | C(6)H(13)N(3)O(7)P(1)Zn(1)    | 335.541 |
| <b>Zn+2_H+1_Cys-2_PO4-3</b>     |     |   |                               |         |
| -2                              | --- | 2 | C(3)H(6)N(1)O(6)P(1)S(1)Zn(1) | 280.500 |
| <b>Zn+2_H+1_Gln-1_PO4-3</b>     |     |   |                               |         |
| -1                              | --- | 1 | C(5)H(10)N(2)O(7)P(1)Zn(1)    | 306.500 |
| <b>Zn+2_H+1_Glu-2_PO4-3</b>     |     |   |                               |         |
| -2                              | --- | 1 | C(5)H(8)N(1)O(8)P(1)Zn(1)     | 306.476 |
| <b>Zn+2_H+1_Gly-1_PO4-3</b>     |     |   |                               |         |
| -1                              | --- | 1 | C(2)H(5)N(1)O(6)P(1)Zn(1)     | 235.421 |
| <b>Zn+2_H+1_His-1_PO4-3</b>     |     |   |                               |         |
| -1                              | --- | 1 | C(6)H(9)N(3)O(6)P(1)Zn(1)     | 315.510 |
| <b>Zn+2_H+1_Histamine_PO4-3</b> |     |   |                               |         |
| 0                               | --- | 1 | C(5)H(10)N(3)O(4)P(1)Zn(1)    | 272.508 |
| <b>Zn+2_H+1_HPO3-2</b>          |     |   |                               |         |
| 1                               | --- | 2 | H(2)O(3)P(1)Zn(1)             | 146.370 |

|                                |     |   |                                |         |
|--------------------------------|-----|---|--------------------------------|---------|
| <b>Zn+2_H+1_Hyp-1_PO4-3</b>    |     |   |                                |         |
| -1                             | --- | 1 | C(5)H(9)N(1)O(7)P(1)Zn(1)      | 291.485 |
| <b>Zn+2_H+1_Ile-1_PO4-3</b>    |     |   |                                |         |
| -1                             | --- | 1 | C(6)H(13)N(1)O(6)P(1)Zn(1)     | 291.528 |
| <b>Zn+2_H+1_Lactic-1_PO4-3</b> |     |   |                                |         |
| -1                             | --- | 1 | C(3)H(6)O(7)P(1)Zn(1)          | 250.432 |
| <b>Zn+2_H+1_Leu-1_PO4-3</b>    |     |   |                                |         |
| -1                             | --- | 1 | C(6)H(13)N(1)O(6)P(1)Zn(1)     | 291.528 |
| <b>Zn+2_H+1_Malic-2_PO4-3</b>  |     |   |                                |         |
| -2                             | --- | 1 | C(4)H(5)O(9)P(1)Zn(1)          | 293.434 |
| <b>Zn+2_H+1_Met-1_PO4-3</b>    |     |   |                                |         |
| -1                             | --- | 1 | C(5)H(11)N(1)O(6)P(1)S(1)Zn(1) | 309.562 |
| <b>Zn+2_H+1_NTA-3_PO4-3</b>    |     |   |                                |         |
| -3                             | --- | 2 | C(6)H(7)N(1)O(10)P(1)Zn(1)     | 349.478 |
| <b>Zn+2_H+1_Oxalic-2_PO4-3</b> |     |   |                                |         |
| -2                             | --- | 1 | C(2)H(1)O(8)P(1)Zn(1)          | 249.381 |
| <b>Zn+2_H+1_P3O10-5</b>        |     |   |                                |         |
| -2                             | --- | 3 | H(1)O(10)P(3)Zn(1)             | 319.306 |
| <b>Zn+2_H+1_P3O10-5(2)</b>     |     |   |                                |         |
| -7                             | --- | 1 | H(1)O(20)P(6)Zn(1)             | 572.222 |
| <b>Zn+2_H+1_Phe-1_PO4-3</b>    |     |   |                                |         |
| -1                             | --- | 1 | C(9)H(11)N(1)O(6)P(1)Zn(1)     | 325.546 |
| <b>Zn+2_H+1_PO4-3</b>          |     |   |                                |         |
| 0                              | --- | 3 | H(1)O(4)P(1)Zn(1)              | 161.362 |

|                                   |     |   |                                |         |
|-----------------------------------|-----|---|--------------------------------|---------|
| <b>Zn+2_H+1_PO4-3(2)</b>          |     |   |                                |         |
| -3                                | --- | 3 | H(1)O(8)P(2)Zn(1)              | 256.333 |
| <b>Zn+2_H+1_Pro-1_PO4-3</b>       |     |   |                                |         |
| -1                                | --- | 1 | C(5)H(9)N(1)O(6)P(1)Zn(1)      | 275.486 |
| <b>Zn+2_H+1_Pyruvic-1_PO4-3</b>   |     |   |                                |         |
| -1                                | --- | 1 | C(3)H(4)O(7)P(1)Zn(1)          | 248.417 |
| <b>Zn+2_H+1_Salicylic-2_PO4-3</b> |     |   |                                |         |
| -2                                | --- | 1 | C(7)H(5)O(7)P(1)Zn(1)          | 297.469 |
| <b>Zn+2_H+1_Ser-1_PO4-3</b>       |     |   |                                |         |
| -1                                | --- | 1 | C(3)H(7)N(1)O(7)P(1)Zn(1)      | 265.447 |
| <b>Zn+2_H+1_Succinic-2_PO4-3</b>  |     |   |                                |         |
| -2                                | --- | 1 | C(4)H(5)O(8)P(1)Zn(1)          | 277.435 |
| <b>Zn+2_H+1_Thr-1_PO4-3</b>       |     |   |                                |         |
| -1                                | --- | 1 | C(4)H(9)N(1)O(7)P(1)Zn(1)      | 279.474 |
| <b>Zn+2_H+1_Trp-1_PO4-3</b>       |     |   |                                |         |
| -1                                | --- | 1 | C(11)H(12)N(2)O(6)P(1)Zn(1)    | 364.582 |
| <b>Zn+2_H+1_Val-1_PO4-3</b>       |     |   |                                |         |
| -1                                | --- | 1 | C(5)H(11)N(1)O(6)P(1)Zn(1)     | 277.502 |
| <b>Zn+2_H+1(2)_Cis-2_PO4-3</b>    |     |   |                                |         |
| -1                                | --- | 1 | C(6)H(12)N(2)O(8)P(1)S(2)Zn(1) | 400.646 |
| <b>Zn+2_H+1(2)_Cys-2_PO4-3</b>    |     |   |                                |         |
| -1                                | --- | 1 | C(3)H(7)N(1)O(6)P(1)S(1)Zn(1)  | 281.508 |
| <b>Zn+2_H+1(2)_His-1_PO4-3</b>    |     |   |                                |         |
| 0                                 | --- | 1 | C(6)H(10)N(3)O(6)P(1)Zn(1)     | 316.518 |

|                                    |     |   |                                       |         |
|------------------------------------|-----|---|---------------------------------------|---------|
| <b>Zn+2_H+1 (2)_HPO3-2 (2)</b>     |     |   |                                       |         |
| 0                                  | --- | 1 | H (4) O (6) P (2) Zn (1)              | 227.358 |
| <b>Zn+2_H+1 (2)_Lys-1_PO4-3</b>    |     |   |                                       |         |
| 0                                  | --- | 1 | C (6) H (15) N (2) O (6) P (1) Zn (1) | 307.551 |
| <b>Zn+2_H+1 (2)_Orn-1_PO4-3</b>    |     |   |                                       |         |
| 0                                  | --- | 1 | C (5) H (13) N (2) O (6) P (1) Zn (1) | 293.524 |
| <b>Zn+2_H+1 (2)_Oxalic-2_PO4-3</b> |     |   |                                       |         |
| -1                                 | --- | 1 | C (2) H (2) O (8) P (1) Zn (1)        | 250.389 |
| <b>Zn+2_H+1 (2)_PO4-3</b>          |     |   |                                       |         |
| 1                                  | --- | 3 | H (2) O (4) P (1) Zn (1)              | 162.370 |
| <b>Zn+2_H+1 (2)_PO4-3 (2)</b>      |     |   |                                       |         |
| -2                                 | --- | 3 | H (2) O (8) P (2) Zn (1)              | 257.341 |
| <b>Zn+2_H+1 (2)_Tyr-2_PO4-3</b>    |     |   |                                       |         |
| -1                                 | --- | 1 | C (9) H (11) N (1) O (7) P (1) Zn (1) | 341.545 |
| <b>Zn+2_H+1 (3)_PO4-3 (2)</b>      |     |   |                                       |         |
| -1                                 | --- | 4 | H (3) O (8) P (2) Zn (1)              | 258.349 |
| <b>Zn+2_H+1 (4)_PO4-3 (2)</b>      |     |   |                                       |         |
| 0                                  | --- | 2 | H (4) O (8) P (2) Zn (1)              | 259.357 |
| <b>Zn+2_H2PO2-1</b>                |     |   |                                       |         |
| 1                                  | --- | 1 | H (2) O (2) P (1) Zn (1)              | 130.371 |
| <b>Zn+2_H2PO2-1 (2)</b>            |     |   |                                       |         |
| 0                                  | --- | 1 | H (4) O (4) P (2) Zn (1)              | 195.359 |
| <b>Zn+2_HPO3-2</b>                 |     |   |                                       |         |
| 0                                  | --- | 1 | H (1) O (3) P (1) Zn (1)              | 145.362 |

|                          |     |    |                                |         |
|--------------------------|-----|----|--------------------------------|---------|
| <b>Zn+2_NTA-3</b>        |     |    |                                |         |
| -1                       | --- | 55 | C (6) H (6) N (1) O (6) Zn (1) | 253.499 |
| <b>Zn+2_OH-1_P2O7-4</b>  |     |    |                                |         |
| -3                       | --- | 2  | H (1) O (8) P (2) Zn (1)       | 256.333 |
| <b>Zn+2_OH-1_P3O10-5</b> |     |    |                                |         |
| -4                       | --- | 2  | H (1) O (11) P (3) Zn (1)      | 335.305 |
| <b>Zn+2_OH-1_PO4-3</b>   |     |    |                                |         |
| -2                       | --- | 1  | H (1) O (5) P (1) Zn (1)       | 177.361 |
| <b>Zn+2_P2O7-4</b>       |     |    |                                |         |
| -2                       | --- | 2  | O (7) P (2) Zn (1)             | 239.326 |
| <b>Zn+2_P2O7-4 (2)</b>   |     |    |                                |         |
| -6                       | --- | 1  | O (14) P (4) Zn (1)            | 413.270 |
| <b>Zn+2_P3O10-5</b>      |     |    |                                |         |
| -3                       | --- | 6  | O (10) P (3) Zn (1)            | 318.298 |
| <b>Zn+2_P3O10-5 (2)</b>  |     |    |                                |         |
| -8                       | --- | 1  | O (20) P (6) Zn (1)            | 571.214 |
| <b>Zn+2_P3O9-3</b>       |     |    |                                |         |
| -1                       | --- | 1  | O (9) P (3) Zn (1)             | 302.299 |
| <b>Zn+2_P4O12-4</b>      |     |    |                                |         |
| -2                       | --- | 1  | O (12) P (4) Zn (1)            | 381.271 |
| <b>Zn+2_P4O12-4 (2)</b>  |     |    |                                |         |
| -6                       | --- | 1  | O (24) P (8) Zn (1)            | 697.160 |
| <b>Zn+2_P4O13-6</b>      |     |    |                                |         |
| -4                       | --- | 1  | O (13) P (4) Zn (1)            | 397.270 |



|                                                                                                              |           |   |                                         |         |
|--------------------------------------------------------------------------------------------------------------|-----------|---|-----------------------------------------|---------|
| <b>Zn+2_P6O18-6</b>                                                                                          |           |   |                                         |         |
| -4                                                                                                           | ---       | 1 | O (18) P (6) Zn (1)                     | 539.215 |
| <b>Zn+2_P8O24-8</b>                                                                                          |           |   |                                         |         |
| -6                                                                                                           | ---       | 1 | O (24) P (8) Zn (1)                     | 697.160 |
| <b>Zn+2_PO4-3</b>                                                                                            |           |   |                                         |         |
| -1                                                                                                           | ---       | 1 | O (4) P (1) Zn (1)                      | 160.354 |
| <b>Zn+2_Pyrid2Azo4DiMeAniline_P3O10-5</b>                                                                    |           |   |                                         |         |
| -3                                                                                                           | ---       | 1 | C (13) H (14) N (4) O (10) P (3) Zn (1) | 544.579 |
| <b>Zn+2 (2)_H+1 (2)_PO4-3 (2)</b>                                                                            |           |   |                                         |         |
| 0                                                                                                            | ---       | 2 | H (2) O (8) P (2) Zn (2)                | 322.723 |
| <b>Zn+2 (2)_OH-1_PO4-3_(s)</b><br>Tarbutite                                                                  |           |   |                                         |         |
| 0                                                                                                            | ---       | 1 | H (1) O (5) P (1) Zn (2)                | 242.743 |
| <b>Zn+2 (2)_OH-1_PO4-3_H2O (1.5)_(s)</b><br>Spencerite                                                       |           |   |                                         |         |
| 0                                                                                                            | ---       | 1 | H (4) O (6.5) P (1) Zn (2)              | 269.766 |
| <b>Zn+2 (2)_P2O7-4_H2O (3)_(s)</b>                                                                           |           |   |                                         |         |
| 0                                                                                                            | ---       | 1 | H (6) O (10) P (2) Zn (2)               | 358.754 |
| <b>Zn+2 (2)_P2O7-4_H2O (5)_(s)</b>                                                                           |           |   |                                         |         |
| 0                                                                                                            | ---       | 1 | H (10) O (12) P (2) Zn (2)              | 394.784 |
| <b>Zn+2 (3)_PO4-3 (2)</b>                                                                                    |           |   |                                         |         |
| 0                                                                                                            | ---       | 1 | O (8) P (2) Zn (3)                      | 386.089 |
| <b>Zn+2 (3)_PO4-3 (2)_(s)</b><br>Zinc phosphate                                                              |           |   |                                         |         |
| 0                                                                                                            | 7779-90-0 | 2 | O (8) P (2) Zn (3)                      | 386.089 |
| <b>Zn+2 (3)_PO4-3 (2)_H2O (4)_(s)</b><br>Hopeite, alpha; Hopeite; Alpha-Hopeite; Zinc phosphate tetrahydrate |           |   |                                         |         |

|                                                                                                                                                    |            |     |                           |         |
|----------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----|---------------------------|---------|
| 0                                                                                                                                                  | 15491-18-6 | 1   | H (8) O (12) P (2) Zn (3) | 458.150 |
| <b>Zn (s)</b><br>Zinc; Zinc, hexagonal                                                                                                             |            |     |                           |         |
| 0                                                                                                                                                  | 7440-66-6  | 131 | Zn (1)                    | 65.3820 |
| <b>ZnO (s)</b><br>Zinc oxide; Flowers of zinc; Philosopher's wool; Zinc white; C.I. Pigment White 4; C.I. 77947; Zinc monoxide, hexagonal; Zincite |            |     |                           |         |
| 0                                                                                                                                                  | 1314-13-2  | 8   | O (1) Zn (1)              | 81.3814 |
| <b>Zr+4</b><br>Zirconium(IV) ion                                                                                                                   |            |     |                           |         |
| 4                                                                                                                                                  | 15543-40-5 | 291 | Zr (1)                    | 91.2420 |
| <b>Zr+4_H+1 (3)_PO4-3 (3)</b>                                                                                                                      |            |     |                           |         |
| -2                                                                                                                                                 | ---        | 1   | H (3) O (12) P (3) Zr (1) | 379.181 |
| <b>Zr (HPO4) 2 . 2H2O (s)</b><br>Zirconium hydrogen phosphate dihydrate                                                                            |            |     |                           |         |
| 0                                                                                                                                                  | ---        | 1   | H (6) O (10) P (2) Zr (1) | 319.232 |
| <b>Zr (HPO4) 2 . H2O (s)</b><br>Zirconium hydrogen phosphate monohydrate                                                                           |            |     |                           |         |
| 0                                                                                                                                                  | ---        | 2   | H (4) O (9) P (2) Zr (1)  | 301.216 |
| <b>Zr (HPO4) 2 (a. , s)</b><br>Zirconium dihydrogen phosphate, alpha                                                                               |            |     |                           |         |
| 0                                                                                                                                                  | ---        | 1   | H (2) O (8) P (2) Zr (1)  | 283.201 |
| <b>Zr (HPO4) 2 (b. , s)</b><br>Zirconium dihydrogen phosphate, beta                                                                                |            |     |                           |         |
| 0                                                                                                                                                  | ---        | 1   | H (2) O (8) P (2) Zr (1)  | 283.201 |
| <b>Zr (s)</b><br>Zirconium; Zirconium, gray-white hcp; Zirconium, alpha                                                                            |            |     |                           |         |
| 0                                                                                                                                                  | 7440-67-7  | 111 | Zr (1)                    | 91.2420 |
| <b>ZrO+2_H+1 (4)_PO4-3 (2)_ (s)</b>                                                                                                                |            |     |                           |         |
| 0                                                                                                                                                  | ---        | 1   | H (4) O (9) P (2) Zr (1)  | 301.216 |

|                                                                  |     |   |                    |         |
|------------------------------------------------------------------|-----|---|--------------------|---------|
| <b>ZrP<sub>2</sub>O<sub>7</sub> (s)</b><br>Zirconium diphosphate |     |   |                    |         |
| 0                                                                | --- | 1 | O (7) P (2) Zr (1) | 265.186 |